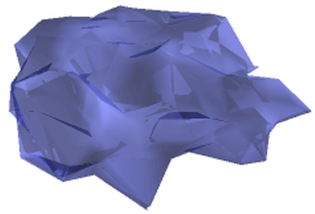
BAYESIAN NONPARAMETRIC GLOTTAL INVERSE FILTERING



by

MARNIX VAN SOOM

A DISSERTATION SUBMITTED IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF

DOCTOR OF PHILOSOPHY

in

THE FACULTY OF SCIENCE

(Computer Science)

VRIJE UNIVERSITEIT BRUSSEL

(Brussels)

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Bayesian Nonparametric Glottal Inverse Filtering

submitted by **Marnix Van Soom** in partial fulfilment of the requirements for the degree of Doctor of Philosophy in Computer Science.

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Cover image: a three-dimensional latent manifold $\mathcal{D}_Q$ of glottal flow waveforms, learned by q-GPLVM from the EGIFA corpus (Chapter 6). Each point on this surface represents the variational posterior $q(\boldsymbol{x}_n) = \text{Normal}(\boldsymbol{\mu}_n^x, \boldsymbol{S}_n^x)$ of a registered derivative glottal flow waveform in $\mathbb{R}^Q$, $Q = 3$.

Abstract

A reliable and accurate white-box model of speech is still something of a holy grail in phonetics, forensics, and clinical medicine. The difficulty is that fitting such models to microphone recordings is a blind identification problem, known to practitioners as glottal inverse filtering, which requires inferring both source signal and filter simultaneously from a single waveform. Because the problem is severely underdetermined, we must lean hard on prior information to constrain the set of mathematically possible solutions: they must adhere to what we already know to be plausible or true. I will show how nonparametric Gaussian processes let us bake such prior knowledge directly into the model by (1) deriving a nonstationary kernel that can model the glottal cycle, (2) specializing that kernel into a surrogate model learned from simulated source signals via interdomain features, and (3) proposing a new and general framework for specifying arbitrary prior information about the filter; and how this improves reliability and accuracy at inference time.

Lay summary

In the past decade, we have made enormous strides in recognizing and synthesizing speech using large "black-box" models trained on massive datasets. Today, these models have learnt to speak and understand major world languages such as English, Chinese, and Spanish with impressive accuracy. Remarkably, they remain largely agnostic to the physical mechanisms underlying speech, relying instead on large-scale pattern recognition and vast quantities of data.

In contrast, "white-box" models grounded in accumulated scientific knowledge do focus explicitly on these physical mechanisms. While they are not remotely competitive with black-box models in speech recognition or synthesis, they do offer the valuable ability to extract biologically meaningful information about the speaker from the signal itself.

This is important. A growing body of evidence shows that quantitative features of voice and speech can act as digital biomarkers of neurological and clinical conditions. In Parkinson's disease, for example, changes in speech and language have been reported to precede diagnosis by as much as a decade, and acoustic measures such as voice quality, articulation rate, and prosody have shown promise for early detection and monitoring. Similarly, alterations in specific acoustic parameters have been linked to mood disorders, suggesting that speech analysis could complement clinical assessment of depression.

Nevertheless, white-box models often lack the flexibility of black-box models needed to deal with real-world speech. I therefore propose a "grey-box" approach that attempts to combine the best of both worlds. By integrating physical insight with data-driven modeling, I derive a class of grey-box models that allows for far more accurate analysis and decomposition of speech into separate contributions from the vocal folds and the vocal tract. These models are also capable of indicating how certain they are of their measurements, while running close to real-time speed on modern hardware.

This was achieved by combining new theoretical insights in glottal flow and vocal tract models with customized machine learning models. I hope that this work can contribute to a more reliable and holistic use of speech analysis in clinical practice, forensic work, and the scientific study of vocalization as a whole.

Preface

Write your preface here. This must include a description of your role in the work as presented, and the role of any collaborators if applicable. Additionally, include a statement on the use of Generative AI tools in producing this work.

Table of contents

Acknowledgments

Write your acknowledgments here. This section is optional, feel free to remove it if you do not wish to include it. Nonetheless, this is the place to thank professional colleagues and people who have given you the most support during the course of your work.

FIGURE HERE

Figure 1 | **Overview of source-filter separation.** The speech waveform (left) is modeled as a glottal source signal (top right) passing through a series of vocal tract filters (bottom left). The task is to find such plausible source-filter pairs (colored samples on the right) that together produce a given speech waveform.

1 Overview

In this thesis we present a new approach to the problem of *glottal inverse filtering* (GIF) called *Bayesian nonparametric glottal inverse filtering* (BNGIF). At the heart of this new approach lies a Bayesian nonparametric *generative model of voiced speech* that reflects general constraints on the voice production process. Though conceived for GIF, the generative model can be conditioned on both raw speech data and high level speech features, which makes it useful for other related tasks such as augmenting or denoising speech data.

Speech can be thought of as the output of a linear system in which an upstream source signal (for example, the rapid opening and closing of the vocal folds) is modulated by a downstream filter (for example, the oral cavity and lips). This linear approximation to speech production, called source-filter theory (Fant, 1960), still underlies most of what is known today as acoustic phonetics (Maurer, 2016), and both source and filter are comparably well understood.

The GIF problem refers to source-filter separation of speech; that is, simultaneously inferring both the glottal source signal and (some of) the vocal tract filter characteristics from the speech signal alone (Figure 1). In the BNGIF approach we systematically take advantage of known constraints on the voice production process to home in on scientifically plausible solutions to the GIF problem. That is the main theme of this thesis.

GIF is an important and difficult problem in acoustic phonetics (Auvinen et al., 2014; Kadiri et al., 2021) and various approaches have been developed over the last 60 years. Paraphrasing Alku (2011), applications mainly come in three broad categories: (i) fundamental research about voice production (for example, quantifying voice quality), (ii) medical applications (such as analysis of pathological voices), and (iii) speech technology applications (for example, voice modification). To this we would add (iv) recent forensic and biometric applications such as evidence-based speaker identification (Bonastre et al., 2015).

Our approach to the GIF problem is mostly relevant to the first category (i) above because we try to model the process of scientific inference – make assumptions, model the data, fit the data, evaluate the model, reiterate – as a whole rather than 'just' modeling the data alone (MacKay, 2005, Ch. 28). It has been shown repeatedly that the unique way to do this is the probabilistic or *Bayesian* framework (Knuth & Skilling, 2012). Within this mathematical framework our generative model of voiced speech ultimately expresses nothing more than our assumptions about voice production in a logically coherent way. Because we insist on modeling scientific inference itself the accuracy of those assumptions will translate directly to the accuracy of our measurements, providing a much-needed means to calibrate uncertainty for use in (i) fundamental research about voice production (Kent & Vorperian, 2018). This is the main benefit of and motivation for using the Bayesian framework for BNGIF, despite its considerable computational expense compared to standard GIF algorithms which are based on traditional signal processing methods.

Another important benefit is the standardized form of Bayesian generative models: posterior inference is mechanical once the prior and likelihood are specified, which allows for a wide variety of generic algorithms to fit the data in true plug-and-play fashion (Murphy, 2022). This standardized form also ensures that generative models are easily modifiable and extendable for more specific applications such as (ii-iv) mentioned above.[1]

Another distinguishing feature of BNGIF is the *nonparametric* nature of our generative model of speech. More precisely, the model contains both nonparametric and parametric components. Figure Figure 1 shows these two neatly divided components on the right hand side: the glottal source signal (top) is modeled nonparametrically with a Gaussian process (Rasmussen & Williams, 2006), while the vocal tract filter (bottom) is expanded into a familiar rational transfer function that is parameterized by an overall gain factor and a vector of poles and zeros.

Nonparametric in this context does not refer to 'distribution-free' methods used in descriptive statistics or nonparametric spectral analysis (Stoica & Moses, 2005); rather it means that the glottal source signal is modeled directly through an infinite-dimensional distribution over functions whose qualitative behavior is specified by a just few global properties like roughness and wiggliness. This is convenient because this is exactly the kind of broad information we possess about the glottal source signal, unlike the vocal tract filter which in source-filter theory is naturally specified in terms of resonances of the vocal tract (Stevens, 2000).

1.1 Problem statement

The central problem of this thesis is boxed in Figure 2.

[1]For example it is expected that our model of human voice production can be useful for mammal vocal communication research (Taylor & Reby, 2010) by simply tweaking some of its prior assumptions. An example of extending the model to downstream tasks is forensic phonetics, since "formant features […] are widely accepted features used in forensic acoustic-phonetic speaker verification" (Becker et al., 2008, p. 1) and "the Bayesian decision framework […] is now the standard paradigm for forensic speaker comparison" (Bonastre et al., 2015, p. 1). This Bayesian decision framework depends on likelihood ratios (Harrison, 2013) that can be calculated with the same computational tools used in this thesis.

Glottal inverse filtering (GIF)

Assume that the source-filter model of speech production (Fant, 1960) holds such that the speech signal at time $t \in \mathbb{R}$ is described by convolving three real-valued functions:

$$s(t) = u(t) * h(t) * r(t). \tag{1}$$

From left to right $s(t)$ is the speech pressure waveform, $u(t)$ is the glottal flow waveform, $h(t)$ is the impulse response of the vocal tract and $r(t)$ is the impulse response of the radiation characteristic, assumed known. Source-filter theory asserts that $s(t)$ can be thought of as the result of the source signal $u(t)$ passing through the filters $h(t)$ and $r(t)$.

The data consists of N noisy samples $\mathbf{d} = d_{1:N}$ of the speech pressure waveform $s(t)$ at times $\mathbf{t} = t_{1:N}$. We assume additive errors $\mathbf{e} = e_{1:N}$ such that

$$\mathbf{d} = \mathbf{s} + \mathbf{e} \implies \quad d_n = s(t_n) + e_n \quad (n = 1 \cdots N). \tag{2}$$

The error components e_n are modeled as white noise with expected power σ_n^2 which means that $e_n \sim \text{Normal}(0, \sigma_n^2)$ for $n = 1 \cdots N$.

The data $(\mathbf{t}, \mathbf{d})$ is possibly augmented with noisy statistics of $s(t)$ such as a list of estimated glottal closing instants (GCIs).

The GIF problem is then to *find plausible estimates of the source* $u(t)$ *and the filter* $h(t)$ that are consistent with the given data.

Figure 2 | **The GIF problem** which we attack in this thesis.

Source-filter theory as expressed by equation (1) is the most important assumption in this thesis because it provides a definite handle on the GIF problem. Even just stating the problem would be difficult without it. In particular, note the four functions (or waveforms) in (1);: each of these play an important role in BNGIF. Before reviewing them in Section 1.2, we first consider how the *plausible estimates* mentioned in the statement above translate within the Bayesian framework adopted in this thesis.

Like most inverse problems, the main difficulty with GIF lies in the fact that it is vastly underdetermined: even in the noiseless case ($\mathbf{d} = \mathbf{s}$), many solutions (that is, pairs of functions $u(t)$ and $h(t)$) are compatible with given data $\mathbf{d}$ (Landau, 1983). It thus appears that next to the data $\mathbf{d}$ itself we need additional criteria to choose between possible solutions of (1);, and this is where the idea of regularization comes in.

Appendix A discusses how in the Bayesian framework regularization is imposed by assigning priors to the unknown functions $u(t)$ and $h(t)$ in such a way that these priors express known constraints on the voice production process. The main idea is that a probability in the Bayesian framework describes a state of knowledge, not necessarily related to an obm(s)erved frequency in the real world, so we can conveniently use prior probabilities to express constraints on the voice production process.

Even with that regularization in place, we will see that the problem is still underdetermined in the sense that we cannot in general expect to find unique solutions: the best we can hope for are *plausible estimates in the form of posterior samples of* $u(t)$ *and* $h(t)$. These are plausible in the sense that they are consistent with both the speech data $\mathbf{d}$ and whatever prior information about voice production we have expressed through the priors for $u(t)$ and $h(t)$.

1.2 Source-filter theory

FIGURE HERE

Figure 3 | Schematical representation of the source-filter model (1) with u, h, r, s and their Fourier transforms.

Source-filter theory[2] models the speech production process as a standard *linear time-invariant* (LTI) system (Antsaklis & Michel, 2006) which is essentially a linear approximation to the underlying nonlinear dynamics of voice production that is roughly valid up to frequencies of 4 or 5 kHz (Doval et al., 2006). The LTI system (1) used in this thesis is illustrated in Figure 3. We proceed with a brief sketch of the important concepts involved.

The *input* to the system is the source $u(t)$. The system has two *transfer functions* which determine how the amplitudes and phase of the frequencies present in the input $\tilde{u}(f)$ are modified or filtered by the system to produce the *output*. That frequency decomposition

$$\tilde{u}(f) = \mathcal{F}[u(t)](f) = \int_{-\infty}^{\infty} u(t) \exp\{-i2\pi f t\} \, \mathrm{d}t \tag{3}$$

is obtained by the Fourier transform $\mathcal{F}$ of $u(t)$. Note that the system's transfer function is assumed to be independent from its input: source-filter theory assumes source and filter to be decoupled.

The vocal tract transfer function $\tilde{h}(f)$ describes the resonances and antiresonances of the vocal tract (comprising the throat, mouth and nasal cavity), while the radiation characteristic $\tilde{r}(f)$ describes the lip radiation effect. These transfer functions can be described equivalently in the time domain by the *impulse responses*

$$\begin{aligned} h(t) = \mathcal{F}^{-1}\left[\tilde{h}(f)\right](t) &= \int_{-\infty}^{\infty} \tilde{h}(f) \exp\{i2\pi f t\} \, \mathrm{d}x \\ r(t) = \mathcal{F}^{-1}[\tilde{r}(f)](t) \quad &= \int_{-\infty}^{\infty} \tilde{r}(f) \exp\{i2\pi f t\} \, \mathrm{d}x \end{aligned} \tag{4}$$

[2]Currently the dominant paradigm in acoustic phonetics (Maurer, 2016), the source-filter theory of voice production is commonly attributed to (Fant, 1960) but appears to date back to the early 19th century (Chen, 2016). Its strength lies in its extremely useful and compact description of the acoustics of phonation, summarized by (Fant, 1960, p. 15) as "the speech wave is the response of the vocal tract filter systems to one or more sound sources." A modern perspective on source-filter theory is given by Svec et al. (2021).

using the inverse Fourier transform $\mathcal{F}^{-1}$. Unlike $u(t)$ and $h(t)$, $r(t)$ is assumed fixed and known: the lip radiation effect is modeled in the GIF literature as a standard 6 dB/octave high pass filter (Stevens, 2000, p. 128) which has transfer function $\tilde{r}(f) = i2\pi x$. This corresponds to differentiation in the time domain as (4) yields $r(t) = \delta'(t)$.[3]

The *output* of the system is the speech pressure signal $s(t)$

$$\begin{aligned} & s(t) = u(t) * h(t) * r(t) && \text{(time domain)} \\ \implies \quad & \tilde{s}(f) = \tilde{u}(f) \quad \tilde{h}(f) \quad \tilde{r}(f) && \text{(frequency domain)} \end{aligned} \tag{5}$$

assumed to be measured in the free field (that is, sufficiently far from the speaker's head) (Stevens, 2000). Filtering in LTI systems happens through multiplication in the frequency domain (5), which is equivalent to convolution in the time domain (1).

The meaning of these equations is best understood in the frequency domain. During speech production, the source drives the air in the vocal tract such that it vibrates with some initial distribution over frequencies $\tilde{u}(f)$. Excited, the vocal tract filter $\tilde{h}(f)$ reacts: frequencies in $\tilde{u}(f)$ close to its resonance frequencies are emphasized,[4] while frequencies in $\tilde{u}(f)$ close to its antiresonance frequencies are (often brutally) suppressed. The result of this spectral imprinting is then radiated from the lips into the free field. This radiation effect is contained within $\tilde{r}(f)$: high frequencies are emphasized, essentially because the lips act like small (passive) tweeters (Schroeder, 1999). We are able to *hear* the spectral imprint of the speaker's vocal tract on the initial energy distribution (Klatt, 1986), because it is what we use to tell the difference between different speech sounds, and ultimately to understand the speaker's message.

Since $r(t)$ is assumed known, we can further simplify (1) by substituting $r(t) = \delta'(t)$:

$$s(t) = u(t) * h(t) * \delta'(t) = [u(t) * \delta'(t)] * h(t) = u(t) * [h(t) * \delta'(t)]. \tag{6}$$

We can differentiate either $u(t)$ or $h(t)$ in (1) because convolution is associative; we choose to differentiate $u(t)$ as is customary in the literature (Doval et al., 2006). This finally yields

$$\boxed{s(t) = u'(t) * h(t)} \tag{7}$$

where $u'(t)$ is the *glottal flow derivative* (DGF) which now acts as the source driving the single filter in (7). The original source $u(t)$ can be recovered through integration:

[3]Convolution of a test function $f(f)$ with the derivative of the Dirac delta function is equivalent to differentiation: $f(t) * \delta'(t) = f'(t)$. This can be justified by distribution theory (Lighthill, 1958).

[4]The vocal tract filter is *passive*, meaning that $|\tilde{h}(f)| < 1$ for all frequencies x (Flanagan, 1965). The resonances of the vocal tract do not actively amplify frequencies; frequencies close to the resonance frequencies are just less dampened relative to neighbouring frequency bands.

$$u(t) = \int_{-\infty}^{t} u'(\tau)\,\mathrm{d}\tau. \tag{8}$$

Equation (7) is the basis for the nonparametric generative model of $s(t)$ which we use to separate speech data [noisy samples of $s(t)$] into source $u'(t)$ [equivalently $u(t)$ via (8)] and filter $h(t)$ [equivalently $\tilde{h}(f)$ via (4)].

With the general picture laid out, we now discuss the $u(t)$, $h(t)$ and $s(t)$ waveforms in further detail, and present a high-level overview of the proposed priors for each of them. Section 1.1 argued that effective regularization of the GIF problem requires realistic priors on $u(t)$ and $h(t)$ and we can show some of their realistic properties already by just sampling them. Note that these priors are nothing but generative models of $u(t)$, $h(t)$ and $s(t)$ – they are 'priors' because we can generate samples from these models according to a probability density that represents our prior information about the speech production process.

1.2.1 The glottal flow waveform $u(t)$

In source-filter theory the glottal flow $u(t)$ is the source component which provides the initial acoustic energy that is subsequently modified by the 'downstream' filter component (comprised of the vocal tract and the lips). This definition is quite general and the actual physical mechanism underlying $u(t)$ depends on the type of phonation employed.

In the GIF literature the type of phonation assumed is *voiced speech*, during which the vocal folds vibrate at the fundamental frequency F_0 while modulating a stream of air expelled from the lungs. The $u(t)$ waveform measures the flow rate (volume velocity, in ml/sec) of this airflow which looks like a quasiperiodic string of pulses, each pulse representing a "puff of air" passing through the glottis (Schroeder, 1999). The quasiperiodicity of $u(t)$ directly transfers to the voiced speech signal $s(t)$ because the vocal articulators move much more slowly than the vibrating vocal folds.

The most widely used model for the DGF $u'(t)$ in voiced speech is the Liljencrants-Fant model of Fant et al. (1985). We use this in Chapter 4 as the basis for a parametric prior for $u(t)$ denoted formally as

$$u(t) \sim \pi_{\mathrm{LF}(u(t))} \tag{9}$$

This parametric prior serves as the basis for a more expressive and nonparametric Gaussian process prior derived in Chapter 5:

$$u(t) \sim \pi_{\mathrm{GP}(u(t))} \tag{10}$$

The GP underlying this prior has a Matérn kernel that is calibrated to π_{LF} using Bayesian transfer learning (Xuan et al., 2021).

FIGURE HERE

Figure 4 | DGF and GF samples from π_{LF} and π_{GP}.

Samples from π_{LF} and π_{GP} are shown in Figure 4. Due to its nonparametric nature the latter is capable of representing a wide variety of $u(t)$ curves both in terms of roughness and overall pulse shape. Therefore, the nonparametric π_{GP} prior can be expected to yield a more realistic model of the glottal flow in real voiced speech compared to the parametric π_{LF} prior.

1.2.2 The impulse response of the vocal tract $h(t)$

Source-filter theory asserts that the physical resonances and antiresonances of the vocal tract show up as local maxima and minima in the envelope of the power spectrum $|\tilde{h}(f)|^2$ of the vocal tract transfer function. These extrema are known as *formants* and *antiformants*, respectively, and their amplitudes, frequencies and bandwidths give a compact description of the resonant and damping ('antiresonant') behavior of the vocal tract. During speech production, we manipulate these (anti)formant frequencies on the fly by skillfully changing the vocal tract configuration (such as rounding the lips, moving the tongue or closing the mouth) in just the right way as to produce the desired vowel or consonant. Humans really are master musicians of the vocal instrument.

The standard LTI assumption and canonical approach in source-filter theory is that $\tilde{h}(f)$ is a *rational transfer function* whose *poles* and *zeros* are identified with the aforementioned formants and antiformants, respectively (Stevens, 2000). We adopt this assumption in this thesis but without the one-to-one correspondence between (formants and pole pairs) and (antiformants and zero pairs). Instead, a single formant may be modeled by one or more poles while some 'spectrum shaping' poles need not correspond to formants at all; and similarly for antiformants and zeros. For this reason we prefer to think of $\tilde{h}(f)$ as a parametric rational *expansion* of the true underlying vocal tract transfer function. The parameters controlling this Padé expansion are its order, an overall gain factor and a vector of poles and zeros. All of these can be inferred from speech data.

A (strictly proper) rational transfer function $\tilde{h}(f)$ in the frequency domain implies through (4) that the impulse response $h(t)$ in the time domain consists of a real-valued superposition of exponentially decaying sinusoids. This real-valued superposition is more convenient mathematically for our purposes than a complex rational function, so in Chapter 3 we express our two priors for $h(t)$ in the time domain.[5]

FIGURE HERE

Figure 5 | Samples from $h(t)$ prior in time and spectral domain.

The first one is a parametric prior over pole-zero expansions of the true underlying vocal tract transfer function of a given order:

$$h(t) \sim \pi_{\text{PZ}(h(t))} \tag{11}$$

[5]This might seem counterintuitive at first, but it is actually a common approach in Bayesian spectrum analysis (Bretthorst, 1988; Turner & Sahani, 2014).

Likewise, the second prior parametrizes all-pole expansions, which, lacking zeros, are a special case of pole-zero expansions.

$$h(t) \sim \pi_{\mathrm{AP}(h(t))} \tag{12}$$

Samples from π_{PZ} and π_{AP} both in the time domain and the frequency domain are shown in Figure Figure 5. The qualitative differences between their power spectra in Figure Figure 5 are due to the fact that pole-zero expansions are more expressive than all-pole expansions. That is, samples from π_{AP} are more constrained because they are essentially samples from π_{PZ} for which the zeros are set up to cancel out exactly.

Without zeros π_{AP} cannot easily model antiformants of the vocal tract, which for example occur with nasal consonants [<u>n</u>ow si<u>ng</u>]. This is often used as an argument against the use of all-pole transfer functions in speech processing (Mehta et al., 2012). Its samples also have an expected spectral tilt that is too steep to be realistic. In contrast π_{PZ} can model a more flexible class of spectra with a wide range of (anti)resonant behavior and possible spectral tilts, and is to be preferred to π_{AP} on these grounds, but at the cost of a roughly doubled amount of parameters.

Nevertheless, in practice all-pole expansions such as the ones represented by π_{AP} are generally thought to be sufficiently expressive to represent the vocal tract filter (Flanagan, 1965; Fulop, 2011) and have been computationally the preferred choice since the very beginning of digital acoustic phonetics (Atal & Hanauer, 1971). Next to theoretical grounds (Titze, 2000) this is mainly because they admit an efficient representation known as linear prediction (Markel & Gray, 1976), which is the traditional workhorse of acoustic analysis.

1.3 Contributions

The relatively long history of research into GIF naturally produced several lines along which the problem may be approached, but, following two recent reviews (Drugman et al., 2019a; Kadiri et al., 2021), it is possible to separate these approaches loosely into two broad classes: inverse filtering and joint source-filter optimization.[6] Before reviewing the specific contributions made by BNGIF to the GIF literature, and what they are hoped to accomplish at what cost, we first situate the approach taken in this thesis within these two classes.

Inverse filtering Miller (1959) founded the basis for the first and larger class of GIF methods, which holds most of the popular GIF methods used today such as closed phase analysis (Wong et al., 1979) and iterative adaptive inverse filtering (Alku, 1992). The inverse filtering approach aims to recover the glottal source signal $u(t)$ by extracting from the speech data $\mathbf{d}$ an estimate of the vocal tract filter $\hat{\hat{h}}(f)$ and then digitally filtering $\mathbf{d}$ by $1/\hat{\hat{h}}(f)$, the inverse of that (hopefully nonzero) estimate. These methods are based on efficient digital

[6]For the sake of the argument we ignore a here smaller third class which uses mixed-phase decomposition methods (Degottex, 2010; Drugman et al., 2019a; Kadiri et al., 2021) such as complex cepstrum decomposition (CCD) (Drugman et al., 2011).

signal processing algorithms. Most of them rely on LP analysis and differ mainly in how the all-pole vocal tract transfer function $\tilde{h}(f)$ is estimated (Kadiri et al., 2021). No explicit parametric form of $u(t)$ is assumed because $u(t)$ is recovered by applying the digital filter $1/\hat{\tilde{h}}(f)$ to a vector of speech samples $\mathbf{d}$.

Joint source-filter optimization Milenkovic (1986) proposed a different approach for the second class of GIF methods, a logical consequence of the increased computing power available in the 1980s. With joint source-filter optimization methods the fit of a parametric model to the speech data $\mathbf{d}$ is optimized to estimate both source $u(t)$ and filter $\tilde{h}(f)$ simultaneously (as opposed to consecutively as in the first class). These methods involve the nonconvex optimization of an objective function to determine the parameters $\boldsymbol{\theta}$ of some assumed functional form of the DGF $u'(t) = f(t; \boldsymbol{\theta})$, typically an approximation of the LF model (Alzamendi & Schlotthauer, 2017; Degottex, 2010; Fu & Murphy, 2006; Schleusing, 2012). This optimization is computationally costly compared to the inverse filtering methods in the first class, and a variety of tricks are used to reduce runtime and improve robustness, such as two-stage optimization (Fu & Murphy, 2006) and regressing $\boldsymbol{\theta}$ to a single parameter (Degottex, 2010).

Although we are not aware of any study that compares the performance of these two approaches systematically, it is possible to make a few general statements. Inverse filtering methods are generally efficient and nonparametric, while joint source-filter optimization methods have the distinct advantage of cleanly separating $u(t)$ and $\tilde{h}(f)$ from the outset. In doing so they avoid having to derive $u(t)$ directly from a filter estimate $\hat{\tilde{h}}(f)$ that inevitably contains pronounced effects from the former (Auvinen et al., 2014), to which Schroeder (1999, p. 94) refers as a “suspiciously circular sounding ‘bootstrap’ method.”

So, where does that leave BNGIF? The main motivation for and innovation of BNGIF is that it mixes something of both classes: it is a joint source-filter optimization method capable of the flexibility of inverse filtering methods through its nonparametric model for $u(t)$. In addition, the Bayesian framework adopted by BNGIF handles uncertainty in a straightforward way through sampling the posterior, which can be seen as generalization of the nonconvex optimization that characterizes (and plagues) the class of joint source-filter optimization methods it belongs to. In contrast, inverse filtering methods pay for their efficiency by having no principled way of dealing with uncertainty, other than operationalising it as input variability.

1.3.1 Contributions and merits of BNGIF

Nonparametric glottal flow model BNGIF utilizes a nonparametric glottal flow model. That is, the unknown function $u(t)$ is modeled nonparametrically with the GP prior (10);, calibrated to the widely used LF model for $u'(t)$. As explained above, this is intended to strike a balance between the nonparametric flexibility of inverse filtering methods and the parametric control of source-filter optimization methods.

Conversely, BNGIF also avoids two particular shortcomings of currently used GIF methods. Inverse filtering methods typically model (a part of) $u(t)$ with only a few poles, either implicitly (such as the CP method mentioned above) or explicitly (such as IAIF, also mentioned above), which has since long been known to be “incapable of representing realistic physiologic glottal volume velocity or area waveforms” (Deller, 1983, p. 62). Source-filter optimization methods, on the other hand, use analytical models for the glottal flow derivative $u'(t)$ (Doval et al., 2006) which are convenient but necessarily “limited in their ability to capture the behavior of the glottal source in natural speech” (Kadiri et al., 2021, p. 1926).

Uncertainty quantification It is commonly agreed that uncertainty analysis and quantification is important for fundamental scientific research and forensic or biometric applications of acoustic phonetics, for example when characterizing quasi-vowels of great apes (Ekström et al., 2023) or calculating the evidence for a proposition in the context of forensic speaker comparison (Bonastre et al., 2015). However, in the specific case of GIF, “a difficult blind separation problem since neither the vocal tract response nor the glottal flow contribution are actually observable” (Degottex et al., 2014, p. 962), we would argue that uncertainty quantification takes on a nearly essential quality, if only as a rough measure of the GIF algorithm’s trust in its own output.

As the reader probably anticipates by now, the natural framework for uncertainty quantification is Bayesian probability theory. We touched upon the reason for this before, which is plain and simple: Bayesian theory prescribes how to reason consistently and quantitatively under uncertainty (that is, how to conduct inference mathematically), and it does so uniquely, as first demonstrated by (Cox, 1946). Uncertainty analysis is automatically incorporated in a Bayesian approach by sampling the posterior for the given problem, and that is of course what BNGIF does,[7] as explained in Section 1.1 and illustrated in Figure 1. Uncertainty quantification of various GIF-derived statistics such as the open quotient (OQ) and first formant frequency (F_1) then reduces to calculating the empirical mean $\pm$ standard deviation of these statistics directly from posterior samples of $u(t)$ and $\tilde{h}(f)$.

Despite their appeal for uncertainty quantification, Bayesian approaches to GIF are rare in the literature, and next to BNGIF we know of only three other probabilistic GIF methods: (Auvinen et al., 2014; Rao & Ghosh, 2018; Wang et al., 2016). (Bleyer et al., 2017) confirm that the first Bayesian inversion approach to GIF is described in (Auvinen et al., 2014), who call it MCMC-GIF. Surprisingly, these methods all produce point estimates of the glottal flow and vocal tract filter and do not seem to use posterior samples for uncertainty analysis.

Adaptive GCI estimation Contrary to the sliding windows approach used by conventional LP analysis for formant estimation, GIF methods are *pitch-synchronous* in nature and require knowledge of the glottal closure instants (GCIs) in order to work properly (Drugman et al., 2019b). This means that the user has to run an independent GCI detection algorithm first before GIF methods can be used.[8] A prominent (and according to (Chien et al., 2017), unique)

[7] Or rather: tries to do, as sampling the posterior is intractable for most real-world problems, and efficient approximations must be used instead.

[8] Note that robust GCI detection “is a very difficult task in practice” (Alzamendi & Schlotthauer, 2017, p. 14) for which many algorithms have been proposed (Drugman et al., 2019b, Sec. 3.3).

exception to this is the aforementioned IAIF algorithm (Alku, 1992), which implements heuristic GCI estimation as part of its iterative structure.

On the other hand, BNGIF infers GCIs alongside and simultaneously with the source $u(t)$ and filter $\tilde{h}(f)$, which allows newly discovered information (during inference) about either one of these to constrain the others. This adaptive strategy helps to 'bootstrap' the GCIs from the speech signal, since GCIs are instrumental in recovering $u(t)$, but ultimately need to be derived from (and are defined in terms of) the latter. Another adaptive aspect of BNGIF's built-in GCI estimation is that noisy GCI estimates may *optionally* be supplied by the user a priori; BNGIF will then use this information to speed up inference, and yield refined estimates as part of its output. These adaptive abilities are quite unique in the GIF literature as far as we are aware.

Modeling inertia and nonstationarity The inertia of the vocal apparatus and articulators imposes constraints on the speed at which the speech signal $s(t)$ can be produced and altered. It is often the case that $s(t)$ appears relatively 'stable' or stationary during short time intervals of about 10 to 50 msec, an observation which underlies the ubiquitous utility and adoption of the short-time Fourier transform (STFT) in acoustic phonetics (Little, 2011). In the case of voiced speech, the providence of GIF, these inertial constraints produce correlations across *dozens* of pitch periods . The source $u(t)$ and filter $\tilde{h}(f)$ constituents of voiced speech are both inherently nonstationary yet strongly correlated at the level of neighbouring pitch periods. We can actually *hear* this fact: pitch and timbre are perceptual renditions of these statistical regularities (Schnupp et al., 2011).

BNGIF can model nonstationary voiced speech which exhibits these inertial correlations, and takes advantage of the latter to speed up inference (since any correlation means more predictability, which often leads to faster exploration of plausible hypotheses). Figure Figure 4 already illustrates the local inertia and nonstationarity of the glottal source. BNGIF also takes into account 'filter correlations'; that is, the constrained variation of $\tilde{h}(f)$ from pitch period to pitch period. A pleasant consequence of this approach is probabilistic (anti)formant tracking, which we discuss just below.

The approach of BNGIF contrasts with many other GIF methods, which work one pitch period at a time and do not model inertial correlations between the pitch periods explicitly. For example, none of the algorithms considered in a recent review by (Chien et al., 2017) of the most important and representative inverse filtering methods use inter-pitch period correlations, and we are not aware of any joint source-filter optimization methods exploiting these correlations either.

Probabilistic (anti)formant tracking Formant tracking refers to estimating from speech data the (smooth) trajectories along which formants move over time, which in turn reflect the movement of the vocal articulators during phonation. Because of their efficiency as low-dimensional 'information bearers', formant tracks are highly important in acoustic phonetics and many algorithms exist for measuring them, mostly based on LP analysis (Van Soom &

de Boer, 2020). These algorithms usually operate on a sliding window approach, estimating formants within each window and then using a post-processing step to join these point estimates into smooth trajectories (Whalen et al., 2022). A prominent example of this strategy is the widely used program (Boersma, 2001), whose default tracking function implements the Viterbi algorithm (Viterbi, 1967) to post-process the formant candidates into smooth trajectories.

In the case of BNGIF, formant tracking happens during inference and is achieved by exploiting the inertial correlations of the filter $\tilde{h}(f)$, as discussed above. That is, pole pairs are subjected to inter-pitch period correlations, and this takes care of 'trajectory smoothing' automatically while avoiding an ad-hoc post-processing step.

The same is true for the zero pairs: antiformant tracking is implicitly enabled in the case of a pole-zero expansion of the rational transfer function $\tilde{h}(f)$. We are not aware of other GIF methods with this ability, since GIF methods almost invariably assume all-pole transfer functions (Alku, 2011; Bleyer et al., 2017; Kadiri et al., 2021), even though spectral zeros occur during the open phase of the glottis due to coupling to the subglottal cavity (Ananthapadmanabha & Fant, 1982; Plumpe et al., 1999).

There are a handful of formant tracking algorithms, however, that implement the same principle – that is, smoothing of formant tracks during inference rather than as a post-processing step – of which the Kalman-based autoregressive moving average (Mehta et al., 2012) method is a relatively recent example. BNGIF and KARMA also have in common the ability to optionally use noisy a priori estimates of the formant trajectories to speed up inference.

Avoiding harmonic attraction Higher values of the fundamental frequency F_0 pose challenges for formant tracking algorithms based on LP analysis due to a phenomenon (Whalen et al., 2022) call "harmonic attraction." This refers to the fact that, as a result of spectral smoothing, the peaks of the estimated envelope obtained through LP are highly biased toward the harmonic partials of a speech spectrum (Yoshii & Goto, 2013), such that the estimated formant frequencies $\boldsymbol{F}$ tend to align with multiples of F_0. For higher values of F_0, this alignment causes the systematic correlations between F_0 and $\boldsymbol{F}$ known as F_0 bias. In the case that $F_0 \approx F_1$, known as $F_0 - F_1$ congruence (Kent & Vorperian, 2018), harmonic attraction can be especially challenging, since LP analysis, as a spectral smoothing device, lacks a conceptual model for separating the two – in contrast to, for example, the auditory processing regions of our brains (Schnupp et al., 2011).

Next to formant tracking, harmonic attraction equally affects GIF methods based on inverse filtering since the F_0-biased formant estimates result in insufficient cancelation of the true vocal tract resonances, which in turn distorts inference of the glottal flow $u(t)$ (Auvinen et al., 2014). This problem is well known, and there is extensive literature on the degradation of inverse filtering methods under increased values of F_0. In fact, 'increased values of F_0' often means 'not a low-pitched male voice' (Airaksinen et al., 2014), meaning that harmonic attraction renders a large majority of cases potentially problematic for inverse filtering methods! We also note that the $F_0 - F_1$ congruence mentioned above is especially problematic for GIF in the case of close rounded vowels, which remains "a major factor that limits the applicability of inverse filtering algorithms to accurate glottal flow estimation from continuous speech" (Chien et al., 2017, p. 17).

BNGIF is a joint-source filter optimization method and thereby automatically avoids harmonic attraction and the problems associated with $F_0 - F_1$ congruence. In addition, the probabilistic approach of BNGIF (as opposed to most GIF methods) is known to improve performance and reduced sensitivity to prior estimation of GCIs (Auvinen et al., 2014; Bleyer et al., 2017).

1.4 Why glottal inverse filtering

A person produces around 16,000 words each day (Andrade-Miranda et al., 2024). Sustained use and neurological insult both leave measurable traces in the glottis. Speech impairment affects up to 90% of patients with Parkinson's disease and is now recognised as a preclinical marker that precedes the defining motor signs by as much as a decade (Cao et al., 2025). The recommended and widely used probe is sustained phonation of /a/: a single breath that exposes vocal fold function and voice quality simultaneously (Cao et al., 2025). From that recording, the central quantities of interest are not the acoustic pressure waveform itself but the *glottal flow* $u(t)$ and the *vocal tract filter* $h(t)$ hidden behind it: the fundamental frequency F_0, the open quotient, jitter, shimmer, the harmonics-to-noise ratio, formant trajectories — all derive from these two signals.

Recovering $u(t)$ and $h(t)$ from $s(t)$ alone is *glottal inverse filtering* (GIF). It is an old and hard problem (Auvinen et al., 2014; Kadiri et al., 2021), and its difficulty is exactly why the clinical signal chain from microphone to biomarker is fragile. This thesis attacks the problem at its root: not better heuristics on top of LP, but a principled probabilistic model of voiced speech in which uncertainty propagates naturally from raw samples all the way to derived clinical features.

1.5 State of the art

Classical signal-processing methods remain the reference standard for GIF. Evaluation studies explicitly benchmark IAIF and QCP as "state-of-the-art GIF techniques" (Freixes et al., 2023), and comparative analyses continue to focus on IAIF, QCP, and LPC-based variants without inclusion of neural alternatives (Palaparthi & Titze, 2020). Clinical workflows still rely on IAIF and QCP as primary estimation tools (Corcoran et al., 2019). Quantitatively,

> QCP-based techniques significantly outperform IAIF-based methods for almost all error metrics. (Freixes et al., 2023)

This positions QCP — specifically its AME-weighted variant — as the strongest method within the classical paradigm. The EGIFA benchmark (Chien et al., 2017) reinforces this: as noted in Dataset 1, the best-performing "WCA" methods in that benchmark correspond in implementation to QCP-style AME weighting rather than the literal WLP formulation described in the paper text. Benchmark results thus confirm the classical framework rather than pointing beyond it.

Despite the success of machine learning in speech processing broadly, there is no neural replacement for GIF in clinical settings. No end-to-end neural inverse filtering method has achieved widespread adoption, and no benchmark has demonstrated neural approaches outperforming IAIF or QCP on glottal waveform recovery. Recent work still proposes model-based or probabilistic extensions rather than neural ones (Shi et al., 2017). The reason is stated plainly in the literature:

> The glottal flow cannot be directly measured… we cannot easily acquire the ground truth. (Sasou & Chen, 2023)

and consequently:

> GIF is hardly realized through supervised deep learning. (Sasou & Chen, 2023)

Machine learning is highly effective in clinical speech analysis, but its role is downstream of GIF rather than a replacement for it (Moell et al., 2025). It operates on acoustic features or on features derived from GIF estimates, and physically-grounded glottal features can outperform learned speech representations in pathology detection — a structured HMM method built on GIF-derived features outperformed self-supervised learned representations in one such setting (Sasou & Chen, 2023). Machine learning has thus largely bypassed GIF rather than replacing it.

Two structural reasons account for the persistence of classical methods. First, GIF is ill-posed: $u'(t)$ must be inferred jointly with $h(t)$ from $s(t)$ alone ((1)), and without strong inductive structure the problem admits many solutions. Second, interpretability is not optional in clinical contexts: parameters such as NAQ, QOQ, H1-H2, and HRF are physiologically meaningful and required for downstream use; black-box models are unsuitable.

There are early signs of movement. Physics-informed neural networks can simultaneously estimate glottal flow and related physiological variables from speech, beginning to integrate physical constraints with data-driven inference (Yokota et al., 2025). DNN-based GIF shows improved accuracy over IAIF in specific constrained settings such as coded speech (Narendra et al., 2019). Advanced model-based methods — quadratic programming for improved closed-phase behavior (Airaksinen et al., 2017), variational EM pole-zero modeling (Shi et al., 2017) — continue to extend the classical paradigm from within. But none of these has yet displaced IAIF or QCP as the default in standard benchmarks or clinical practice.

The central bottleneck is ground truth: synthetic corpora generated by physics-based synthesizers like VocalTractLab (Dataset 1) provide the only controlled setting where the reference $u'(t)$ is known exactly. BNGIF uses those reference signals not just as evaluation targets but as training data for the nonparametric DGF prior, directly addressing the supervision deficit that limits neural approaches.

1.6 BNGIF: the narrative arc

The thesis builds BNGIF as a converging series of prior refinements, each step motivated by a concrete deficiency of the previous.

Step 1: IKLP (Chapter 2, Algorithm 1). Yoshii & Goto (2013) showed that the LP excitation prior can be replaced by a mixture of GP kernels, with kernel weights inferred by CAVI under a gamma process prior. This single move simultaneously infers F_0, voiced/unvoiced status, and the spectral envelope — replacing three separate heuristic DSP stages with one probabilistic model. Chapter 2 introduces the engine at the conceptual level and shows that the vanilla priors still lose badly to a QCP variant on EGIFA. Algorithm 1 then gives the CAVI derivation, Woodbury-Cholesky reformulation, and prior reparameterizations used later.

Step 2: AR prior (Chapter 3). The LP coefficients $\mathbf{a}$ themselves carry prior structure: stability, physical pole locations, spectral constraints. An informative Gaussian prior on $\mathbf{a}$ derived from stability theory and spectral constraints replaces the isotropic default, sharpening the filter estimate and regularizing the GIF problem further.

Step 3: DGF kernel prior — PACK (Chapter 5). The kernel bank in IKLP is populated not with ad hoc periodic kernels but with kernels *calibrated to real DGF data* from VocalTractLab via the PRISM algorithm. PACK (periodic arc-cosine kernel), QPACK, and g-QPACK are the resulting kernel family, encoding quasiperiodicity, spectral rolloff, and the open/closed phase asymmetry respectively.

Step 4: Learned surrogate prior — q-GPLVM (Chapter 6). A quantized Gaussian process latent variable model compresses the PRISM amplitude object $\mathcal{D}_R$ into the compact mixture $\mathcal{M}_R$. That object is the operational nonparametric prior for BNGIF: it supplies the DGF archetypes and their weights that seed IKLP inference.

Step 5: BNGIF assembled (Chapter 7). With IKLP, AR prior, and q-GPLVM prior in place, the full model is assembled and evaluated on EGIFA and VTRFormants.

1.7 Chapter outline

The thesis develops BNGIF from the bottom up:

- Chapter 2: IKLP as Bayesian LP; early EGIFA preview and the need for better priors.
- Algorithm 1: IKLP CAVI, Woodbury-Cholesky reformulation, prior reparameterizations.
- Chapter 3: informative AR prior from stability theory and spectral constraints.

- Chapter 4: the LF model as the basis for the DGF prior; analytical spectrum of glottal flow models.
- Chapter 5: PACK kernel family calibrated to EGIFA via PRISM.
- Chapter 6: q-GPLVM compression from $\mathcal{D}_R$ to the compact surrogate prior $\mathcal{M}_R$.
- Chapter 7: assembled BNGIF model; evaluation on EGIFA and VTRFormants.
- Chapter 8: contributions, limitations, future directions.

Supporting material:
- Dataset 1 (dataset), Dataset 2, Dataset 3, Dataset 4.
- Algorithm 1 (IKLP inference), Algorithm 2 (LBGID), Algorithm 3 (PRISM), Algorithm 4 (t-PRISM), Algorithm 5 (q-GPLVM).
- Appendix A, Appendix E, Appendix G.

2 Infinite kernel linear prediction

The inference engine at the heart of BNGIF is infinite kernel linear prediction (IKLP), the nonparametric Bayesian extension of linear prediction proposed by Yoshii & Goto (2013). The point of this chapter is conceptual. We want to see why IKLP matters for GIF, what it buys over ordinary LP, and why the vanilla model is still not enough. The full CAVI derivation, Woodbury-Cholesky reformulation, and prior reparameterizations are given separately in Algorithm 1.

The story is simple. LP works by modeling the vocal-tract envelope and treating everything else as white excitation. That is already useful, but it breaks in a characteristic way on voiced speech. IKLP fixes the engine by replacing white excitation with a bank of kernel hypotheses. What remains, and what drives the rest of the thesis, is the question of what those hypotheses should be.

2.1 Linear prediction

Let $\mathbf{x} = (x_1, ..., x_M)^\top \in \mathbb{R}^M$ be M consecutive speech samples drawn from a short analysis frame. The autoregressive (AR) model of order P asserts that each sample is a linear combination of the P most recent ones plus an excitation residual:

$$x_m = \sum_{p=1}^{P} a_p x_{m-p} + \varepsilon_m. \tag{13}$$

The vector $\mathbf{a} = (a_1, ..., a_P)^\top \in \mathbb{R}^P$ collects the predictor coefficients and $\boldsymbol{\varepsilon} = (\varepsilon_1, ..., \varepsilon_M)^\top$ is the excitation. In source-filter terms the two play distinct roles: $\mathbf{a}$ encodes the resonance structure of the vocal tract, while $\boldsymbol{\varepsilon}$ carries the source.

Two matrices appear repeatedly. The AR operator $\mathbf{\Psi}(\mathbf{a}) \in \mathbb{R}^{M \times M}$ is approximately lower-triangular with unit diagonal and sub-diagonal entries given by $-a_1, ..., -a_P$; it satisfies $\boldsymbol{\varepsilon} = \mathbf{\Psi}\mathbf{x}$, or equivalently $\mathbf{x} = \mathbf{\Psi}^{-1}\boldsymbol{\varepsilon}$. The lag matrix $\mathbf{X} \in \mathbb{R}^{M \times P}$ has m-th row $(x_{m-1}, ..., x_{m-P})$ and is the design matrix for the normal equations.

The classical probabilistic LP model assumes white Gaussian excitation,

$$\boldsymbol{\varepsilon} \sim \texttt{Normal}(\mathbf{0}, \nu\mathbf{I}), \tag{14}$$

which yields the likelihood

$$\mathbf{x} \sim \texttt{Normal}\left(\mathbf{0}, \nu\mathbf{\Psi}^{-1}\mathbf{\Psi}^{-\top}\right). \tag{15}$$

Maximum-likelihood estimation under (15) gives the familiar normal equation

$$\mathbf{X}^\top\mathbf{X}\mathbf{a} = \mathbf{X}^\top\mathbf{x}, \tag{16}$$

which is what standard LPC implementations solve.

The model is not useless on voiced speech; in practice it often works surprisingly well. But its failure mode is clear. During voiced phonation the excitation is periodic, not white, and fitting (15) to a pitched signal forces the estimated envelope to track the harmonic partials of the spectrum: the poles of the all-pole filter chase the pitch harmonics rather than the formants. This is the defect that motivates everything that follows.

2.2 Kernel linear prediction

The natural fix is to replace the white-excitation model by a richer one. We model the excitation as a continuous function $\varepsilon(t)$ over time, approximated by a weighted sum of basis functions plus a broadband residual:

$$\varepsilon(t) = \boldsymbol{\varphi}(t)^\top \mathbf{w} + \eta(t), \tag{17}$$

where $\boldsymbol{\varphi}(t) = \left(\varphi_1(t), ..., \varphi_{J(t)}\right)^\top$ and $\eta(t)$ is the residual. Sampling at times $t_1, ..., t_M$ and writing $\mathbf{\Phi} \in \mathbb{R}^{M \times J}$ for the design matrix with rows $\boldsymbol{\varphi}(t_m)^\top$ gives $\boldsymbol{\varepsilon} = \mathbf{\Phi}\mathbf{w} + \boldsymbol{\eta}$ in vector form.

With independent Gaussian priors

$$\mathbf{w} \sim \texttt{Normal}(\mathbf{0}, \nu_w \mathbf{I}), \quad \boldsymbol{\eta} \sim \texttt{Normal}(\mathbf{0}, \nu_e \mathbf{I}), \tag{18}$$

marginalizing over $\mathbf{w}$ yields

$$\boldsymbol{\varepsilon} \sim \texttt{Normal}(\mathbf{0}, \nu_w \mathbf{K} + \nu_e \mathbf{I}), \quad \mathbf{K} = \mathbf{\Phi}\mathbf{\Phi}^\top. \tag{19}$$

The matrix $\mathbf{K}$ is a kernel matrix with entries $K_{mm'} = \boldsymbol{\varphi}(t_m)^\top \boldsymbol{\varphi}(t_{m'})$, and the kernel trick lets us specify it directly as $K_{mm'} = k(t_m, t_{m'})$ without ever constructing the basis functions explicitly.

Substituting (19) into $\mathbf{x} = \mathbf{\Psi}^{-1}\boldsymbol{\varepsilon}$ gives the kernelized LP likelihood,

$$\mathbf{x} \sim \texttt{Normal}\left(\mathbf{0}, \mathbf{\Psi}^{-1}(\nu_w \mathbf{K} + \nu_e \mathbf{I})\mathbf{\Psi}^{-\top}\right). \tag{20}$$

Classical LP (15) is the special case $\mathbf{K} = \mathbf{I}$ with $\nu = \nu_w + \nu_e$. A periodic kernel with period T concentrates excitation power at harmonics and thereby decorrelates the envelope estimate from the pitch structure. That is Yoshii's key move.

Excitation semantics

It is worth being precise about how the structured and broadband parts of $\boldsymbol{\varepsilon}$ map onto GIF. We write the excitation as $\varepsilon(t) = s(t) + \eta(t)$, where $s(t) = \boldsymbol{\varphi}(t)^\top \mathbf{w}$ is the structured GP component and $\eta(t)$ is broadband noise, and we distinguish this from $u'(t)$, the true physical DGF. In this thesis $u'(t)$ is the theoretical DGF in the source-filter picture, and inferred $\varepsilon(t)$ is our practical DGF estimate. During voiced regions these are close but not equal, and we always report $\varepsilon(t)$ rather than $s(t)$ alone.

This is deliberate. Even ordinary LPC — the $s = 0$ limit where all excitation comes from $\eta(t)$ — already recovers recognizable DGF estimates from long enough analysis frames, because $\eta(t)$ is free to inject the sharp broadband transients needed to excite all formant resonances simultaneously. Those transients also model the broadband energy injections associated with glottal closure. IKLP enriches what $s(t)$ can model, but it leaves $\eta(t)$ active and useful.

2.3 Infinite kernel linear prediction

The kernelized model (20) requires choosing one kernel $\mathbf{K}$ to match the excitation structure. For voiced speech the fundamental period T is itself unknown, so this is exactly the wrong place to commit too early. The Bayesian response is multiple kernel learning: prepare a bank of candidate kernels $\mathbf{K}_1, ..., \mathbf{K}_I$ and define the excitation kernel as their weighted sum,

$$\mathbf{K} = \sum_{i=1}^{I} \theta_i \mathbf{K}_i, \quad \theta_i \geq 0. \tag{21}$$

Each weight θ_i quantifies the degree to which hypothesis i explains the excitation. In the original formulation of Yoshii & Goto (2013), the bank indexes candidate fundamental periods $T_1, ..., T_I$ on a grid, so the weights encode a posterior over F0 candidates.

That is the right starting point, but too narrow as a final interpretation. The index i is really a generic hypothesis axis: it can index candidate periods, candidate glottal-flow morphologies, candidate phonation modalities, or any structured excitation model that can be expressed as a kernel. Later chapters exploit this generality directly. The bank will no longer contain ad hoc periodic kernels but learned GP priors over DGF shape.

The resulting multiple-kernel likelihood is

$$\mathbf{x} \sim \texttt{Normal}\left(\mathbf{0}, \mathbf{\Psi}^{-1}\left(\nu_w \sum_{i=1}^{I} \theta_i \mathbf{K}_i + \nu_e \mathbf{I}\right)\mathbf{\Psi}^{-\top}\right). \tag{22}$$

MAP estimation of $\boldsymbol{\theta}$ under a regularizing prior does not yield truly sparse weights. Yoshii & Goto (2013) solve this by taking $I \to \infty$ and placing a gamma-process prior on the infinite weight sequence: for any finite truncation level I,

$$\theta_i \sim \texttt{Gamma}(\alpha/I, \alpha), \tag{23}$$

and as $I \to \infty$ the vector $\boldsymbol{\theta}$ converges to a draw from a GaP with concentration α. The effective number of active elements remains almost surely finite.

The full IKLP likelihood is therefore

$$\mathbf{x} \sim \texttt{Normal}\left(\mathbf{0}, \mathbf{\Psi}^{-1}\left(\nu_w \sum_{i=1}^{\infty} \theta_i \mathbf{K}_i + \nu_e \mathbf{I}\right)\mathbf{\Psi}^{-\top}\right). \tag{24}$$

This single model simultaneously infers the spectral envelope through $\mathbf{a}$, the dominant excitation hypothesis through $\boldsymbol{\theta}$, and voiced/unvoiced status through the ratio $\mathbb{E}[\nu_w]/\mathbb{E}[\nu_w + \nu_e]$.[9]

2.4 What IKLP buys

IKLP turns LP into a hypothesis-testing engine. The kernel bank is the only interface between the inference engine and the prior on the excitation. Once that interface is in place, changing the source model does not require changing the variational architecture. This is why the method remains central throughout the thesis even though its priors change completely.

It also removes a great deal of DSP staging. Pitch, voiced/unvoiced structure, and envelope estimation are no longer separate heuristic steps but consequences of a single probabilistic model. That does not make the problem easy. It does make the assumptions explicit.

The technical price is the variational machinery itself. That price is paid in Algorithm 1: the mean-field family, the auxiliary lower bound, the CAVI updates, the Woodbury-Cholesky rewrite, and the hyperparameter reparameterizations all live there. Here it is enough to keep the high-level lesson in view: IKLP gives exactly the kind of modular Bayesian engine we want.

Figure 6 is worth keeping in mind when reading the benchmark below. The vanilla periodickernel model already does the right *kind* of Bayesian work: it uses the kernel bank to localize a plausible F_0, transfers that periodicity into the excitation estimate, and keeps the tract and source coupled in one fit. So the problem is not that IKLP has the wrong inference architecture. The problem is that the periodic hypothesis family is still too crude.

2.5 Early benchmark on EGIFA

Before moving on, it is worth looking at what the vanilla engine can actually do. Table 1 gives one early benchmark preview on EGIFA. The exact evaluation protocol is developed more carefully later in Dataset 1 and Appendix G; for now only the ranking matters. Higher is better.

The message is not subtle. Yoshii's periodic prior is a real improvement over white excitation, so the direction is correct. Figure 7 shows exactly that: the periodic-kernel density is shifted to the right of the white-excitation density. But the shift is small. The QCP-variant density sits in a completely different region of score space.

Figure 8 makes the same point in waveform space. The three rows are, respectively, a modal frame at 1414 Pa, a pressed frame at 1414 Pa, and a slightly breathy frame at 2000 Pa. The periodic kernel often cleans up the gross periodic structure and can materially outperform white excitation on individual frames. But it still leaves too much broadband mismatch

[9] Setting $P = 0$ is also valid: the AR operator becomes $\boldsymbol{\Psi} = \mathbf{I}$ and IKLP reduces to a pure multi-kernel Bayesian inference engine over the excitation process with no envelope component.

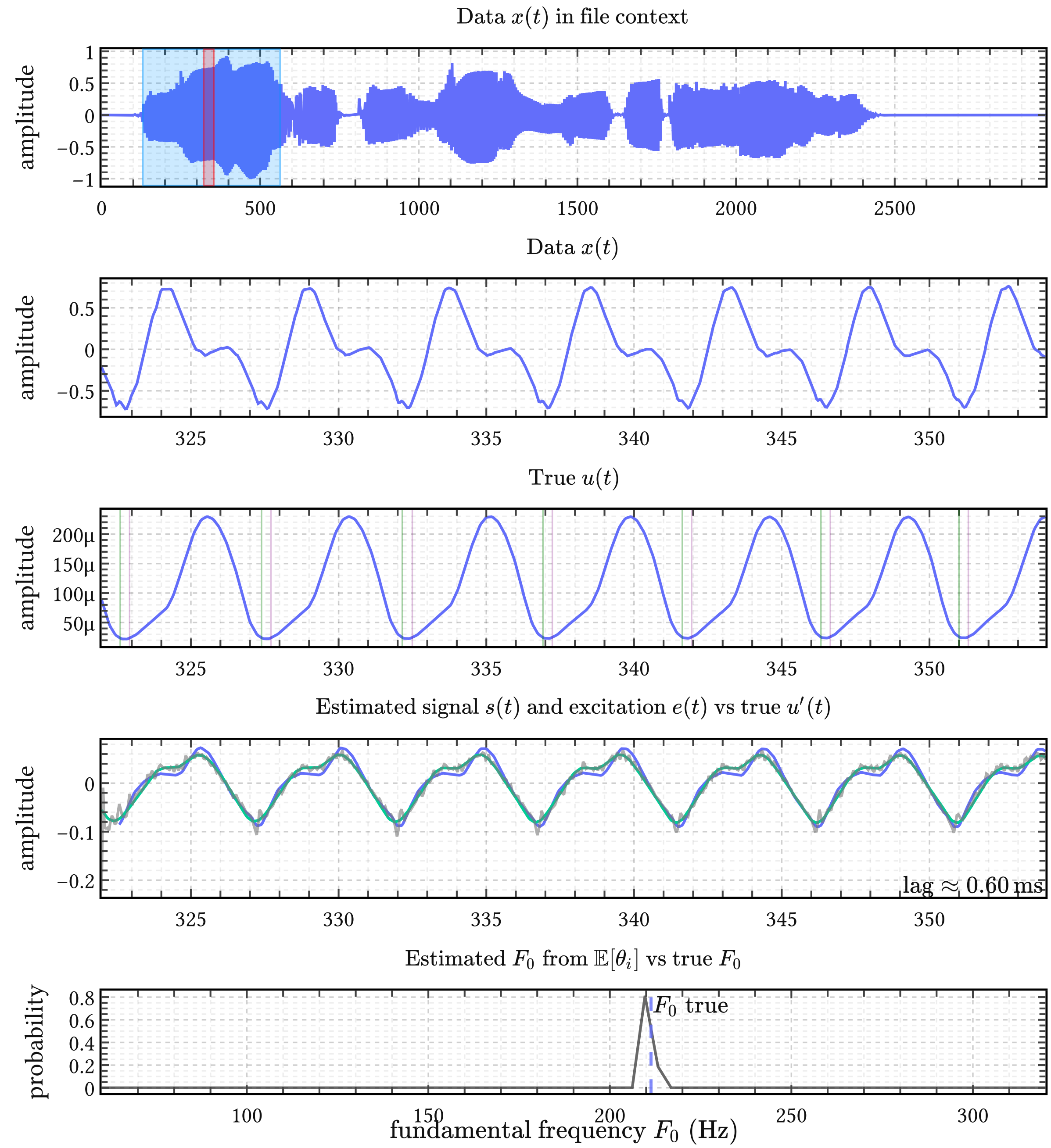


Figure 6 | **One periodickernel IKLP fit in detail.** Top: the analyzed EGIFA speech file and the selected 32 ms frame. Middle: the frame-local speech signal, the aligned true DGF $u'(t)$, and the inferred structured signal $s(t)$ together with the full excitation estimate $\varepsilon(t)$. Bottom: the posterior over candidate F_0 values derived from $\mathbb{E}[\theta_i]$. The intended mechanism is already visible: weight mass concentrates near one fundamental frequency and the excitation estimate is pulled toward that periodic structure while the tract envelope is fitted jointly.

around the main closures, and that is precisely where the QCP variant remains far stronger. The gap is therefore not a quirk of one summary score: it is visible in the estimated DGFs themselves.

Method	Vowel median	Speech median
White excitation	0.325	0.338
Periodic kernel	0.344	0.349
QCP variant	0.957	0.910

Table 1 | **Early EGIFA preview for vanilla IKLP.** The first two rows come from the `whitenoise` and `periodickernel` IKLP sweeps in `experiments/iklp/vanilla`. The third row is the AME-weighted QCP variant from the classical EGIFA benchmark sweep in `experiments/iklp/chien`, using the LBGID backend. Values are median centered cosine similarity on the aligned excitation waveform.

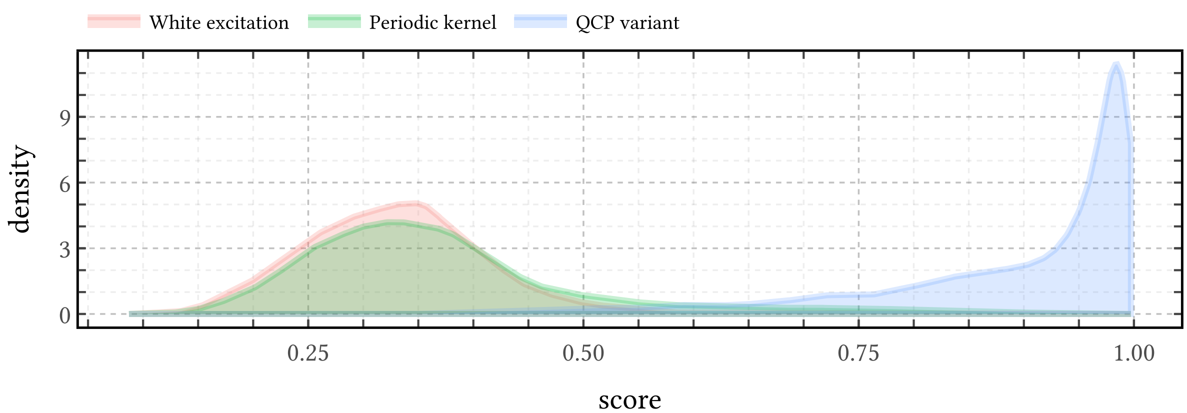


Figure 7 | **Score distributions on the early EGIFA preview.** The periodic kernel shifts the white-excitation distribution slightly to the right, confirming that structured periodic excitation is the correct direction. But the shift is small, while the QCP-variant distribution is concentrated much further to the right.

Vanilla IKLP is therefore not yet a competitive GIF method simply by being Bayesian.

This is also why the thesis does not stay in the world of hand-designed periodic kernels. The engine is sound. The priors are not.

2.6 What is missing

Two deficiencies are already visible. On the excitation side, the periodic kernel is too crude: it can express smooth periodicity, but not the quasiperiodicity, open/closed asymmetry, or learned morphology of real DGF waveforms. On the filter side, the isotropic AR prior is too weak to push the tract estimate toward physically plausible regions when the source-filter tradeoff becomes difficult.

The rest of the thesis fixes exactly these two problems. Chapter 3 sharpens the filter side by replacing the isotropic AR prior with an informative Gaussian one. Chapter 5 and Chapter 6 then replace Yoshii's periodic hypotheses by calibrated and learned GP priors over DGF shape. Only after those two repairs does IKLP become the BNGIF inference engine of Chapter 7.

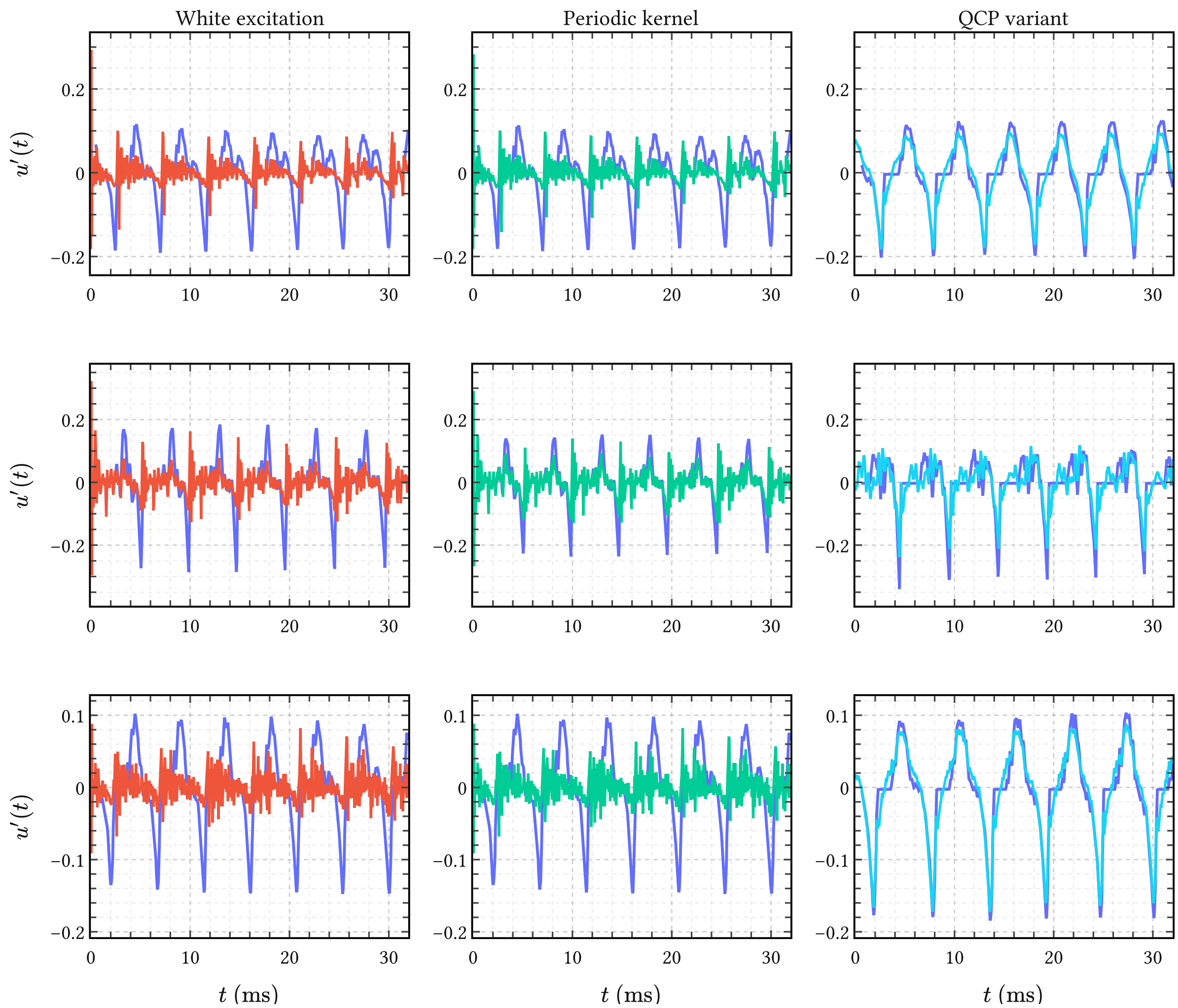


Figure 8 | **Three representative EGIFA speech frames.** In each panel, the blue curve is the aligned true DGF $u'(t)$ and the colored curve is the estimated excitation. Left: white excitation. Center: Yoshii's periodic kernel. Right: the QCP variant. The first row shows the typical case: periodic structure helps, but only modestly. The second row shows a frame where the periodic kernel already gives a clearly better voiced fit than white excitation. The third row shows the real failure mode: both vanilla IKLP variants still miss badly, while the QCP variant locks onto the closure pattern.

3 Priors for autoregressive filters

The vocal tract filter is modeled throughout this thesis as an $\mathrm{AR}(P)$ process: a rational all-pole transfer function whose P coefficients $\mathbf{a} = (a_1, ..., a_P)^\top$ encode the resonant structure of the vocal tract. Estimating $\mathbf{a}$ from speech data is done traditionally by LPC methods, or in our case, by IKLP. Like any Bayesian model, IKLP requires a prior $p(\mathbf{a})$, and throughout their paper Yoshii & Goto (2013) simply assert

$$p(\mathbf{a}) = \mathtt{Normal}(\mathbf{0}, \lambda \mathbf{I}_P), \quad \text{with a fixed } \lambda := 0.1. \tag{25}$$

This choice, which we will refer to as the *isotropic prior*, is clearly something more of a blunt regularizer than an informative prior. It's there to prevent the normal matrix in (204) from becoming singular, not to encode anything known a priori about the vocal tract.

While it gets the job done, two things about (25) are perhaps unsatisfying. First, the isotropic prior with $\lambda = 0.1$ implies a standard deviation of about 3.16 per coefficient. This is a very loose prior under which most induced AR filters will almost surely be unstable. Second, we actually know a great deal about vocal tract filters: their resonances are spaced roughly 500 Hz apart, their spectral tilt is on the order of $-6\ \mathrm{dB/octave}$, and, plain from the physics of speech production, the filter must be at least be passive and causal. Except for causality none of this is encoded in (25).

This chapter then develops two priors that attempt to do better in this regard. The first, the *Monahan prior*, is the closest Gaussian to the uniform distribution over stable AR coefficient vectors; at first sight, this sounds like exactly the thing to do, but we show that it doesn't work. The second, the *soft spectral shaping prior*, is a new construction that encodes a number of spectral features useful in speech – resonances and spectral rolloff – directly into the prior mean and covariance through a closed-form projection. Both priors remain Gaussian and are therefore compatible with IKLP's variational inference without modification, and we test their merit in Chapter 7.

3.1 The isotropic prior

3.1.1 Expected output power

To understand the inductive bias that $\lambda = 0.1$ represents, consider what an isotropic Gaussian prior

$$\mathtt{Isotropic}(\lambda) := \mathtt{Normal}(\mathbf{0}, \lambda \mathbf{I}_P) \tag{26}$$

does to the expected output power of an $\mathrm{AR}(P)$ process driven by white noise with variance ν as in equations (13) and (14) from the previous chapter:

$$x_m = \sum_{p=1}^{P} a_p x_{m-p} + \varepsilon_m, \quad \text{where} \quad \begin{cases} \mathbf{a} & \sim \mathtt{Normal}(\mathbf{0}, \lambda \mathbf{I}_P) \\ \varepsilon_m & \sim \mathtt{Normal}(0, \nu) \end{cases} \tag{27}$$

Up to scale, then, the output power is determined by the frequency response

$$H(e^{i\omega}) = \frac{1}{A(e^{i\omega})} \quad \text{where} \quad A(z) = 1 - \sum_{p=1}^{P} a_p z^{-p}. \tag{28}$$

To second order, the Maclaurin expansion of $|H(e^{j\omega})|^2$ in a_p gives

$$|H(e^{j\omega})|^2 \approx 1 + 2\sum_{p=1}^{P} a_p \cos(p\omega) - \sum_{p=1}^{P}\sum_{\ell=1}^{P} a_p a_\ell \cos((p-\ell)\omega) \tag{29}$$

$$+ \; 4\sum_{p=1}^{P}\sum_{\ell=1}^{P} a_p a_\ell \cos(p\omega)\cos(\ell\omega) + \ldots \tag{30}$$

Taking the expectation over the zero-mean isotropic prior (25), the linear terms and the cross-terms $p \neq \ell$ vanish and the expected spectral power becomes

$$\mathbb{E}_{\mathbf{a}}[|H(e^{j\omega})|^2] \approx 1 - P\lambda + 4\lambda \sum_{p=1}^{P} \cos^2(p\omega). \tag{31}$$

Integrating over frequency and applying Parseval's identity gives the expected output variance of (27):

$$\mathbb{E}[\mathrm{var}(x_m)] \approx \nu(1 + P\lambda). \tag{32}$$

With $\nu := 1$, $P = 30$ and $\lambda = 0.1$ as in Yoshii & Goto (2013), this gives an expected standard deviation of $\sqrt{1+3} = 2$, which is indeed $O(1)$. Nevertheless, power scales linearly with P, which contradicts the prevalent idea that increasing P just prepares the model for more fine-structured data, not "louder" data. The lesson from (32) is thus: to keep power approximately controlled under an isotropic prior, one needs $\lambda \propto 1/P$. This rule is a known heuristic in signal processing literature (Rabiner et al., 1993, for example; Sørbye & Rue, 2017), here we have derived it from a probabilistic argument.

3.1.2 Stability

Beyond raw power, there is a more fundamental problem: `Isotropic`(λ) places most of its mass outside the stability region $\mathcal{A}_P \subset \mathbb{R}^P$, which is the set of coefficient vectors for which all roots of $A(z) = 1 - \sum_p a_p z^{-p}$ lie strictly inside the unit disk. Physically, *stability* means the filter's impulse response decays over time: the vocal tract is a passive resonator that dissipates energy, so its transfer function cannot have poles on or outside the unit circle. An unstable AR filter would predict an impulse response that grows without bound, which is physically impossible for a tube of flesh and air (the vocal tract).

For $P = 1$ the stability region is the interval $(-1, 1)$, and a unit Gaussian already places about 32% of its mass outside. For larger P the stability region is a complicated manifold and the fraction of unstable draws tends to grow rapidly – see Figure 9.

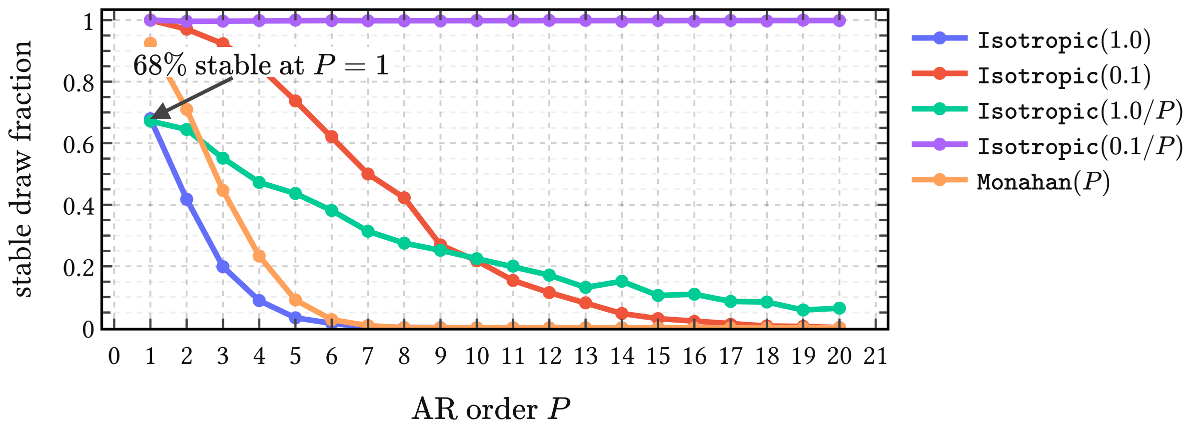


Figure 9 | **Probability that a prior draw lies in the stable AR region $\mathcal{A}_P$** for different values of λ with the isotropic prior (26). Stable prior mass collapses quickly under isotropic Gaussian AR priors, unless the $\lambda \propto 1/P$ rule is used with a $O(0.1)$ proportionality constant; the Monahan prior (36) performs underwhelmingly.

This figure also reveals that the proportional constant in the $\lambda \propto 1/P$ rule does seem to matter, because a Gaussian prior for $\mathbf{a}$ entangles expected power output and stability. At one extreme, $\lambda \to 0$ yields a filter than is trivially stable but will only produce signals with vanishing power. However, turning up expected power output to $O(1)$ by increasing λ then gradually decreases stability. The `Isotropic`$(0.1/P)$ prior is seen to strike a reasonable balance here. But can't we encode our preference for stability directly?

3.2 The Monahan prior

3.2.1 The Monahan map

A classical result from time series analysis (Monahan, 1984) is that the partial autocorrelation function (PACF) parameters $\boldsymbol{\varphi} = (\varphi_1, ..., \varphi_P)^\top$ provide a smooth bijection between the hypercube $(-1, 1)^P$ and the stability region $\mathcal{A}_P$:

$$T : (-1, 1)^P \to \mathcal{A}_P, \quad \mathbf{a} = T(\boldsymbol{\varphi}). \tag{33}$$

The map T is defined by the so-called Levinson recursion which we illustrate below in Figure 10, but whose details are unimportant for our purpose. Its main value is that drawing $\varphi_p \overset{\text{iid}}{\sim}$ `Uniform`$(-1, 1)$ produces only stable AR coefficient vectors. The push-forward of this uniform through T defines a density

$$g(\mathbf{a}) = 2^{-P} \; |\det \nabla_{\mathbf{a}} T^{-1}(\mathbf{a})|, \quad \mathbf{a} \in \mathcal{A}_P, \tag{34}$$

on the stability region, where 2^{-P} is just the density of the uniform base measure on $(-1, 1)^P$. This g is the natural "uniform over stable filters" measure: it assigns equal mass to equal volumes in PACF space, mapped into coefficient space by T.

Since tractable inference in AR models usually requires Gaussian priors, which (34) is not, the best we can do then is to approximate g by a Gaussian $p(\mathbf{a}) = \mathtt{Normal}(\boldsymbol{\mu}_P, \mathbf{S}_P)$ that envelops $\mathcal{A}_P$ in some optimal way.

3.2.2 Deriving the prior

We want the Gaussian to encompass $g(\mathbf{a})$, such that it is neither overconfident about any subregion of $\mathcal{A}_P$, nor wasteful in placing mass far outside it. We thus minimize $D_{\mathrm{KL}}\left(g \parallel \mathtt{Normal}(\boldsymbol{\mu}_P, \mathbf{S})_P\right)$ over all Gaussians and solve for $\boldsymbol{\mu}_P$ and $\mathbf{S}_P$. This is a standard convex problem whose unique solution is the moment-matched Gaussian:

$$\boldsymbol{\mu}_P = \mathbb{E}_g[\mathbf{a}] = \mathbf{0}, \quad \mathbf{S}_P = \mathbb{E}_g\left[\mathbf{a}\mathbf{a}^\top\right]. \tag{35}$$

Surprisingly, the mean is identically zero by the sign-symmetry of the uniform PACF law and the fact that the Monahan map is odd in each coordinate. The Monahan prior is therefore

$$\mathtt{Monahan}(P) \coloneqq \mathtt{Normal}(\mathbf{0}, \mathbf{S}_P), \tag{36}$$

where $\mathbf{S}_P$ captures the full geometry of the stability region in P dimensions within the approximation of a Gaussian family.

$\mathbf{S}_P$ has no closed form but is easy to estimate by Monte Carlo: draw N samples $\boldsymbol{\varphi}^{(n)} \overset{\text{iid}}{\sim} \mathtt{Uniform}(-1,1)^P$, compute $\mathbf{a}^{(n)} = T\left(\boldsymbol{\varphi}^{(n)}\right)$, and average the outer products,

$$\hat{\mathbf{S}}_P = \frac{1}{N}\sum_{n=1}^{N} \mathbf{a}^{(n)}\mathbf{a}^{(n)\top} \to \mathbf{S}_P. \tag{37}$$

This converges at standard $\mathcal{O}\left(N^{-1/2}\right)$ rate and can be precomputed for any desired P. At first sight this all appears promising: the Monahan map guarantees stability by construction, and (36) provides a convenient, tractable prior for IKLP that one might hope captures the shape of $\mathcal{A}_P$ sufficiently well.

3.2.3 Why it does not work

Sampling from $\mathtt{Monahan}(P)$, however, leads to a rapidly vanishing fraction of stable filters as P increases, as shown by the last curve in Figure 9. The issue is geometric: g is supported entirely on $\mathcal{A}_P$, which is a highly non-elliptical subset of $\mathbb{R}^P$, while any nondegenerate Gaussian has full support on $\mathbb{R}^P$. Figure 10 illustrates this in one and two dimensions.

What if we tuck $\mathtt{Monahan}(P)$ inside the support of $g(\mathbf{a})$ and make sure every draw leads to a stable filter? This would be done by minimizing the reverse projection $D_{\mathrm{KL}}(\mathtt{Normal}(\boldsymbol{\mu}_P, \mathbf{S}_P) \parallel g)$. Unfortunately, it is unbounded for any nondegenerate Gaussian since g assigns zero density outside $\mathcal{A}_P$. There is thus no Gaussian approximation that is simultaneously tractable and faithful to the stability constraint: the stable set is simply too irregular to be well-approximated by an ellipsoid.

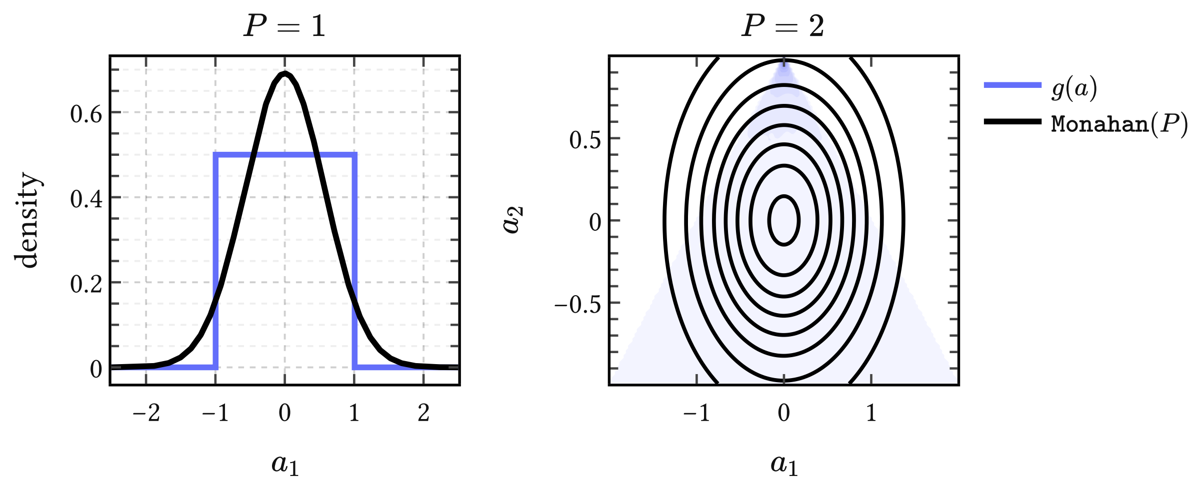


Figure 10 | **Geometry mismatch.** For $P = 1$ the Monahan map is just $a_1 = \varphi_1$ and $\mathcal{A}_1 = (-1, 1)$. For $P = 2$ it gives $(a_1, a_2) = (\varphi_1(1 - \varphi_2), \varphi_2)$, so uniform PACF draws induce a density $g(\mathbf{a})$ supported on the triangular stable region. The Monahan prior $\texttt{Monahan}(P)$ matches the first two moments of g, but it cannot reproduce the bounded, non-elliptical shape of that support faithfully, and mass leaks out of the region of stable filters.

3.3 Spectral shaping priors

Setting worries about power and stability aside for a moment, suppose we expect that the AR filter has a resonance near 500 Hz, or that its spectrum rolls off at a particular rate, or that it has a formant structure typical of a particular vowel. All of these are relevant for acoustic phonetics. How can we “fold” such beliefs into a the first and second moments of a simple Gaussian prior?

The spectral shaping priors that we introduce in this section tackle this problem in some generality. We present a recipe for encoding “spectral beliefs” into the prior mean of $p(\mathbf{a})$ through a single closed-form projection (50). This works by shifting the burden on the user to encode their desired *spectral features* as *roots of* $A(z)$, which are then imposed on average.

3.3.1 Spectral features as root constraints

Features of an AR filter’s spectral envelope $|H(e^{i\omega})|^2 = 1/\,|A(e^{i\omega})|^2$ can be traced to the pole structure of $H(z)$, or, equivalently, roots of $A(z)$. That is, controlling where $A(z)$ has its roots is an operational handle on the spectral shape of the filter.

For example, a conjugate pair of roots close to the unit circle at angle ω_0 produces a resonance peak at that frequency; the closer the roots sit to the unit circle, the sharper the peak. A real root near $z = 1$ boosts low frequencies and therefore induces negative spectral tilt, which is exactly the speech-like case of interest. These and other examples of spectral features are collected in Table 2.

One interesting way to encode such spectral features is through *polynomial divisibility*. Saying “we want $A(z)$ to have a root at z_0” is the same as saying “the factor $(1 - z_0^{-1} z)$ divides

Spectral feature	$Q(z)$	deg
Resonance at (ρ, ω_0)	$1 - 2\rho\cos(\omega_0)z + \rho^2 z^2$	2
Real pole at ρ	$1 - \rho z$	1
Negative spectral tilt $-6k$ dB/oct	$(1 - \rho z)^k$	k
Seasonal pole of period s	$1 - z^s$	s
Repeated resonance at (ρ, ω_0)	$\left(1 - 2\rho\cos(\omega_0)z + \rho^2 z^2\right)^k$	$2k$

Table 2 | **Spectral features and their polynomial encodings.** Each feature corresponds to a polynomial $Q(z)$ whose roots impose the desired pole structure on $|H(z)|^2 = 1/\ |A(z)|^2$.

$A(z)$." A collection of desired roots defines a polynomial $Q(z)$, and the spectral constraint becomes

$$Q(z) \mid A(z), \tag{38}$$

which is read as "Q divides A". For instance, a resonance near frequency ω_0 with damping $1 - \rho$ corresponds to the conjugate pair $z = \rho e^{\pm i\omega_0}$, giving the quadratic

$$Q(z) = 1 - 2\rho\cos(\omega_0)z + \rho^2 z^2. \tag{39}$$

Generalizing, assume $K \geq 1$ features are desired simultaneously. The individual $\{Q_k(z)\}_{k=1}^K$ are combined into a single *constraint polynomial*

$$M(z) = \texttt{lcm}\{Q_1(z), ..., Q_K(z)\}. \tag{40}$$

Over $\mathbb{C}$ this amounts to collecting the union of desired roots with their maximum multiplicities, and the effective constraint degree is $L_{\text{eff}} = \deg M$. For example, a prior that expects both (1) a resonance and (2) a 18 dB/octave negative rolloff uses $M(z) = Q_1(z) \cdot Q_2(z)$ with $L_{\text{eff}} = 2 + 3 = 5$, where the degrees can be read off the last column of Table 2.

3.3.2 Divisibility is linear

The observation that makes (38) a tractable ansatz for encoding inductive bias towards given spectral features is that polynomial divisibility is a *linear* condition on the coefficients of A. $M(z) \mid A(z)$ if and only if the remainder of the polynomial division $A \bmod M$ vanishes, and this remainder depends linearly on $\mathbf{a}$.

More precisely, for a constraint polynomial $M(z)$ of degree L_{eff}, there exists a matrix $\mathbf{F} \in \mathbb{R}^{L_{\text{eff}} \times P}$ and a vector $\mathbf{c} \in \mathbb{R}^{L_{\text{eff}}}$ such that the remainder

$$\mathbf{f}(\mathbf{a}; M) \coloneqq \mathbf{F}\mathbf{a} + \mathbf{c} = \mathbf{0} \quad \Leftrightarrow \quad M \mid A. \tag{41}$$

The matrix $\mathbf{F}$ and offset $\mathbf{c}$ are determined by the coefficients of $M(z)$ and the monic leading term $a_0 = 1$ of $A(z)$. The set of AR(P) coefficient vectors compatible with the constraint is therefore an affine subspace of $\mathbb{R}^P$ of dimension $P - L_{\text{eff}}$.

Worked example To see this concretely, consider an AR(4) polynomial

$$A(z) = 1 - a_1 z - a_2 z^2 - a_3 z^3 - a_4 z^4 \tag{42}$$

and suppose we want to encode a resonance at (ρ, ω_0) via the quadratic $Q(z) = 1 - cz + dz^2$ where $c = 2\rho\cos(\omega_0)$ and $d = \rho^2$. Divisibility means there exists a quotient $R(z) = 1 - r_1 z - r_2 z^2$ such that $A(z) = Q(z)R(z)$. Expanding and matching the coefficients of z, z^2, z^3, z^4 gives

$$a_1 = c + r_1, \qquad a_2 = r_2 - cr_1 - d, \tag{43}$$

$$a_3 = dr_1 - cr_2, \quad a_4 = dr_2. \tag{44}$$

The last two equations pin down the quotient: $r_2 = \frac{a_4}{d}$ and $r_1 = \frac{a_3 + ca_4/d}{d}$. Substituting back into the first two gives two linear equations in $(a_1, ..., a_4)$ alone,

$$\underbrace{\begin{pmatrix} 1 & 0 & -\frac{1}{d} & -\frac{c}{d^2} \\ 0 & 1 & \frac{c}{d} & \frac{c^2}{d^2} - \frac{1}{d} \end{pmatrix}}_{\mathbf{F}} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} + \underbrace{\begin{pmatrix} -c \\ d \end{pmatrix}}_{\mathbf{c}} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \tag{45}$$

from which the matrix $\mathbf{F}$ and vector $\mathbf{c}$ in (41) are read off. The resonance constraint thus carves out a $(P - L_{\text{eff}}) = 2$-dimensional affine subspace of $\mathbb{R}^4$.

3.3.3 The soft spectral shaping prior

We can now state derive the soft spectral shaping prior for $\mathbf{a}$. Let $p_0(\mathbf{a}) = \texttt{Normal}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$ represent our initial state of knowledge of the AR coefficients and let $M(z)$ be the constraint polynomial (40) encoding the desired spectral features. For example, the assignments

$$p_0(\mathbf{a}) \coloneqq \texttt{Isotropic}(0.1/P), \quad M(z) \coloneqq Q(z), \tag{46}$$

where $Q(z)$ is the resonance defined in (39), express a state of knowledge where we expect the AR filter to have unit power output, be stable, and have a single resonance.

We now update $p_0(\mathbf{a})$ to $p^*(\mathbf{a})$ by imposing the new piece of information that the statistic $\mathbf{f}(\mathbf{a}; M)$ has zero expectation under the new state of knowledge expressed by p^* – that is,

$$\mathbb{E}_{p^*}[\mathbf{f}(\mathbf{a}; M)] = \mathbf{0}. \tag{47}$$

Knuth & Skilling (2012) have shown that the unique variational potential for updating information is the Kullback-Leibler divergence, so our task is to

$$\text{find} \quad p^* = \operatorname*{argmin}_p D_{\text{KL}}(p \parallel p_0) \quad \text{given} \quad \mathbb{E}_p[\mathbf{f}(\mathbf{a}; M)] = \mathbf{0}. \tag{48}$$

Since $p_0(\mathbf{a})$ is Gaussian and $\mathbf{f}(\mathbf{a}; M) = \mathbf{F}\mathbf{a} + \mathbf{c}$ is linear in $\mathbf{a}$, we can solve this in closed form:

$$p^*(\mathbf{a}) = \texttt{Normal}(\boldsymbol{\mu}_M, \boldsymbol{\Sigma}_M), \tag{49}$$

which is the unique normalized density closest to p_0 under the constraint (47). The solution keeps $\mathbf{\Sigma}_0$ unchanged, so $\mathbf{\Sigma}_M = \mathbf{\Sigma}_0$, but it does shift the mean to

$$\boldsymbol{\mu}_M = \boldsymbol{\mu}_0 - \mathbf{\Sigma}_0\mathbf{F}^\top\left(\mathbf{F}\mathbf{\Sigma}_0\mathbf{F}^\top\right)^{-1}(\mathbf{F}\boldsymbol{\mu}_0 + \mathbf{c}). \tag{50}$$

Wrapping up, we write the resulting prior formally as

$$\texttt{SoftSpectral}(\boldsymbol{\mu}_0, \mathbf{\Sigma}_0; M) := \texttt{Normal}(\boldsymbol{\mu}_M, \mathbf{\Sigma}_0) \equiv p^*(\mathbf{a}), \tag{51}$$

and refer to it as the *soft spectral shaping prior.*

An example is shown in the left and middle panels of Figure 11, where we encode three resonances that have center frequency and bandwidth typical for the first three vowel formants. The prior spectrum induced by the base prior (left panel) has no discernable resonance structure. The middle panel shows the prior spectrum obtained after updating our state of information per (48). We can see the three resonances clearly imprinted, but what is more interesting is that the prior is noncommital with respect to their actual locations: the zoom on the F_1 region shows clearly that the actual peak locations "happen around" the mean: this is the maximum entropy principle at work. There is also some interesting tail structure which indicates that the prior is anticipating extra resonances at higher frequencies.

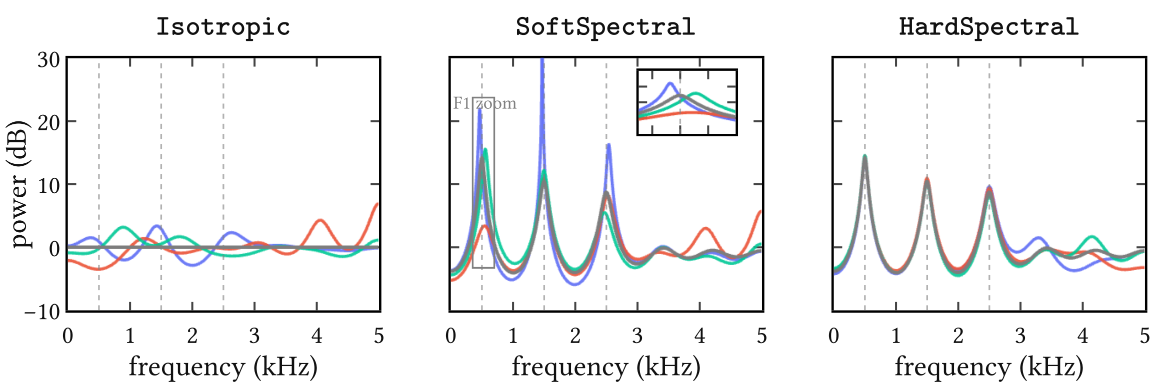


Figure 11 | **Example of imposing spectral features** on an isotropic base prior leading to the soft and hard spectral shaping priors from (51) and (54). We used a constraint polynomial $M(z) = Q_1(z) \cdot Q_2(z) \cdot Q_3(z)$ to encode the spectral features for this example. These are three resonances at $(F_1, F_2, F_3) = (500, 1500, 2500)$ Hz as indicated by the dashed vertical lines. Each panel shows the spectrum mean (grey) together with three samples (colors) of the associated prior. From left to right, these are (1) the base prior $p_0(\mathbf{a}) := \texttt{Isotropic}(0.1/P)$ with $P = 12$; (2) the soft spectral shaping prior $\texttt{SoftSpectral}(\mathbf{0}, (0.1/P)\mathbf{I}_P; M)$ with an inset that zooms in on the F_1 region; and (3) the hard spectral shaping prior $\texttt{HardSpectral}(\mathbf{0}, (0.1/P)\mathbf{I}_P; M)$.

3.3.4 The hard spectral shaping prior

If one prefers to enforce divisibility $M \mid A$ exactly rather than in expectation – for instance when a particular formant location is known with certainty – this can derived to by solving a related D_{KL} problem. However, since we already know $p^*(\mathbf{a})$ to be Gaussian, we can just condition it on the observation

$$\mathbf{f}(\mathbf{a}; M) = \mathbf{0}. \tag{52}$$

Then the standard conditioning formula gives the hard version of (51): the same mean shift, but the covariance is additionally deflated along the constrained directions,

$$\boldsymbol{\Sigma}_M = \boldsymbol{\Sigma}_0 - \boldsymbol{\Sigma}_0 \mathbf{F}^\top \left(\mathbf{F} \boldsymbol{\Sigma}_0 \mathbf{F}^\top\right)^{-1} \mathbf{F} \boldsymbol{\Sigma}_0, \tag{53}$$

and we write the *hard spectral shaping prior* as

$$\texttt{HardSpectral}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0; M) := \texttt{Normal}(\boldsymbol{\mu}_M, \boldsymbol{\Sigma}_M). \tag{54}$$

Since we are rarely certain of spectral features in advance, this prior might be overconfident so less useful for applications.

An example is shown in the last panel of Figure 11. Compared to the soft spectral shaping prior spectra (middle panel), it is seen that the hard spectral shaping prior collapses the uncertainty around the first three resonances, which is allowed to increase again towards the tail end of the spectrum.

3.4 Modeling or expanding?

Jaynes (1991) recounts in a memorable passage the confusion of the renowned experimental physicist R. W. Wood, who in 1911 proposed optical experiments to determine whether the sidebands $\omega \pm \nu$ in the amplitude-modulated wave

$$(1 + m \cos \nu t) \cos \omega t = \cos \omega t + \frac{m}{2} \cos(\omega + \nu)t + \frac{m}{2} \cos(\omega - \nu)t \tag{55}$$

are “physically real frequencies actually present,” or whether “there is in reality only one frequency ω, but with varying amplitude.” Today it is clear that the above is a mathematical identity; the two descriptions are two ways of saying the same thing, and “no experiment could have any bearing on it,” as Jaynes wrote.

$\mathrm{AR}(P)$ filters raise similar questions. The polynomial $A(z)$ has P roots, and each conjugate pair near the unit circle produces a resonance peak in the spectral envelope $|H(e^{i\omega})|^2$. Now, the vocal tract genuinely has resonances – the formants $F_1, F_2, F_3, \ldots$ with their bandwidths $B_1, B_2, B_3, \ldots$ – and these are as physically real as anything in acoustics: they correspond to standing wave patterns in the cavities of the vocal tract and show up as broad peaks in the spectrum.

When P is chosen small, say $P \approx 2N_F$ where N_F is the number of formants in the analysis bandwidth, each conjugate pole pair tends to line up with one formant, and the correspondence is fairly direct. In this regime, $A(z)$ is acting as a *physical model* of the vocal tract, and it makes sense to constrain it: stability is a physical requirement (the vocal tract is a passive resonator), spectral tilt reflects the glottal source coupling, and the formant locations themselves are knowable a priori within a vowel class (Peterson & Barney, 1952).

But this is not the only way to read $A(z)$. In signal processing practice, P is routinely set much larger than $2N_F$ – the well-known rule of thumb $P \approx f_s/1000 + 2$ for a sampling rate f_s gives $P = 2$ at 10 kHz, which is generous enough that many poles will have no individual formant counterpart. These “extra” poles shape spectral tilt, model subglottal coupling, smooth over source harmonics, or simply approximate fine spectral structure that no small set of resonances could capture on its own.[10] In this regime, $A(z)$ is no longer a model but just a *rational expansion* of whatever the true transfer function happens to be – a Padé-type approximant whose individual poles are artifacts of the expansion order, not of the physics (Press et al., 1992).

This distinction underlies frequent confusions between poles and formants in acoustic phonetics (Titze et al., 2015). It also matters from the Bayesian point of view. If we are *modeling*, then encoding formant structure into the prior is encoding real physics and we should expect it to help inference. If we are *expanding*, then the same prior could easily turn into a straitjacket on what is essentially a flexible basis, and we should expect it to hurt inference. These two situations map cleanly on whether we assume informative or noninformative priors.

The same tension applies to the concept of stability itself. In the modeling regime, stability is a genuine physical constraint and enforcing it through the prior is well motivated. But in the expansion regime, a root landing slightly outside the unit circle need not signal physical instability – it may simply mean that the rational approximation of order P is locally imperfect, and penalizing it amounts to penalizing the approximation for being an approximation.

3.4.1 Modeling *and* expanding?

Where does the soft spectral shaping prior (51) stand on all this? In theory at least, rather well, because it operates at what one might call the *suggestion* level.

The constraint polynomial $M(z)$ encodes beliefs about the few poles that *are* likely to have physical counterparts – the formants F_1, F_2, F_3 with their bandwidths, perhaps a spectral tilt factor – and (50) shifts the prior mean to a coefficient vector $\boldsymbol{\mu}_M$ that is compatible with these features in expectation. But note that the covariance $\boldsymbol{\Sigma}_0$ is left untouched. Indeed, the remaining $(P - L_{\text{eff}})$ degrees of freedom, corresponding to the “expansion” poles that have

[10] A single broad formant, for instance, can easily be shaped by two or three poles acting together rather than a single conjugate pair (Van Soom & de Boer, 2021). Conversely, a pole may sit far from the unit circle and contribute little more than a nearly flat spectral factor. The correspondence between poles and formants is thus many-to-many, not one-to-one, except approximately in the parsimonious regime.

no individual physical meaning, are entirely free to go wherever the data require. So we are simultaneously modeling *and* expanding.

Nevertheless, even this soft, suggestive encoding of spectral structure may not always help. The $\lambda \propto 1/P$ scaling rule derived in (32) keeps expected power controlled under an isotropic prior, and does so without imposing any spectral shape at all. Whether the soft spectral shaping prior (51) actually improves inference over this simple baseline – or whether the mean shift inadvertently constrains the expansion poles in ways that hurt the fit – is ultimately an empirical question that depends on the data, the phonation type, and the model order P.

3.5 Summary

The IKLP machinery from the previous chapter requires us to choose a prior $p(\mathbf{a})$ for the AR coefficients. In this chapter we discuss three families of such priors, each attempting to encode progressively more structure about the vocal tract filter.

We began with the *isotropic prior* and showed that it entangles two things one would rather control independently: expected output power, which requires $\lambda \propto 1/P$ to stay bounded, and stability, which tends to collapse with P. In practice, the choice $\lambda := 0.1/P$ appears to be a good comprise, however.

The *Monahan prior* attempts to address the stability issue in a more principled way by approximating the uniform distribution over stable AR coefficient vectors, but the geometry of the stability region is too irregular to be faithfully captured by any Gaussian, and the attempt does not survive increasing P.

The *soft* and *hard spectral shaping priors* take a different approach entirely: rather than constraining where the coefficients may live, they encode what the resulting spectrum should roughly look like. These priors answer the question of how to anticipate complex speech-like spectra with something as simple as a Gaussian prior.

The proposed construction rests on the idea that an operational handle on "$A(z)$ should have certain roots" is given by the concept of polynomial divisibility. That is, we can encode the desired spectral features as roots of $A(z)$, and ask that they divide the latter "on average." This turns out to be a linear constraint on the coefficients $\mathbf{a}$, and we can solve the associated D_{KL} update in closed form. We present soft and hard flavors of the principle, where the former merely suggests spectral features and the latter insists on them.

This is, as far as we know, a new possibility to have it both ways in general AR modeling: *model* what you know, *expand* where you don't. The soft and hard spectral shaping priors have broad applicability beyond speech processing and can be used for modeling any time series, whether finance or EEG signals. Because they are derived from a well-posed maximum entropy problem, they form the bridge between work in the literature on *model-based* (informative) priors on the one hand, and *expansion-based* (uninformative) priors on the other (Cuevas et al., 2021; Gibson, 2018; Sørbye & Rue, 2017).

4 From parametric to nonparametric glottal flow models

Glottal flow models (GFMs) describe the source signal $u(t)$ as it drives the vocal tract during voiced speech. These models operate in the time domain because the delicate phase characteristics of the *glottal cycle* are an integral part of vocal communication.[11]

Decades of empirical work have over time produced a "model zoo" of GFMs: handcrafted waveforms of $u(t)$ with handfuls of carefully chosen parameters. Because these are *parametric* models, they all share the same basic trait: interpretable, but inevitably inflexible (Schleusing, 2012). Their parametric nature limits the range of time domain information they can encode. This motivates the construction of a family of *nonparametric* models, which are more expressive as they are able to grow "parametric capacity" depending on the data at hand.

With the many glottal flow models to follow, it is all too easy to lose connection to the actual physical phenomenon being studied. After a short look at the glottal cycle, we begin with a review of the classic Liljencrants-Fant model. Then we revisit the parametric piecewise polynomial models and generalize them to the nonparametric regime, where they are identified with the *arc cosine kernel* of Cho & Saul (2009). The latter is argued to combine the advantages of both worlds: the interpretability of parametric models with the flexibility of nonparametric ones. This then sets the stage for Chapter 5, which derives the *(quasi)periodic arc cosine kernel* as a learnable surrogate model of synthetic glottal flow data.

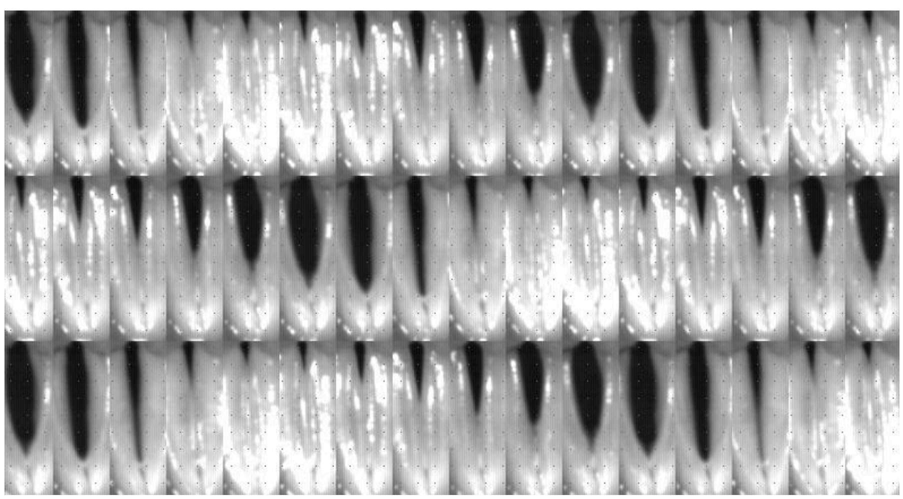


Figure 12 | **A look at the glottal cycle** via laryngeal stroboscopy of the vocal folds. Each frame here is a picture taken at intervals spanning different cycles to give the illusion of steady-state behavior. The glottis is the opening between the folds. After Chen (2016); Jopling (2009).

[11]For example, plosives: glottal stops are micro-events at the millisecond level that underlie semantic information two orders of magnitude higher up the scale. The timing of the vocal fold movement is so precisely controlled by our brain that the slightest deviations are studied as biomarkers for neurodegenerative diseases like Parkinson's (Ma et al., 2020).

The glottal cycle is the periodic opening and closing of the vocal folds, depicted in Figure 12. During one *pitch period*, a single cycle is completed, in this case spanning $T = 6$ ms. GFMs describe the rate of airflow $u(t)$ [typically expressed in cm³/s] through the opening between the vocal folds called the *glottis*, visible as a black slit in Figure 12. Importantly, it is not at maximum glottis aperture that the acoustic output is greatest; it is at the sudden *glottal closure instant* (GCI), as first shown by Miller (1959) in modern literature. At this point, the moving air column in the vocal tract is abruptly interrupted, and kinetic energy is converted efficiently into acoustic energy which then excites the vocal tract much like plucking a harp or clapping in a resonant room (Chen, 2016, Section 4.6). The sharpness of this transition governs much of the clarity and perceived strength of voiced speech (Fant, 1979). Thus, efficiency demands that the glottal cycle be strongly asymmetric: a slow buildup of flow during the open phase, followed by a rapid, almost impulsive closure to the closed phase.

4.1 The Liljencrants-Fant model

A model that has been very useful in describing how $u(t)$ varies during the glottal cycle is the *Liljencrants-Fant (LF) model* proposed by Fant et al. (1985). It has been the GFM of choice for many joint-inverse filtering approaches[12] but also been studied in its own right.[13]

The model states that $u'(t)$ in a single pitch period of length T consists of three parts, or phases. The *open phase* (O) is modeled as a rising and falling exponential sinusoid part, and transitions to the *closed phase* (C) via an exponential return during the *return phase* (R). The latter is thus part of (C). Mathematically, this is defined piecewise as

$$u'_{\mathrm{LF}}(t) = \begin{cases} -E_e e^{a(t-t_e)} \frac{\sin(\pi t/t_e)}{\sin(\pi t_e/t_p)} & 0 \leq \ t \leq t_e \qquad \text{(O)} \\ -\frac{E_e}{\varepsilon T_a}\left(e^{-\varepsilon(t-t_e)} - e^{-\varepsilon(t_c-t_e)}\right) & t_e < t \leq t_c \qquad \text{(C, R)} \\ 0 & t_c \leq t \leq T \qquad \text{(C)} \end{cases} \tag{56}$$

where $\boldsymbol{\theta}_{\mathrm{LF}} = \left\{E_e, t_p, t_e, t_c, T_a, T\right\}$ are the six model parameters explained below in (59) and $\{a, \varepsilon\}$ are determined implicitly from the *closure constraint*

$$\int_0^T u'(t)\,\mathrm{d}t = 0 \quad \text{such that} \quad u(0) = u(T), \tag{57}$$

which holds for any GFM, and the continuity constraint

$$\lim_{\substack{t \to t_e \\ <}} u'(t) = \lim_{\substack{t \to t_e \\ >}} u'(t) \tag{58}$$

which is peculiar to GFMs with smooth return phases rather than hard, discontinuous GCI events (Veldhuis, 1998). Note that like most GFMs, the LF model is defined in $u'(t)$ not $u(t)$

[12] See **Joint source-filter optimization** (page 9).

[13] See Degottex (2010, Section 2.4) for a discussion. Research into the LF model falls mainly into study of its spectral characteristics (Doval et al., 2006) and (re)parametrization, notably Fant (1995); Perrotin et al. (2021).

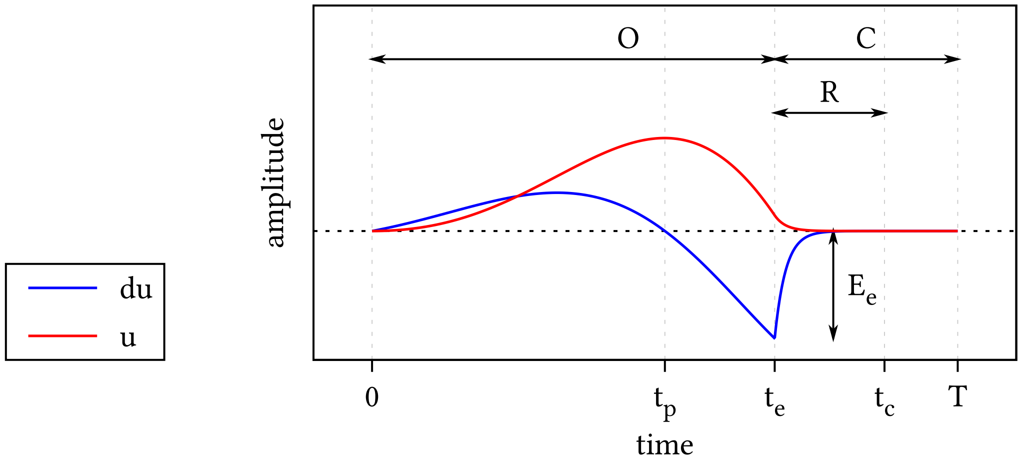


Figure 13 | **The Liljencrants-Fant model** for —— $u'_{\mathrm{LF}}(t)$ and —— $u_{\mathrm{LF}}(t)$ during a single period of length T. The open phase (O), closed phase (C) and return phase (R) are marked at the top and are determined by the changepoints $\{t_p, t_e, t_c, T\}$. The amplitude at peak excitation t_e is E_e. Not shown here is time constant of the return phase T_a, which during (R) determines the steepness of exponential decay towards zero in $u'(t)$.

due to the radiation effect (7), though $u(t)$ can be recovered easily via integration (8). Figure 13 shows the waveforms for $u'_{\mathrm{LF}}(t)$ and $u_{\mathrm{LF}}(t)$ for a typical setting of $\boldsymbol{\theta}_{\mathrm{LF}}$. When varying $\boldsymbol{\theta}_{\mathrm{LF}}$ the LF waveforms trace a family of broad to narrow asymmetric pulses with smooth to abrupt closure encoding positive ($u(t) \geq 0$) glottal flow.

Parameters of the LF model The LF model achieved broad adoption because it strikes a balance between simplicity and empirical accuracy. After several iterations, Fant et al. (1985) arrived at this parametrization:

$$\boldsymbol{\theta}_{\mathrm{LF}} = \begin{cases} E_e & = \min_t u'(t) = u'(t_e) \qquad \text{peak excitation amplitude} \\ t_p & = \operatorname{argmax}_t u(t) \qquad \text{instant of peak glottal flow} \\ t_e & = \operatorname{argmin}_t u'(t) \qquad \text{instant of peak excitation} \\ t_c & \qquad \text{instant of glottal closure (GCI)} \\ T_a & \qquad \text{time constant of the return phase} \\ T & \qquad \text{fundamental period} \end{cases} \tag{59}$$

which turned out to be effective in capturing the glottal cycle in both modal and more extreme phonations.[14] Note that, apart from E_e, which sets the vertical scale, these are all

[14]Actually, Fant et al. present LF as a “four-parameter model”. They ignore E_e and set $T := t_c$ for convenience, thus arriving at a count of four not six as in (59). In their unifying theory of common GFMs, Doval et al. (2006) also collapse T to t_c. It is common too in applications (for example, O Cinneide, 2012). Of course, nothing is really lost because a closed phase of any length can be appended to any GFM by piecewise concatenation. We choose to keep t_c distinct from T explicitly because this becomes significant for short return phases and is experimentally observed to be important, especially in clinical scenarios (Kreiman et al., 2007; Rosenberg, 1971).

time domain parameters which determine *changepoints* in the glottal cycle in Figure 12. These tend to covary significantly and can in fact be regressed into a single convenient parameter R_d (Fant, 1995).

From the modeling perspective, the most important changepoints are t_e and t_c: when these lie close together and return is swift (T_a small and vanishing return phase), the glottal cycle is at peak efficiency and $\tilde{u}(t)$ has broadband spectral energy. This results in a clear voice and sharp features in the speech waveform data $\mathbf{d}$. Conversely, a smooth return (T_a medium to large) results in a longer return phase with t_c tending to T and the GCI becoming ill-defined. The result here is a more mellow voice and more smooth features in $\mathbf{d}$. Both modes are common in voiced speech (Maurer, 2016) and GIF algorithms must be able to handle both cases.

Implementation In practice, the LF model is unwieldy to compute (Gobl, 2017; Veldhuis, 1998). The closure and continuity constraints in (57) and (58) lack an analytical solution, meaning that for each evaluation of $u'(t)$ one must numerically solve for the auxiliary constants $\{a, \varepsilon\}$ by a bisection or root-finding routine. This seems like a high price to pay for an analytical model, and worse, this optimization is brittle. As noted by Perrotin et al. (2021), simplified LF models exist that are perceptually equivalent to LF and easier to work with.

Nevertheless, it remains the de facto standard in parametric source-filter modeling and we therefore use it here. Our code in Appendix C implements the LF equations in `JAX` (Bradbury et al., 2020), so the bisection routines are compiled to native code for a speedup and the entire model is differentiable and easily batcheable. The code also implements four LF parametrizations, including the previously mentioned R_d formulation of Fant (1995).

4.1.1 One of many: the glottal flow "model zoo"

Over the years, a large number of glottal flow models have been proposed in acoustic phonetics. A sample of these is shown in Figure 14, already an impressive lineup by 1986. Despite their visual differences, these models are, essentially, variations on a single theme: they all describe a glottal pulse that tracks the periodic opening and closing of the glottis.

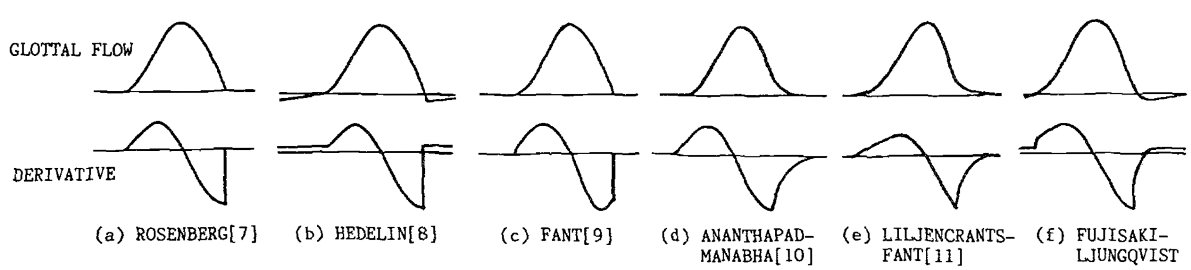


Figure 14 | **A lineup of GF models** back from 1986 (handdrawn?), together with their derivatives (DGFs). These models are visually very similar. From Fujisaki & Ljungqvist.

[15] These are LF, KLGLOTT88 (Klatt & Klatt, 1990), Rosenberg C (Rosenberg, 1971), R++ (Veldhuis, 1998).

Doval et al. (2006) demonstrated that the most common GFMs[15] share a common mathematical structure and can be unified under a single framework; the $\boldsymbol{\theta}_{\mathrm{LF}}$ parameters in (59) can in fact express all of them. These GFMs all model a glottal flow $u(t)$ that is positive or null, continuous, and typically differentiable except at the glottal closure instant (GCI). Within each period, $u(t)$ rises during the opening phase, falls during the closing phase, and returns to zero during the closed phase, possibly through a short return phase that smooths the closure. The derivative $u'(t)$ alternates positive, zero, and negative in the expected order and integrates to zero over one period, satisfying the closure constraint (57). It is continuous in case of a nonempty return phase and discontinuous otherwise (meaning a jump at t_c as in Figure 15), known as hard closure.

Allow partial closure? The strict closure constraint in (57) is of course just a convenient modeling choice and must be expected to be violated to varying degrees in reality, as real folds often leak, especially in breathy or pathological voices. On the one hand, it clearly is a defining feature[16] of the periodic nature of the glottal cycle. On the other, empirical data show the limits of this hard constraint. For example, Kreiman et al. (2007, p. 601) observe that

> in many cases in our data, returning flow derivatives to 0 at the end of the cycle conflicted with the need to match the experimental data and conflicted with the requirement for equal areas under positive and negative curves in the flow derivative. [...] These conflicts between model features, constraints, and the empirical data were handled by abandoning the equal area constraint.[17]

The approach we take in Chapter 5 is to learn a softened version of the closure constraint (57) from data rather than set it a priori. This means we learn both whether the flow typically returns to zero and, if not, to what extent this constraint is plausibly violated.

Allow negative flow? Most GFMs enforce a positive glottal flow and assert $u(t) \geq 0$. But there are exceptions such as Fujisaki & Ljungqvist (1986). They allow for transient negative flow which could represent a lowering of the vocal folds after GCI (presumably when air is sucked back into the lungs). As with the closure constraint, we will learn this effect from data in Chapter 5 and do not forbid $u(t) < 0$ a priori.

[16]The pioneering study on GIF by Miller (1959) had to deconvolve taped speech waveforms with analogue resistor networks and manual optimization. A review article on GIF methods notes the part played by the defining quality of the closure constraint in this search for $u(t)$: "the fine-tuning of the resistors of the inverse network was conducted by searching for such values that yielded zero flow after the instant of glottal closure" (Alku, 2011, p. 626).

[17]That "equal area constraint" is just another name for the closure constraint (57).

Degree d	Name	Reference
0	Triangular pulse model	Alku et al. (2002)
1	—	Titze (2000); Verdolini & Titze (1995)
2	KLGLOTT88	Klatt & Klatt (1990) Chang-Sheng Yang & Kasuya (1998)
3	R++	Veldhuis (1998)
3	FL model	Fujisaki & Ljungqvist (1986)
6	—	Childers & Hu (1994)

Table 3 | **Classic piecewise polynomial models** from the literature. Of these, KLGLOTT88 and R++ remain popular today (Doval et al., 2006).

4.2 Classic piecewise polynomial models

Polynomial GFMs occupy an interesting corner of the "model zoo." They fit into the same piecewise framework we have been using for the LF model, but with each part modeled as a polynomial of degree d. We list several GFMs proposed in the literature in Table 3.

This class of GFMs mainly appeal to simplicity and spectral strength. Piecewise polynomial models admit an analytical solution to the closure constraint while remaining computationally cheap. More importantly, the sharp transitions generated by low-order polynomials have bright spectra with slow decay, which is exactly what it takes to model hard GCIs (Childers & Hu, 1994) and score well on perceptual tests of synthetic speech (Rosenberg, 1971).

To get a feel for these models, we now turn to the simplest case: $d = 0$.

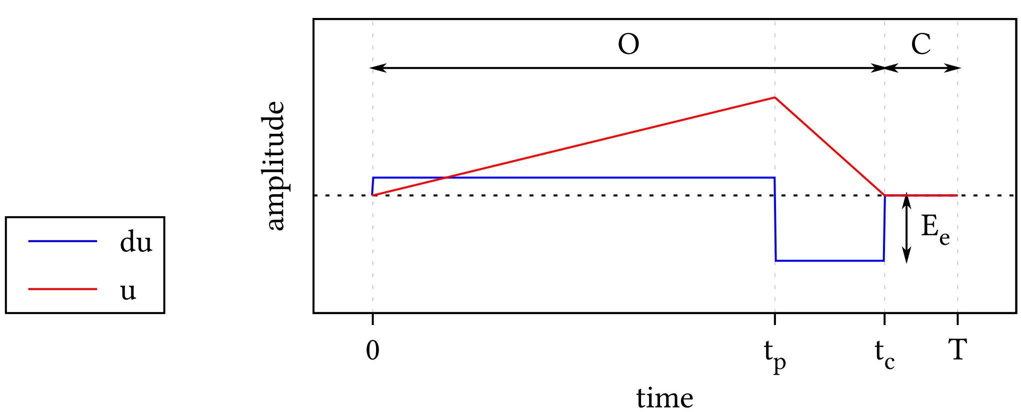


Figure 15 | **The triangular pulse model** for —— $u'_\Delta(t)$ and —— $u_\Delta(t)$ during a single period of length T. The parameters used in this model are identical to Figure 13, except for the lack of a return phase and that the 'instant' of maximum excitation t_e now spans the interval $[t_p, t_c]$.

4.2.1 The triangular pulse model

The simplest polynomial model is the *triangular pulse model* proposed by Alku et al. (2002). It takes $u(t)$ piecewise linear and $u'(t)$ piecewise constant, as shown in Figure 15. Mathematically, the triangular pulse model is given by:

$$u'_{\Delta}(t) = \begin{cases} +E_e\left(\frac{t_c}{t_p} - 1\right) & 0 \ < t \leq t_p \qquad \text{(O)} \\ -E_e & t_p < t \leq t_c \qquad \text{(O)} \\ 0 & t_c < t \leq T \qquad \text{(C)} \end{cases} \tag{60}$$

The model has no return phase and only four parameters: $\boldsymbol{\theta}_{\Delta} = \left\{E_e, t_p, t_c, T\right\}$. Note that the amplitude of the first case in (60) is fixed because the closure constraint (57) removes one degree of freedom. Alku et al. (2002) point out that this enables a difficult-to-measure time domain quantity to be expressed as the ratio of two easy-to-measure quantities in the amplitude domain and exploit this fact to measure the open quotient more robustly.

4.3 Parametric piecewise polynomial models

The triangular pulse model (60) contains two jumps in the derivative $u'_{\Delta}(t)$, so we can write the latter more compactly as a linear combination of two Heaviside functions

$$u'_{\Delta}(t) = \begin{cases} a_1(t-b_1)^0_+ + a_2(t-b_2)^0_+ & 0 \ < t \leq t_c \qquad \text{(O)} \\ 0 & t_c < t \leq T \qquad \text{(C)} \end{cases} \tag{61}$$

where $(t-b)^0_+ = \texttt{Heaviside}(t-b)$ and

$$\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = E_e \begin{pmatrix} \frac{t_c}{t_p} - 1 \\ -\frac{t_c}{t_p} \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} 0 \\ t_p \end{pmatrix}. \tag{62}$$

During the open phase, we can interpret the rewritten triangular pulse model as a *parametric piecewise polynomial model* of *degree* $d = 0$ and *order* $H = 2$. It has parameters $\boldsymbol{\theta}_2 = \{\mathbf{a}, \mathbf{b}\}$ which describe $H = 2$ jumps with amplitudes $\mathbf{a}$ at locations $\mathbf{b}$. Note that in this formulation the amplitudes $\mathbf{a}$ differ from the piecewise amplitudes in (60) but are still linearly dependent given the changepoints $\mathbf{b}$ and t_c.

Continuing, nothing stops us from generalizing both H and d. The open phase part of (61) becomes:

$$u'_H(t) = \sum_{h=1}^{H} a_h(t-b_h)^d_+ \qquad 0 < t \leq t_c \qquad \text{(O)} \tag{63}$$

This is the general parametric piecewise polynomial model of degree d and order H. Again, its parameters $\boldsymbol{\theta}_H = \{\mathbf{a}, \mathbf{b}\}$ are the amplitudes $\mathbf{a} = (a_1, ..., a_H)^\top \in \mathbb{R}^H$ and the changepoints $\mathbf{b} = (b_1, ..., b_H)^\top \in \mathbb{R}^H$. These scale and shift H piecewise monomials of degree d:

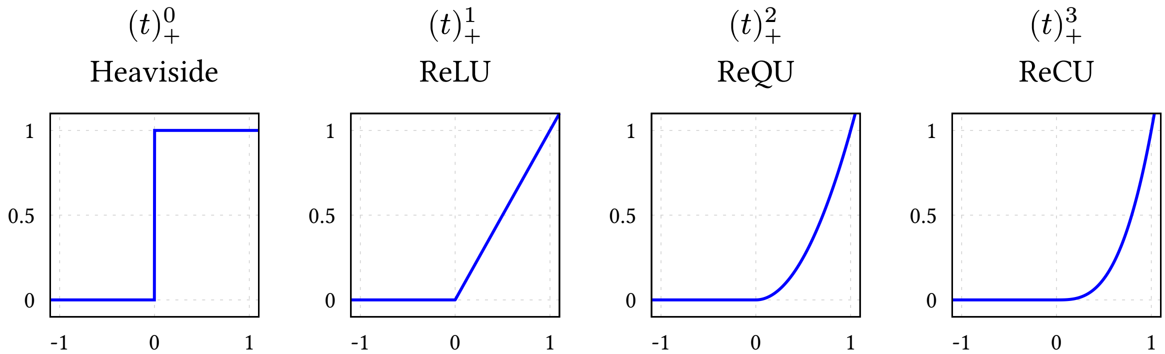


Figure 16 | **The RePU activation functions** —— $(t)^d_+$ for $d \in \{0, 1, 2, 3\}$.

$$(t-b)^d_+ = (t-b)^d \; \texttt{Heaviside}(t-b) = \begin{cases} 0 & t < b \\ (t-b)^d & t \geq b \end{cases} \tag{64}$$

We recognize (64) as the family of activation functions called *rectified power units* (RePUs). These include Heaviside for $d = 0$ and the influential ReLU function for $d = 1$, both shown among higher degree variants in Figure 16. It may sound like a stretch to relate polynomial GFMs to neural nets, but this view of things will pay off greatly in Section 4.4 when we take the limit $H \to \infty$.

4.3.1 Regression view

GFMs ultimately play the role of regression models in the GIF problem. From this point of view, the general parametric model in (63) is just a standard *linear model*, and we will now develop it into a Bayesian regression model following MacKay (1998).

Suppose we observe N time samples $\boldsymbol{t} = \{t_n\}_{n=1}^N$ during one glottal cycle, and fix H basis functions of the form

$$\phi_h(t) = (t - b_h)^d_+, \tag{65}$$

where the changepoints $\mathbf{b}$ and t_c are treated as given. Collecting these into the design matrix $\mathbf{\Phi} \in \mathbb{R}^{N \times H}$ with

$$[\mathbf{\Phi}]_{nh} = \phi_h(t_n), \tag{66}$$

the general parametric model during the open phase becomes simply

$$\mathbf{u}' = \mathbf{\Phi}\mathbf{a}. \tag{67}$$

This is linear in the amplitudes $\mathbf{a}$ given $\mathbf{\Phi}$, so the model is rather constrained at this point. Indeed, $\mathbf{a}$ is currently the only source of variability and power in this model because we assume all other parameters fixed. Which prior then for $\mathbf{a}$ should we choose? If we want to maximize model support while adhering to the known additivity and power constraints, the optimal choice is the canonical Gaussian prior over amplitudes (Bretthorst, 1988):

$$p(\mathbf{a} \mid \mathbf{b}) = \texttt{Normal}(\mathbf{a} \mid \mathbf{0}, \sigma_a^2 \mathbf{I}). \tag{68}$$

Moving on to data space, the induced prior probability of $\mathbf{u}'$ is also Gaussian,

$$p(\mathbf{u}' \mid \mathbf{b}) = \int \delta(\mathbf{u}' - \mathbf{\Phi}\mathbf{a})\, p(\mathbf{a} \mid \mathbf{b})\, \mathrm{d}\mathbf{a} = \texttt{Normal}(\mathbf{u}' \mid \mathbf{0}, \sigma_a^2 \mathbf{\Phi}\mathbf{\Phi}^\top), \tag{69}$$

due to closure of the Gaussian under linear combinations, and

$$\mathbb{E}_{\mathbf{a}|\mathbf{b}}[\mathbf{u}'] = \mathbf{\Phi}\, \mathbb{E}_{\mathbf{a}|\mathbf{b}}[\mathbf{a}] = 0, \tag{70}$$

$$\mathbb{E}_{\mathbf{a}|\mathbf{b}}\left[\mathbf{u}'\mathbf{u}'^\top\right] = \mathbf{\Phi}\, \mathbb{E}_{\mathbf{a}|\mathbf{b}}[\mathbf{a}\mathbf{a}^\top]\, \mathbf{\Phi}^\top = \sigma_a^2 \mathbf{\Phi}\mathbf{\Phi}^\top. \tag{71}$$

Because $\operatorname{rank}(\mathbf{\Phi}\mathbf{\Phi}^\top) \leq H$, prior samples of $\mathbf{u}'$ are intrinsically confined to an H-dimensional subspace of $\mathbb{R}^N$. Assuming $H < N$, we thus see that linear models like (67) are inherently low-rank. Their expressivity grows with H and saturates when H reaches N.

Closure constraint The closure constraint (57) becomes

$$\int_0^T u'_H(t)\, \mathrm{d}t = \sum_{h=1}^H a_h \int_0^{t_c} \phi_h(t)\, \mathrm{d}t = \sum_{h=1}^H a_h r_h = 0, \tag{72}$$

with $r_h = (t_c - b_h)^{d+1}/(d+1)$. Observe that the prior (68) already satisfies the closure constraint in expectation:

$$\mathbb{E}_{\mathbf{a}|\mathbf{b}}\left[\sum_{h=1}^H a_h r_h\right] = \sum_{h=1}^H \mathbb{E}_{\mathbf{a}|\mathbf{b}}[a_h]\, r_h = 0. \tag{73}$$

This motivates setting the prior mean of the $\mathbf{a}$ to zero, otherwise than symmetry of ignorance around $\mathbf{a} = \mathbf{0}$. We can also enforce it:[18]

$$\sum_{h=1}^H a_h r_h = 0 \Longrightarrow p(\mathbf{a} \mid \texttt{closure}, \mathbf{b}) = \texttt{Normal}(\mathbf{a} \mid \mathbf{0}, \sigma_a^2(\mathbf{I} - \mathbf{q}\mathbf{q}^\top)) \tag{74}$$

where $\mathbf{r} = (r_1, ..., r_H)^\top$ and $\mathbf{q} = \frac{\mathbf{r}}{\|\mathbf{r}\|}$. A convenient way to a sample from this updated prior makes use of the fact that this is a rank-one projection:

$$\mathbf{a} = (\mathbf{q}\mathbf{q}^\top)\mathbf{a} + (\mathbf{I} - \mathbf{q}\mathbf{q}^\top)\mathbf{a} \quad \sim \quad p(\mathbf{a} \mid \mathbf{b}) \tag{75}$$

$$\Longrightarrow \mathbf{a}_\perp = \qquad (\mathbf{I} - \mathbf{q}\mathbf{q}^\top)\mathbf{a} \quad \sim \quad p(\mathbf{a} \mid \texttt{closure}, \mathbf{b}). \tag{76}$$

How does this look in data space? By the same calculation (70), marginalizing over $\mathbf{a}$ yields

$$p(\mathbf{u}' \mid \texttt{closure}, \mathbf{b}) = \texttt{Normal}(\mathbf{u}' \mid \mathbf{0}, \sigma_a^2 \mathbf{\Phi}(\mathbf{I} - \mathbf{q}\mathbf{q}^\top)\mathbf{\Phi}^\top). \tag{77}$$

[18]The general solution is given by $\operatorname{argmin}_f D_{\mathrm{KL}}[f(\mathbf{a}) \parallel p(\mathbf{a} \mid \mathbf{b})]$ under the constraint (72), but the closure under linear conditioning of Gaussian distributions allow us to take a shortcut here.

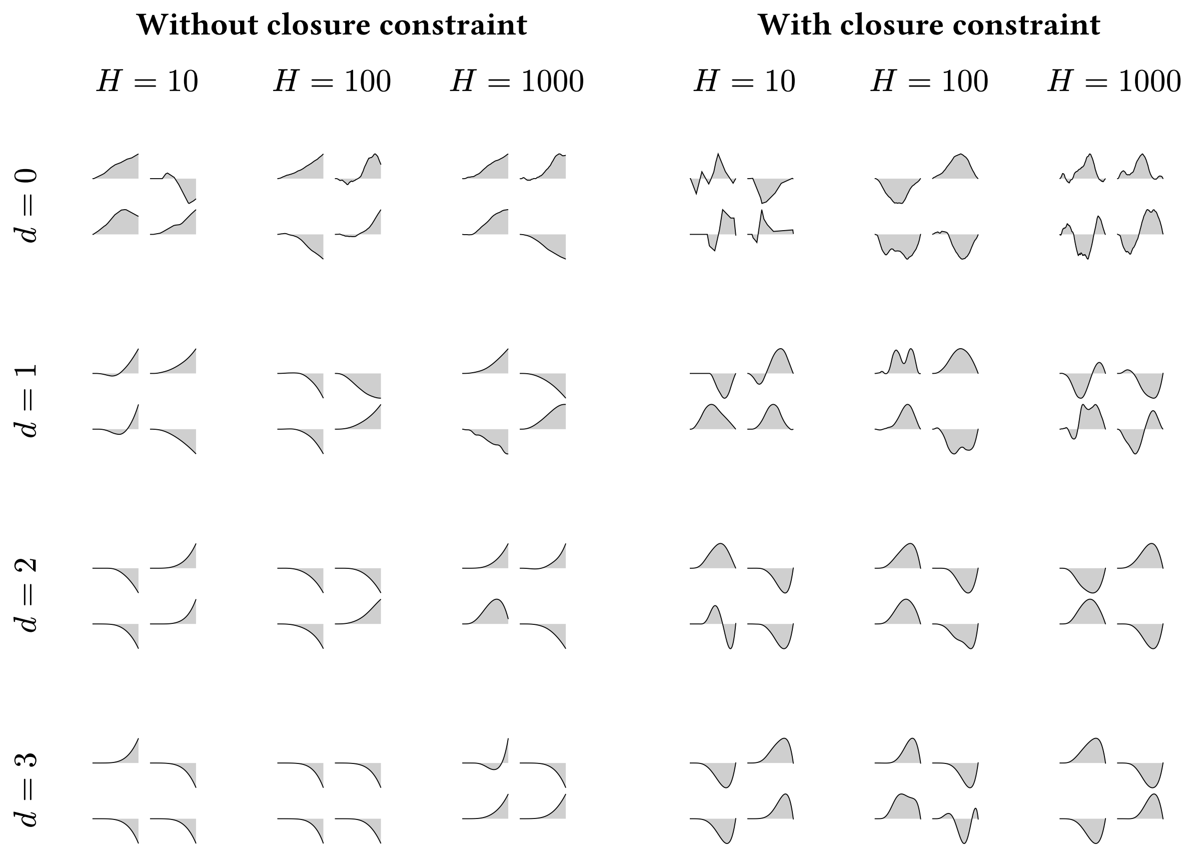
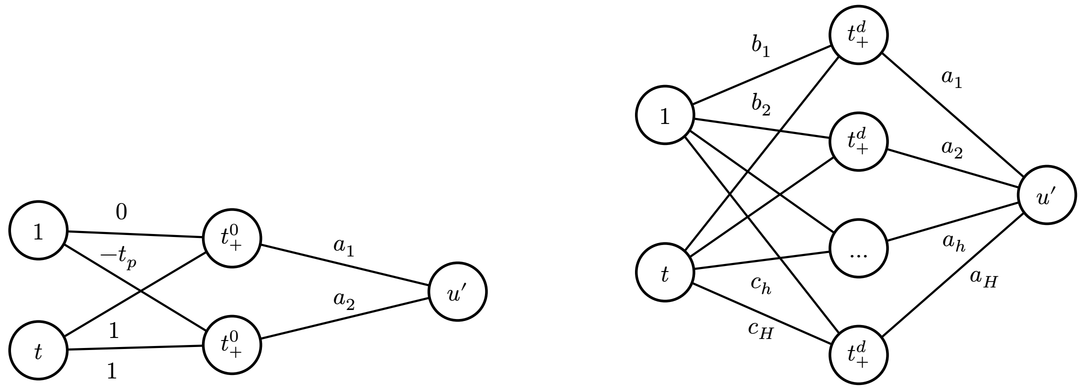


Table 4 | **Samples of $\boldsymbol{u(t)}$** from the general parametric piecewise polynomial model (63) with or without the closure constraint (72). For each combination of degree d and order H, a total of H changepoints $b_h \sim \texttt{Uniform}(0, t_c)$ were drawn randomly, and four instances of $\mathbf{u}'$ conditioned on these were sampled either from (69) [**Without closure constraint**] or from (77) [**With closure constraint**]. Then $u(t) = \int_0^t u'(\tau)\,\mathrm{d}\tau$ was obtained by quadrature of $\mathbf{u}'$.

We can then obtain $\mathbf{u}$ from $\mathbf{u}'$ via integration using (8). Compared to samples from the isotropic prior (69), the anisotropic prior (77) induces glottal flows $u(t)$ that tend to look more pulse-like, as shown in Table 4. Indeed, moving to the bottom right corner, it is seen that increasing H and d tends to produce LF-like pulses out of the box!

Regression with (77) ensures that posterior mass vanishes at solutions $\mathbf{u}$ that violate the closure constraint. This illustrates how linear constraints on the amplitudes $\mathbf{a}$ in the linear model (67) can encode GFM properties at the cost of just a single rank-one downdate per constraint. Moving on from linear models to Gaussian processes, these linear constraints become linear functionals known as *interdomain features*, which may be used to impose structure directly in function space, without touching rank, but more challenging mathematically.

Connection to classic polynomial models Observe that the analytical solution (74) to the closure constraint is valid for any degree $d \geq 0$ and order $H \geq 1$, so each classic polynomial



(a) (b)

Figure 17 | **Neural network view.** (a) The triangular pulse model as a tiny neural network with a single hidden layer, linear readout and t_+^0 activation. (b) The general parametric polynomial model of degree d with order H with hidden weights $\{\mathbf{b}, \mathbf{c}\} \in \mathbb{R}^{2H}$, output weights $\mathbf{a} \in \mathbb{R}^H$ and RePU activation function t_+^d.

model listed in Table 3 can be expressed in our linear model framework given suitable choices of $\mathbf{b}$ and t_c, without specialized derivations.

4.3.2 Neural network equivalence

Figure 17 illustrates that the triangular pulse model of Section 4.2.1 can be seen as a tiny neural network. Likewise, the general parametric piecewise polynomial model (63) can be recast as a single hidden layer of width H with a RePU activation and a linear readout:

$$u'_{\text{NN}}(t) = \sum_{h=1}^{H} a_h (c_h t - b_h)_+^d. \tag{78}$$

Each unit computes $\phi_h(t) = \left(\mathbf{w}_h^\top \mathbf{x}_t\right)_+^d$ with hidden weights $\mathbf{w}_h = (c_h, -b_h)^\top$ and $\mathbf{x}_t = (1, t)^\top$, exactly like a neuron with a bias input. The $3H$ parameters of the neural net are thus $\mathbf{a}$ and $\mathbf{b}$ (as before) and the newly acquired degrees of freedom $\mathbf{c} = (c_1, ..., c_H)^\top$.

Seen from this point of view, a natural symmetry appears. We argued in Section 4.3.1 that in linear regression context the output weights $\mathbf{a}$ are best assigned an uninformative Gaussian prior, so

$$p(\mathbf{a}) = \texttt{Normal}(\mathbf{a} \mid \mathbf{0}, \sigma_a^2 \mathbf{I}). \tag{79}$$

But the neural network equivalence makes very clear that the hidden weights $\{\mathbf{b}, \mathbf{c}\}$ actually play similar roles: they enter linearly inside the activation, so the same argument applies, and we are encouraged to assign Gaussians again:

$$p(\mathbf{b}) = \texttt{Normal}(\mathbf{b} \mid \mathbf{0}, \sigma_b^2 \mathbf{I}), \quad p(\mathbf{c}) = \texttt{Normal}(\mathbf{c} \mid \mathbf{0}, \sigma_c^2 \mathbf{I}). \tag{80}$$

This step is not obvious from the viewpoint of any of the GFMs we've discussed before, where changepoints $\mathbf{b}$ were nonlinear hyperparameters fixed a priori; here they become "active" and "linear" degrees of freedom. Assuming t_c is known, the remaining hyperparameters to describe the open phase are thus the three scales $\boldsymbol{\theta}_{\text{NN}} = \{\sigma_a, \sigma_b, \sigma_c\}$ controlling amplitude, typical changepoint location, and horizontal stretch, respectively.

Turning to data space again and having assigned priors to all parameters, we can now do a full marginalization unlike the conditional in (69). Updating the design matrix $\mathbf{\Phi}$ to

$$[\mathbf{\Phi}]_{nh} = \phi_h(t_n) = (c_h t_n - b_h)_+^d, \tag{81}$$

and assuming the isotropic priors (79) and (80) we have

$$p(\mathbf{u}') = \int \delta(\mathbf{u}' - \mathbf{\Phi}\mathbf{a})\, p(\mathbf{a})\, p(\mathbf{b})\, p(\mathbf{c})\, \mathrm{d}\mathbf{a}\, \mathrm{d}\mathbf{b}\, \mathrm{d}\mathbf{c} \tag{82}$$

$$= \int \texttt{Normal}\big(\mathbf{u}' \mid \mathbf{0}, \sigma_a^2 \mathbf{\Phi}\mathbf{\Phi}^\top\big)\, p(\mathbf{b})\, p(\mathbf{c})\, \mathrm{d}\mathbf{b}\, \mathrm{d}\mathbf{c}. \tag{83}$$

This is a Gaussian mixture with varying covariances (not means) as $\mathbf{\Phi}$ depends on the hidden weights $\{\mathbf{b}, \mathbf{c}\}$. A closed form for this integral exists only for $H \to \infty$ as $p(\mathbf{u}')$ converges to a Gaussian process, as we will see in Section 4.4.

Closure constraint The closure constraint (57) given $\{\mathbf{b}, \mathbf{c}\}$ remains a linear condition on $\mathbf{a}$, so we can define

$$p(\mathbf{a}, \mathbf{b}, \mathbf{c} \mid \texttt{closure}) = p(\mathbf{a} \mid \texttt{closure}, \mathbf{b}, \mathbf{c})\, p(\mathbf{b})\, p(\mathbf{c}), \tag{84}$$

where $p(\mathbf{a} \mid \texttt{closure}, \mathbf{b}, \mathbf{c})$ is identical to (74) but with $r_h = (c_h t_c - b_h)_+^{d+1} / c_h(d+1)$. This prior yields a mixture in data space that is similar to (82):

$$p(\mathbf{u}' \mid \texttt{closure}) = \int \texttt{Normal}\big(\mathbf{u}' \mid \mathbf{0}, \sigma_a^2 \mathbf{\Phi}(\mathbf{I} - \mathbf{q}\mathbf{q}^\top)\mathbf{\Phi}^\top\big)\, p(\mathbf{b})\, p(\mathbf{c})\, \mathrm{d}\mathbf{b}\, \mathrm{d}\mathbf{c}. \tag{85}$$

Here both $\mathbf{q} = \frac{\mathbf{r}}{\|\mathbf{r}\|}$ and $\mathbf{\Phi}$ depend on $\{\mathbf{b}, \mathbf{c}\}$. This integral has no closed form for $H \to \infty$, but we can still impose the closure constraint on (82) indirectly in the spectral domain via interdomain features (Chapter 5).

Connection to LF and other GFMs The neural network view of the parametric polynomial model also clarifies expressivity. It is well known that a single RePU layer of sufficient width can approximate any continuous waveform on a bounded interval (Hornik, 1993). Therefore, with its analytical closure constraint (84) and the ability to represent sharp changepoint behavior, the parametric piecewise polynomial model in the form (78) effectively unifies the entire family of GFMs encountered in the literature, from triangular pulses to LF-type exponentials and sinusoids, given sufficiently large H.

4.4 Nonparametric piecewise polynomial models

The correspondence of piecewise polynomials GFMs with a single hidden layer shows that their expressivity depends mostly on the order H, as model order is equivalent to layer width in the neural network picture. Indeed, *parametric* linear models have a finite representation capacity determined primarily by rank because the modeled function can only explore the subspace spanned by the basisfunctions.[19] *Nonparametric* models grow capacity as the number of datapoints increases; in the case of Gaussian processes, posterior inference given N datapoints is equivalent to Bayesian linear regression with N linearly independent basisfunctions, hence full-rank (Rasmussen & Williams, 2006).

It thus makes sense to consider the limit $H \to \infty$ in our search for more expressive GFMs. Continuing where we left, we already saw in (82) that $p(\mathbf{u}')$ is a nontrivial Gaussian covariance mixture weighed by the isotropic priors defined in (80). This can be made explicit:

$$p(\mathbf{u}') = \int \texttt{Normal}(\mathbf{u}' \mid \mathbf{0}, \sigma_a^2 \mathbf{\Phi}\mathbf{\Phi}^\top)\, p(\mathbf{b})\, p(\mathbf{c})\, \mathrm{d}\mathbf{b}\, \mathrm{d}\mathbf{c} \tag{86}$$

$$= \int \texttt{Normal}(\mathbf{u}' \mid \mathbf{0}, \mathbf{Q})\, p(\mathbf{Q})\, \mathrm{d}\mathbf{Q}, \tag{87}$$

where we defined the push forward

$$p(\mathbf{Q}) = \int \delta(\mathbf{Q} - \sigma_a^2 \mathbf{\Phi}\mathbf{\Phi}^\top)\, p(\mathbf{b})\, p(\mathbf{c})\, \mathrm{d}\mathbf{b}\, \mathrm{d}\mathbf{c} \tag{88}$$

which essentially counts the number of $\{\mathbf{b}, \mathbf{c}\}$ configurations that give rise to a given $\mathbf{Q}$. Thus each random sample of $\{\mathbf{b}, \mathbf{c}\} \in \mathbb{R}^{2H}$ drawn from (80) induces in data space a random covariance matrix $\mathbf{Q} \in \mathbb{R}^{N \times N}$,

$$\{\mathbf{b}, \mathbf{c}\} \mapsto \mathbf{Q} = \sigma_a^2 \mathbf{\Phi}\mathbf{\Phi}, \qquad \mathrm{rank}(\mathbf{Q}) \le H \tag{89}$$

and hence a Gaussian component with weight $p(\mathbf{Q})$ in the mixture (86). Below we will show that by choosing $\sigma_a \propto 1/\sqrt{H}$ and letting $H \to \infty$, the density of states (88) degenerates to

$$p(\mathbf{Q}) \longrightarrow \delta(\mathbf{Q} - \mathbf{K}), \qquad \mathrm{rank}(\mathbf{K}) = N \text{ almost surely}, \tag{90}$$

for some $\mathbf{K} \in \mathbb{R}^{N \times N}$ derived below in (126), causing the mixture to collapse to a single Gaussian and thus completing the transition to an indexed Gaussian process with covariance matrix $\mathbf{K}$.

[19] More precisely: a higher rank naturally enlarges the subspace of attainable functions, but the prior over amplitudes still constrains where probability mass is placed within that subspace. Geometrically, the model explores an H-dimensional ellipsoid like (69) or (77) rather than the entire hyperplane. When data accumulate, the prior constraints are eventually overwhelmed and the model can move freely on that plane. Random kitchen sinks (Rahimi & Recht, 2008) take these two ideas to the extreme: forgo specialized modeling; just use H *random* basis functions with uninformative priors and rely on the fact that an entire H-dimensional hyperplane is reachable with high probability.

4.4.1 Mean and covariance description

While the mixture (86) has no closed form for finite H, progress can be made by computing its mean and covariance as before in (70):

$$\mathbb{E}[\mathbf{u}'] = \mathbb{E}_{\mathbf{b},\mathbf{c}}\left[\mathbb{E}_{\mathbf{a}|\mathbf{b},\mathbf{c}}[\mathbf{u}']\right] = \mathbf{0}, \tag{91}$$

$$\mathbb{E}\left[\mathbf{u}'\mathbf{u}'^{\top}\right] = \mathbb{E}_{\mathbf{b},\mathbf{c}}\left[\mathbb{E}_{\mathbf{a}|\mathbf{b},\mathbf{c}}\left[\mathbf{u}'\mathbf{u}'^{\top}\right]\right] = \sigma_a^2\,\mathbb{E}_{\mathbf{b},\mathbf{c}}[\mathbf{\Phi}\mathbf{\Phi}^{\top}], \tag{92}$$

The mean is zero everywhere due to our zero-mean prior $p(\mathbf{a})$. The covariance matrix, however, is nontrivial. To investigate, define

$$\mathbf{C} := \mathbb{E}\left[\mathbf{u}'\mathbf{u}'^{\top}\right] \in \mathbb{R}^{N\times N} \tag{93}$$

with elements

$$[\mathbf{C}]_{nm} = \sigma_a^2\,\mathbb{E}_{\mathbf{b},\mathbf{c}}\left[\sum_{h=1}^{H}\phi_h(t_n)\phi_h(t_m)\right], \tag{94}$$

where $\phi_h(t) = (c_h t - b_h)_+^d$ for the neural network model $u'_{\text{NN}}(t)$ was defined in (81). Since the $\phi_h(t)$ are statistically equivalent under i.i.d. priors like the ones we adopted in (80), define a single “mother wavelet” *without index h* as

$$\phi(t; b, c) = (ct - b)_+^d, \quad b \sim \texttt{Normal}(0, \sigma_b^2), \quad c \sim \texttt{Normal}(0, \sigma_c^2). \tag{95}$$

Then the elements of $\mathbf{\Phi}\mathbf{\Phi}^{\top}$ become sums of H i.i.d. random variables, and their expectation factorizes across h:

$$\begin{aligned}
[\mathbf{C}]_{nm} &= \sigma_a^2 \sum_{h=1}^{H} \mathbb{E}_{\mathbf{b},\mathbf{c}}[\phi_h(t_n)\phi_h(t_m)] && (96)\\
&= \sigma_a^2 \sum_{h=1}^{H} \mathbb{E}_{b,c}[\phi(t_n; b, c)\phi(t_m; b, c)] && (97)\\
&= \sigma_a^2\, H\, \mathbb{E}_{b,c}[\phi(t_n; b, c)\phi(t_m; b, c)] && (98)\\
&= \sigma_a^2\, H\, \texttt{TACK}(t_n, t_m). && (99)
\end{aligned}$$

Here we anticipate the *temporal arc cosine kernel* of degree d, defined below in (114):

$$\begin{aligned}
\texttt{TACK}(t, t') &:= \mathbb{E}_{b,c}[\phi(t; b, c)\phi(t'; b, c)] && (100)\\
&\equiv \int \phi(t; \mathbf{w})\phi(t'; \mathbf{w})\ \texttt{Normal}(\mathbf{w} \mid \mathbf{0}, \mathbf{\Sigma})\,\mathrm{d}\mathbf{w}, && (101)
\end{aligned}$$

where $\mathbf{\Sigma} := \begin{pmatrix}\sigma_b^2 & 0\\ 0 & \sigma_c^2\end{pmatrix}$ describes the covariance of the hidden weights $\mathbf{w} = (-b, c)^{\top}$. For legibility we suppress the dependence of TACK, written $\texttt{TACK}(t, t')$, on both the degree d and the covariance matrix $\mathbf{\Sigma}$.

To recapitulate, the first and second central moments of the mixture (86) are known for any H:

$$\mathbb{E}[\mathbf{u}'] = \mathbf{0}, \tag{102}$$

$$\mathbb{E}\left[\mathbf{u}'\mathbf{u}'^\top\right] \equiv \mathbf{C} = \sigma_a^2 \, H \, \mathtt{TACK}(t, t'). \tag{103}$$

The third, fourth, ... central moments are in general nonzero and difficult to compute. They will vanish, however, when $H \to \infty$ and we choose $\sigma_a \propto 1/\sqrt{H}$. Nevertheless, even before taking any limit, the first two moments of the prior already mirror something kernel-like. Note that this result hinges critically on the independence assumption for $\{b_h, c_h\}$ in (95). The closure-constrained prior of (85) would couple these parameters nonlinearly, destroying that independence and making the derivation intractable, which is why we temporarily set it aside.

4.4.2 The temporal arc cosine kernel

Of course, I wouldn't have had the courage to attempt this whole calculation had I not already known that a very similar limit has been evaluated succesfully in the neural-network-as-a-Gaussian-process literature. The marginalization we performed above is, in fact, a one-dimensional variant of the classic infinite-width limit of feedforward networks studied by Neal (1996); Williams (1998) and later generalized by Cho & Saul (2009).

In these works, the expectation over random weights $\mathbf{w}$ with $\mathtt{Normal}(\mathbf{w} \mid \mathbf{0}, \mathbf{\Sigma})$ priors gives rise to a family of kernels which describe an infinitely wide Bayesian network, and which differ only with respect to the activation function used and the imposed a priori covariance $\mathbf{\Sigma}$. This same reasoning will now allow us to identify our expected covariance (100) with a time-domain specialization of the *arc cosine kernel family* of degree d.

Table 5 organizes the two kernels we will introduce shortly. From here on we refer to these kernels in prose simply as ACK and TACK, and suppress their dependence on the listed parameters for legibility.

The arc cosine kernel (ACK) of degree d proposed by Cho & Saul (2009) is defined as the expected inner product between infinite-dimensional random RePU features of D-dimensional inputs $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^D$:

Name	**Form**	**Parameters**	**Eq.**
arc cosine kernel	$\mathtt{ACK}(\mathbf{x}, \mathbf{x}') = \frac{1}{\pi} \, \Vert\mathbf{x}\Vert^d \, \Vert\mathbf{x}'\Vert^d \, J_d(\theta)$	d	(105)
temporal arc cosine kernel	$\mathtt{TACK}(t, t') = \frac{1}{2\pi} \, \Vert\mathbf{\Sigma}^{\frac{1}{2}}\mathbf{x}_t\Vert^d \, \Vert\mathbf{\Sigma}^{\frac{1}{2}}\mathbf{x}_{t'}\Vert^d \, J_d(\theta)$	$d, \mathbf{\Sigma}$	(114)

Table 5 | **Arc cosine kernels** introduced in this section. The ACK defined on arbitrary input dimension $D \geq 1$ was originally proposed by Cho & Saul (2009). The TACK is a simple application of the ACK for 1D time series.

$$\mathtt{ACK}(\mathbf{x}, \mathbf{x}') = 2 \int (\mathbf{w}^\top \mathbf{x})_+^d (\mathbf{w}^\top \mathbf{x}')_+^d \ \mathtt{Normal}(\mathbf{w} \mid \mathbf{0}, \mathbf{I}_D) \, \mathrm{d}\mathbf{w}, \tag{104}$$

where the hidden weights $\mathbf{w} \in R^D$, and $\mathbf{I}_D$ is the $D \times D$ identity matrix. This integral representation makes it clear that this is indeed a valid positive definite kernel in any input dimension $D \geq 1$. It admits a beautiful closed form:

$$\mathtt{ACK}(\mathbf{x}, \mathbf{x}') = \frac{1}{\pi} \ \|\mathbf{x}\|^d \ \|\mathbf{x}'\|^d \ J_d(\theta), \tag{105}$$

where

$$\theta = \arccos\left(\frac{\mathbf{x}^\top \mathbf{x}'}{\|\mathbf{x}\| \ \|\mathbf{x}'\|}\right) \in [0, \pi] \tag{106}$$

is the positive angle between $\mathbf{x}$ and $\mathbf{x}'$, and $J_d(\theta)$ is given by Cho & Saul (2009, Eq. 4) as the generator expression

$$J_d(\theta) = (-1)^d (\sin\theta)^{2d+1} \left(\frac{1}{\sin\theta} \frac{\mathrm{d}}{\mathrm{d}\theta}\right)^d \left(\frac{\pi - \theta}{\sin\theta}\right). \tag{107}$$

The first few instances of which are

$$J_0(\theta) = \pi - \theta, \tag{108}$$

$$J_1(\theta) = \sin\theta + (\pi - \theta)\cos\theta, \tag{109}$$

$$J_2(\theta) = 3\sin\theta\cos\theta + (\pi - \theta)(1 + 2\cos^2\theta), \tag{110}$$

$$J_3(\theta) = 15\sin\theta - 11\sin^3\theta + (\pi - \theta)(9\cos\theta + 6\cos^3\theta). \tag{111}$$

Compared to the familiar Matérn kernels, the arc-cosine kernel is a rather atypical kernel, and probably only sparingly used in machine learning applications. It is, for one, blatantly nonstationary, and unlike stationary kernels such as the RBF, it can represent asymptotic behavior: its posterior mean need not revert to the prior mean outside the data regime, much like a neural network's response. The closed form (105) shows this structure explicitly: the integral representation factors into a polynomial term $\|x\|^d \ \|x'\|^d$, which controls the overall dynamic range and absorbs asymptotic variance, and an angular term $J_d(\theta)$, which captures the nonlinear thresholding of the RePU activation and thus allows for modeling the kind of changepoint structure we know is essential for GFMs.

The temporal arc cosine kernel (TACK) is the kernel which describes $\mathbf{C} \equiv \mathbb{E}\left[\mathbf{u}'\mathbf{u}'^\top\right]$ in (91). We can cast it as an affine shift of the bias augmented ACK on the input dimension $D = 2$. Define the auxiliary vectors

$$\mathbf{x}_t = \begin{pmatrix} 1 \\ t \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} -b \\ c \end{pmatrix}. \tag{112}$$

and the covariance matrix for the hidden weights $\{b, c\}$

$$\mathbf{\Sigma} = \begin{pmatrix} \sigma_b^2 & 0 \\ 0 & \sigma_c^2 \end{pmatrix}. \tag{113}$$

Then, by substituting these into (105) and integrating with respect to the Gaussian weight prior $\texttt{Normal}(\mathbf{w} \mid \mathbf{0}, \mathbf{\Sigma})$, we obtain the TACK:

$$\begin{aligned} \texttt{TACK}(t, t') &= \mathbb{E}_{b,c}[\phi(t; b, c)\phi(t'; b, c)] && (114) \\ &= \int (\mathbf{w}^\top \mathbf{x}_t)_+^d (\mathbf{w}^\top \mathbf{x}_{t'})_+^d \; \texttt{Normal}(\mathbf{w} \mid \mathbf{0}, \mathbf{\Sigma}) \, \mathrm{d}\mathbf{w} && (115) \\ &= \frac{1}{2} \, \texttt{ACK}\left(\mathbf{\Sigma}^{\frac{1}{2}} \mathbf{x}_t, \mathbf{\Sigma}^{\frac{1}{2}} \mathbf{x}_{t'}\right) && (116) \\ &= \frac{1}{2\pi} \, \|\mathbf{\Sigma}^{\frac{1}{2}} \mathbf{x}_t\|^d \, \|\mathbf{\Sigma}^{\frac{1}{2}} \mathbf{x}_{t'}\|^d \; J_d(\theta), && (117) \end{aligned}$$

where the factor $1/2$ compensates for the convention used in Cho & Saul (2009). Here,

$$\theta = \arccos\left(\frac{\mathbf{x}_t^\top \mathbf{\Sigma} \mathbf{x}_{t'}}{\sqrt{\mathbf{x}_t^\top \mathbf{\Sigma} \mathbf{x}_t} \sqrt{\mathbf{x}_{t'}^\top \mathbf{\Sigma} \mathbf{x}_{t'}}} \right), \tag{118}$$

is the angle between $\mathbf{x}_t$ and $\mathbf{x}_{t'}$ in the geometry induced by $\mathbf{\Sigma}$. This anisotropic variant was also used in the context of D-dimensional feature extraction by Pandey & Dukkipati (2014).

Figure 18 makes the role of the degree parameter visible before we move on to the infinite-width limit. Across all three covariance settings, low degrees produce rougher draws with sharper local bends, whereas higher degrees regularize these bends into smoother open-phase trajectories. The covariance parameters then control where this structure is concentrated: smaller σ_b imply a more localized changepoint location and lead to sharp changepoint

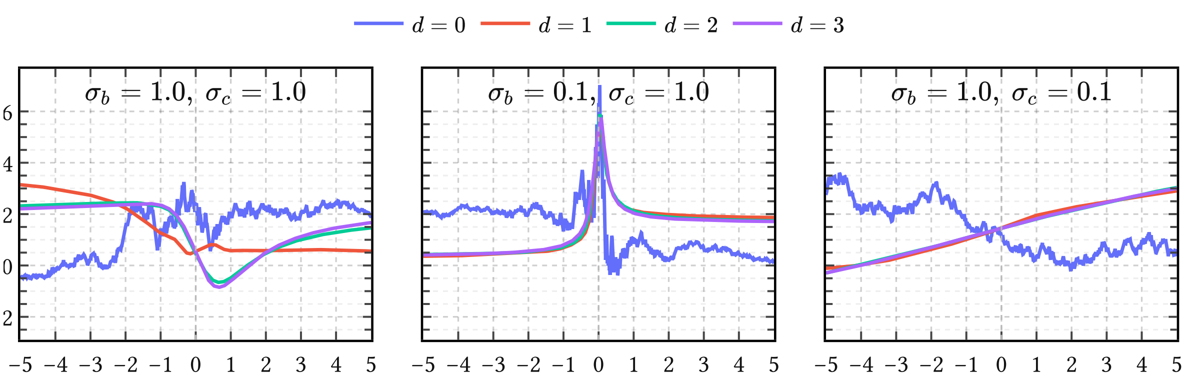


Figure 18 | **Prior samples from TACK for different degrees d.** Each column fixes a different covariance setting (σ_b, σ_c). Within each column the four curves correspond to different values of d. Increasing d progressively softens the hingepoint-like behavior, while decreasing σ_b relative to σ_c makes the prior more localized around the center.

behavior. Smaller values of σ_c just smear out time, which have the effect of longer timescales. Note that $d = 2$ and $d = 3$ are practically indistinguishable.

4.4.3 Taking the Gaussian-process limit

We turn again to the mixture (86). Recall from (89) that each random draw $\{\mathbf{b}, \mathbf{c}\} \in \mathbb{R}^{2H}$ from the prior (80) produces a random covariance matrix $\mathbf{Q} \in \mathbb{R}^{N \times N}$, which elementwise decomposes as a sum of H i.i.d. terms:

$$[\mathbf{Q}]_{nm} = \sigma_a^2 \sum_{h=1}^{H} \phi_h(t_n)\phi_h(t_m). \tag{119}$$

This sum tends to grow as $\mathcal{O}(H)$, since from (96)

$$\mathbb{E}_{\mathbf{b},\mathbf{c}}[\mathbf{Q}] = \mathbf{C} \propto H. \tag{120}$$

Such growth of the marginal variance of $p(\mathbf{u}')$ makes the $u'_{\text{NN}}(t)$ model (78) useless as a prior when we want strong inductive bias even for large H, which we do. Therefore, we couple σ_a and H by choosing $\sigma_a \propto 1/\sqrt{H}$, in accordance with the variance-preserving initialization principle (Glorot & Bengio, 2010). Thus, substituting $\sigma_a \to \sigma_a/\sqrt{H}$ in (119), we get

$$[\mathbf{Q}]_{nm} = \sigma_a^2 \times \left[\frac{1}{H}\sum_{h=1}^{H} \phi_h(t_n)\phi_h(t_m)\right]. \tag{121}$$

Observe that the quantity between brackets is a Monte Carlo estimate of the TACK (114). Therefore, by the strong law of large numbers we conclude that

$$\left[\frac{1}{H}\sum_{h=1}^{H} \phi_h(t_n)\phi_h(t_m)\right] \longrightarrow \mathbb{E}_{b,c}[\phi(t;b,c)\phi(t';b,c)] \qquad \text{as } H \to \infty, \tag{122}$$

with deviations vanishing as $\mathcal{O}\left(1/\sqrt{H}\right)$. Therefore from (100)

$$[\mathbf{Q}]_{nm} \longrightarrow \sigma_a^2\, \texttt{TACK}(t_n, t_m) \qquad \text{as } H \to \infty. \tag{123}$$

Equivalently, the induced density $p(\mathbf{Q})$ of covariance matrices collapses to

$$p(\mathbf{Q}) \longrightarrow \delta(\mathbf{Q} - \mathbf{K}) \qquad \text{as } H \to \infty, \tag{124}$$

which in turn collapses the mixture (86) to

$$p(\mathbf{u}') \longrightarrow \texttt{Normal}(\mathbf{0}, \mathbf{K}) \qquad \text{as } H \to \infty. \tag{125}$$

Here the *kernel matrix* is the symmetric PSD matrix $\mathbf{K} \in \mathbb{R}^{N\times N} \propto \mathbf{C}$ given by

$$[\mathbf{K}]_{nm} = \sigma_a^2\, \texttt{TACK}(t_n, t_m). \tag{126}$$

In other words, the marginal $p(\mathbf{u}')$ converges entirely to a Gaussian and the first and second moments in (102) become *sufficient statistics* (Jaynes, 2003). All higher central moments vanish as $\mathcal{O}\left(1/\sqrt{H}\right)$ because the covariance fluctuations in (122) themselves decay at that rate. Since this argument holds for any finite collection of times $\boldsymbol{t} = \{t_n\}_{n=1}^N$, the limiting process is indeed officially a Gaussian process:

$$u'_{\text{NN}}(t) \sim \texttt{GaussianProcess}\left(0, \sigma_a^2 \, \texttt{TACK}(t, t')\right). \tag{127}$$

Except for H, it has the same hyperparameters as the neural network model of Section 4.3.2: the scales $\boldsymbol{\theta}_{\text{NN}} = \{\sigma_a, \sigma_b, \sigma_c\}$ describing overall power, typical changepoint location and horizontal stretch, respectively; and the two higher-level hyperparameters $\{d, t_c\}$, which determine the polynomial degree and the instant of glottal closure, respectively.

Integrating out changepoints Note that t_c is the only changepoint parameter that survived the Gaussian-process limit. All other changepoints [each changepoint being defined by a pair of amplitude and location parameters $\{a_h, b_h\}$] have been analytically integrated out. That act of marginalization removed the parametric bottleneck carried by finite rank models like (56), (60), (63), (78) completely. This may come as a surprise, because we pointed out before that changepoint parameters as in (59) constitute the main "control points" of any GFM expressed in the time domain: what previously required a discrete arrangement of control points is now averaged into a single continuous covariance function capable of learning the "timing relationships [which] are very important for modelling the glottal flow signal" (Doval et al., 2006, p. 1).

More aspects of (127) are discussed in greater detail in Appendix D.

4.5 Summary

Great care was taken to argue that any *parametric* glottal flow model of the open phase of the glottal cycle can be expressed as a RePU network with a single hidden layer of width H, given H large enough.

Then we showed that in a Bayesian regression context with noninformative priors, the limit $H \to \infty$ corresponds to a *nonparametric* glottal flow model: a zero-mean GP with the temporal arc cosine kernel. This well-known limit, borrowed from classical GP literature, is applied here to unify the two dominant approaches to GIF. These are [parametric $\Leftrightarrow$ joint source-filter estimation] and [nonparametric $\Leftrightarrow$ inverse filtering], and we attempt here to combine the best of both worlds: parametric interpretability and nonparametric expressivity, respectively.

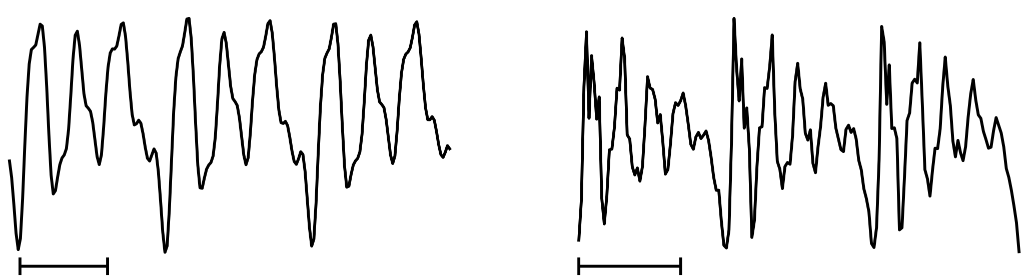
That is, the proposed probabilistic model has support for any parametric glottal flow model while retaining capacity to "let the data speak for itself" if need be. In the next chapters we will refine this initially tiny support gradually into a strong inductive bias by learning features from synthetic glottal flow simulations.

5 The periodic and quasiperiodic arc cosine kernel

Chapter 4 established parametric and nonparametric glottal flow models for a *single period* of the glottal cycle. In this chapter, our goal is to extend this basic building block to *multiple periods.* Recordings of voiced speech often visually exhibit self-similarity on timescales of $O(10 \text{ ms})$ because in normal speech the glottal cycle often reaches steady-state, and a clear pitch is perceived as a consequence.[20] An illustrative example for the case of steady-state vowels is given in Figure 19.

To model this steady-state source behavior we proceed in three steps. We begin with an idealized periodic kernel, the *periodic arc cosine kernel* (PACK), which is stationary and perfectly periodic. We then relax this to the *quasiperiodic arc cosine kernel* (QPACK), which retains the same within-cycle carrier while allowing slow amplitude variation and residual baseline structure across cycles. Finally we introduce *gated variants* (g-PACK and g-QPACK) in which the closed phase is incorporated explicitly. This last step breaks stationarity, but it turns out to be a useful one in practice.

The second half of the chapter turns from construction to learning. It is well known from functional data analysis that learning a single shared basis for a large corpus of training data requires aligning the data points, the process of which is called *registration.* (Ramsay & Silverman, 1997). We show how to do this accurately for the EGIFA data by aligning the



(a) (b)

Figure 19 | **Examples of steady-state vowels** during three pitch periods. Here (a) is perfectly periodic because it was synthesized with (56); and (b) is quasiperiodic because it is taken from a real recording.[21] The timescale is indicated by the horizontal bars ⊢—⊣ below the waveforms, each bar having a length of 5 msec.

[20]Results from clinical trials (Little et al., 2007) indicate that normal (non-pathological) voices are usually something called Type I (Titze, 1995); that is, phonation results in nearly periodic sounds, which supports the notion that uttered vowels in normal speech typically reach a steady-state before "moving on". Thus when a vowel is perceived with a clear and constant pitch, it is reasonable to assume that the vowel attained steady-state at some point (though perceptual effects forbid a one-to-one correspondence). In practice, steady-state vowels are identified simply by looking for such quasiperiodic strings, which typically consist of about 3 to 5 pitch periods in everyday speech (Rabiner & Schafer, 2007).

[21]Taken from a recorded steady-state instance of the vowel /æ/ at a fundamental frequency of 138 Hz. Source: `bdl/arctic\_a0017.wav:0.51-0.54sec` (Kominek & Black, 2004).

examples to GCI (and optional GOI) events. This also allows us to extract voiced groups from the data, which then serve as high-quality training data to *calibrate* the global kernel hyperparameters. These hyperparameters determine the shared basis seen by the whole corpus, and we learn and evaluate them directly from the registered data.

Three algorithms do the practical work of the chapter. LBGID (Algorithm 2) detects the GCI and GOI events needed for registration; PRISM (Algorithm 3) calibrates a single global basis on the registered corpus; and t-PRISM (Algorithm 4) robustly filters raw period tracks into clean voiced groups before calibration. For the sake of brevity their full details and development have been deferred to the algorithm chapters; here we just accept them as given tools and only motivate their usage.

5.1 The periodic arc cosine kernel

The TACK or temporal arc cosine kernel derived in (114) provides a nonstationary covariance structure for the open phase of a *single* glottal cycle. That covariance reflects the change-point geometry inherited from piecewise polynomial GFMs, and it does so in a principled, nonparametric manner. But a single pitch period cannot exist in isolation: voiced speech is characterized by the imperfect repetition of this basic unit, and so we must now try to extend TACK to periodic signals.

Arguably the most natural route to periodicity is to replace the linear time coordinate in the bias-augmented input of TACK, namely $\mathbf{x}_t = (1, t)^\top$, by a set of *harmonic features* that are periodic simply by construction. This is the classical idea of embedding time on a circle (MacKay, 1998): functions of the embedded coordinates are automatically periodic in the original time variable. The construction proceeds in three steps: define the embedding, apply the arc cosine kernel, and normalize.

5.1.1 Harmonic embedding

Let $\omega_0 = 2\pi/T$ denote the fundamental angular frequency for a period $T > 0$, and let $J \geq 1$ be the *harmonic order*, which controls the richness of the embedding. Define the harmonic feature map

$$\boldsymbol{\varphi}(t) = \begin{pmatrix} \sigma_b \\ \sigma_{c,1} \cos(\omega_0 t) \\ \sigma_{c,1} \sin(\omega_0 t) \\ \vdots \\ \sigma_{c,J} \cos(J\omega_0 t) \\ \sigma_{c,J} \sin(J\omega_0 t) \end{pmatrix} \in \mathbb{R}^{2J+1}, \tag{128}$$

where $\sigma_b > 0$ and $\sigma_{c,j} > 0$ for $j = 1, ..., J$ are scale parameters playing analogous roles to σ_b and σ_c in Section 4.3.2, but act on the embedded space rather than linear time. These scale parameters modulate the harmonic basis (128) and thereby shape the angular similarity seen by the kernel. The constant coordinate σ_b is not a DC component of the waveform itself;

after normalization it becomes the zeroth harmonic weight and shifts the carrier toward or away from perfect self-alignment. The remaining scales control how strongly each harmonic pair participates in that comparison. In any case, where TACK lived in $\mathbb{R}^2$, we now work in $\mathbb{R}^{2J+1}$ with $(J+1)$ independent harmonic scales, which gives the kernel much greater spectral flexibility, while still inheriting (some of) the changepoint behavior from its ancestor.

An interesting property of this particular embedding is that its squared norm is constant in t:

$$\|\varphi(t)\|^2 = \sigma_b^2 + \sum_{j=1}^{J} \sigma_{c,j}^2 \left(\cos^2(j\omega_0 t) + \sin^2(j\omega_0 t)\right) = \sigma_b^2 + \sum_{j=1}^{J} \sigma_{c,j}^2 := R^2. \tag{129}$$

Every point $\varphi(t)$ therefore lies on a sphere of radius R in $\mathbb{R}^{2J+1}$. For the inner product between two embedded points, the cosine addition formula $\cos\alpha\cos\beta + \sin\alpha\sin\beta = \cos(\alpha - \beta)$ collapses the harmonic pairs:

$$\varphi(t)^\top \varphi(t') = \sigma_b^2 + \sum_{j=1}^{J} \sigma_{c,j}^2 \cos(j\omega_0(t - t')). \tag{130}$$

This depends on t and t' only through their difference, and is manifestly periodic with period T. Define θ as the angle between the two embedded points,

$$\theta(t - t') := \arccos\left(\frac{\varphi(t)^\top \varphi(t')}{R^2}\right), \tag{131}$$

which takes values in $[0, \pi]$ and satisfies $\theta(0) = 0$.

5.1.2 Definition

Applying the ACK (105) to $\varphi(t)$ and $\varphi(t')$ gives

$$\texttt{ACK}(\varphi(t), \varphi(t')) = \frac{1}{\pi} R^{2d} J_d(\theta(t - t')), \tag{132}$$

which depends on t and t' only through their difference since R is constant. The prefactor R^{2d}/π couples the overall differentiability of sample paths (controlled by d) to the output scale. To circumvent this, we apply standard normalization and define the *periodic arc cosine kernel of degree d* as

$$\texttt{PACK}(t, t') := \sigma_a^2 \frac{J_d(\theta(t - t'))}{J_d(0)}, \tag{133}$$

where $J_d(\theta)$ was defined in (107) and $J_d(0) = \pi(2d-1)!!$ is the value of the angular kernel at zero angle. Positive definiteness is preserved because composing a positive definite kernel with a deterministic feature map yields another positive definite kernel. By construction, the variance $\texttt{PACK}(t, t) = \sigma_a^2$, so σ_a is the only parameter that controls the overall amplitude

Symbol	Name	Role
σ_a	amplitude	marginal standard deviation
σ_b	bias amplitude	DC component of the embedding
$\sigma_{c,1}, \ldots, \sigma_{c,J}$	harmonic scales	modulate each harmonic pair
T	period	fundamental period of the kernel
d	degree	smoothness class (RePU order)
J	harmonic order	number of harmonics in embedding

Table 6 | **Hyperparameters of PACK.** Of these, σ_a, σ_b and T are positive scalar, the harmonic scales $\sigma_{c,j}$ are J-dimensional, and d and J are discrete.

scale. Note that for legibility we generally suppress the dependence of $\texttt{PACK}(t, t')$ on its hyperparameters; we list them in Table 6 for an oversight.

How come PACK is stationary even though ACK and TACK were not? This is simply because we assigned equal weight to each cosine-sine pair in (130). It is a practical choice to ease training the kernel, and does not impair learning nonstationary phenomena. The thresholding behavior of the RePU activations is still present, but it is now replicated periodically.[22] This means that PACK can still express very sharp within-cycle structure even though it is now stationary and periodic.

5.1.3 Hyperparameters and the weight simplex

The definition (133) involves $J+1$ scale parameters (σ_b and $\sigma_{c,1}, \ldots, \sigma_{c,J}$) that only enter through the angle θ in (131). Since θ depends on the normalized inner product $\varphi(t)^\top \varphi(t')/R^2$, only the *ratios* of these scales matter. This motivates introducing the *harmonic weights*

$$w_0 = \frac{\sigma_b^2}{R^2}, \quad w_j = \frac{\sigma_{c,j}^2}{R^2} \quad (j = 1, \ldots, J), \tag{134}$$

which satisfy the simplex constraint $w_0 + \sum_{j=1}^J w_j = 1$ with $w_j \geq 0$. In terms of these weights, the cosine of the angle (131) becomes

$$\cos\theta(t-t') = w_0 + \sum_{j=1}^{J} w_j \cos(j\omega_0(t-t')), \tag{135}$$

and the kernel (133) can be evaluated by computing $\theta(t-t')$ from (135) and substituting into J_d. We learn the harmonic weights directly in the simplex space to avoid degeneracies.

[22]The neural network interpretation of Section 4.3.2 carries over directly. PACK is the $H \to \infty$ limit of a single hidden layer with RePU activations applied to the $(2J+1)$-dimensional harmonic input (128) rather than the 2-dimensional bias-augmented input.

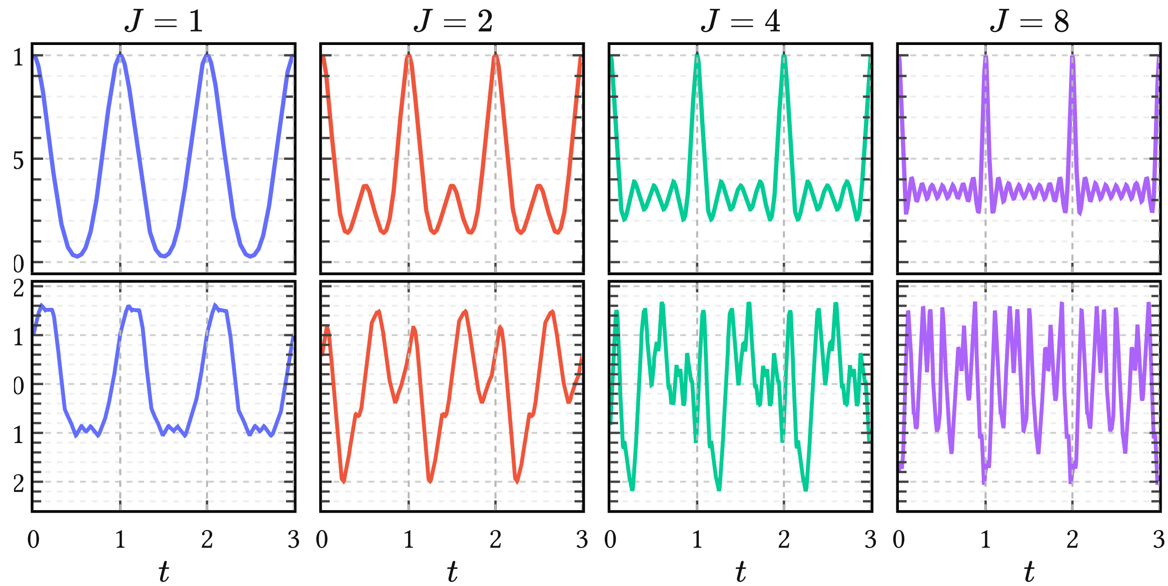


Figure 20 | **Effect of harmonic order J on PACK.** The top row shows the kernel section $\texttt{PACK}(t, 0)$ and the bottom row a prior sample. At fixed degree $d = 1$, increasing J decreases the effective within-cycle lengthscale and yields progressively wigglier covariance structure and sample paths. The remaining hyperparameters are fixed at $T = 1$, $\sigma_a^2 = 1$, $w_0 = 0.2$, and uniform harmonic weights $w_1, \ldots, w_J$ over the remaining mass.

The zeroth order harmonic weight w_0 plays an interesting role. It descends directly from the bias amplitude σ_b in the TACK construction, but its behavior is altered nontrivially by the harmonic embedding. When $w_0 = 1$ (all harmonic weights zero), the angle θ is identically zero and the kernel degenerates to a constant σ_a^2, describing a DC signal. Intermediate values of w_0 let the cosines oscillate about zero in (130) so that θ can reach $\pi/2$ and beyond, where $J_d(\theta)$ is able to produce steep, changepoint-like transitions of the type we need to model sharp glottal closures. If $w_0 = 0$, the autocovariance $\texttt{PACK}(t, 0)$ is allowed to reach zero periodically, which for higher d effectively creates kernels with compact support, although it isn't immediately clear what this implies for modeling speech signals.

At least J has a clear geometric effect. Increasing J inserts higher harmonics directly into the embedding and therefore decreases the effective within-cycle lengthscale of the process. In prior draws this appears as progressively finer oscillatory detail, as seen in Figure 20. Practically, increasing J while keeping all $w_j \sim O(1/J)$ gives rise to smaller effective lengthscales.

5.1.4 Spectral decay and smoothness

Because PACK is both stationary and periodic with period T, its covariance is an even T-periodic function that admits a Fourier cosine expansion

$$\texttt{PACK}(t, t') = a_0 + \sum_{m=1}^{\infty} a_m \cos(m\omega_0(t - t')), \tag{136}$$

with coefficients

$$a_0 = \frac{1}{T}\int_0^T \texttt{PACK}(t,0)\,\mathrm{d}t, \quad a_m = \frac{2}{T}\int_0^T \texttt{PACK}(t,0)\cos(m\omega_0 t)\,\mathrm{d}t \quad (m \geq 1). \tag{137}$$

By Bochner's theorem, positive definiteness of PACK is equivalent to $a_m \geq 0$ for all $m \geq 0$, so we can interpret these coefficients as encoding power. The Fourier expansion (136) has a direct interpretation as a *Bochner line spectrum* with spectral density given by

$$\tilde{k}(\xi) = 2\pi a_0\,\delta(\xi) + \pi\sum_{m=1}^{\infty} a_m[\delta(\xi - m\omega_0) + \delta(\xi + m\omega_0)]. \tag{138}$$

The decay of the harmonic weights a_m determines both the spectral character of the prior and the mean square differentiability of GP sample paths. For stationary kernels, this is governed by the behavior of the covariance near the origin: the more derivatives $k(t) = k(t, 0)$ has at $t = 0$, the more mean square derivatives the process admits. A familiar example is the Matérn family, whose sample paths are r times mean square differentiable for every integer $r < \nu$. Now, it can be shown that the sample paths of a degree-d PACK kernel are exactly d times mean square differentiable, independent of J. This implies that the harmonic weights decay in power as

$$a_m = \mathcal{O}\big(m^{-(2d+2)}\big), \tag{139}$$

corresponding to an approximate rolloff in magnitude of $-6(d+1)$ dB per octave, as illustrated for the $d = 1$ case in Figure 22.

This behavior is also visible in Figure 21: increasing d removes the cusp at the origin and produces progressively smoother kernels with faster spectral decay. Low degrees ($d = 0$ or $d = 1$) therefore yield broad spectra with slowly decaying harmonics, appropriate for modeling sharp glottal closure events.

Classical source models report a spectral tilt of roughly -12 to -18 dB per octave for the glottal flow $u(t)$ itself (Doval et al., 2006; Rosenberg, 1971). Differentiation increases spectral slope by $+6$ dB per octave, so the corresponding glottal flow derivative $u'(t)$ exhibits a tilt of approximately -6 to -12 dB per octave, which corresponds to PACK kernels of degree $d = 0$ and $d = 1$.

5.1.5 Comparison to the standard periodic kernel

The standard periodic kernel in GP literature is the benign periodic squared exponential, which MacKay (1998) obtained by composing the RBF kernel with a circular embedding:

$$k_{\text{per}}(t, t') = \sigma^2 \exp\left(-2\sin^2\left(\pi\frac{t - t'}{T}\right)/\ell^2\right). \tag{140}$$

Its exact cosine expansion is (Riutort-Mayol et al., 2020, Eq. B.2):

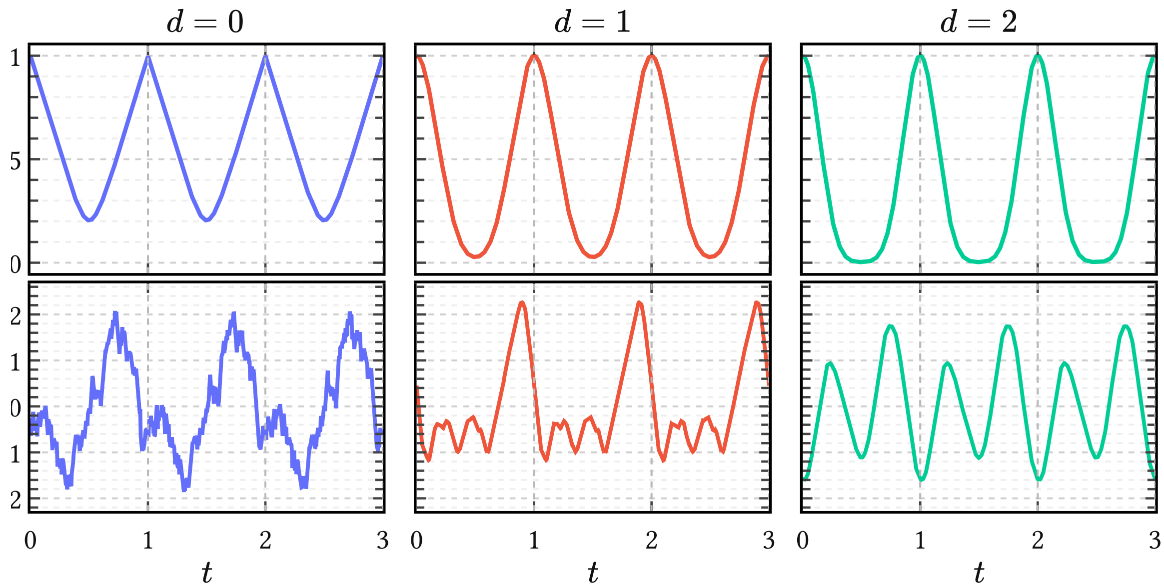


Figure 21 | **Effect of degree d on within-cycle smoothness.** The top row shows the kernel section $\texttt{PACK}(t, 0)$ and the bottom row a single prior sample. Holding J and the harmonic weights fixed, increasing d softens the changepoint-like geometry and produces smoother periodized waveforms. The remaining hyperparameters are fixed at $J = 1$, $T = 1$, $\sigma_a^2 = 1$, $w_0 = 0.2$, and $w_1 = 0.8$.

$$k_{\text{per}}(t, t') = \sigma^2 e^{-\frac{1}{\ell^2}} \left[I_0(\ell^{-2}) + 2 \sum_{m=1}^{\infty} I_{m(\ell^{-2})} \cos(m\omega_0(t - t')) \right]. \tag{141}$$

Thus its entire harmonic envelope is controlled by a single lengthscale ℓ. Changing ℓ only broadens or narrows a smooth low-pass spectrum.

Figure 22 illustrates how PACKs are different in two ways. First, the harmonic weights w_j can redistribute mass freely across harmonics rather than merely broadening a fixed envelope. Second, the coupling between degree d and the asymptotic decay allows the model to learn more realistic, non-exponential spectral rolloff.

5.2 The quasiperiodic arc cosine kernel

Of course, perfect periodicity is too rigid for actual glottal flow; even synthesized glottal flow contains all sorts of drifts and interesting imperfections, which add organic warmth to the synthesized voice. Therefore the next step is to relax PACK into a quasiperiodic variant.

We thus define the *quasiperiodic arc cosine kernel* by

$$\texttt{QPACK}(t, t') := \texttt{env}(t, t') \cdot \texttt{PACK}(t, t') + \texttt{base}(t, t'). \tag{142}$$

This is a composite kernel modeling amplitude modulation of a periodic carrier, plus a second term that adds a small low pass correction. The full parameterization of (142) is listed below

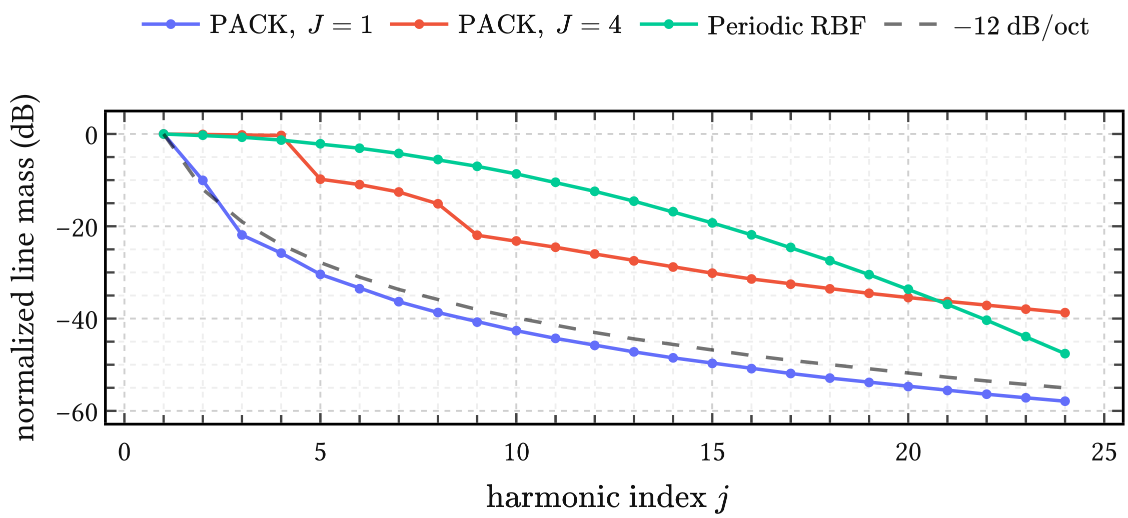


Figure 22 | **PACK versus the periodic squared exponential in the line spectrum.** The curves show the harmonic weights a_m from (138) in dB. PACK with $d = 1$ exhibits the expected -12 dB per octave tail, while the periodic squared exponential traces out a one-parameter low-pass envelope. The PACK curves use $T = 1$, $\sigma_a^2 = 1$, and $w_0 = 0.2$, with uniform weights over the remaining harmonic mass; the periodic squared exponential uses lengthscale $\ell = 0.20$. The dashed reference line indicates -12 dB per octave.

in Table 7. In what follows, we choose the envelope $\texttt{env}(t, t')$ and baseline kernel $\texttt{base}(t, t')$ to be stationary kernels. This makes sure that QPACK is still stationary such that it has a well defined Bochner spectrum. We give an example of the latter in Figure 23 and now turn to each of the three components that make up QPACK.

Carrier kernel The carrier is a PACK of degree d from the previous section. The amplitude modulation introduced by the envelope does not materially alter the high-frequency rolloff, so the degree d still governs closure sharpness and spectral slope, while the weights and harmonic order J determine the detailed within-cycle geometry.

Envelope kernel The envelope kernel, written $\texttt{env}(t, t')$, modulates this carrier on a much longer timescale ℓ_{env} than the fundamental period T – typically on the order of a dozen periods. We use a unit-variance RBF,

$$\texttt{env}(t, t') := \exp\left(-\frac{(t - t')^2}{2\ell_{\text{env}}^2}\right), \tag{143}$$

for the envelope because this does not interfere with the differentiability class set by d. In the frequency domain, the product of a slowly varying RBF envelope with the PACK carrier broadens each of the harmonic lines into narrow bands like the ones on display in Figure 23.

Baseline kernel The baseline kernel, written $\texttt{base}(t, t')$, is a small additive correction. We choose the rational quadratic (RQ) kernel

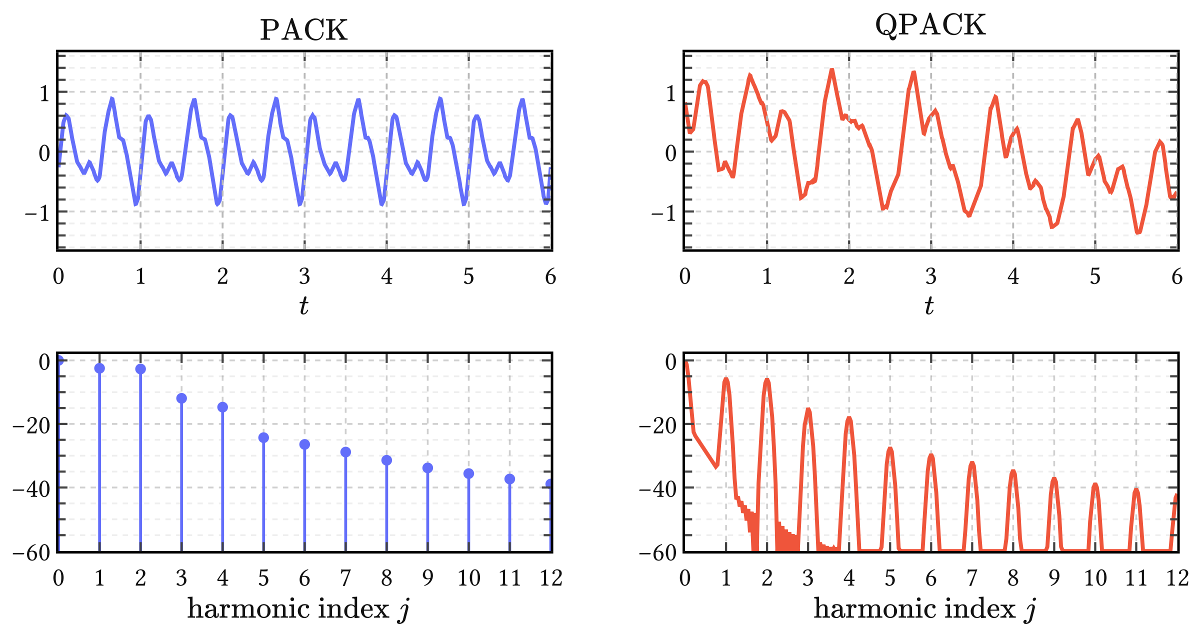


Figure 23 | **From PACK to QPACK in time and frequency.** The top row shows prior samples over six periods; the bottom row shows the corresponding normalized spectra in dB against harmonic index j. The shared carrier hyperparameters are $d = 1, J = 2, T = 1, \sigma_a^2 = 1$, and $w = (0.2, 0.4, 0.4)$. For QPACK the envelope is a unit-variance RBF with lengthscale $\ell_{\text{env}} = 3$, and the baseline is a rational quadratic kernel with $\sigma_{\text{base}}^2 = 0.02$, $\ell_{\text{base}} = 0.5$, and $\alpha_b = 1$.

$$\texttt{base}(t, t') := \sigma_{\text{base}}^2 \left(1 + \frac{(t - t')^2}{2\alpha_b \ell_{\text{base}}^2}\right)^{-\alpha_b}, \tag{144}$$

with scale $\sigma_{\text{base}} \ll \sigma_a$, lengthscale $\ell_{\text{base}} \ll \ell_{\text{env}}$, and a general shape parameter $\alpha_b > 0$. Like the RBF, the RQ kernel is infinitely differentiable and will not influence the differentiability of the composite kernel. Its role is to capture weak residual structure which is not phase-locked enough to belong in the carrier but also not unstructured enough to be dismissed as white noise. This includes open-phase ripple caused by glottal-vocal-tract interaction (Ananthapadmanabha & Fant, 1982), as well as violations of the closure condition that occur frequently in more irregular or pathological voices (Kreiman et al., 2007). In spectral terms the baseline adds a broad low-frequency "pedestal" underneath the broadened carrier bands.

5.3 Gated variants

PACK and QPACK are manifestly stationary kernels. After conditioning on data they can produce phase-dependent (nonstationary) posterior means, but *a priori* they assign the same variance to every phase of the cycle, whereas we know that the excitation energy depends strongly on the phase of the glottal cycle.

Symbol	Component	Role
σ_a	carrier	PACK amplitude scale; fixed to 1 in the whitened calibration experiments
$w_0, \ldots, w_J$	carrier	harmonic weight simplex inherited from PACK
d	carrier	within-cycle smoothness and spectral slope
J	carrier	harmonic order; effective within-cycle lengthscale control
T	carrier	fundamental period
ℓ_{env}	envelope	timescale of cycle-to-cycle amplitude modulation
σ_{base}	baseline	amplitude of the additive correction
ℓ_{base}	baseline	correlation lengthscale of the additive correction
α_b	baseline	rational quadratic shape parameter

Table 7 | **Hyperparameters of QPACK.** QPACK inherits the full PACK carrier and adds a long-scale envelope and a weak baseline. In the EGIFA calibration experiments the envelope is taken to have unit variance, so only its lengthscale is learned.

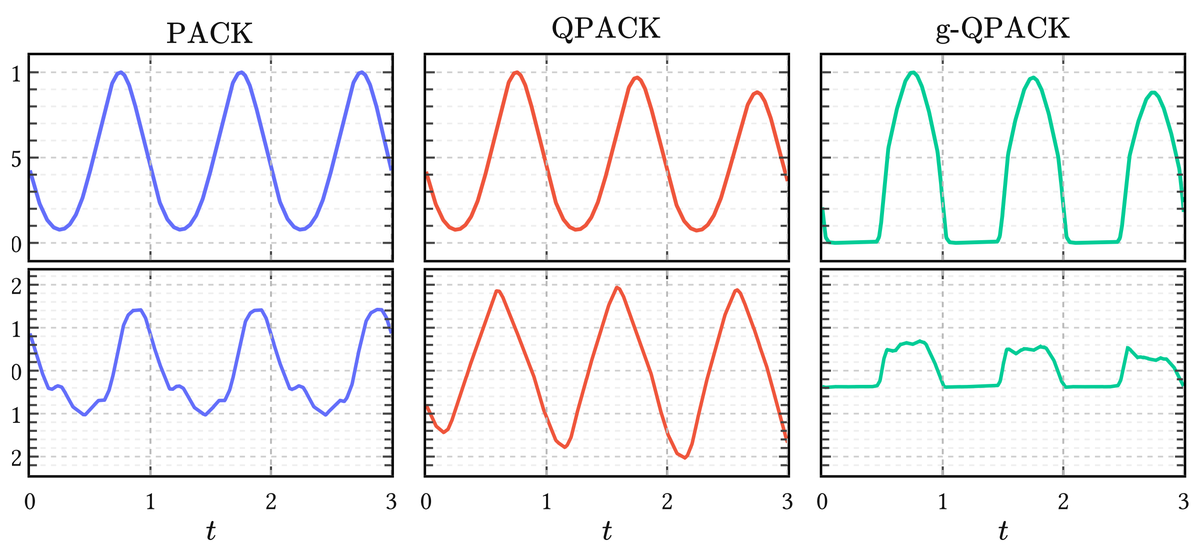


Figure 24 | **From PACK to QPACK to g-QPACK.** The top row shows kernel sections $k(t, t')$ at anchor time $t' = 0.75$, which is halfway the first open phase. The bottom row shows prior samples, all with matched carrier hyperparameters. QPACK relaxes strict periodicity by adding a slow envelope; g-QPACK then suppresses covariance in the closed phase of the cycle. The shared carrier settings are $d = 1$, $J = 1$, $T = 1$, and $w = (0.2, 0.8)$. To emphasize the closed phase the baseline correction was suppressed by setting $\sigma_{\text{base}} = 0$. The envelope is a unit-variance RBF with $\ell_{\text{env}} = 4$, and the gate sharpness of g-QPACK is $\beta = 12$.

The next step in taking advantage of prior information is therefore to encode the open/closed asymmetry directly in the kernel itself. This can be done easily by gating the kernel covariance (Alvarado & Stowell, 2016). Define therefore the two *gated variants*

$$\texttt{g-PACK}(t, t') \coloneqq g_\beta(t)\, g_\beta(t')\, \texttt{PACK}(t, t'), \tag{145}$$

$$\texttt{g-QPACK}(t, t') \coloneqq g_\beta(t)\, g_\beta(t')\, \texttt{QPACK}(t, t'). \tag{146}$$

These inherit all properties and hyperparameters of their respective base kernel except the property of stationarity.[23] The periodic gate is defined as

$$g_\beta(t) = \sigma(-\beta \sin(2\pi t / T)), \tag{147}$$

where $\beta > 0$ is a new hyperparameter that controls the sharpness and $\sigma(z) = (1 + e^{-z})^{-1}$ is the standard logistic sigmoid. This is a smooth approximation to a square wave: over one half-cycle $g_\beta(t) \approx 0$, over the other $g_\beta(t) \approx 1$, and the transitions are smooth.

Figure 24 illustrates the behavior of g-QPACK and compares it to its ancestors. As the gate $g_\beta(t)$ is itself periodic, g-PACK remains a proper extension of PACK (and incidentally a rare example of a periodic-but-nonstationary kernel). In practice, however, there is little reason to give up on the extra flexibility of QPACK, so the real model of interest is g-QPACK, not g-PACK, and we will focus on the former from here on.

Note that (147) implies that the first half of the period is closed phase and the second half open phase, with the two subintervals equally long. This means that g-QPACK always generates GF waveforms with fixed open quotient $\text{OQ} = 0.5$ (unless $\beta \to 0$). We relax this in the next section by warping physical time t into normalized time τ which can stretch and compress the phases and achieve any value of interest for OQ.

5.4 Registration with EGIFA

In the next section, we intend to learn the hyperparameters of the kernels introduced so far from the EGIFA dataset (Dataset 1). EGIFA provides a large number of high-quality glottal flow waveforms $\{u(t)\}$ simulated from a physics-based articulatory speech production model called VocalTractLab (Birkholz, 2013), which is the current open-source state of the art (Gao et al., 2024). We can use the generated $\{u(t)\}$ waveforms as examplars to calibrate the PACK family on, but this requires aligning them beforehand – a procedure known as *registration* in functional data analysis, hence the title of this section.

In order to maximize learning signal, we therefore

1. extract only clean voiced groups from the raw $\{u(t)\}$ waveforms;
2. calculate their derivatives $\{u'(t)\}$ in a physiologically grounded way;[24]
3. normalize power across the DGF exemplars;

[23]This also means that the gated variants lack well-defined Bochner spectra. They are still in the family of harmonizable kernels, however (Altamirano & Tobar, 2022).

[24]See Appendix E for details.

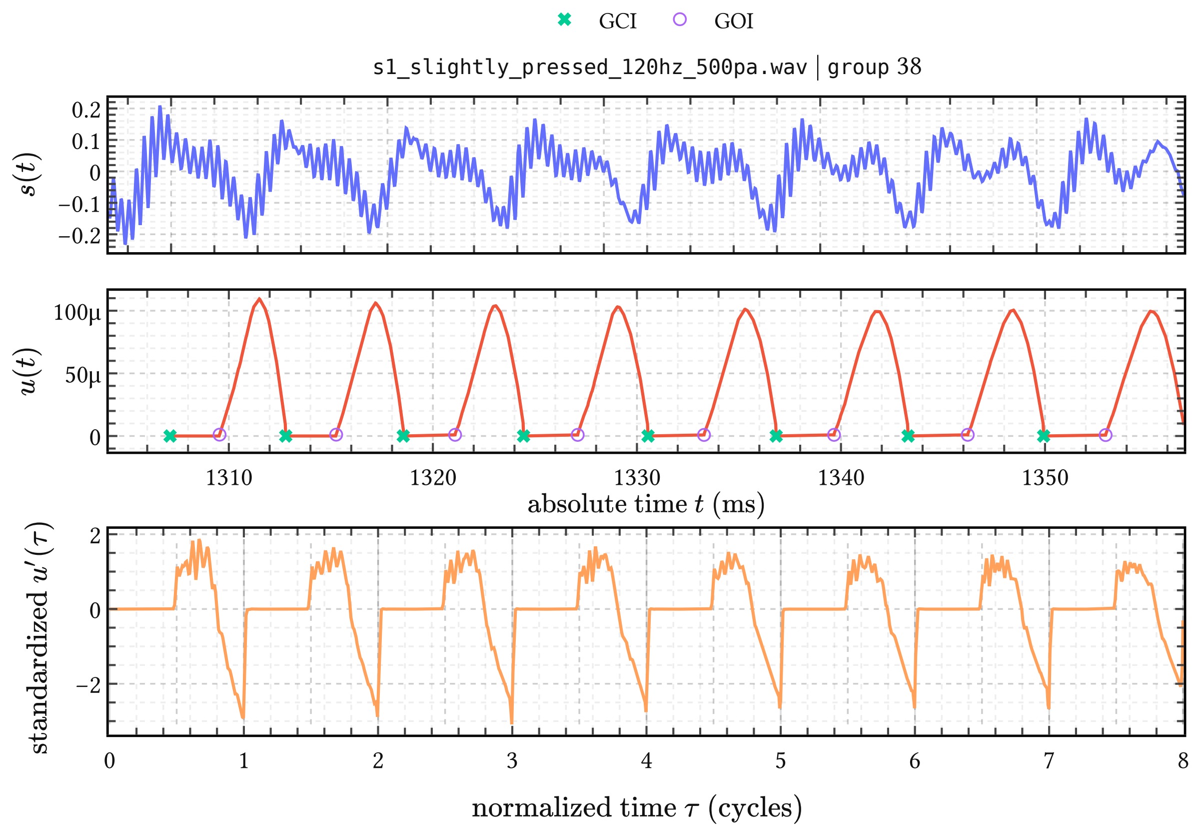


Figure 25 | **Registration of one retained EGIFA voiced group.** Top: the raw speech waveform $s(t)$ in physical time, extracted from the indicated `.wav` file and voiced group. Middle: the corresponding raw glottal flow $u(t)$, again in physical time, with LBGID-detected GCI and GOI landmarks. Bottom: the same group after differentiation, amplitude normalization and registration to normalized time, now shown as $u'(\tau)$. In this representation the GCIs land at integers $\tau = 0, 1, 2, \ldots$ and the GOIs at half-integers $\tau = 0.5, 1.5, 2.5, \ldots$.

4. and finally align them in time such that the GCI and GOI events across different $u'(t)$ waveforms align in *normalized time* τ.

The procedure is illustrated in Figure 25, which follows a single EGIFA group through the registration process. With regard to step 4, time alignment, it is seen that the upper two rows live in physical time t, where period length and OQ vary continually and the GCIs and GOIs appear as irregularly spaced events. The bottom row lives in "subdued" normalized time τ, where the glottal closing and opening instants have been mapped to integer and half integer values of τ, respectively, and $\mathrm{OQ} = 0.5$ everywhere.

Rather unfortunately, steps 1 and 4 above are nontrivial data preprocessing steps, and difficult to get right consistently, even for 1D time series. Because data quality is crucial, and with the prospect of reusing these transformations for future large scale benchmarks, we have had to resort to custom algorithms in order to guarantee meaningful alignment. LBGID (Algorithm 2) is a simple heuristic to reliably extract GCIs and GOIs from ground truth glottal flow $u(t)$.

$\{(\mathbf{t}_n, \mathbf{u}_n)\}_{n=1}^{N}$

raw voiced groups in physical time

$\xrightarrow{\text{registration}}$

$\mathcal{D} = \{(\boldsymbol{\tau}_n, \mathbf{u}'_n)\}_{n=1}^{N}$

registered DGF corpus in shared cycle time

$\xrightarrow{\text{calibration}}$

$\mathcal{D}_R = \{(\mathbf{m}_n, \mathbf{S}_n)\}_{n=1}^{N}$

PRISM posteriors in amplitude space

Figure 26 | **From EGIFA waveforms to the calibrated PRISM object.** Left: retained voiced groups in physical time before differentiation, normalization, and registration. Center: the registered DGF corpus $\mathcal{D}$ used for PACK calibration in shared cycle time. Right: the calibrated PRISM output $\mathcal{D}_R$, that is, one Gaussian amplitude posterior per registered waveform. The compression from $\mathcal{D}_R$ to $\mathcal{D}_Q$, $\mathcal{M}_Q$, and finally $\mathcal{M}_R$ is continued in Figure 31.

With these in hand, t-PRISM (Algorithm 4) then clusters consistent progressions of such GCIs into voiced groups; that is, it segments the raw $\{u(t)\}$ waveforms into regions where the glottal cycle has reached something resembling steady state.

In this manner, most of the amplitude and phase variations in the raw data $\{u(t)\}$ are normalized to an orderly collection of waveforms with reliable signal to learn from.

We leave a detailed discussion of the registration procedure to Dataset 1 and now proceed to the task of *calibration*; that is, learning the hyperparameters of the PACK kernels from the aligned $\{u'(\tau)\}$ exemplars.

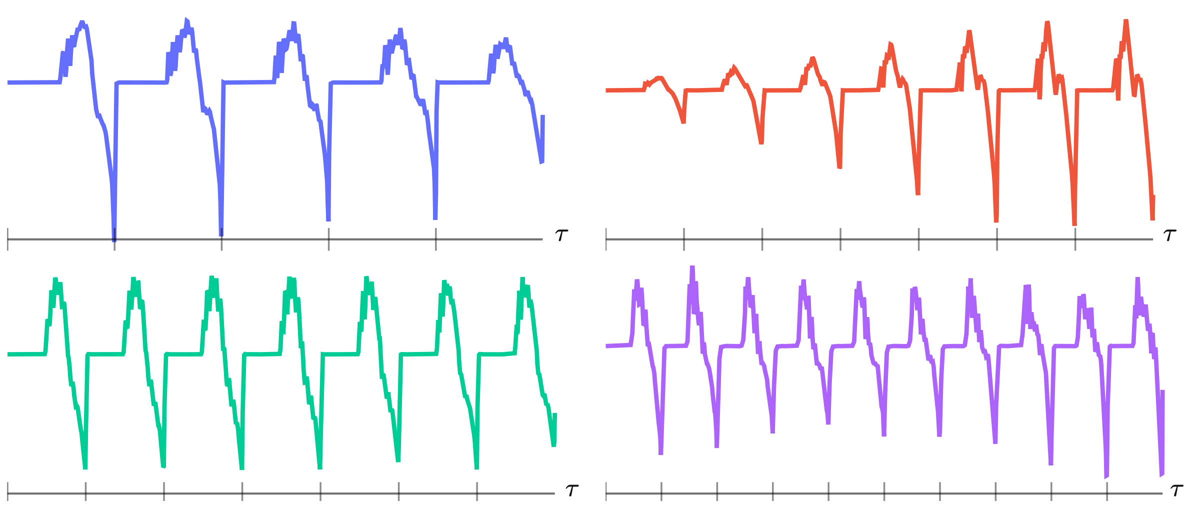


Figure 27 | **Four samples from the registered EGIFA corpus.** From left to right and top to bottom these are waveforms with index $n = 19, 601, 1452, 5892$ out of a total $N = 7697$, with $\left(M_n, \tau_{nM_n}\right) = (875, 5), (884, 7), (860, 7), (759, 10)$.

5.5 Calibration on EGIFA

In ordinary GP regression we have one data set, learn a distribution over latent functions parametrized by the hyperparameters, and predict in the context of that same data set. Our setting here is quite different: we are given a large collection of N aligned DGF waveforms

$$\mathcal{D} = \{(\boldsymbol{\tau}_n, \mathbf{u}'_n)\}_{n=1}^{N}, \tag{148}$$

where waveform n has its own registered grid

$$\boldsymbol{\tau}_n = \left(\tau_{n1}, ..., \tau_{nM_n}\right) \tag{149}$$

and standardized DGF samples

$$\mathbf{u}'_n = \left(u'_{n1}, ..., u'_{nM_n}\right). \tag{150}$$

As Figure 27 illustrates, each waveform (150) is a data set in its own right, with its own length M_n, its own irregularly sampled grid (149) and its own number of completed cycles τ_{nM_n}.

Given one of the above kernels $k_{\boldsymbol{\theta}}(\tau, \tau')$, the task is then to learn a single set of kernel hyperparameters $\boldsymbol{\theta}$ that maximizes the support for $\mathcal{D}$. To distinguish from ordinary GP inference, we call this process *calibration*: learning a common basis from the corpus without yet "finetuning" on individual data points.

This is similar to basis learning in functional data analysis, but here we incorporate GP features like kernel regularization; minibatching; principled handling of irregularly sampled data. Algorithm 3 gives the derivation and further details. The basic idea is to extend the well-known sparse variational inference machinery of Titsias (2009) from a single function to many independent waveforms. We refer to this extension as PRISM, for *process-induced surrogate modeling*, and use it to accomplish the calibration task. The output is a learned shared feature map $\boldsymbol{\varphi}(\tau)$ together with the amplitude-space collection

$$\mathcal{D}_R = \{(\mathbf{m}_n, \mathbf{S}_n)\}_{n=1}^{N}, \tag{151}$$

which is the exact object handed on to refinement in Chapter 6.

Quantile	0%	25%	50%	75%	95%
Cycles covered	5	5	6	8	9

Table 8 | **Quantiles of covered pitch cycles at width $W = 2048$** across the registered EGIFA corpus. At the 95th percentile, the padded width accommodates about 9 cycles per frame. At the standard IKLP frame length of 32 ms and a 99th-percentile F_0 of roughly 250 Hz, the maximum expected number of pitch periods per frame is about $32 \text{ ms} \times 0.25 \text{ kHz} = 8$; this table confirms that $W = 2048$ covers 95% of observed cycle counts comfortably.

5.5.1 Experimental setup

All calibration experiments use the registered EGIFA corpus from Dataset 1. The raw objects there are registered voiced groups; the calibration unit here is a *frame*. Each voiced group is first resampled to 20 kHz, smoothed, differentiated, and standardized within group, and is then cut without overlap into frames of width $W = 2048$. Only incomplete terminal frames are padded, purely for GPU batching, and padded positions are masked so they do not contribute to the PRISM objective. At this width, Table 8 shows that 95% of retained frames cover at most about nine glottal cycles $\tau_{\max} \sim 9$.

The framing step produces $N = 7697$ registered frames in total, from which we retain 6500 for training and a disjoint held-out set of 1000 for testing. For ungated models only GCIs are aligned to integers; for gated models GOIs are additionally aligned to half-integers, so that the gate transition in (147) remains phase-consistent across the corpus.

Each run maximizes the minibatch PRISM objective (250), that is, a stochastic estimate of the collapsed corpus objective (240). We optimize with Adam (Kingma & Ba, 2015) at learning rate 10^{-3} for 2500 iterations with a batch size of $B = 128$ complete frames. Learning is stable as judged from ELBO curves, and we optimize on the GPU in float32 with Cholesky jitter 10^{-5} throughout.

All runs use the same RBF envelope (143) and rational quadratic baseline (144) inside QPACK (142) or g-QPACK (145). What varies is the carrier, whether that carrier is gated, the harmonic order J, and the inducing rank R. Here `pack:0` and `pack:1` mean that the carrier is the $d = 0$ or $d = 1$ PACK kernel (133), while `periodic` replaces that carrier by the standard periodic kernel (140), which acts as a baseline. The values of J and R investigated here were chosen from early exploratory runs, and the present grid is meant to decide which prior structure is worth carrying forward given limited compute.

Concretely, we compare carrier names `pack:0`, `pack:1`, and `periodic`, harmonic orders $J \in \{16, 32\}$ and inducing basis sizes $R \in \{128, 192\}$. Each configuration is run from 32 random initializations. For evaluating model performance we use the held-out reduced-rank log evidence per effective sample

$$\mathcal{S}_{\text{test}}^{c(k)} := \frac{1}{|\mathcal{D}_{\text{test}}|} \sum_{n \in \mathcal{D}_{\text{test}}} \frac{1}{M_n^{\text{eff}}} \log p_{\text{RR}}(\mathbf{u}'_n \mid \boldsymbol{\tau}_n, k), \tag{152}$$

where k denotes the kernel being tested, M_n^{eff} denotes the number of non-padded samples in held-out frame n, and

$$p_{\text{RR}}(\mathbf{u}'_n \mid \boldsymbol{\tau}_n, k) = \texttt{Normal}\Big(\mathbf{u}'_n \mid \mathbf{0}, \mathbf{Q}_{nn} + \sigma^2 \mathbf{I}_{M_n^{\text{eff}}}\Big) \tag{153}$$

is exactly the reduced-rank Gaussian term from (239) in Algorithm 3, evaluated on the held-out grid $\boldsymbol{\tau}_n$ at the fitted global parameters of kernel k. It should be noted that good

carrier	R	J	QPACK	g-QPACK
`pack:1`	192	16	0.451 ± 0.109	0.560 ± 0.071
`pack:1`	192	32	0.298 ± 0.136	0.538 ± 0.083
`pack:1`	128	16	0.353 ± 0.121	0.480 ± 0.091
`pack:1`	128	32	0.221 ± 0.145	0.412 ± 0.169
`pack:0`	192	16	0.161 ± 0.081	0.349 ± 0.086
`pack:0`	192	32	0.040 ± 0.055	0.255 ± 0.064
`pack:0`	128	16	0.169 ± 0.059	0.305 ± 0.096
`pack:0`	128	32	0.011 ± 0.104	0.161 ± 0.094
`periodic`	192	1	−0.107 ± 0.284	0.138 ± 0.240
`periodic`	128	1	0.020 ± 0.245	0.239 ± 0.251

Table 9 | **Held-out calibration scores on EGIFA.** Each entry reports mean ± sd of (152) across 32 independent random restarts. Indicated are the QPACK's carrier, rank R and the harmonic order J, which is 1 for the standard periodic kernel.

performance on this test is not necessarily indicative of good performance in the GIF task, though we expect these to correlate.

As a sanity check, every trained model is also compared to a white noise model with kernel $k_{\text{null}}(\tau, \tau') = \sigma^2 \delta_{\tau\tau'}$ with matched variance σ^2 and observation noise scale. On the scale of (152), the best null models achieve scores $\mathcal{S}_{\text{test}}^{c(\text{null})} \approx -6$, which indicates that the calibration fits pick up significant structure from the data.

5.5.2 Results

The results are summarized in Table 9, which reports (152) across random restarts.

The strongest mean configuration is `g-QPACK`, `pack:1`, `R=192`, `J=16`, with a held-out score of 0.560 ± 0.071. Its ungated counterpart, `QPACK`, `pack:1`, `R=192`, `J=16`, reaches 0.451 ± 0.109. Across the matched grid, g-QPACK scores higher mean scores than QPACK for every carrier and basis size reported here, and also has a smaller deviation around that mean. Therefore the gated variant of QPACK seems to be worthwile, and Chapter 7 will confirm this for GIF.

The effects of basis size R and harmonic order J are as follows: a higher R increases performance for the g-QPACK kernels, but not always for the QPACK kernels, and never for the periodic carrier. A similar tempering effect is seen for J: at least on this test, $J = 32$ seems to be overfitting. This is comparable to kernels with too short lengthscales overfitting data and generalizing badly. It thus appears that $J = 16$ is a better choice here.

The held-out fits in Figure 28 show that the strongest mean configuration has learnt bases that are able to extrapolate beyond the observed data (left column) and model very sharp GCI spikes (right column). It is also evident in the top left pane that QPACK loses memory of the

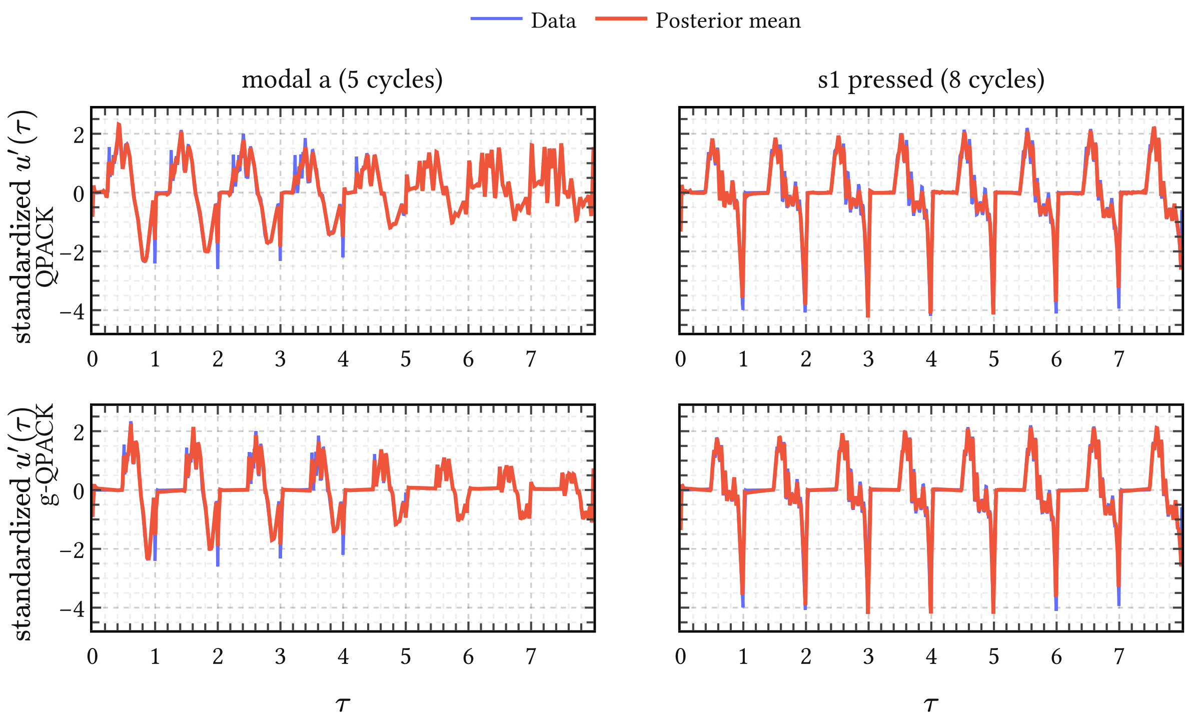


Figure 28 | **Held-out fits from the strongest matched QPACK and g-QPACK configurations.** The best QPACK and g-QPACK restarts from the `pack:1`, `R=192`, `J=16` family fit held-out data and extrapolate from $\tau = 4$ up until $\tau_{\max} = 8$ in the left column.

closed phase quickly, while g-QPACK is able to retain it. Note also the amplitude modulation by the envelope in the bottom left pane modeling a decrease in volume.

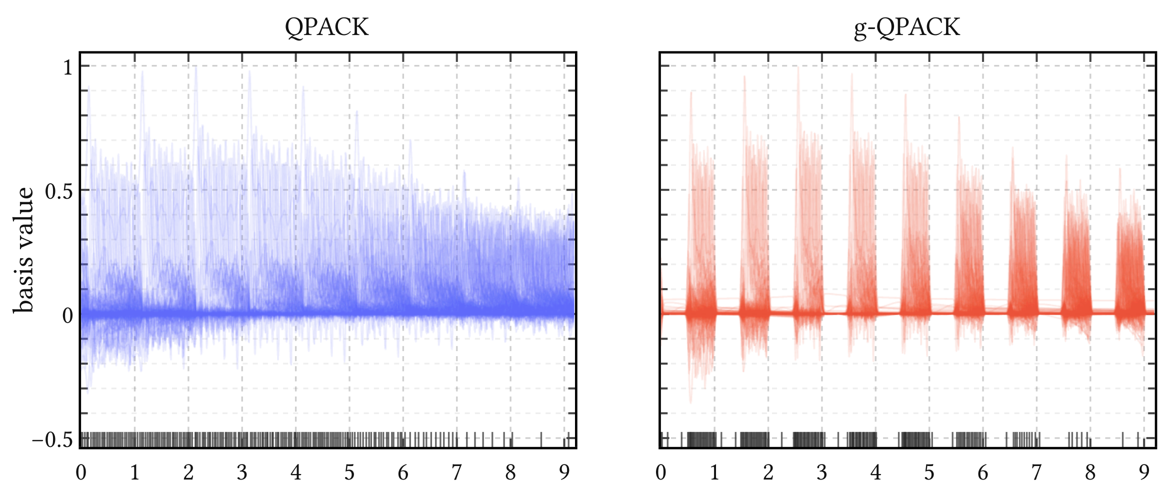


Figure 29 | **Learned bases for the strongest matched QPACK and g-QPACK configurations.** Left: the best QPACK restart within the `pack:1`, `R=192`, `J=16` family. Right: the best g-QPACK restart within the same family. Each panel shows the learned basis functions $\varphi(\tau) \in \mathbb{R}^R$ on a τ-grid. The locations of the inducing points are also shown as a carpet plot.

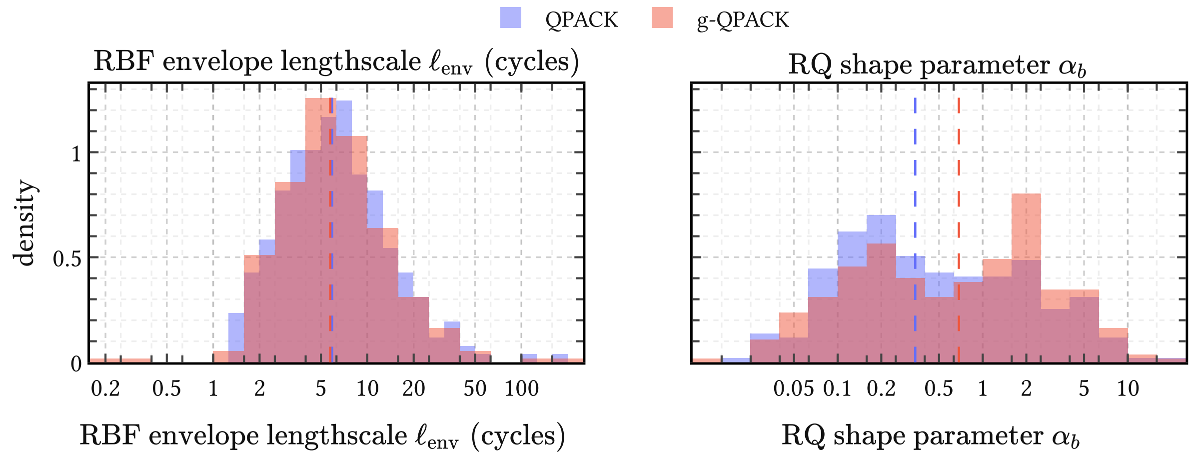


Figure 30 | **Distributions of learned long-scale hyperparameters across restarts.** Left: the RBF envelope lengthscale ℓ_{env}, which measures cycle-to-cycle coupling in units of glottal cycles. Right: the rational quadratic shape parameter α_b of the additive baseline.

An impression of the learned bases in Figure 29 make the effect of gating more tangible. In QPACK, both the basis functions and the inducing locations are spread across the full cycle. In g-QPACK, covariance is suppressed in the closed phase, and the inducing locations are correspondingly pushed out of it and packed more densely into the open phase. So gating not only improves the aggregate score; it redistributes basis resolution towards the more dynamic phase of the glottal cycle.

Some future work One obvious extension would be to relax the envelope itself. The histograms in Figure 30 show a broad distribution of learned envelope lengthscales across random restarts, which suggests that the simple RBF envelope may still be too rigid and that an RQ envelope may be worth testing. The current grid is also incomplete: `pack:2` is still missing, though early pilot runs and the known tail decay of the DGF suggest it is unlikely to beat `pack:0` or `pack:1`; and the ranges $J = 1, 2, 4, 8$ and $R = 16, 32, 64$ remain to be checked in order to quantify where the present gains first appear. We did not push these directions further here simply because of limited compute.

5.6 Summary

We began with PACK, the periodic extension of TACK obtained by replacing the linear time coordinate with a harmonic embedding on the circle. Its stationarity and exact line spectrum made it a convenient starting point, and its harmonic weights provide spectral flexibility that the standard periodic squared exponential cannot match: where the latter traces a one-parameter low-pass envelope, PACK can redistribute mass freely across harmonics while the degree d independently sets the asymptotic rolloff.

QPACK then relaxed strict periodicity to more realistic quasiperiodic behavior by modulating the carrier with a slow RBF envelope and adding a weak rational quadratic baseline. The decisive step, however, was gating: g-QPACK suppresses covariance in the closed phase directly through a periodic sigmoid gate rather than relying on posterior inference alone. This breaks stationarity, but the calibration experiments show it is worth the trade: g-QPACK consistently outperforms its ungated counterpart across all carriers and basis sizes, and redistributes inducing resolution from the closed phase into the more informative open phase. This comes with the downside of requiring GOI information next to GCI information.

To make calibration possible at all, the raw EGIFA waveforms first had to be registered. LBGID and t-PRISM turned a large and heterogeneous collection of simulated glottal flows into aligned voiced groups in shared cycle time, compressing the corpus into a form amenable to learning. On that aligned corpus, PRISM then calibrated a single low-rank surrogate basis shared across thousands of waveforms, with stable optimization throughout.

The resulting basis extrapolates cleanly beyond observed data and reproduces sharp GCI spikes, confirming that the registration and calibration pipeline does extract meaningful structure from the corpus. This calibrated basis is the endpoint of the present chapter; the next chapter takes it as input and refines it into richer latent models of the DGF.

6 Surrogate glottal flow models

In the last two chapters we have been working towards building a suitable DGF prior $p(\mathbf{u}' \mid \boldsymbol{\tau}, \mathcal{M}_R)$. Chapter 4 derived ACK and TACK from changepoints in classical glottal flow models, in order to move from the parametric to the nonparametric regime. Chapter 5 specialized the TACK to PACK, QPACK and g-QPACK and calibrated the resulting kernels on registered EGIFA waveforms. That calibration step gave us a useful global basis, able to extrapolate succesfully, but not yet a compact generative model of the DGF.

More precisely, after learning, PRISM yields a global basis of rank R that is efficient in representing a large corpus of DGF examplars. Each examplar can then be projected on that basis via (253) to give the amplitude-space collection

$$\mathcal{D}_R = \{(\mathbf{m}_n, \mathbf{S}_n)\}_{n=1}^N, \tag{154}$$

whose means $\{\mathbf{m}_n\}$ are coordinates in *amplitude space*. PRISM thus maps the registered corpus

$$\mathcal{D} = \{(\boldsymbol{\tau}_n, \mathbf{u}'_n)\}_{n=1}^N \tag{155}$$

from variable-size data space to one shared fixed-dimensional amplitude space in $\mathbb{R}^R$.

This further manipulation is where the current chapter comes in. q-GPLVM compresses $\mathcal{D}_R$ into $K \ll N$ compact components in a nonlinear *manifold space* of dimension $Q \ll R$. Each k-th component thus captures a discovered DGF “mode” or archetype. We shall see below that these modes mainly capture smooth variation in the open quotient (OQ) for the QPACK kernels, and additional fine structure for the g-QPACK kernels. We then push each k-th component from manifold space via amplitude space to data space directly, and thus end up with a mixture of K sparse GP models that define the generative prior $p(\mathbf{u}' \mid \boldsymbol{\tau}, \mathcal{M}_R)$ for DGF waveforms. We call these sparse GP models *surrogate glottal flow models* because they cheaply emulate a corpus of expensive-to-simulate glottal flow examples.

We explore $p(\mathbf{u}' \mid \boldsymbol{\tau}, \mathcal{M}_R)$ in this chapter via generative samples, combining it with *time warping* and the $\mathrm{AR}(P)$ prior from Chapter 3 for generating short voiced speech fragments. In the next chapter, we use it for GIF by plugging the mixture directly into IKLP, which then knows about what kind of DGFs to expect in real-world speech, and how to regularize accordingly.

6.1 Refinement

Refinement on EGIFA (Dataset 1) is the last step after registration (Section 5.4) and calibration (Section 5.5) on that same corpus.

Figure 31 shows that the full pipeline has an hourglass, VAE-like structure:

- PRISM compresses variable-length waveforms to a fixed amplitude space in $\mathbb{R}^R$;

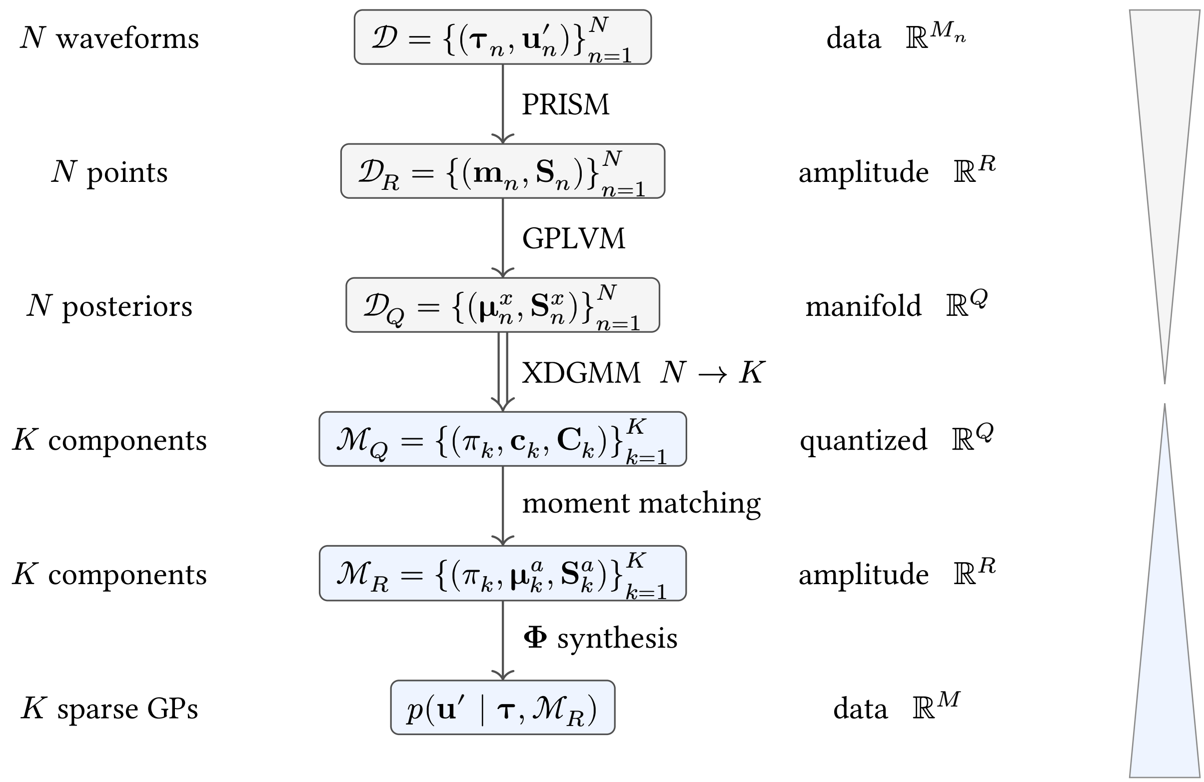


Figure 31 | **From examplars to surrogate.** The hourglass on the right tracks the imploding and expanding dimensionalities. The orders of magnitude involved are

$$M_n \sim M \sim 10^3 > R \sim 10^2 > K \sim 10^1 > Q \sim 10^0. \tag{156}$$

Top half: compression of N variable-length DGF waveforms to N posteriors in a small Q-dimensional manifold. The double arrow marks the $N \to K$ quantization step. Bottom half: synthesis of K sparse GP surrogates in data space from the quantized manifold components.

- q-GPLVM discovers a low-dimensional manifold in $\mathbb{R}^Q$ at the bottleneck;
- XDGMM quantizes that manifold into K Gaussian mixture components;
- and those components are pushed back through amplitude space to data space as sparse GPs with again variable dimension M.

We now proceed to define each object below. Refer to Algorithm 3 and Algorithm 5 for detailed derivations and more details on PRISM and q-GPLVM, respectively.

6.1.1 The three spaces

Data space Each registered DGF waveform $\mathbf{u}'_n \in \mathbb{R}^{M_n}$ lives in a potentially different-dimensional space because cycle counts and sampling grids vary across utterances. PRISM (Algorithm 3) absorbs this variability by learning a shared feature map $\boldsymbol{\varphi}(t)$ of rank R, under which any waveform is synthesized as

$$u'(t) \approx \boldsymbol{\varphi}(t)^\top \mathbf{a}, \quad \mathbf{a} \in \mathbb{R}^R. \tag{157}$$

Amplitude space Projecting each waveform onto this basis yields the amplitude-space object

$$\mathcal{D}_R = \{(\mathbf{m}_n, \mathbf{S}_n)\}_{n=1}^{N}. \tag{158}$$

By interpreting the projection (157) as inference, this induces the empirical density

$$p(\mathbf{a} \mid \mathcal{D}_R) = \frac{1}{N}\sum_{n=1}^{N} \texttt{Normal}(\mathbf{a} \mid \mathbf{m}_n, \mathbf{S}_n). \tag{159}$$

This is not yet a compact model: it has N components and remains tied to the training set. Note that, unlike similar adaptor approaches in functional data analysis, PRISM actually infers a latent *density* in the amplitude space, being the extension of a well-posed Bayesian inference problem (Titsias, 2009). We talk about point clouds $\{\mathbf{m}_n\}$ here and discard the uncertainty in the $\{\mathbf{S}_n\}$ because q-GPLVM demands point estimates.

Manifold space q-GPLVM (Algorithm 5) discovers a Q-dimensional nonlinear manifold within the amplitude cloud. Each training waveform n receives a variational posterior, collected as

$$\mathcal{D}_Q = \{(\boldsymbol{\mu}_n^x, \mathbf{S}_n^x)\}_{n=1}^{N}, \tag{160}$$

equivalently

$$q(\mathbf{X} \mid \mathcal{D}_Q) = \prod_{n=1}^{N} \texttt{Normal}(\mathbf{x}_n \mid \boldsymbol{\mu}_n^x, \mathbf{S}_n^x). \tag{161}$$

This is the same collection viewed in a shared manifold coordinate system. Note that the S_n^x covariance matrices are diagonal. We set $Q := 3$, motivated by the six parameters of the LF model (59) with amplitude, open quotient and period already normalized at registration time, leaving three residual degrees of freedom. This also keeps the manifold directly visualizable.

6.1.2 Quantization and pushforward

The manifold posteriors (160) are quantized into $K \ll N$ components by XDGMM (Algorithm 5.4.):

$$\mathcal{M}_Q = \{(\pi_k, \mathbf{c}_k, \mathbf{C}_k)\}_{k=1}^{K}, \tag{162}$$

equivalently

$$q(\mathbf{x} \mid \mathcal{M}_Q) = \sum_{k=1}^{K} \pi_k \, \texttt{Normal}(\mathbf{x} \mid \mathbf{c}_k, \mathbf{C}_k), \quad \mathbf{c}_k \in \mathbb{R}^Q, \quad \mathbf{C}_k \in \mathbb{R}^{Q\times Q}. \tag{163}$$

Because q-GPLVM returns Gaussian posteriors rather than point estimates, XDGMM accounts for the latent uncertainties $\mathbf{S}_n^x$ directly: this time we do not have to discard it.

Each quantized component is then pushed forward through the learned GPLVM map to amplitude space via moment matching ((303), (308)), giving the compact object

$$\mathcal{M}_R = \{(\pi_k, \boldsymbol{\mu}_k^a, \mathbf{S}_k^a)\}_{k=1}^K, \quad \boldsymbol{\mu}_k^a \in \mathbb{R}^R, \quad \mathbf{S}_k^a \in \mathbb{R}^{R\times R}. \tag{164}$$

and hence the amplitude-space density

$$p(\mathbf{a} \mid \mathcal{M}_R) = \sum_{k=1}^{K} \pi_k \, \texttt{Normal}(\mathbf{a} \mid \boldsymbol{\mu}_k^a, \mathbf{S}_k^a). \tag{165}$$

This is where *uncertainty calibration* enters. q-GPLVM returns uncertain manifold posteriors $\mathcal{D}_Q$, XDGMM fits $\mathcal{M}_Q$ against both their means and covariances, and moment matching then pushes that spread forward into $\mathcal{M}_R$. The resulting surrogate is therefore intentionally less rigid than a pure point-estimate fit, which matters both for generation here and for avoiding overconfident priors later in Chapter 7.

6.1.3 The DGF prior

Before refinement, the same synthesis map already turns the empirical PRISM density $p(\mathbf{a} \mid \mathcal{D}_R)$ into the unrefined data-space prior

$$p(\mathbf{u}' \mid \boldsymbol{\tau}, \mathcal{D}_R) = \frac{1}{N} \sum_{n=1}^{N} \texttt{Normal}\big(\mathbf{u}' \mid \boldsymbol{\Phi}(\boldsymbol{\tau})\mathbf{m}_n, \boldsymbol{\Phi}(\boldsymbol{\tau})\mathbf{S}_n\boldsymbol{\Phi}(\boldsymbol{\tau})^\top + \sigma^2\mathbf{I}\big). \tag{166}$$

Refinement replaces this N-component empirical prior by the compact K-component surrogate below.

Composing $\mathcal{M}_R$ with the PRISM feature map $\boldsymbol{\Phi}(\boldsymbol{\tau})$ yields the sought object: a mixture of K sparse Gaussian process priors over DGF waveforms,

$$p(\mathbf{u}' \mid \boldsymbol{\tau}, \mathcal{M}_R) = \sum_{k=1}^{K} \pi_k \, \texttt{Normal}(\mathbf{u}' \mid \boldsymbol{\mu}_k^u, \boldsymbol{\Sigma}_k^u) \tag{167}$$

$$\text{where} \qquad \boldsymbol{\mu}_k^u = \boldsymbol{\Phi}(\boldsymbol{\tau})\boldsymbol{\mu}_k^a, \tag{168}$$

$$\boldsymbol{\Sigma}_k^u = \boldsymbol{\Phi}(\boldsymbol{\tau})\mathbf{S}_k^a\boldsymbol{\Phi}(\boldsymbol{\tau})^\top + \sigma^2\mathbf{I}. \tag{169}$$

Each component is a sparse (degenerate) GP: its covariance $\boldsymbol{\Phi}\mathbf{S}_k^a\boldsymbol{\Phi}^\top$ has rank at most R, inherited from the PRISM basis, plus isotropic noise $\sigma^2\mathbf{I}$. The k-th component's mean function $\boldsymbol{\Phi}\boldsymbol{\mu}_k^a$ is a learned DGF archetype; its covariance encodes the plausible variation around that archetype.

6.1.4 Learning

Refinement starts from a retained calibrated PRISM source on EGIFA. That source fixes the learned feature map $\boldsymbol{\varphi}(\tau)$ and the amplitude-space object $\mathcal{D}_R$.

We then fit a Bayesian GPLVM at fixed $Q = 3$, obtaining $\mathcal{D}_Q$, quantize its latent posteriors by XDGMM at $K \in \{1, 3, 5, 7\}$ to obtain $\mathcal{M}_Q$, and push those components back to amplitude space as $\mathcal{M}_R$ and then to data space as the surrogate prior (167). The two optimizations are separate:

- The GPLVM stage optimizes the variational bound (293) by full-batch Adam on all 6500 training waveforms in $\mathcal{D}_R$ at once, with learning rate $5 \cdot 10^{-2}$ for 4000 iterations and otherwise standard settings.
- The XDGMM stage then fits $\mathcal{M}_Q$ by a separate EM optimization of the standard extreme-deconvolution likelihood (Algorithm 5.4.), using the implementation of (Holoien et al., 2017) for 100 EM iterations.

As in the calibration experiments, all runs were carried out on the GPU in float32 and both stages proved uneventful and predictable across the full sweep.

For one held-out EGIFA waveform $(\boldsymbol{\tau}_n, \mathbf{u}'_n)$, the refinement score is the q-GPLVM log evidence per effective sample,

$$\mathcal{S}_{\text{test}}^{q(\mathcal{M}_R)} = \frac{1}{N_{\text{test}}} \sum_{n=1}^{N_{\text{test}}} \frac{1}{M_n} \log p(\mathbf{u}'_n \mid \boldsymbol{\tau}_n, \mathcal{M}_R), \tag{170}$$

where M_n is the number of observed samples in waveform n. Here $N_{\text{test}} = 1000$. This score gauges how realistic the refined surrogate prior is, but it does not guarantee strong performance when the same surrogate is used for IKLP inference in Chapter 7.

6.2 Experiments

We refine the top-5 best-scoring PRISM calibration fits on EGIFA for each family (QPACK, g-QPACK) and EGIFA collection (vowel, speech). After deduplication this gives 17 calibrated sources in total, 8 from the QPACK family and 9 from the g-QPACK family, each compressed at $Q = 3$ and $K \in \{1, 3, 5, 7\}$, for 68 q-GPLVM fits overall. Table 10 shows the result. Every refinement improves the held-out q-GPLVM score (170) over the score attained by its PRISM source. In other words, refinement helps in modeling more realistic DGF waveforms. That is encouraging, but it remains only a refinement score. The important question is still deferred: whether these surrogates actually help GIF once plugged into IKLP.

Since most refinement runs behave very similarly we choose two representative runs for illustration below: one QPACK source shown at $K = 5$, and one g-QPACK source at $K = 3$. Table 11 lists the precise configurations used.

6.2.1 QPACK

The performance and nature of the surrogate prior $p(\mathbf{u}' \mid \boldsymbol{\tau}, \mathcal{M}_R)$ depend on which kernel family PRISM used to produce the amplitude-space object $\mathcal{D}_R$. We begin with the representative QPACK refinement at $K = 5$ in Figure 32.

Family	Source runs	q-GPLVM fits	Improved	Median gain	Best q-GPLVM score
`QPACK`	8	32	32/32	+0.057	0.579
`g-QPACK`	9	36	36/36	+0.086	0.588

Table 10 | **Refinement sweep on EGIFA. Source runs** counts distinct calibrated PRISM sources; **q-GPLVM fits** counts all refinements over $K \in \{1, 3, 5, 7\}$. **Median gain** is the median increase in held-out score (170) over the corresponding PRISM score, and **Best q-GPLVM score** is the best absolute value attained within the family.

Family	Carrier	R	J	Iteration	K
`QPACK`	`pack:0`	128	16	15	5
`g-QPACK`	`pack:1`	128	32	31	3

Table 11 | **Configurations of representative refinements.** Both use $Q = 3$.

The top-left panel shows the posterior means $\{\boldsymbol{\mu}_n^x\}_{n=1}^N$ from $\mathcal{D}_Q$ as a diffuse manifold cloud, colored by ground-truth OQ. That diffuse appearance matters: once q-GPLVM has returned $\mathcal{D}_Q$, the subsequent XDGMM fit and moment-matching pushforward preserve uncertainty rather than collapsing to rigid point templates. The manifold is organized mainly by OQ, with only a small number of diffuse outliers around its edge.

Figure 33 shows that the same ordering becomes strongly nonlinear once it is pushed back to data space. The component means $\boldsymbol{\mu}_k^u = \boldsymbol{\Phi}(\boldsymbol{\tau})\boldsymbol{\mu}_k^a$ implied by $\mathcal{M}_R$ trace a smooth progression from more open and rounded cycles to sharper, peakier closure patterns. So the dominant OQ variation is real, but it does not appear as a simple linear direction in amplitude space: the same progression also sharpens the GCI region substantially.

The bottom row of Figure 32 makes the surrogate mixture explicit. The leftmost **prior** panel shows draws from the unrefined PRISM-induced prior $p(\mathbf{u}' \mid \boldsymbol{\tau}, \mathcal{D}_R)$ in (166). Each numbered panel instead draws

$$\mathbf{a} \sim \texttt{Normal}(\boldsymbol{\mu}_k^a, \mathbf{S}_k^a) \tag{171}$$

from component k of $p(\mathbf{a} \mid \mathcal{M}_R)$ in (165) and maps it to data space through (167). The slightly noisy appearance of these curves is exactly the uncertainty calibration just described: the surrogate retains posterior spread instead of collapsing to overconfident waveform templates.

6.2.2 g-QPACK

The g-QPACK manifold in Figure 34 is again ordered mainly by OQ. Since OQ is effectively normalized away during registration, this ordering means that there must be strong correlations between OQ and other features, which is incidentally a feature of the LF model (56). Its components are visibly tighter in the open-phase region because the gate function builds that suppression into the prior itself. The closed phase is correspondingly cleaner, though

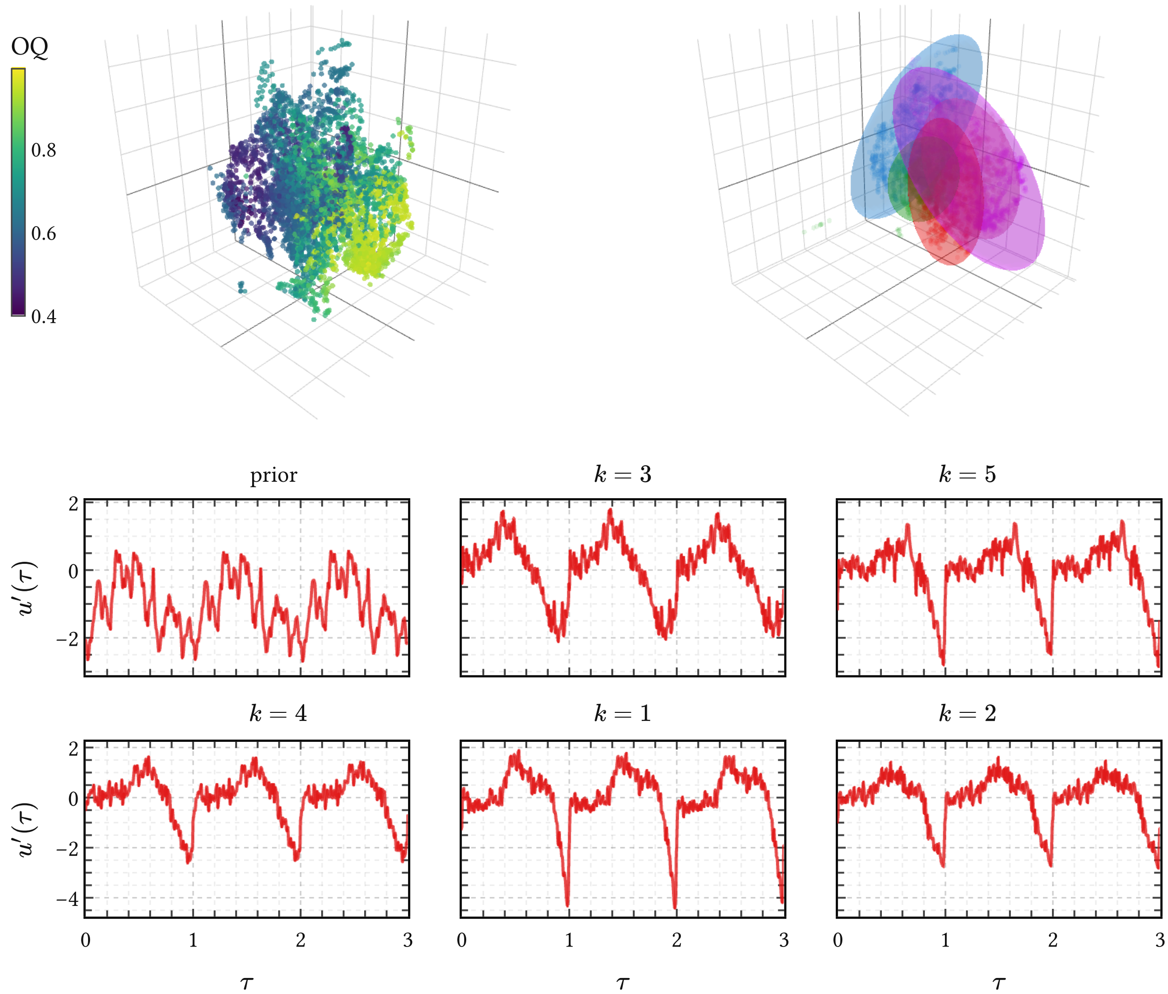


Figure 32 | **QPACK surrogate: manifold space, quantization, and data-space samples.** Top left: posterior means $\{\boldsymbol{\mu}_n^x\}_{n=1}^N$ from $\mathcal{D}_Q$, shown as a diffuse manifold cloud and colored by ground-truth OQ. Top right: the $K = 5$ XDGMM ellipsoids of $\mathcal{M}_Q$ fit to that uncertain manifold representation. Bottom: the left panel samples the unrefined PRISM prior $p(\mathbf{u}' \mid \boldsymbol{\tau}, \mathcal{D}_R)$ from (166); the numbered panels sample the component conditionals of $p(\mathbf{u}' \mid \boldsymbol{\tau}, \mathcal{M}_R)$ induced by (167). OQ is the dominant visible ordering variable, but a few diffuse outliers remain around the edge of the manifold.

no longer literally flat once that uncertainty calibration has been pushed back to data space. QPACK reaches much the same phonation organization without GOI information, but must learn it from the amplitude distribution $p(\mathbf{a} \mid \mathcal{D}_R)$ instead.

For generation, QPACK is the natural choice. The g-QPACK gate is defined relative to GOI positions, which must be available at evaluation time. QPACK needs only GCI positions, which the t-PRISM pitch model (Section 6.3, Algorithm 4) provides directly — no auxiliary GOI estimation step is required.

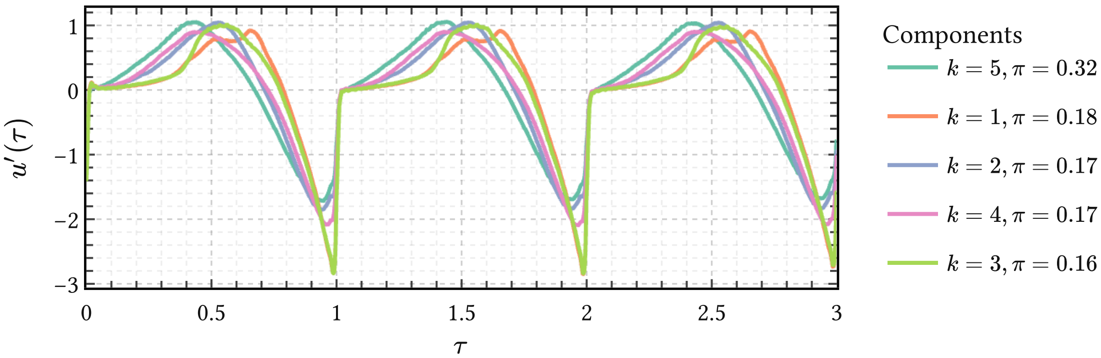


Figure 33 | **Learned QPACK component means in data space.** Shown are the data-space means $\boldsymbol{\mu}_k^u = \boldsymbol{\Phi}(\boldsymbol{\tau})\boldsymbol{\mu}_k^a$ implied by $\mathcal{M}_R$ in (167). Their smooth progression in OQ corresponds to a strongly nonlinear trajectory in amplitude space and sharpens closure from rounder to peakier cycles.

This is also the bridge to Chapter 7. Each surrogate component becomes one IKLP kernel hypothesis,

$$\mathbf{K}_k = \boldsymbol{\Phi}\mathbf{S}_k^a\boldsymbol{\Phi}^\top + \sigma^2\mathbf{I}, \tag{172}$$

with the corresponding GaP weight initialized from π_k. No extra codebook or adaptor layer is needed.

6.3 Phase warping

Static DGF surrogates are not enough for voiced speech. Even if the shape of one cycle is modeled well, realistic utterances still require the cycles to arrive at plausible times. That is the role of a pitch prior. The prior itself is learned on APLAWD (Dataset 2) using t-PRISM (Algorithm 4); here we use only the result. For a voiced sequence with GCI sample indices $n_{\mathrm{GCI},0}, \ldots, n_{\mathrm{GCI},K}$, the local periods

$$T_k = \frac{n_{\mathrm{GCI},k+1} - n_{\mathrm{GCI},k}}{f_s}, \quad k = 0, \ldots, K-1, \tag{173}$$

form a trajectory whose logarithm $\ell_k = \log T_k$ is modeled by a shared GP. Working in log-period does two things at once: it keeps periods positive and turns multiplicative pitch variation into additive GP structure. Prior draws from the learned model already produce smooth period sequences on the timescale of human speech, which is exactly the object needed to warp surrogate cycles into physical time (Figure 35).

6.3.1 Instantaneous phase and pitch

Much of the nonstationary variation of voiced speech can be captured by a single monotone phase function $\tau(t)$ satisfying

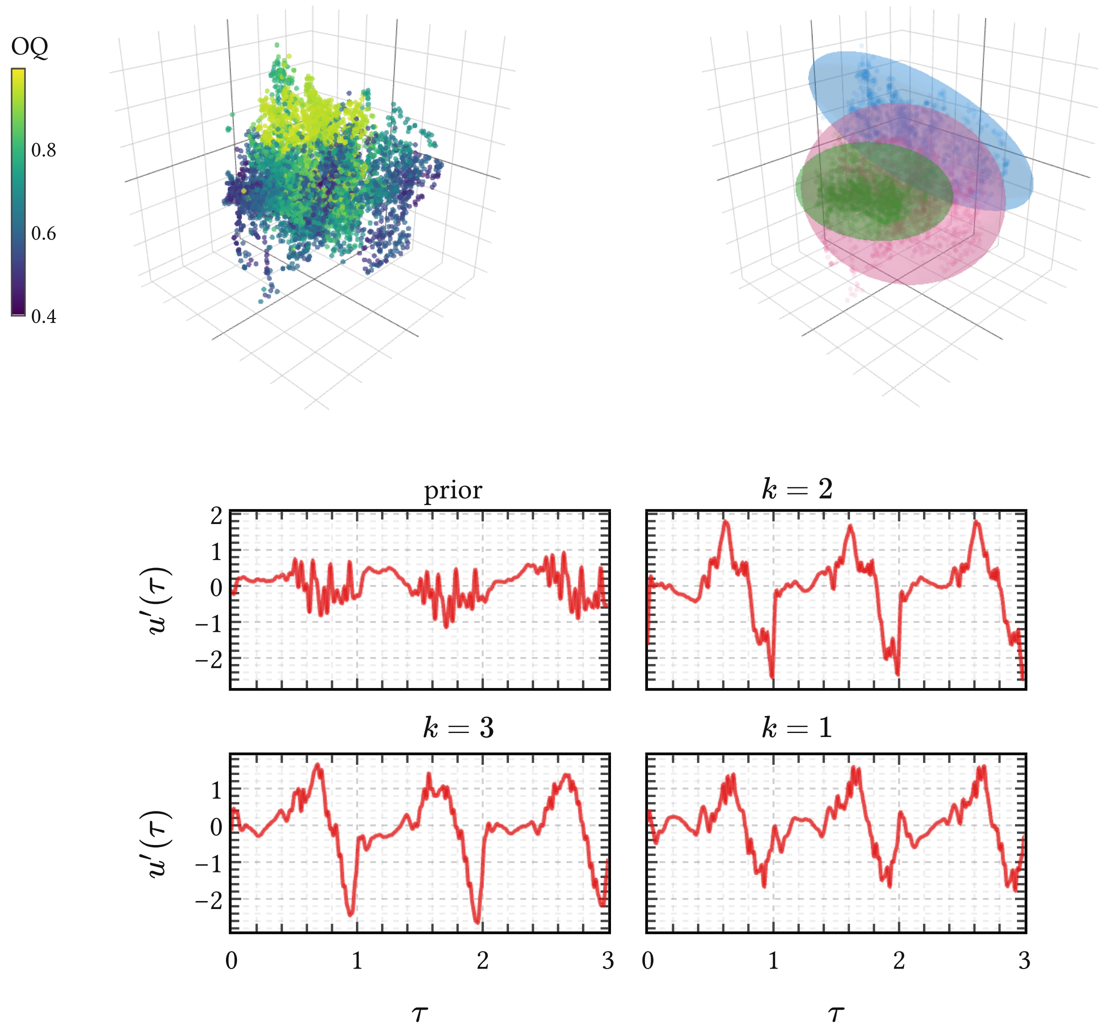


Figure 34 | **g-QPACK surrogate: manifold space, quantization, and data-space samples.** Same three-step layout as Figure 32, but here the fitted quantization uses only $K = 3$ components. The GOI-aware gate suppresses open-phase variance structurally; compare the tighter bottom-row spread with Figure 32. After uncertainty calibration the closed phase is no longer perfectly flat, but its variance is still substantially suppressed by the gate.

$$\frac{\mathrm{d}\tau}{\mathrm{d}t} = f_0(t), \tag{174}$$

where $f_0(t)$ is instantaneous pitch (Hz) and τ counts cycles (dimensionless). The useful view is the inverse one: express physical time as a function of phase, not the other way around.

For constant pitch the result is just $t(\tau) = t_0 + T\tau$. In general, defining the instantaneous period $T(\tau) := \frac{1}{f_0}(\tau)$ and integrating (174) gives

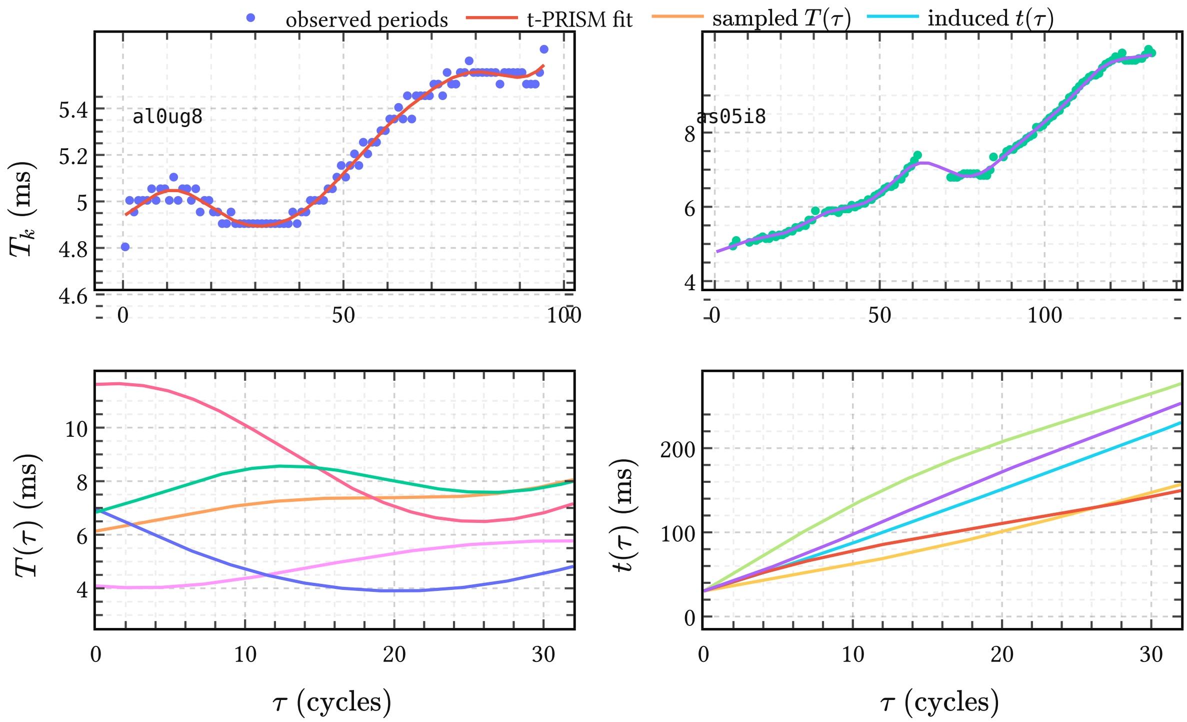


Figure 35 | **From learned pitch prior to time warp.** Top: two retained APLAWD period tracks together with their t-PRISM posterior means. Bottom left: prior samples of the instantaneous period function $T(\tau)$ from the learned APLAWD model. Bottom right: the corresponding induced warps $t(\tau) = t_0 + \int_0^\tau T(u)\,\mathrm{d}u$. This is the object needed later to turn refined DGF surrogates in cycle time into voiced signals in physical time.

$$t(\tau) = t_0 + \int_0^\tau T(u)\,\mathrm{d}u. \tag{175}$$

A GP prior on $T(\tau)$ therefore induces a prior over monotone warps directly. Sample a smooth period function, integrate it, and a physically plausible phase map follows automatically. This is why the pitch model belongs here: it does not provide labels or segmentation anymore, but the continuous-time warp that stretches normalized DGF cycles into speech time.

6.4 Generative voiced speech

With $\mathcal{M}_R$ providing data-space DGF cycles in normalized time τ and the pitch prior of Section 6.3 converting those cycles to physical time t, both ingredients for a voiced source model are in hand. Adding an AR filter closes the loop to speech.

6.4.1 DGF synthesis

To sample a voiced source fragment, first draw a smooth period trajectory $T(\tau)$ over the whole segment and integrate (175) to obtain the physical-time warp. Then, cycle by cycle, draw a component $k \sim \text{Categorical}(\pi_1, ..., \pi_K)$ and amplitude $\mathbf{a} \sim \texttt{Normal}(\boldsymbol{\mu}_k^a, \mathbf{S}_k^a)$ from

$\mathcal{M}_R$ (164), evaluate the PRISM synthesis map (281) on a dense within-cycle grid $\tau \in [0, 1)$, and place the resulting normalized DGF cycle on the sampled physical-time grid.

This is where QPACK becomes the practical downstream choice. The phase story supplies GCIs through the pitch warp, but not GOIs. QPACK needs only those GCIs; g-QPACK would require an auxiliary GOI track as well.

Concatenating the warped cycles assembles $u'(t)$. Shape variation comes from the Gaussian spread inside each component, timing variation from the period draws $T(\tau)$, and resampling k across cycles allows transitions between phonation modes.

6.4.2 Voiced speech

A speech fragment follows from filtering the generated source with a sampled vocal tract. The AR coefficients are drawn from the SoftSpectral prior (51) conditioned on a target formant structure:

$$\mathbf{a}_{\mathrm{AR}} \sim \texttt{SoftSpectral}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0; M), \tag{176}$$

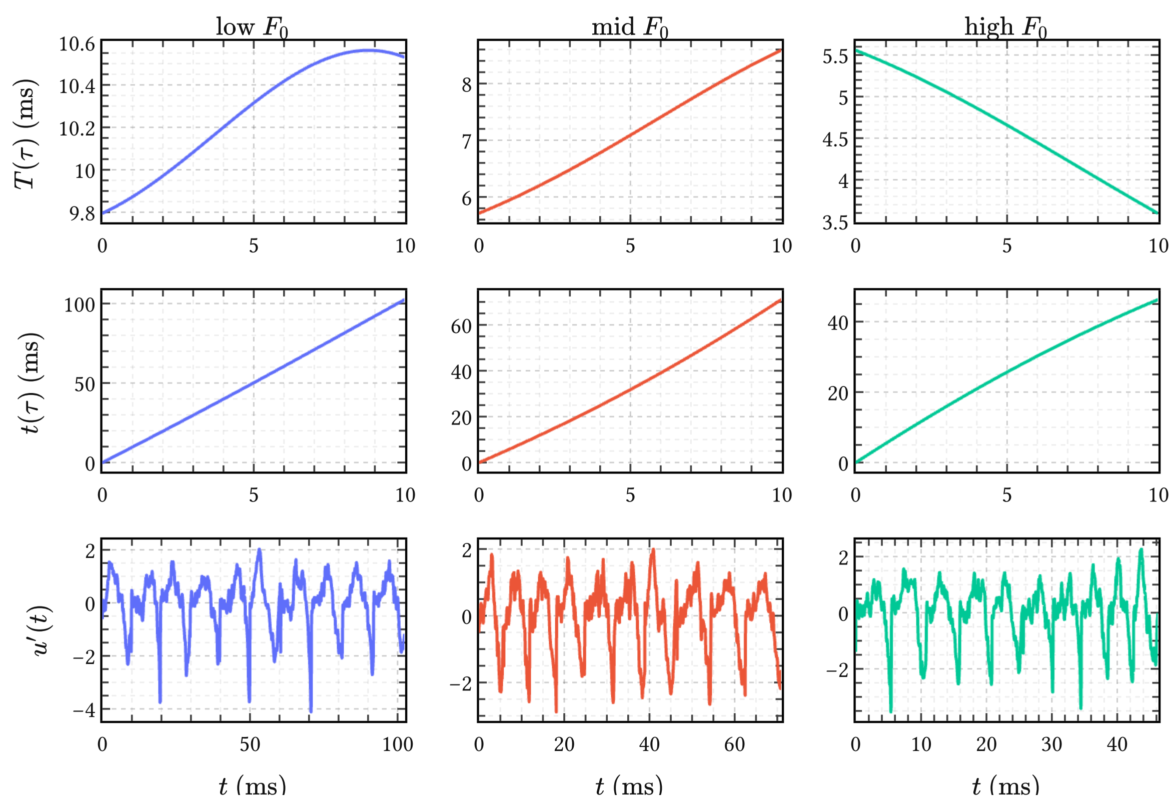


Figure 36 | **From data-space surrogate to voiced source.** Top: three sampled instantaneous period trajectories $T(\tau)$ from the t-PRISM prior, representing low, mid, and high F_0 regimes. Center: the induced time warps $t(\tau) = t_0 + \int_0^\tau T(u)\,\mathrm{d}u$ from (175). Bottom: piecewise DGF waveforms $u'(t)$ assembled from QPACK data-space surrogate draws and stretched to physical time; each spans ten pitch periods.

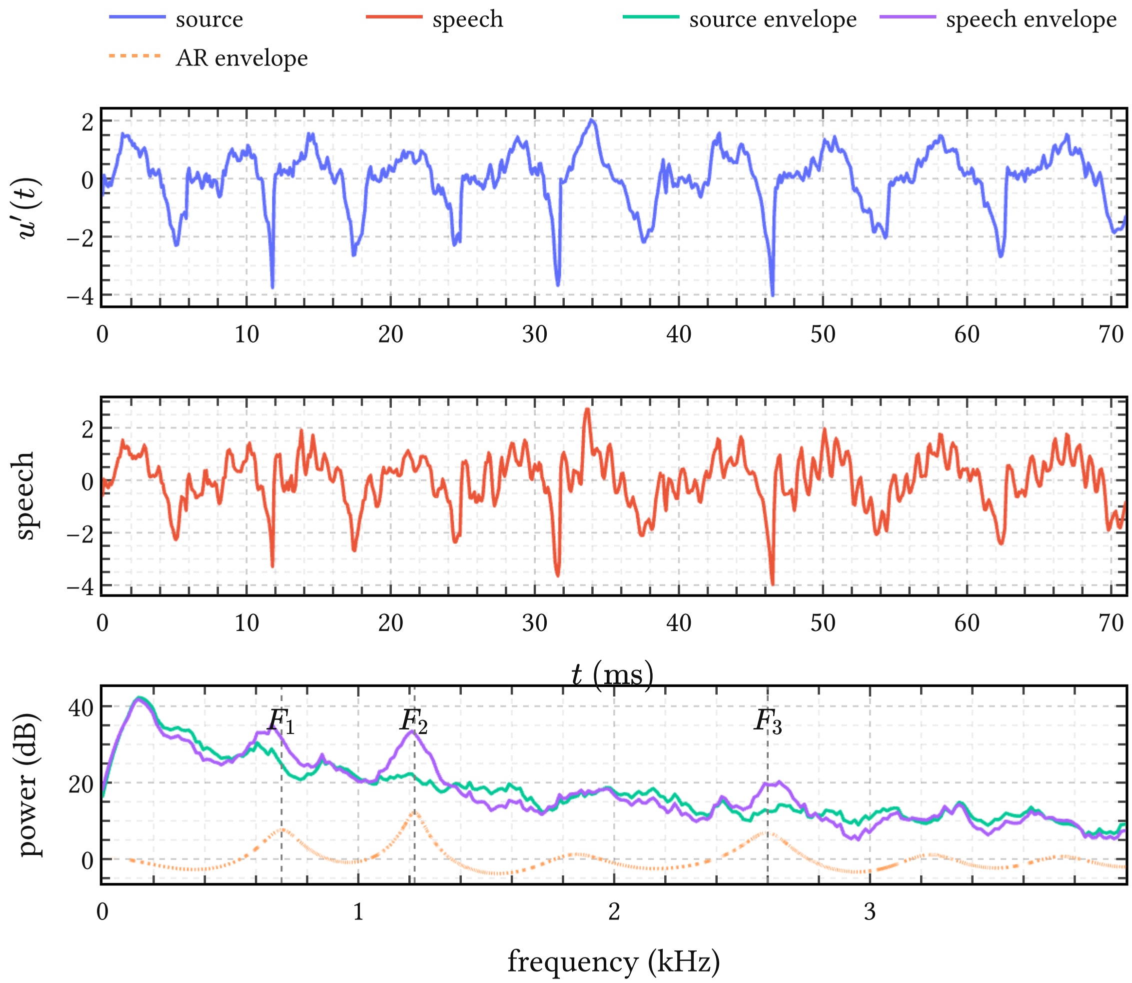


Figure 37 | **Generated voiced speech fragment.** Top: data-space DGF source $u'(t)$ drawn from the QPACK surrogate (167). Middle: speech waveform after applying the AR filter with coefficients from (176), conditioned on an open-vowel formant pattern. Bottom: spectral envelopes of source and speech together with the AR envelope; F_1, F_2, F_3 targets are marked.

where the constraint polynomial $M(z)$ ((50)) encodes resonances at the desired (F_1, F_2, F_3) frequencies and bandwidths. Applying the AR operator $\mathbf{\Psi}(\mathbf{a}_{\text{AR}})$ to $u'(t)$ produces the speech waveform.

Generated sources and vowel-filtered fragments can be heard at the surrogate audio page.

The idea of data-driven voice-source modeling is not new: (Thomas et al., 2009) built a codebook of prototype inverse-filtered waveforms from APLAWD. The key difference here is that the surrogate components are already in the Gaussian form required by IKLP. The same $\mathcal{M}_R$ that generates speech here also serves as the GIF prior in Chapter 7 — there is no separate codebook to maintain.

6.5 Summary

PRISM leaves behind a rich but unwieldy object: the amplitude-space collection $\mathcal{D}_R$ together with its induced density $p(\mathbf{a} \mid \mathcal{D}_R)$. q-GPLVM distills that object into the compact manifold mixture $\mathcal{M}_Q$ and then the compact amplitude mixture $\mathcal{M}_R$, organized mainly by open quotient and pushed back to data space as cheap surrogate DGF models. QPACK is the practical downstream choice because it captures the same broad phonation variation as g-QPACK while requiring only GCIs.

Combined with the t-PRISM pitch prior and the SoftSpectral AR filter (176), this compact Gaussian object is already a workable generative model of voiced speech. In Chapter 7 it stops being illustrative and becomes operative: the same surrogate components enter IKLP directly as prior source hypotheses. That is the whole point of refinement.

7 Bayesian nonparametric glottal inverse filtering

7.1 Recap of all the ingredients

Every preceding chapter has supplied one component. The AR operator $\mathbf{\Psi}(\mathbf{a})$ and its prior $\mathbf{a} \sim \text{Normal}(\boldsymbol{\mu}_a, \mathbf{\Sigma}_a)$ come from Chapter 3. Calibration in Chapter 5 and PRISM (Algorithm 3) turn the registered EGIFA corpus into the shared basis $\boldsymbol{\varphi}(\tau)$ and amplitude-space object $\mathcal{D}_R$. Refinement in Chapter 6 and q-GPLVM (Algorithm 5) compresses that further to the compact mixture

$$\mathcal{M}_R = \{(\pi_i, \boldsymbol{\mu}_i^a, \mathbf{S}_i^a)\}_{i=1}^{I}. \tag{177}$$

This is still a mixture of nonzero-mean Gaussian laws in amplitude space. IKLP, however, consumes zero-mean low-rank kernels of the form $\mathbf{K}_i = \mathbf{\Phi}_i \mathbf{\Phi}_i^\top$ (205). So each surrogate component undergoes a *mean-folding* step: the nonzero mean is absorbed into the covariance so that the second moment is preserved under a zero-mean law, and the result is converted to a feature matrix by

$$\mathbf{L}_i \mathbf{L}_i^\top = \mathbf{S}_i^a + \boldsymbol{\mu}_i^a (\boldsymbol{\mu}_i^a)^\top, \quad \mathbf{\Phi}_i = \mathbf{\Phi}(\boldsymbol{\tau}) \mathbf{L}_i. \tag{178}$$

Then $\mathbf{K}_i = \mathbf{\Phi}_i \mathbf{\Phi}_i^\top$ is one source hypothesis in the IKLP kernel bank. This is not the original nonzero-mean surrogate distribution anymore, but it is the correct zero-mean proxy once IKLP is taken as the downstream target. The isotropic nugget of the surrogate is not carried over separately; it is already represented by the broadband term $\nu_e \mathbf{I}$ in (179). The mixture weights π_i then provide the natural initialization for the IKLP kernel weights θ_i.

IKLP (Chapter 2, Algorithm 1) then assembles these into the full BNGIF likelihood

$$\mathbf{x} \sim \text{Normal}\left(\mathbf{0}, \mathbf{\Psi}^{-1} \left(\nu_w \sum_{i=1}^{I} \theta_i \mathbf{K}_i + \nu_e \mathbf{I}\right) \mathbf{\Psi}^{-\top}\right), \tag{179}$$

which is (24) with the concrete kernel bank and AR prior substituted in. CAVI inference on (179) is the Woodbury-Cholesky algorithm of Algorithm 1.

With every marginalized amplitude — in PRISM, in q-GPLVM, in IKLP — and retained posterior uncertainty we automatically regularize and calibrate, drawing on what (Jaynes, 2003, Sec. 17.8.1) calls the "Bayesian safety device." Seen from above, BNGIF is just the final narrowing of support. TACK was already a physically shaped but still very broad source prior. The PACK family made that prior quasiperiodic and phase-aware. PRISM then narrowed it to the EGIFA amplitude cloud $\mathcal{D}_R$, and q-GPLVM compressed that cloud to the small mixture $\mathcal{M}_R$. IKLP finally conditions those source hypotheses on the speech frame at hand. The point of all this was not to kill uncertainty, but to spend it on realistic DGF structure.

7.1.1 Runtime

Memory scales as $\mathcal{O}(BMr + Ir^2)$ and compute as $\mathcal{O}(Mr^2 + Ir^2)$ where r is the rank of the inducing-point approximation and B is the batch size; both are linear in frame length M for fixed r. At 20 kHz (native EGIFA sampling rate) and x32 precision, BNGIF runs at 5× realtime on a single RTX 2080; at x64, at 0.5× realtime. The full EGIFA corpus processes in approximately 3 minutes at x32, excluding JAX tracing and I/O.

NaN events from GIG distribution numerics occur in approximately 1% of `EGIFA/vowel` frames and 0.1% of `EGIFA/speech` frames at x32; random restarts reduce the vowel rate below 0.001% with no effect on performance.

7.2 Classic GIF methods

All five classic methods evaluated in Chapter 7 follow from (181): they minimize a weighted LP residual $\mathbf{x}^\top \boldsymbol{\Psi}^\top \mathbf{W} \boldsymbol{\Psi} \mathbf{x}$ and return $\boldsymbol{\varepsilon} = \boldsymbol{\Psi}(\hat{\mathbf{a}})\mathbf{x}$ as the DGF estimate. The key structural distinction from BNGIF is this:

- Classic methods act through the *likelihood* ($\mathbf{W}$ shapes the data fit); the DGF estimate is a deterministic function of $\hat{\mathbf{a}}$.
- BNGIF acts through the *prior* ($\mathbf{K}_i$ encodes DGF structure); the DGF estimate is a posterior mean with calibrated uncertainty.

The closed-phase weighting of CPCA/SLP/WLP-QCP is thus the complement of g-QPACK gating: both exploit the open/closed asymmetry, but one penalizes open-phase residuals while the other suppresses open-phase prior variance.

7.2.1 Weighted LP variants

The family CPCA → SLP → WLP/QCP forms a continuum of increasingly smooth, cycle-aware weight functions w_m (see Table 14). Weighted LP was first motivated by (Ma et al., 1993) to reduce coupling between source harmonics and filter resonances. QCP (Airaksinen et al., 2014) is the strongest classical method, using the AME (attenuated main excitation) window — a smooth, GCI-anchored envelope that downweights the open phase without a hard mask. As shown in Dataset 1, the WCA implementations in (Chien et al., 2017) use AME weighting, so the EGIFA "WLP" baseline is effectively QCP.

7.2.2 IAIF

IAIF (Alku, 1992) is the only method requiring no GCI input: two LP passes at different orders iteratively peel apart glottal and tract contributions. It is pitch-asynchronous and therefore the most robust to GCI estimation error, at the cost of weaker spectral separation.

7.2.3 CCD

CCD (Drugman et al., 2011) works in the complex cepstral domain, extracting the maximum-phase anticausal component as the tract estimate. It selects a different gauge representative than LP-based methods (see Appendix G), which explains the large systematic shift in its scores under fixed-delay evaluation in (Chien et al., 2017) and its relative improvement under frame-level affine alignment.

7.3 EGIFA benchmark

The dataset and registration are described in Dataset 1; the gauge-invariant evaluation metric is defined in Appendix G ((334), (336)). All results below use LBGID-estimated GCIs, frame-level affine+lag alignment, and centered cosine similarity as the score.

7.3.1 Results

7.3.2 Failure modes

Two failure modes appear at the low-score tail:

$\boldsymbol{F_1 \approx F_0}$ congruence. When the first formant frequency is close to F_0, the AR filter begins sculpting the broadband noise component of the excitation into a quasi-harmonic structure instead of modeling the true DGF. The DGF estimate shrinks below the noise floor, though it often still traces the gross pulse shape. This is the classical GIF limitation noted in (Chien et al., 2017) and Chapter 1; it affects all methods and is the main remaining open problem. AR prior information about expected formant locations may partially remedy it.

Genuine alternative solutions. At very low F_1, BNGIF occasionally finds a structurally different posterior mode — a plausible low-noise explanation that is not the ground truth. This is a consequence of the inherent nonidentifiability of GIF (Appendix G) rather than an inference failure. Variational restarts can detect this case.

7.3.3 F_0 dependence

7.3.4 GCI backend ablation

7.3.5 Window size

7.4 Evaluation on VTRFormants

BNGIF is not merely a GIF method: it is a full grey-box generative model of voiced speech from which all frame-level source and filter statistics follow simultaneously. In particular, the posterior over $\mathbf{a}$ induces a posterior over formant frequencies and bandwidths, making BNGIF a principled formant tracker as a byproduct of GIF inference, without any dedicated tracking post-processing step.

VTRFormants (Deng et al., 2006) provides hand-corrected F1–F3 trajectories for 538 TIMIT sentences at 10 ms frame resolution (Dataset 4). We evaluate formant RMSE against these reference tracks using QuickGCI for GCI detection, overlap-add projection to 10 ms frames, and $P = 20$ LP poles — the setting that gave best results in ablation. The baselines are KARMA (Mehta et al., 2012), WaveSurfer, and Praat, as reported in (Mehta et al., 2012).

Method	F1 (Hz)	F2 (Hz)	F3 (Hz)	Overall (Hz)
KARMA (Mehta et al., 2012)	114	226	320	220
Praat (Mehta et al., 2012)	185	254	303	247
WaveSurfer (Mehta et al., 2012)	170	276	383	276
BNGIF ($P = 16$, whitenoise)	90	172	278	189
BNGIF ($P = 16$, QPACK)	97	172	257	183
BNGIF ($P = 20$, whitenoise)	97	184	216	171
BNGIF ($P = 20$, QPACK)	**89**	**182**	**210**	**167**

Table 12 | **Formant RMSE on VTRFormants (Hz).** BNGIF with $P = 20$ LP poles and a QPACK kernel prior beats KARMA (Mehta et al., 2012) across all three formants and reduces overall RMSE from 220 Hz to 167 Hz (-24%). The improvement is largest on F3 (-35%): 210 vs 320 Hz. "Whitenoise" is IKLP with a flat (white) excitation prior; QPACK is IKLP with the calibrated PACK kernel from Chapter 5. GCI detection uses QuickGCI throughout; windows are 128 ms with triangular overlap-add projection to 10 ms frames.

BNGIF at $P = 20$ with a QPACK kernel achieves F1/F2/F3/overall RMSE of 89/182/210/167 Hz, versus KARMA's 114/226/320/220 Hz. The largest gain is on F3 (-35%), where classical LP-based trackers are most affected by spectral smoothing artifacts from the excitation structure. The QPACK kernel helps most on F3 relative to whitenoise ($P = 20$: 210 vs 216 Hz); the F1 and F2 improvement over whitenoise is modest at this LP order, consistent with F1–F2 being largely determined by the AR coefficients once P is large enough. Increasing P from 16 to 20 improves overall RMSE by about 9% and is the dominant lever. The AR prior strategy (Yoshii-lambda vs scaled) has negligible effect.

7.5 Summary

8 Conclusion

8.1 Summary of contributions

This thesis built BNGIF as a converging series of prior refinements, each step resolving a concrete deficiency of the previous.

IKLP (Chapter 2, Algorithm 1) replaced the white-noise LP excitation prior with an infinite mixture of GP kernels, with weights inferred by CAVI under a gamma process prior ((23)). A Woodbury-Cholesky reformulation (Algorithm 1) eliminates all $M \times M$ matrix inversions, enabling 5× realtime throughput at x32 on commodity hardware — bringing a Bayesian treatment to a domain where MCMC alternatives required 2.5 hours per 25 ms frame (Auvinen et al., 2014).

AR prior (Chapter 3) replaced the isotropic default on the LP coefficients $\mathbf{a}$ with an informative Gaussian prior $\mathbf{a} \sim \mathrm{Normal}(\boldsymbol{\mu}_a, \boldsymbol{\Sigma}_a)$ ((190)) derived from stability theory and spectral constraints. This sharpens the vocal tract filter estimate and regularizes the GIF problem independently of kernel choice.

PACK kernel family (Chapter 5) populated the IKLP kernel bank with periodic arc-cosine kernels calibrated to VocalTractLab DGF waveforms via PRISM (Algorithm 3). PACK, QPACK, and g-QPACK encode quasiperiodicity, spectral rolloff, and the open/closed phase asymmetry respectively. Calibration on the EGIFA corpus (Dataset 1) yields the amplitude-space object $\mathcal{D}_R$ under a shared GP basis, reflecting the full factorial variation of voice quality, F_0, and phonation pressure present in VocalTractLab synthesis.

q-GPLVM surrogate prior (Chapter 6, Algorithm 5) compressed the PACK basis into a discrete mixture prior over DGF archetypes. The quantized GPLVM manifold is visibly organized by open quotient, and the fitted compact mixture $\mathcal{M}_R$ provides kernel-weight initialization that seeds IKLP inference with physiologically structured priors.

BNGIF assembled (Chapter 7) achieved On VTRFormants it achieved The strongest classical baseline on EGIFA is QCP-style WLP (= AME weighting, (Chien et al., 2017)), consistent with (Freixes et al., 2023).

Registration pipeline (LBGID + t-PRISM, Algorithm 2, Algorithm 4) provides principled voiced-group extraction from any synthesizer corpus, avoiding the heuristic area-based GCI detectors bundled with EGIFA.

Gauge-invariant evaluation (Appendix G) established frame-level affine+lag-invariant scoring as the correct unit of comparison for GIF methods, replacing the cycle-level fixed-delay evaluation of (Chien et al., 2017).

8.2 Clinical significance

A person produces around 16,000 words each day (Andrade-Miranda et al., 2024). Speech impairment affects up to 90% of patients with Parkinson's disease and precedes motor diagnosis by as much as a decade (Cao et al., 2025). The recommended clinical probe is sustained phonation of /a/ (Cao et al., 2025) — a single breath from which GIF can in principle extract F_0, jitter, shimmer, open quotient, HNR, and formant trajectories simultaneously.

Classical pipelines produce point estimates of these quantities with no propagated uncertainty. BNGIF's posterior produces calibrated distributions over all of them jointly, enabling proper errors-in-variables inference downstream and making confound separation — between measurement noise, session variability, and disease-related change — tractable within a single coherent model.

8.3 Limitations

LTI assumption. All GIF methods, including BNGIF, assume a time-invariant vocal tract filter within the analysis frame. During the open phase, source-filter coupling slightly shifts formant locations and widens bandwidths (Walker & Murphy, 2005); this variation ends up as ripple on the open-phase DGF estimate. The gated g-QPACK prior partially mitigates this by suppressing open-phase prior variance, but does not eliminate the LTI approximation.

$\boldsymbol{F_1 \approx F_0}$ congruence. When the first formant is close to F_0, the AR filter shapes broadband noise into a quasi-harmonic structure and the DGF estimate deteriorates. This is the known major failure mode of all LP-based GIF methods (Chien et al., 2017) and it persists in BNGIF, particularly for close rounded vowels /o/ and /u/. An informative AR prior with formant-location constraints is the natural next step.

Synthetic training data. The DGF prior is calibrated on VocalTractLab synthesis. VocalTractLab 2.1 cannot simulate pathological phonation (Chien et al., 2017), so the prior may not cover all clinically relevant voice qualities. Extending the calibration corpus to include measured glottal area waveforms (Andrade-Miranda et al., 2024) or higher-fidelity numerical simulations (Schoder et al., 2024; Zhang et al., 2020) is high priority.

8.4 Future work

End-to-end optimization. BNGIF currently uses a multi-stage pipeline: LBGID ⟶ t-PRISM ⟶ PRISM ⟶ q-GPLVM ⟶ IKLP. IKLP is fully differentiable end-to-end (at x64), so future work should optimize all parameters jointly on the speech likelihood, eliminating the staged approximations.

Learning the time warp from data. LBGID + t-PRISM require ground-truth $u(t)$ for registration. Learning the GCI/GOI time warp directly from $s(t)$ would remove this dependence (Kazlauskaite et al., 2018).

Generating training databases. The AR prior and surrogate DGF model can generate large quantities of realistic synthetic speech with known ground truth, at far lower computational cost than VocalTractLab synthesis. This enables supervised neural GIF training data at scale — currently the central bottleneck for deep learning approaches (Sasou & Chen, 2023) — with richer source variation than LF-based corpora (Airaksinen et al., 2014; Alku et al., 2019).

ARMA and pole-zero models. The spectral prior machinery of Chapter 3 is agnostic to numerator polynomials; extending it to ARMA models would allow modeling antiresonances from nasal coupling and subglottal cavities (Ananthapadmanabha & Fant, 1982).

Online learning. The posterior on $\mathbf{a}$ and the GIG kernel weights from frame t are natural informative priors for frame $t + 1$. Propagating these across frames would give BNGIF inter-frame memory analogous to KARMA (Mehta et al., 2012) without separate smoothing.

IKLP extensions. Natural extensions include: mixture of AR priors (voiced/unvoiced/creaky), relaxing the MAP approximation on $\mathbf{a}$ to a full variational Gaussian, and full spectral-domain inference.

Datasets

Dataset 1. EGIFA

EGIFA is the synthetic benchmark introduced by (Chien et al., 2017). It pairs radiated speech waveforms $s(t)$ with matched glottal flow waveforms $u(t)$ generated by a physics-based articulatory synthesizer. In this thesis it plays two roles: in Chapter 5 it provides the corpus on which the PACK family is calibrated, and in Chapter 7 it provides the controlled benchmark on which inferred source signals are compared against reference glottal flow. These two roles place different demands on how the data are processed, which is why they are treated separately.

1.1. Corpus

VocalTractLab (Birkholz, 2013) is an open-source articulatory speech synthesizer and the current state of the art in physics-based speech production (Gao et al., 2024). Its glottis model is a self-oscillating, two-mass triangular system coupled to a transmission-line vocal tract; it simulates time-domain finite-difference acoustic wave propagation including nonlinear source-filter interaction. Because $u(t)$ is a direct output of the simulator alongside $s(t)$, every utterance comes with a ground-truth glottal flow.

The EGIFA corpus contains two data sets synthesized with VocalTractLab 2.1.

The vowel subset has a fully crossed factorial structure that makes it well-suited for studying how algorithm performance varies with voice quality, pitch, and phonation pressure. The speech subset adds realistic continuous phonatory dynamics and coarticulation. Close rounded vowels /o/ and /u/ are empirically the most challenging case for inverse filtering, as noted in (Chien et al., 2017), because their narrow first formants bring F_1 close to F_0, making the source-filter boundary hard to locate.

Subset	Files	Factors	Duration	Native f_s
vowel	750	6 vowels {/i/, /e/, /ɛ/, /ä/, /o/, /u/} × 5 voice qualities × 5 F_0 levels × 5 subglottal pressures	0.6 s each	44.1 kHz
speech	125	5 median F_0 × 5 median pressures × 5 voice qualities	5–10 s each	44.1 kHz

Table 13 | **EGIFA corpus structure.** Voice qualities are pressed, slightly pressed, modal, slightly breathy, and breathy. F_0 levels are 90, 120, 150, 180, and 210 Hz; subglottal pressures are 500, 708, 1000, 1414, and 2000 Pa. The speech subset is derived from a prototype score for the German sentence "Lea und Doreen mögen Bananen" with controlled continuous variation of the glottal parameters. All files are resampled to 20 kHz for processing.

1.2. Registration

For learning, we turn the raw $u(t)$ waveforms into a corpus of DGF exemplars $u'(\tau)$ in shared normalized time τ. The pipeline extracts GCI and GOI instants from $u(t)$ via LBGID (Algorithm 2), segments the resulting sequence into stable voiced groups via t-PRISM (Algorithm 4), warps each group from physical time t to normalized time τ, and finally smooths, differentiates, and power-normalizes to obtain $u'(\tau)$. The subsections below treat each stage in turn.

GCI and GOI detection

LBGID (level-based glottal instant detection, Algorithm 2) extracts (GCI, GOI) pairs directly from the continuous $u(t)$ waveform. It works by identifying low-flow regions via a relative amplitude threshold and then selecting, within each such closed phase, the index pair (i, j) that maximizes $E(i, j) = -u'(i)\, u'(j)$ ((222)), i.e., the pair of maximally opposed changes flanking the closure.

EGIFA provides no ground-truth GCI or GOI annotations. We use LBGID rather than the GCI estimation built into EGIFA's own evaluation scripts. The EGIFA `area_gci` path segments cycles from the synthesizer's glottal area signal, extracts GCIs via local minima, and then applies a hard offset of $+13$ samples before feeding the result to a separate GOI detector. This produces clustered triggers with no refractory guard, making the implied cycle lengths noisy. LBGID operates directly on $u(t)$ without any area signal or hard offset, and is better suited to the dense factorial variation of the corpus.

Voiced group segmentation

Raw EGIFA recordings contain transient regions — onset glides, pressure ramps, the beginning and end of each utterance — where the glottal cycle is not stationary and the resulting $u(t)$ waveform is not representative of any particular phonatory mode. Including these transients in the learning corpus would corrupt the prior.

t-PRISM (Algorithm 4) segments the detected GCI sequence into voiced groups by fitting a GP pitch-track model to the implied period sequence $T_{\{0,k\}} = \left(n_{\{\mathrm{GCI},k+1\}} - n_{\{\mathrm{GCI},k\}}\right)/f_s$ and thresholding on the predictive probability. Cycles whose period falls below a weight threshold are discarded as outliers; maximal contiguous runs of accepted cycles form the voiced groups used for learning.

Smoothing, differentiation, and normalization

Before differentiating, $u(t)$ is smoothed on a physiologically meaningful timescale of 0.1 ms (Appendix E). At 20 kHz this acts mainly as regularization. The derivative $u'(t)$ is then computed by finite differences, registered to τ via (175), and standardized within each group to unit power. The per-group normalization removes inter-utterance amplitude variation that would otherwise dominate the learning signal.

Implied corpus statistics

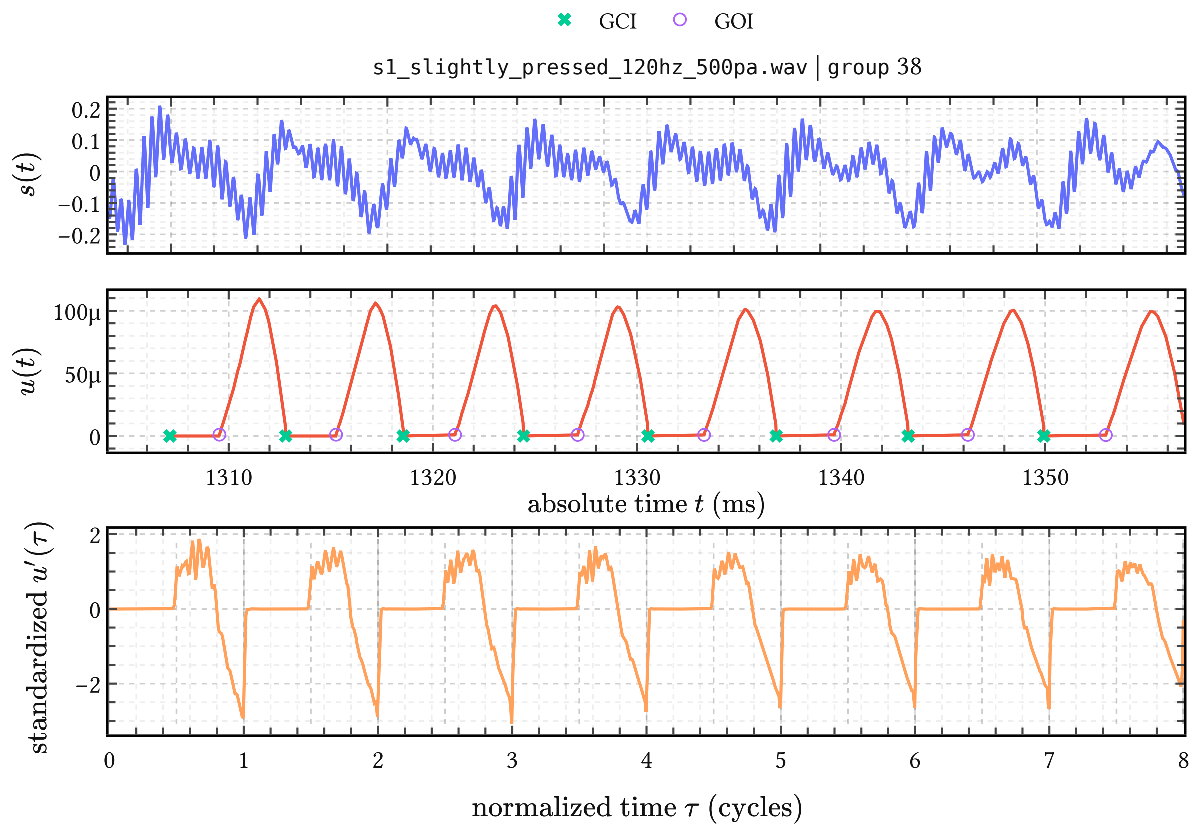


Figure 38 | **Registration of one EGIFA voiced group.** Top: raw speech $s(t)$ in physical time. Middle: raw glottal flow $u(t)$ with LBGID-detected GCI and GOI landmarks. Bottom: the group after differentiation, normalization, and warp to normalized time τ, shown as $u'(\tau)$; GCIs at integers, GOIs at half-integers.

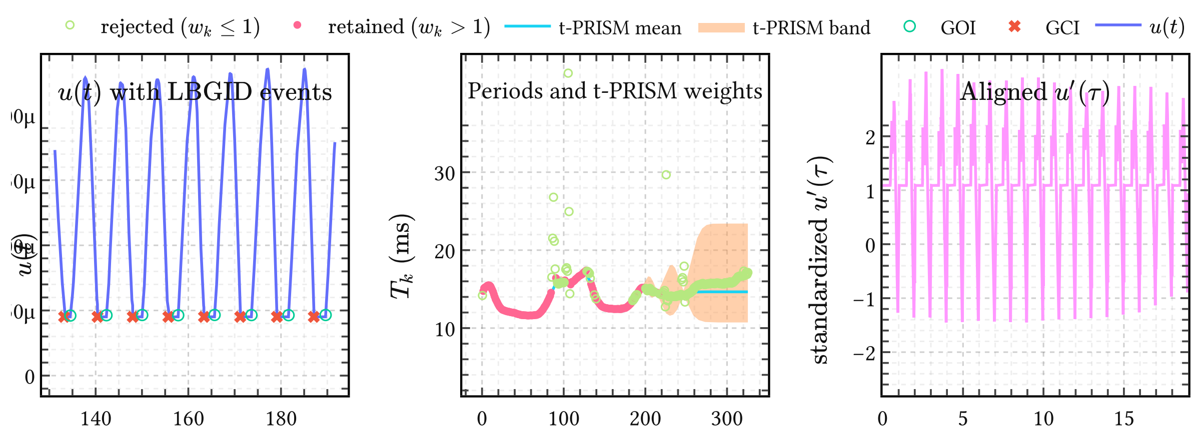


Figure 39 | **The EGIFA registration pipeline on one recording.** (a) Raw glottal flow with LBGID GCI and GOI events. (b) Implied period sequence over the full recording; t-PRISM posterior mean and uncertainty band with retained inliers. (c) One retained voiced group in shared cycle time τ after smoothing, differentiation, GOI-aware registration, and per-group normalization.

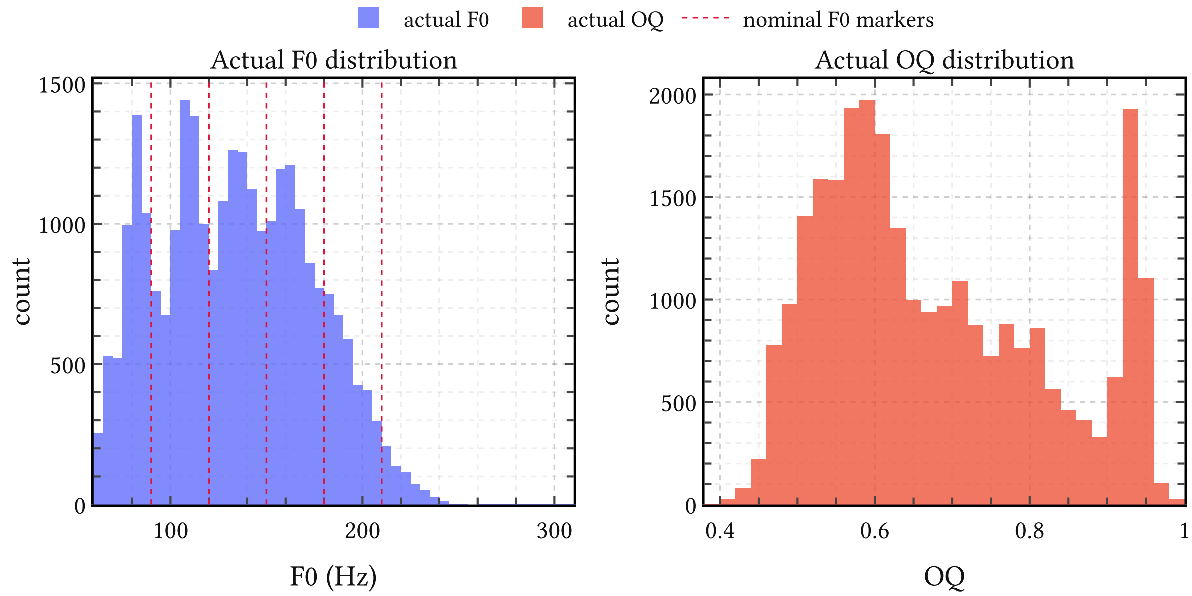


Figure 40 | **Implied F_0 and OQ distributions from LBGID on EGIFA.** Left: frame-level F_0 implied by detected GCIs, with dashed lines at EGIFA's nominal pitch settings. Right: frame-level OQ implied by detected GCI/GOI pairs.

The detected GCI and GOI landmarks imply per-cycle pitch and open quotient values:

$$F_{0,k} = f_s / (n_{\mathrm{GCI},k+1} - n_{\mathrm{GCI},k}), \quad \mathrm{OQ}_k = (n_{\mathrm{GOI},k} - n_{\mathrm{GCI},k}) / (n_{\mathrm{GCI},k+1} - n_{\mathrm{GCI},k}) \tag{180}$$

The F_0 modes line up with the corpus's nominal settings and the OQ values stay in a physiologically plausible range, confirming that LBGID produces consistent landmarks across the full factorial spread.

1.3. The benchmark

The registration pipeline above transforms the corpus into learning data by starting from $u(t)$. The benchmark poses the inverse problem: given only $s(t)$, infer $u'(\tau)$ and compare against the registered reference.

Classic GIF methods

All five methods from (Chien et al., 2017) operate frame-by-frame on the speech signal $\mathbf{x} = (x_1, ..., x_M)^\top$ (= discretized $s(t)$, Chapter 2). Four of the five estimate the vocal tract AR coefficients $\mathbf{a}$ by minimizing a weighted LP residual:

$$\hat{\mathbf{a}} = \operatorname*{argmin}_{\mathbf{a}} \mathbf{x}^\top \mathbf{\Psi}(\mathbf{a})^\top \mathbf{W} \mathbf{\Psi}(\mathbf{a}) \mathbf{x}, \tag{181}$$

where $\mathbf{\Psi}(\mathbf{a})$ is the AR operator from (13) and $\mathbf{W} = \mathrm{diag}(w_1, ..., w_M) \geq 0$ encodes how much each sample influences the estimate; the inferred DGF is then the LP residual $\boldsymbol{\varepsilon} = \mathbf{\Psi}(\hat{\mathbf{a}})\mathbf{x}$. The methods differ only in $\mathbf{W}$:

Method	Weight w_m	GCI/GOI required
CPCA	$\mathbb{1}[m \in \text{closed phase}]$ (hard mask)	yes (both)
SLP	$1 - \kappa \sum_l \exp\left(-(m-\gamma_l)^2 / \left(2(\sigma f_s)^2\right)\right)$	yes (GCI)
WLP/QCP	AME: quasi-closed phase envelope, piecewise smooth	yes (GCI)
IAIF	$\mathbf{W} = \mathbf{I}$ (two LP passes, iterative)	no
CCD	— (cepstral, not LP-based)	yes (GCI)

Table 14 | **Classic GIF methods in EGIFA, unified under (181).** CPCA uses a hard binary closed-phase mask; SLP uses soft Gaussian attenuation of closed-phase samples; WLP in (Chien et al., 2017)‘s implementation uses an AME (attenuated main excitation) window, which corresponds to QCP weighting (Alku et al., 2013). IAIF uses uniform weights but applies two LP passes iteratively. CCD is not a weighted LP method: it separates the maximum-phase component of the speech cepstrum and is entirely independent of $\mathbf{W}$.

A practically important detail: the WLP implementation in (Chien et al., 2017) uses the AME weighting scheme introduced for formant estimation of high-pitched vowels (Alku et al., 2013). AME is structurally identical to QCP (quasi-closed phase analysis), which is the current state-of-the-art classical GIF method (Freixes et al., 2023). So the strongest classical baseline in EGIFA is effectively QCP, not the literal piecewise-linear WLP of (Airaksinen et al., 2014).

IAIF (Alku, 1992) stands apart in that it requires no GCI input: two LP passes at different orders iteratively separate glottal and tract contributions. CCD (Drugman et al., 2011) stands apart in a different way: it works in the complex cepstral domain, extracting the anticausal maximum-phase component of the speech spectrum as the tract estimate and leaving the causal residual as ε.

Evaluation

The original evaluation in (Chien et al., 2017) aligns estimated and reference DGFs per glottal cycle with a fixed 0.65 ms propagation delay and measures the normalized waveform error within each cycle. The arguments of Appendix G show why cycle-level alignment with a fixed delay is inappropriate: different algorithms select different gauge representatives from the source-filter ambiguity, so comparing raw cycle-level samples conflates algorithmic style with algorithmic accuracy.

Throughout this thesis we therefore evaluate all methods — classical and BNGIF — under a single frame-level affine-plus-lag-invariant metric ((334), (336)). This measures shape, not gauge convention, so improvements in the metric transfer directly to downstream tasks.

Finding high quality voiced groups

The key idea is to use an explicit pitch-track model — t-PRISM (Algorithm 4) on top of LBGID landmarks — rather than ad hoc heuristics. Several practical choices make the fit robust across the full corpus:

- **Normalization.** Period sequences are mapped to $(0, 1)$, freezing the GP mean to 0 and kernel variance to 1.
- **Lengthscale initialization.** Starting at ≈ 10 (close to the converged value of ≈ 15) saves iterations without affecting the result.
- **Inducing point placement.** Trajectory lengths are heavy-tailed, so inducing points are initialized from the inverse empirical CDF to concentrate predictive power where data are dense.
- **Degrees of freedom.** ν is set from event statistics rather than optimized (see below); results are insensitive to the exact value as long as $\nu < 10$.

Remaining hyperparameters are optimized with Adam at default settings.

1.3.3.1. Heuristic for setting ν

We set ν from a tail-probability argument. After normalization the GP has unit variance and the learned noise scale is $\sigma \approx 0.15$. Typical spikes have amplitude ≈ 5, giving residuals $r \approx \frac{5}{0.15} \approx 30$–$35$ noise standard deviations. These should be absorbed by the Student-t tails, not by the GP bending to explain them.

The Student-t tail satisfies $P(|X| > r) \approx Cr^{-\nu}$ for large r. Matching this to the empirical spike rate $p_{\text{spike}} \approx 1\%$ gives

$$\nu \approx -\frac{\log(p_{\text{spike}})}{\log(r)}. \tag{182}$$

For $r \approx 35$ and $p_{\text{spike}} \approx 0.01$ this yields $\nu \approx 1$–2, a Cauchy-like regime. Extreme residuals are then downweighted rather than forcing the GP into excessive curvature.

Dataset 2. APLAWD

APLAWD is a speech corpus with synchronized speech, electroglottograph (EGG), differentiated EGG (DEGG), and hand-corrected GCI reference markings (Naylor et al., 2007). It has been used both for data-driven voice-source modeling (Thomas et al., 2009) and for large-scale GCI detection evaluation (Serwy, 2017). Serwy reports a hand-verified markings database over 10,944 waveforms, which is exactly what makes the corpus useful here: APLAWD supplies trustworthy landmark sequences from which a prior over smooth pitch-period trajectories can be learned.

In this thesis APLAWD is not a glottal-flow learning corpus. It enters only through the geometry of pitch. The pitch-track prior of Algorithm 4 is trained on APLAWD and then reused later for voiced-group selection on EGIFA and CommonVoice. That role is narrower than EGIFA's role in Chapter 5, but it is also cleaner. For learning a pitch prior, accurate GCI landmarks matter more than matched source waveforms.

The manual markings are important because the quantity of interest is not the speech waveform itself but the inter-GCI geometry it induces. For recording n with GCI sample indices $n_{\mathrm{GCI},n,0}, \dots, n_{\mathrm{GCI},n,M_n}$ at sampling rate f_s, define the period sequence

$$T_{nm} = 1000\frac{n_{\mathrm{GCI},n,m+1} - n_{\mathrm{GCI},n,m}}{f_s} \quad (m = 0, \dots, M_n - 1), \tag{183}$$

measured in milliseconds. We then work with the log-periods

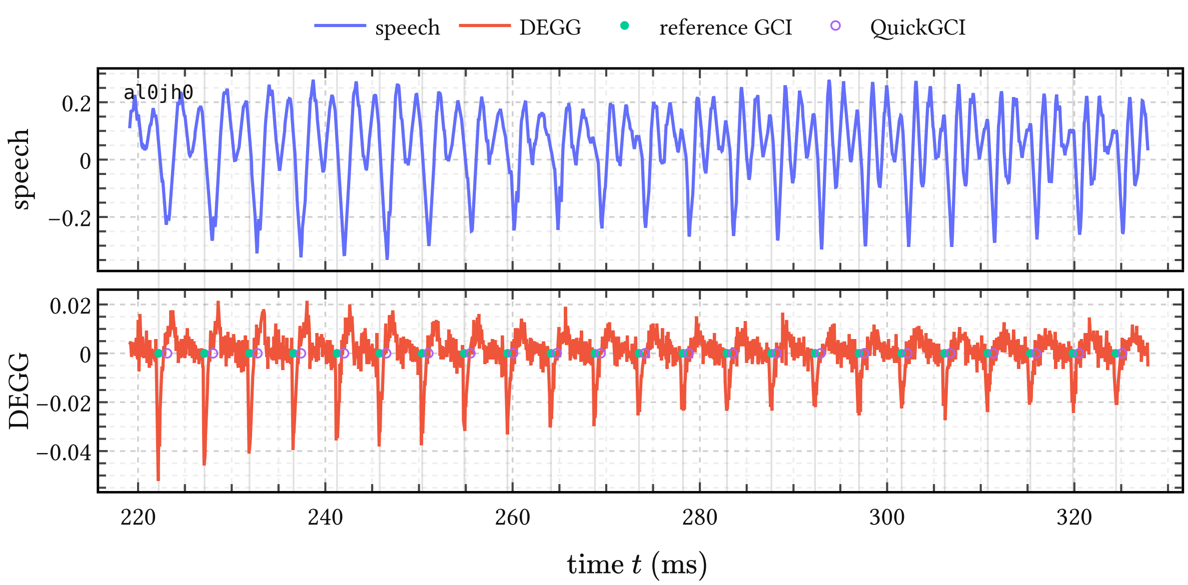


Figure 41 | **One APLAWD excerpt with DEGG landmarks.** Top: the acoustic waveform. Bottom: the synchronized DEGG trace together with the hand-corrected GCI references and QuickGCI estimates. The database provides this kind of alignment throughout the corpus, which is why it is the right place to learn a pitch prior.

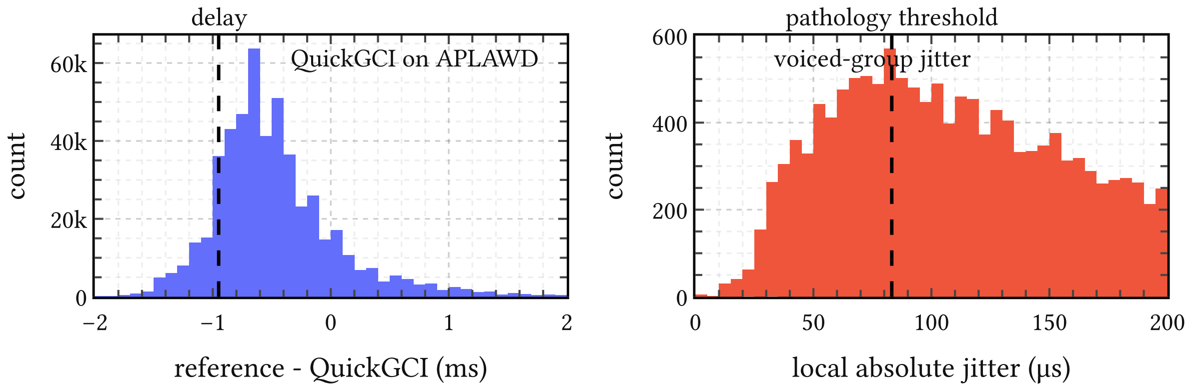


Figure 42 | **Two APLAWD diagnostics.** Left: QuickGCI timing errors against the hand-corrected GCI references. The mode is centered close to the expected larynx-to-microphone delay. Right: local absolute jitter measured over retained voiced groups. Most of the mass lies well below the usual pathology threshold.

$$\ell_{nm} = \log_{10} T_{nm}, \tag{184}$$

which stabilize the scale and turn multiplicative pitch variation into additive variation. The training data for Algorithm 4 is simply the collection of these sequences over the corpus, after a global whitening transform.

2.1. Landmark quality

Because APLAWD contains both speech and hand-corrected GCI references, it also provides a natural place to sanity-check the geometry we later ask the pitch model to explain. Serwy's Hilbert-phase detectors already showed that QuickGCI is close to the stronger GADFLI detector on this corpus (Serwy, 2017). For the thesis the main point is simpler: the detector errors are tightly localized, and the local jitter measured over voiced groups remains in a healthy range.

This does not prove that every landmark is perfect, and it need not. The thesis only needs a corpus where the dominant pitch geometry is trustworthy enough to learn a smooth prior from it. APLAWD meets that standard.

2.2. Learning the pitch prior

The goal is to learn a single GP prior that captures how pitch typically rises and falls over a voiced stretch of speech. This is not a source model. It is a model of period trajectories. The target is therefore the collection of log-period series (184), not the speech waveform.

We train the model with t-PRISM from Algorithm 4. The Student-t likelihood is useful here because even in APLAWD some period sequences contain long silent gaps, detector slips, or locally erratic closures. The robust weights absorb such observations instead of forcing the prior to explain them. Model selection is done by held-out log likelihood per effective test point.

Kernel	M	Held-out log likelihood	Lengthscale	Obs. std.
`rbf`	64	$2.117 + -0.012$	15.25	0.137
`matern:52`	64	$2.070 + -0.009$	17.35	0.144
`rbf`	32	$2.051 + -0.011$	15.28	0.147
`matern:32`	64	$2.029 + -0.008$	18.65	0.148
`matern:52`	32	$2.008 + -0.010$	17.23	0.153

Table 15 | **Held-out model selection on APLAWD.** Values are mean test log likelihood per effective test point with the uncertainty estimate from `experiments/svi/aplawd/stats.R`. The winning model is an RBF prior with $M = 64$ inducing points.

The result is clean. An RBF prior with $M = 64$ inducing points wins decisively enough to become the default pitch-track prior used throughout the thesis. The learned lengthscale is about 15 cycles, which is the quantity of real interest here: it sets the timescale on which the model expects pitch to vary smoothly. Neither the rational-quadratic alternative nor the learned-ν Student-t extension improves on this winner in held-out score.

2.3. Voiced-group segmentation

Once learned, the pitch prior can be turned around and used as a robust filter on new GCI-annotated period tracks. Given a period sequence $\{T_{nm}\}_{m=0}^{M_n-1}$, t-PRISM returns local weights

$$w_{nm} = \mathbb{E}[\lambda_{nm}], \tag{185}$$

and in this thesis we use the conservative rule

$$w_{nm} > 1 \tag{186}$$

to declare period m an inlier. Contiguous inlier runs are then taken as voiced groups. This is the segmentation rule later reused on EGIFA and CommonVoice.

This is where the APLAWD chapter meets the rest of the thesis. APLAWD provides the corpus on which the pitch prior is learned and selected. Later chapters reuse that prior as a robust notion of normal voiced pitch, but they do not need to repeat its training story.

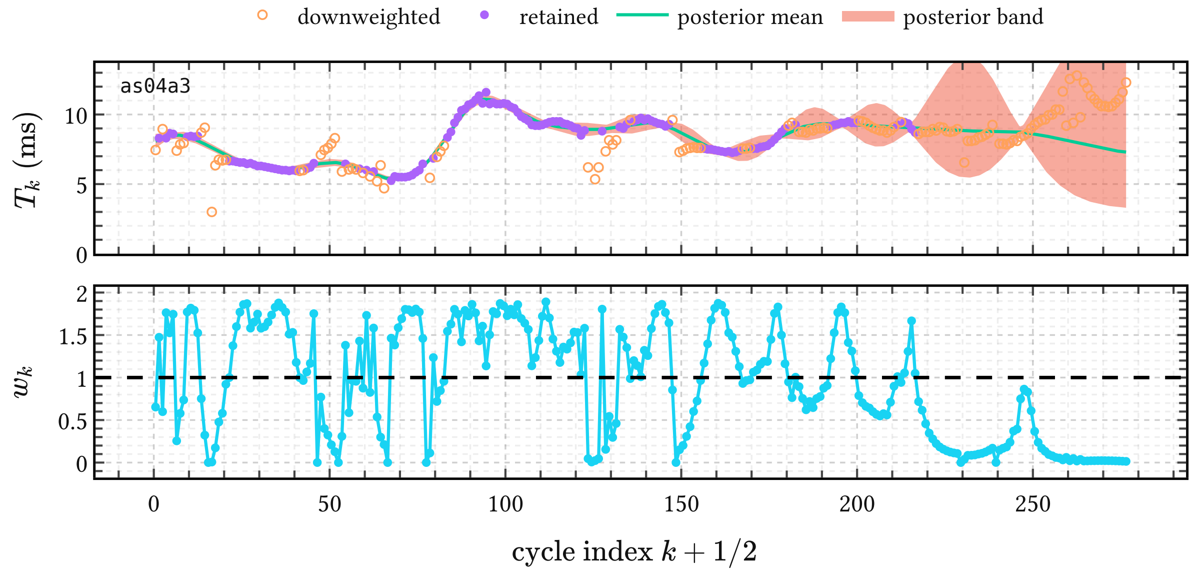


Figure 43 | **Voiced-group selection on one APLAWD pitch track.** Top: the observed inter-GCI periods together with the t-PRISM posterior mean and uncertainty band. Bottom: the inferred weights $w_k = \mathbb{E}[\lambda_k]$. Weight collapses carve the period track into retained voiced runs and rejected gaps.

Dataset 3. CommonVoice

The CommonVoice corpus used here pairs speech recordings with matched glottal-flow waveforms derived from an articulatory resynthesis pipeline. Relative to EGIFA the material is longer, less orderly, and closer to the large-corpus setting the registration pipeline was ultimately designed for.

CommonVoice is mainly a scaling target. The LBGID + t-PRISM + registration pipeline is developed and stress-tested on EGIFA, where synthetic control is tighter; the same machinery is then reused on CommonVoice to build a larger registered corpus for source modeling beyond the benchmark setting.

(Gao et al., 2024; Sun et al., 2022)

(Birkholz, 2013; Krug et al., 2021)

PAULE and CommonVoice

PAULE is a neural model that inverts speech waveforms to the control parameter (CP) trajectories that drive VocalTractLab (VTL), effectively reconstructing what the articulators did to produce the observed speech. Resynthesis quality is poor (as its authors acknowledge), but the goal here is not faithful reconstruction: it is to explore the space of physically plausible glottal flows that VTL can generate.

We forked VTL and patched it to expose the glottal flow output, then ran the PAULE-inverted CP trajectories through the modified synthesizer. This produced roughly 21k paired speech and GF waveforms, most of which are nondescript muffled babblings.

The registration pipeline proceeds as on EGIFA: LBGID extracts GCI and GOI landmarks, t-PRISM detects stable voiced regions, and the DGF is computed after removing the large DC offset caused by unphysical breathiness in the synthesizer.[25]

Synthesis quality correlates with utterance length: short outputs tend to collapse to the same muffled grunt and are filtered out.

From the surviving waveforms we measure T and OQ. The aim is to cover enough OQ variation to learn the closed phase and spectral rolloff, which are heavily correlated. Because the synthesizer does not reliably produce full closure, we additionally enforce a clean closed phase, consistent with physiological observations of complete glottal closure during modal phonation.

[25] An alternative would be to absorb the low-frequency baseline with an RBF kernel in PRISM-VFF directly, but this bias would then need to be unlearned during BNGIF inference.

Dataset 4. VTRFormants

VTRFormants (Deng et al., 2006) is the TIMIT-derived corpus of hand-corrected formant and bandwidth tracks, distributed as frame-wise FB files aligned with the original TIMIT waveforms and phone labels (Garofolo et al., 1993).

Its role is to provide reference resonance trajectories for later evaluation. EGIFA asks whether a model can recover glottal source structure when reference flow is available. VTRFormants asks whether the same probabilistic machinery yields useful formant estimates on real speech.

4.1. Spectral tilt and TIMIT

The same TIMIT vowel material used to construct the reference formant tracks also gives a useful estimate of spectral rolloff.[26] What we want is not just the tilt of voiced speech itself, but an estimate of the tilt contributed by the *vocal tract filter*. I have not been able to find direct empirical estimates of that quantity in the literature.

Now, broadband slope is not uniquely attributable in source-filter decompositions: depending on the assumed glottal model and smoothing procedure, some of the slope can be assigned to the source and some to the filter (Fant, 1960). Nevertheless, physical models of the glottal cycle and lip radiation do imply a certain range of plausible tilt for these stages of speech production. We know that:

1. the glottal flow $u(t)$ itself has a rolloff of about -12 dB/oct for abrupt closure, and can be steeper, up to about -18 dB/oct, for smoother return phases (Doval et al., 2006);
2. lip radiation contributes about $+6$ dB/oct (Schroeder, 1999);
3. what remains to be estimated is the filter-side term; a quantity that should be interpreted as an effective residual tilt rather than a directly observed property of the vocal tract, but still not entirely unidentifiable;
4. these add up to the spectral tilt of the observed speech signal.

So by measuring 4 and subtracting 1 and 2, we can estimate 3.

Estimating the spectral tilt of voiced speech can be done with the marked vowel segments on which the VTRFormants reference tracks are based. So the data are essentially quasi-periodic vowels.

One could try to estimate rolloff from two frequencies an octave apart, but that is too noisy in practice because the FFT power spectrum is strongly modulated by harmonic structure. Instead, we smooth by regression. For each voiced vowel segment, we estimate the FFT magnitude spectrum of the speech signal in dB and fit the linear trend

$$P(f) = a \log_2 f + b. \tag{187}$$

[26] Note that 'rolloff', 'spectral tilt', and 'spectral slope' all loosely mean the same thing, although some authors reserve rolloff for asymptotic high-frequency decay and tilt for an overall spectral slope.

Then

$$P(2f) - P(f) = a. \tag{188}$$

so the coefficient a itself is the estimated rolloff in dB per octave.

The fit is only performed above a cutoff of about 200 Hz. The reason is that the additive rolloff picture only really makes sense beyond the glottal-formant region; below that, the low-frequency source structure dominates. Both Doval et al. (2006) and Fulop & Disner (2011) point to about 200 Hz as a reasonable cutoff for this regime.

Figure 44 shows four representative fits. Averaging over the full set of usable TIMIT vowel segments, the resulting voiced-speech tilt is about $-8(2)$ dB per octave. Combining this with the source-side heuristics above gives an effective residual filter tilt of about $-2(2)$ dB per octave.

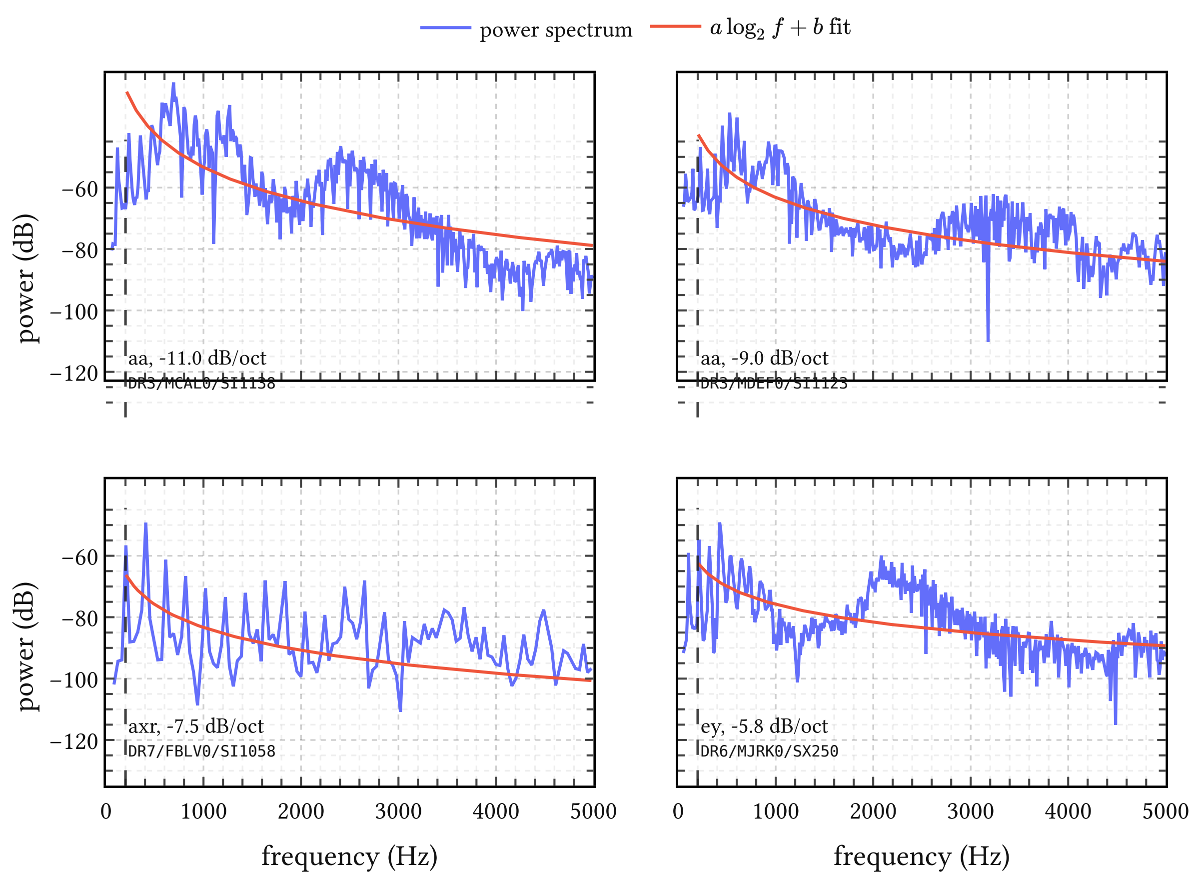


Figure 44 | **Example of spectral tilt fits on TIMIT vowel segments.** Each panel shows the empirical power spectrum together with the fitted trend $a \log_2 f + b$. The dashed vertical line marks the 200 Hz cutoff above which the fit is performed. The four examples span a representative range from steeper to flatter observed rolloff.

This is consistent with mild tilt as predicted by lossy acoustic tubes (de Boer, 2008) and with Chen (2013)'s theoretical estimate of the decay rate of formants (Eq. 24) which is $-3\ \mathrm{dB/oct}$, as opposed to Fant's suggestion of $-6\ \mathrm{dB/oct}$ (Eq. 26).

Algorithms

Algorithm 1. IKLP

Chapter 2 introduces *infinite kernel linear prediction* (IKLP) as the conceptual extension of linear prediction (LP) and multiple kernel linear prediction (MKLP). This chapter gives the technical side deferred there: the variational family, the tractable lower bound, the closed-form CAVI updates, the Woodbury-Cholesky rewrite used in this thesis, and the prior reparameterizations used later. The base model follows Yoshii & Goto (2013) the Woodbury-Cholesky reformulation and generalized AR-prior interface are the thesis-side additions.

1.1. Variational inference

Priors and variational family

To complete the model we place priors on the positive scale parameters and on $\mathbf{a}$. Following Yoshii & Goto (2013), the scales receive gamma priors,

$$\nu_w \sim \texttt{Gamma}(a_w, b_w), \quad \nu_e \sim \texttt{Gamma}(a_e, b_e). \tag{189}$$

For the AR coefficients we generalize the isotropic prior $\mathbf{a} \sim \texttt{Normal}(\mathbf{0}, \lambda \mathbf{I})$ of the original paper to a full multivariate Gaussian,

$$\mathbf{a} \sim \texttt{Normal}(\boldsymbol{\mu}_a, \boldsymbol{\Sigma}_a), \quad \mathbf{Q} := \boldsymbol{\Sigma}_a^{-1}, \tag{190}$$

with precision matrix $\mathbf{Q}$. The isotropic case is $\boldsymbol{\mu}_a = \mathbf{0}$, $\mathbf{Q} = \lambda^{-1}\mathbf{I}$. We need the full form in Chapter 3, where informative priors derived from stability theory and spectral constraints replace the isotropic default.

Exact posterior inference is intractable, so we use a mean-field variational family,

$$q(\boldsymbol{\theta}, \mathbf{a}, \nu_w, \nu_e) = q(\mathbf{a})\, q(\nu_w)\, q(\nu_e) \prod_{i=1}^{I} q(\theta_i), \quad q(\mathbf{a}) = \delta(\mathbf{a} - \mathbf{a}^*), \tag{191}$$

with $\mathbf{a}$ treated as a MAP point estimate. Each remaining factor is updated by coordinate ascent on the ELBO $\mathcal{L}$:

$$\log p(\mathbf{x}) \geq \underbrace{\mathbb{E}[\log p(\mathbf{x} \mid \boldsymbol{\theta}, \mathbf{a}, \nu_w, \nu_e)]}_{\text{expected log-likelihood}} + \sum_z \mathbb{E}[\log p(z)] - \sum_z \mathbb{E}[\log q(z)] =: \mathcal{L}. \tag{192}$$

Compared to MCMC-based GIF approaches — where Auvinen et al. (2014) report runtimes on the order of 2.5 hours of single-CPU time per 25-millisecond analysis frame — variational inference keeps the Bayesian treatment while reducing inference to a sequence of closed-form updates. CAVI is convergence-guaranteed, parallelizable over batches of frames, and compatible with modern autodiff and GPU workflows even when the update steps themselves remain analytic.

Tractable lower bound via matrix inequalities

The expected log-likelihood involves $\mathbb{E}[\log|\mathbf{K}|]$ and $\mathbb{E}[\mathbf{x}^\top \boldsymbol{\Psi}^\top \mathbf{K}^{-1} \boldsymbol{\Psi} \mathbf{x}]$, where $\mathbf{K} = \nu_w \sum_i \theta_i \mathbf{K}_i + \nu_e \mathbf{I}$, and neither expectation is tractable in closed form. Two matrix-variate inequalities provide bounds.

The log-determinant $\log|\mathbf{V}|$ is concave, so a first-order Taylor expansion around any PSD matrix $\boldsymbol{\Omega}$ gives

$$\log|\mathbf{V}| \le \log|\boldsymbol{\Omega}| + \operatorname{tr}(\boldsymbol{\Omega}^{-1}\mathbf{V}) - M. \tag{193}$$

The quadratic form $\mathbf{z}^\top \mathbf{V}^{-1} \mathbf{z}$ is convex, and an inequality of Sawada et al. (2012) gives

$$\mathbf{z}^\top \left(\sum_i \mathbf{V}_i \right)^{-1} \mathbf{z} \le \sum_i \mathbf{z}^\top \boldsymbol{\Upsilon}_i^\top \mathbf{V}_i^{-1} \boldsymbol{\Upsilon}_i \mathbf{z}, \tag{194}$$

where $\{\boldsymbol{\Upsilon}_i\}$ are auxiliary matrices summing to the identity.

Applying both inequalities to the expected log-likelihood and optimizing the auxiliary quantities in closed form gives the tractable lower bound $\mathcal{L}' \le \mathcal{L}$, with optimal values

$$\boldsymbol{\Omega} = \mathbb{E}[\nu_w] \sum_i \mathbb{E}[\theta_i] \mathbf{K}_i + \mathbb{E}[\nu_e] \mathbf{I}, \tag{195}$$

$$\boldsymbol{\Upsilon}_i = \tilde{w}_i \mathbf{K}_i \mathcal{S}^{-1}, \quad \boldsymbol{\Upsilon}_0 = \tilde{\nu}_e \mathcal{S}^{-1}, \tag{196}$$

where the *quadratic operator* $\mathcal{S}$ and harmonic-mean weights are defined by

$$\mathcal{S} = \sum_i \tilde{w}_i \mathbf{K}_i + \tilde{\nu}_e \mathbf{I}, \quad \tilde{\nu}_e = 1/\mathbb{E}[\nu_e^{-1}], \quad \tilde{w}_i = 1/(\mathbb{E}[\nu_w^{-1}]\mathbb{E}[\theta_i^{-1}]). \tag{197}$$

Both $\boldsymbol{\Omega}$ and $\mathcal{S}$ have the form $\nu \mathbf{I} + \sum_i w_i \mathbf{K}_i$; they differ only in which moment or harmonic-mean expectations enter as weights. This shared algebraic structure is what the Woodbury-Cholesky reformulation below exploits.

GIG posteriors and CAVI updates

The sufficient statistics of a gamma prior are $\log z$ and z; the bounds above introduce terms in z and $1/z$. The optimal variational posteriors for θ_i, ν_w, ν_e therefore all take the form of a *generalized inverse Gaussian*,

$$\texttt{GIG}(z \mid \gamma, \rho, \tau) \propto z^{\gamma-1} \exp(-1/2(\rho z + \tau/z)), \quad z > 0, \tag{198}$$

with moments

$$\mathbb{E}[z] = \sqrt{\tau/\rho}\, K_{\gamma+1}(\sqrt{\rho\tau})/K_{\gamma(\sqrt{\rho\tau})}, \quad \mathbb{E}[z^{-1}] = \sqrt{\rho/\tau}\, K_{\gamma-1}(\sqrt{\rho\tau})/K_{\gamma(\sqrt{\rho\tau})}, \tag{199}$$

where K_γ is the modified Bessel function of the second kind.

With the LP residual $\mathbf{r} := \boldsymbol{\Psi}(\mathbf{a})\mathbf{x}$, the weighted residual $\mathbf{w} := \mathcal{S}^{-1}\mathbf{r}$, and the per-kernel quadratic statistics

$$\mathbf{u}_i := \mathbf{\Phi}_i^\top \mathbf{w}, \quad q_i := \|\mathbf{u}_i\|^2, \quad q_0 := \|\mathbf{w}\|^2, \tag{200}$$

the CAVI parameter updates take the compact form. For the kernel weights:

$$\gamma_{\theta_i} = \alpha / I, \quad \rho_{\theta_i} \leftarrow 2\alpha + \mathbb{E}[\nu_w] \ \mathrm{tr}(\mathbf{\Omega}^{-1}\mathbf{K}_i), \quad \tau_{\theta_i} \leftarrow (1/\mathbb{E}[\nu_w^{-1}])\left(1/\mathbb{E}[\theta_i^{-1}]^2\right) q_i. \tag{201}$$

For the structured scale:

$$\gamma_w = a_w, \quad \rho_w \leftarrow 2b_w + \sum_i \mathbb{E}[\theta_i] \ \mathrm{tr}(\mathbf{\Omega}^{-1}\mathbf{K}_i), \quad \tau_w \leftarrow \sum_i \left(1/\mathbb{E}[\nu_w^{-1}]^2\right)(1/\mathbb{E}[\theta_i^{-1}]) q_i. \tag{202}$$

For the broadband scale:

$$\gamma_e = a_e, \quad \rho_e \leftarrow 2b_e + \mathrm{tr}(\mathbf{\Omega}^{-1}), \quad \tau_e \leftarrow \left(1/\mathbb{E}[\nu_e^{-1}]^2\right) q_0. \tag{203}$$

The AR update is MAP under the generalized prior (190), giving the normal equation

$$(\mathbf{X}^\top \mathcal{S}^{-1}\mathbf{X} + \mathbf{Q})\mathbf{a}^* = \mathbf{X}^\top \mathcal{S}^{-1}\mathbf{x} + \mathbf{Q}\boldsymbol{\mu}_a. \tag{204}$$

All updates are iterated until the ELBO converges or a maximum iteration count is reached.

The dominant costs per iteration are the I trace terms $\mathrm{tr}(\mathbf{\Omega}^{-1}\mathbf{K}_i)$ and the operator actions $\mathcal{S}^{-1}\mathbf{r}$ and $\mathcal{S}^{-1}\mathbf{X}$, all requiring solutions to linear systems in matrices of the form $\nu\mathbf{I} + \sum_i w_i \mathbf{K}_i$. The next section reduces all of these to triangular solves on a single small matrix.

1.2. Woodbury-Cholesky reformulation

Both $\mathbf{\Omega}$ and $\mathcal{S}$ are instances of the abstract operator

$$\mathbf{S}(\nu, \mathbf{w}) := \nu \mathbf{I}_M + \sum_{i=1}^{I} w_i \mathbf{K}_i \in \mathbb{R}^{M \times M}. \tag{205}$$

The direct approach materializes this $M \times M$ matrix, factors it by Cholesky ($\mathcal{O}(M^3)$), and solves systems against it ($\mathcal{O}(M^2)$ each). For the frame lengths relevant to GIF this is expensive; for large windows it is prohibitive.

Suppose each kernel admits a low-rank factorization $\mathbf{K}_i = \mathbf{\Phi}_i \mathbf{\Phi}_i^\top$ with $\mathbf{\Phi}_i \in \mathbb{R}^{M \times R_i}$. In the settings we care about this is exact: the periodic kernels of the original paper are parameterized by Fourier features, and the arc-cosine GP priors of later chapters are defined by their feature matrices $\mathbf{\Phi}_i$. Form the weighted stacked factor

$$\tilde{\mathbf{\Phi}} := \left[\sqrt{w_1}\mathbf{\Phi}_1, ..., \sqrt{w_I}\mathbf{\Phi}_I\right] \in \mathbb{R}^{M \times L}, \quad L = \sum_i R_i, \tag{206}$$

so that $\mathbf{S}(\nu, \mathbf{w}) = \nu \mathbf{I}_M + \tilde{\mathbf{\Phi}}\tilde{\mathbf{\Phi}}^\top$. The Woodbury matrix identity gives its inverse as

$$\mathbf{S}^{-1}\mathbf{v} = \nu^{-1}\mathbf{v} - \nu^{-2}\tilde{\mathbf{\Phi}}\mathbf{A}^{-1}\tilde{\mathbf{\Phi}}^\top \mathbf{v}, \quad \mathbf{A} := \mathbf{I}_L + \nu^{-1}\tilde{\mathbf{\Phi}}^\top\tilde{\mathbf{\Phi}} \in \mathbb{R}^{L \times L}. \tag{207}$$

The $L \times L$ *core matrix* $\mathbf{A}$ is the only object that needs to be factored. Its Cholesky decomposition $\mathbf{A} = \mathbf{R}^\top \mathbf{R}$ — computed once per operator instance per CAVI iteration — makes the following quantities available through triangular solves alone.

Operator solves To compute $\mathbf{S}^{-1}\mathbf{v}$: form $\mathbf{t} = \tilde{\mathbf{\Phi}}^\top \mathbf{v} \in \mathbb{R}^L$, solve the triangular systems $\mathbf{R}^\top \mathbf{y} = \mathbf{t}$ then $\mathbf{R}\mathbf{u} = \mathbf{y}$ to get $\mathbf{u} = \mathbf{A}^{-1}\mathbf{t}$, then return $\nu^{-1}\mathbf{v} - \nu^{-2}\tilde{\mathbf{\Phi}}\mathbf{u}$. No explicit inverse is formed.

Log-determinant By the matrix determinant lemma, $|\mathbf{S}| = \nu^M \ |\mathbf{A}|$, so

$$\log|\mathbf{S}| = M \log \nu + 2 \sum_{\ell=1}^{L} \log R_{\ell\ell}. \tag{208}$$

Trace of inverse

$$\operatorname{tr}(\mathbf{S}^{-1}) = (M - L)/\nu + \nu^{-1} \ \operatorname{tr}(\mathbf{A}^{-1}). \tag{209}$$

Per-kernel trace Define $\mathbf{B}_i := \tilde{\mathbf{\Phi}}^\top \mathbf{\Phi}_i \in \mathbb{R}^{L \times R_i}$. Then

$$\operatorname{tr}(\mathbf{S}^{-1}\mathbf{K}_i) = \nu^{-1} \ \|\mathbf{\Phi}_i\|_F^2 - \nu^{-2} \ \|\mathbf{R}^{-\top}\mathbf{B}_i\|_F^2, \tag{210}$$

where $\|\mathbf{R}^{-\top}\mathbf{B}_i\|_F^2$ is the squared Frobenius norm of the solution to $\mathbf{R}^\top \mathbf{Z} = \mathbf{B}_i$.

AR normal equation With $\mathcal{S}^{-1}$ available as a triangular-solve operator, the $P \times P$ Gram matrix $\mathbf{G} = \mathbf{X}^\top \mathcal{S}^{-1} \mathbf{X}$ is assembled column-by-column, and (204) is then a single $P \times P$ Cholesky solve of $\mathbf{G} + \mathbf{Q}$.

Every dense operation in the CAVI loop has been reduced to triangular solves on matrices of size at most $\max(L, P) \ll M$.

Complexity

Let M be the frame length, P the AR order, I the number of kernels, R the rank per kernel, $L = IR$ the total factor dimension, and B the number of frames in a batch.

In the dense baseline, building $\mathbf{S}$ costs $\mathcal{O}(IM^2R)$, factoring it costs $\mathcal{O}(M^3)$, each solve costs $\mathcal{O}(M^2)$, and storing it requires $\mathcal{O}(M^2)$ memory.

In the Woodbury-Cholesky regime, the core matrix $\mathbf{A}$ is assembled in $\mathcal{O}(ML^2)$ and factored in $\mathcal{O}(L^3)$. Each operator solve costs $\mathcal{O}(ML + L^2)$, the log-determinant costs $\mathcal{O}(L)$ after factorization, the per-kernel traces cost $\mathcal{O}(L^2R)$ each, and memory drops to $\mathcal{O}(ML + L^2)$: the $M \times M$ covariance is never stored. The crossover is favorable whenever $L \ll M$, precisely the regime of interest.

For B frames the loop vectorizes linearly in B. Since no dense $M \times M$ matrix is ever formed, the formulation is insensitive to its conditioning. Running in 32-bit floating point gives a substantial throughput gain on accelerators while retaining numerical stability through the Cholesky factorization and a small diagonal jitter on $\mathbf{A}$.

1.3. Prior reparameterizations

The priors (23) and (189) are written in shape-rate form, natural for conjugate analysis but opaque for prior elicitation. We reparameterize both into quantities with direct physical interpretations.

Kernel sparsity: α-calibration

Under (23), define $T := \sum_i \theta_i$ and the normalized weights $p_i = \theta_i / T$. For any finite truncation I, the sum $T \sim \texttt{Gamma}(\alpha, \alpha)$ has $\mathbb{E}[T] = 1$ and $\mathrm{sd}(T) = \alpha^{-\frac{1}{2}}$, while the proportions follow a symmetric Dirichlet:

$$(p_1, ..., p_I) \sim \texttt{Dirichlet}(\alpha/I, ..., \alpha/I). \tag{211}$$

Small α concentrates mass toward the vertices of the probability simplex — one dominant hypothesis — while large α spreads it uniformly.

A natural summary of effective sparsity is the exponentiated Shannon entropy,

$$I_{\text{eff}} := \exp(\mathbb{E}[H(\mathbf{p})]), \quad H(\mathbf{p}) = -\sum_i p_i \log p_i, \tag{212}$$

which counts the number of equally probable hypotheses that would produce the same entropy. Since the Dirichlet expectation of H is known analytically,

$$\mathbb{E}[H(\mathbf{p})] = \psi(\alpha + 1) - \psi(1 + \alpha/I), \tag{213}$$

we obtain

$$I_{\text{eff}(\alpha;I)} = \exp(\psi(\alpha + 1) - \psi(1 + \alpha/I)). \tag{214}$$

As $I \to \infty$ with α fixed, I_{eff} saturates at the constant $\exp(\psi(\alpha + 1) + \gamma_{\text{EM}})$ where γ_{EM} is the Euler-Mascheroni constant. At $\alpha = 1$ and large I this gives $I_{\text{eff}} \to e \approx 2.72$, meaning roughly a handful of simultaneously active hypotheses on average. This matches what Yoshii & Goto (2013) observe empirically at $\alpha = 1$, $I = 400$.

To calibrate α to a desired effective count I^0_{eff}, one solves (213) for α; the solution $\alpha^*(I)$ approaches 1 rapidly for $I^0_{\text{eff}} = e$ as I grows. The local sensitivity of I_{eff} to α near this calibration is described by the log-elasticity

$$b(\alpha; I) := \frac{\partial \log I_{\text{eff}}}{\partial \log \alpha} = \alpha[\psi_1(\alpha + 1) - I^{-1} \psi_1(1 + \alpha/I)], \tag{215}$$

where ψ_1 is the trigamma function. As $I \to \infty$ this approaches $b \to \psi_1(2) = \pi^2/6 - 1 \approx 0.645$, giving the local approximation

$$I_{\text{eff}(\alpha;I)} \approx e(\alpha/\alpha^*(I))^{0.645}. \tag{216}$$

So α behaves like a concentration parameter whose effect on sparsity is governed by a fixed exponent near the default calibration.

Excitation power decomposition: the (p, s, κ)-reparameterization

When $b_w = b_e =: b$, the total excitation power $\nu_w + \nu_e \sim \texttt{Gamma}(a_w + a_e, b)$ and the structured fraction $\pi := \nu_w / (\nu_w + \nu_e) \sim \texttt{Beta}(a_w, a_e)$ are independent of each other. This motivates the reparameterization

$$a_w = \kappa p, \quad a_e = \kappa(1 - p), \quad b_w = b_e = \kappa / s, \tag{217}$$

under which the total power has prior $\nu_w + \nu_e \sim \texttt{Gamma}(\kappa, \kappa/s)$ with

$$\mathbb{E}[\nu_w + \nu_e] = s, \quad \mathrm{Var}(\nu_w + \nu_e) = s^2/\kappa, \tag{218}$$

and the structured fraction has prior $\pi \sim \texttt{Beta}(\kappa p, \kappa(1 - p))$ with

$$\mathbb{E}[\pi] = p, \quad \mathrm{Var}(\pi) = p(1 - p)/(\kappa + 1). \tag{219}$$

The parameter κ controls concentration only — it tightens both distributions simultaneously without moving their means. The parameters (p, s) directly control the prior expected values: s is calibrated to the expected dynamic range of the analysis frame, and p reflects prior beliefs about phonation type. A frame expected to be clearly voiced might receive $p = 0.8$; an unvoiced prior might use $p = 0.1$.

The default of Yoshii & Goto (2013), $(a_w, a_e, b_w, b_e) = (1, 1, 1, 1)$, corresponds to $(p, s, \kappa) = (1/2, 2, 2)$: equal prior probability of structured and broadband excitation, total power expected at 2, with moderate concentration. A unit-power neutral default is $(p, s, \kappa) = (1/2, 1, 1)$.

1.4. Summary

The technical IKLP picture is now complete. The model itself is the sparse kernel-mixture LP likelihood of Chapter 2; the inference layer is a CAVI scheme with closed-form GIG updates; and the Woodbury-Cholesky rewrite removes the cubic bottleneck by reducing every dense step to triangular solves on matrices of size $\max(L, P) \ll M$.

This is why IKLP is useful later in the thesis. The inference engine itself stays fixed while the kernel bank changes. Replacing Yoshii's periodic kernels by the arc-cosine GP priors of later chapters requires no change to the CAVI architecture — only the feature matrices $\mathbf{\Phi}_i$ change.

Algorithm 2. LBGID

LBGID stands for *level-based glottal instant detection.*

Glottal closure and opening instants (GCIs and GOIs) define the pseudoperidocity of speech and are important speech features with many practical applications (Drugman et al., 2012). Within the thesis, GCIs and GOIs are used as temporal anchors for the kernels modeling the glottal cycle. Learning these kernels from example $\{u(t)\}$ glottal flow waveforms requires these instants to be known. The task LBGID attempts to solve is to extract consistent (GCI, GOI) pairs from a given $u(t)$, which can then be used downstream for aligning a set of $\{u(t)\}$ waveforms.

In a physics-driven high-quality synthesizer like VocalTractLab (Birkholz, 2013; Krug et al., 2021), glottal closure is not given as a symbolic event and often does not appear as a sharp isolated spike. Instead, it appears as an emergent soft low-flow regime embedded in a continuous waveform whose baseline, amplitude, and degree of breathiness can all vary over time. LBGID extracts the glottal instants based on the following assumption: closure happens where the flow $u(t)$ is below a certain threshold, and the glottal instants are the strongest *opposing changes* in $u(t)$ that flank it.

2.1. Adaptive envelope normalization

A difficulty with glottal airflow signals is that absolute amplitude is rarely interpretable on its own. Slow baseline drift and physical effects like variation in subglottal pressure all alter amplitude to various degrees. To measure effective flow rather than simulator artifacts, LBGID tracks adaptive upper and lower envelopes, denoted by $\mathrm{roof}(t)$ and $\mathrm{floor}(t)$, by detecting peaks and troughs and interpolating between them with a standard peak picking algorithm. These define a smoothly varying local amplitude range,

$$A(t) = \mathrm{roof}(t) - \mathrm{floor}(t) \geq 0, \tag{220}$$

which sets the local scale for the threshold. Note that we assume the polarity of $u(t)$ to be positive; that is, $u(t) \geq 0$ everywhere.

2.2. Level-based closed phase detection

With the local amplitude frame in hand, candidate closed phases are identified via a simple relative threshold,

$$\mathrm{level}(t) = \mathrm{floor}(t) + \alpha(\mathrm{roof}(t) - \mathrm{floor}(t)), \tag{221}$$

for a small constant α, and points satisfying $u(t) \leq \mathrm{level}(t)$ are treated as belonging to a closed phase. This yields contiguous low-flow segments that naturally partition the signal into bounded candidate intervals. In the thesis we use $\alpha = 0.01$ on EGIFA and $\alpha = 0.10$ on CommonVoice. The latter is needed because the inferred glottal flow there tends to retain a more persistent breathy floor. The three functions in (221) are illustrated in Figure 45 and Figure 46.

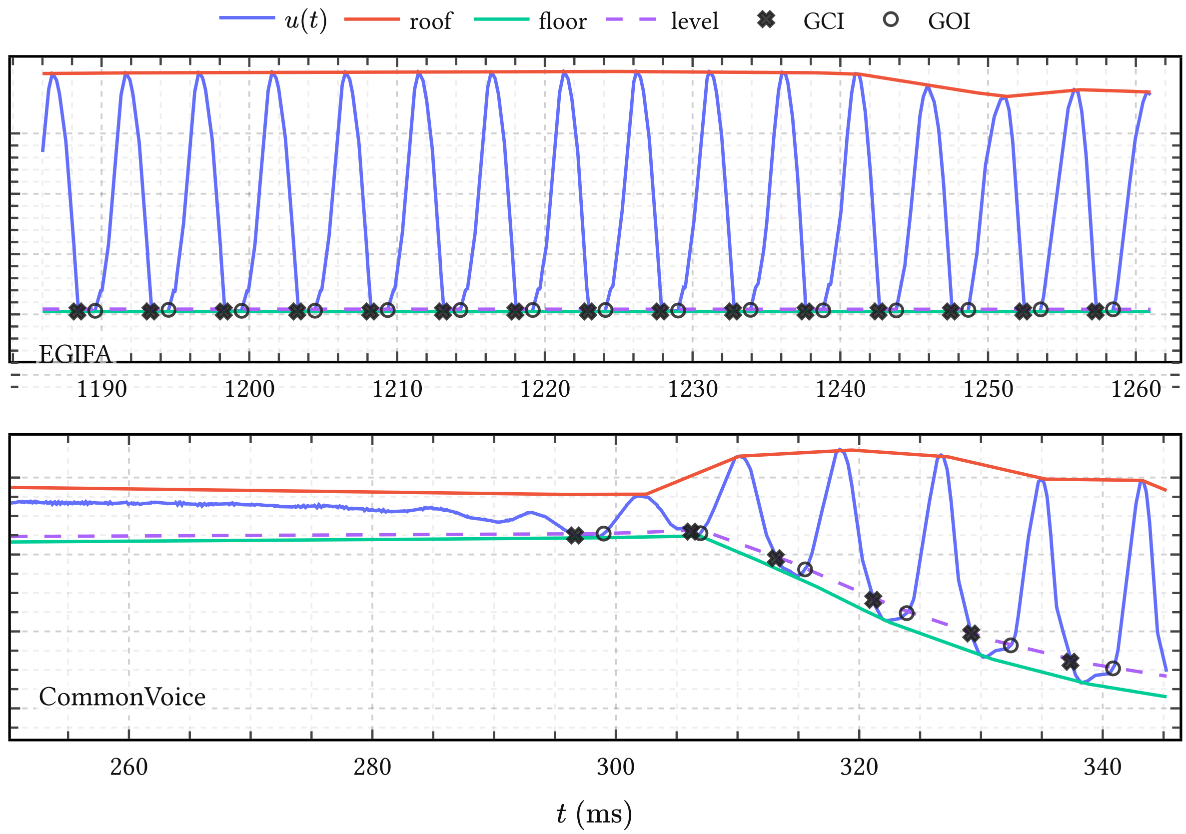


Figure 45 | **LBGID on EGIFA and CommonVoice.** Each panel shows $u(t)$ together with $\mathrm{roof}(t)$, $\mathrm{floor}(t)$, the relative threshold $\mathrm{level}(t)$, and the detected (GCI, GOI) pairs. On EGIFA we use $\alpha = 0.01$; on CommonVoice we use $\alpha = 0.10$. Note how the level function rejects the non-voiced part (roughly before 300 ms) in the bottom panel.

2.3. Dynamic pairing inside the closed phase

Within each low-flow segment, LBGID searches for a pair of maximally opposed changes in $u(t)$. First, the derivative $u'(t)$ is computed by finite differences. Then algorithm evaluates all pairs of sample indices (i, j), where $i < j$, within the segment by the score

$$E(i, j) = -u'(i)\, u'(j). \tag{222}$$

Among these admissible ordered pairs, the one maximizing $E(i, j)$ is selected. The sample indices of the optimal pair are then output as the glottal closure instant (GCI) and glottal opening instant (GOI), respectively. Equation (222) is thus just a cheap proxy for quantifying the "abruptness" of a candidate glottal instant pair.

Figure 45 shows the result on an excerpt from EGIFA (Dataset 1) and one from CommonVoice (Dataset 3). EGIFA has better defined closed phases than CommonVoice due to synthesis quality (even boasting handcrafted control trajectories for the synthesizer) and the choice of α is relatively inconsequential. By contrast, CommonVoice is much more challenging, and the

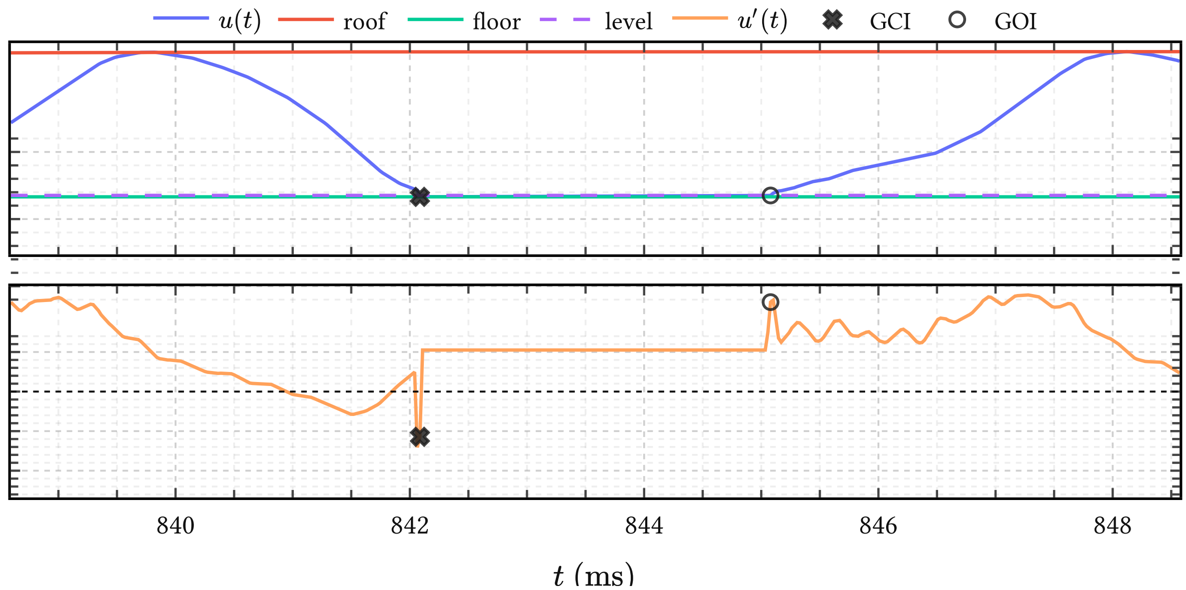


Figure 46 | **Zoom-in of a single closed phase.** Top: $u(t)$ together with $\text{roof}(t)$, $\text{floor}(t)$ and $\text{level}(t)$, with the selected low-flow region shaded. Bottom: $u'(t)$ on the same window, with the chosen GCI and GOI marked.

choice of α must balance accuracy (which requires α small) and false positives like splitting up closed phases when a bump happens in the middle (which happens happens when α is small).

Figure 46 shows a more detailed view of the algorithm for a given located closed phase. The algorithm first finds the candidate closed phase by amplitude, and then locates the strongest negative-to-positive dynamic pair inside it.

2.4. Evaluation

There is no direct ground truth for the GCI and GOI estimates themselves, except for manually checking every output (Serwy, 2017). On EGIFA, a proxy for LBGID's performance is simply to look at the implied quantities

$$F_{0,k} = \frac{f_s}{n_{\text{GCI},k+1} - n_{\text{GCI},k}}, \quad \text{OQ}_k = \frac{n_{\text{GOI},k} - n_{\text{GCI},k}}{n_{\text{GCI},k+1} - n_{\text{GCI},k}}, \tag{223}$$

whose distributions over frames in EGIFA are shown in Figure 40. These look reasonable and show a healthy amount of variation.

2.5. Summary

LBGID estimates (GCI, GOI) pairs from given $u(t)$ waveforms via level-based glottal instant detection.

It does this with two heuristics: closed phases correspond to intervals where flow is relatively small, and the GCI and GOI events happen with "maximally opposite abruptness."

These simple rules appear to be sufficient for the registration of the EGIFA and CommonVoice datasets.

Algorithm 3. PRISM

PRISM stands for *process-induced surrogate modeling*.

PRISM is an adapter from variable-length datasets to Gaussians in a common amplitude space. The idea is not to fit one GP per waveform and stop there, but to learn a single shared kernel-induced basis and express each dataset in that finite-dimensional basis with calibrated uncertainty. Each dataset is modeled by its own latent function corrupted independently by globally scaled noise.

The name signifies the following. "Process-induced" means that each observation is indexed by an arbitrary input grid and may have arbitrary length, so no ad-hoc resampling or interpolation is required. "Surrogate modeling" refers to the fact that the learned basis defines a cheaper probabilistic surrogate for the original collection of functions, to be optionally refined or compressed by the user. In the language of sparse GPs, PRISM is therefore best viewed as a Nyström-style inducing-feature approximation with analytic Bayesian linear-regression projection (Titsias, 2009; Wilk et al., 2020).

In Chapter 5 this shared-basis view is used for *calibration*: one global kernel is fit to a large registered EGIFA corpus of over 7k waveforms and 6.6M samples. Later chapters use the same representation for downstream surrogate modeling. PRISM is therefore best thought of as the bridge from datasets of functions to tractable Gaussian coordinates in an amplitude space of fixed dimension R.

The mechanism that makes this tractable is an extension of the classic collapsed variational bound proposed by Titsias (2009). For Gaussian likelihoods, the local inducing-variable posteriors can be optimized analytically and substituted back into the objective, so only global quantities need to be trained. Projection to per-dataset Gaussians then becomes a cheap post-training step. This chapter develops that construction explicitly.

3.1. Data and model

We have N independent datasets collected as

$$\mathcal{D} = \{(\mathbf{t}_n, \mathbf{y}_n)\}_{n=1}^{N}. \tag{224}$$

Dataset n consists of an input grid

$$\mathbf{t}_n = \left(t_{n1}, ..., t_{nM_n}\right) \tag{225}$$

and observations

$$\mathbf{y}_n = \left(y_{n1}, ..., y_{nM_n}\right), \tag{226}$$

where M_n may vary freely with n. For simplicity the inputs are written as scalar times, but nothing below depends on that; the same construction works for arbitrary input dimension. Indeed, any model compatible with Titsias (2009) can be handled by PRISM. An important restriction is the assumption of a Gaussian likelihood, which is what makes the local states collapse analytically.

Each dataset is modeled by an independent latent function

$$f_n(.) \sim \texttt{GaussianProcess}(0, k_\theta(., .)), \tag{227}$$

sharing the same kernel hyperparameters θ across all n but with independent draws of f_n. The kernel k_θ in (227) ties the draws together and acts as a global regularizer in function space.[27] The likelihood is given by

$$\mathbf{y}_n \mid f_n \sim \texttt{Normal}\left(f_n(\mathbf{t}_n), \sigma^2 \mathbf{I}_{M_n}\right) \tag{228}$$

which expresses the assumption that observations are conditionally independent given f_n, and imposes additional regularization through the global noise scale σ.

3.2. From kernel to shared basis

Choose R inducing locations

$$\mathbf{z} = (z_1, ..., z_R) \tag{229}$$

in the input domain and treat them as global learnable parameters. More general inducing *features* are also possible (Wilk et al., 2020), but vanilla inducing point evaluations are convenient for more general nonstationary kernels such as g-QPACK (145). For dataset n, define the inducing variables

$$\mathbf{u}_n = f_n(\mathbf{z}) \in \mathbb{R}^R. \tag{230}$$

Since f_n is a GP,

$$\mathbf{u}_n \sim \texttt{Normal}(\mathbf{0}, \mathbf{K}_{zz}), \tag{231}$$

where $\mathbf{K}_{zz} \in \mathbb{R}^{R \times R}$ with $(\mathbf{K}_{zz})_{rj} = k_\theta\left(z_r, z_j\right)$ is the same for all n. The GP conditional on $\mathbf{u}_n$ is

$$f_n(\mathbf{t}_n) \mid \mathbf{u}_n \sim \texttt{Normal}(\mathbf{K}_{nz}\mathbf{K}_{zz}^{-1}\mathbf{u}_n, \mathbf{K}_{nn} - \mathbf{Q}_{nn}), \tag{232}$$

where $\mathbf{K}_{nz} \in \mathbb{R}^{M_n \times R}$ has (m, r) entry $k_\theta(t_{nm}, z_r)$ and

$$\mathbf{Q}_{nn} = \mathbf{K}_{nz}\mathbf{K}_{zz}^{-1}\mathbf{K}_{zn} \tag{233}$$

is the Nyström approximation to $\mathbf{K}_{nn}$.

We emphasize that the column space of $\mathbf{K}_{nz}$ is shared across all datasets because the inducing locations $\mathbf{z}$ are shared. That is the common basis PRISM learns. The local variables $\mathbf{u}_n$ only say how dataset n "sits" in that basis.

[27] Since functions are infinite dimensional, some prior notion of smoothness is typically needed to make the problem well-posed (Sivia & Skilling, 2006).

3.3. Collapsed variational bound

The cost of keeping local parameters A direct sparse variational treatment would introduce one Gaussian variational factor per dataset,

$$q_n(\mathbf{u}_n) = \texttt{Normal}(\boldsymbol{\mu}_n, \boldsymbol{\Sigma}_n), \tag{234}$$

and maximize a separate ELBO for every n. That is workable for a single dataset, but for a large corpus it means carrying a synchronized local Gaussian state for every dataset during training. PRISM avoids this by keeping the local variational problem only long enough to solve it analytically, and then substituting the solution back into the objective.

Collapsed bound Following (Titsias, 2009), the per-dataset variational bound can be written as

$$\begin{aligned}\mathcal{L}_n(q_n) = \mathbb{E}_{q_n}&\left[\log \texttt{Normal}\left(\mathbf{y}_n \mid \mathbf{K}_{nz}\mathbf{K}_{zz}^{-1}\mathbf{u}_n, \sigma^2\mathbf{I}_{M_n}\right)\right] \\ &-D_{\mathrm{KL}}(q_n(\mathbf{u}_n) \parallel p(\mathbf{u}_n)) - \frac{1}{2\sigma^2}\operatorname{tr}(\mathbf{K}_{nn} - \mathbf{Q}_{nn}).\end{aligned} \tag{235}$$

For Gaussian likelihoods the optimal factor is available in closed form:

$$q_n(\mathbf{u}_n) = \texttt{Normal}(\boldsymbol{\mu}_n, \boldsymbol{\Sigma}_n), \tag{236}$$

with

$$\mathbf{C}_n = \mathbf{K}_{zz} + \sigma^{-2}\mathbf{K}_{zn}\mathbf{K}_{nz}, \tag{237}$$

$$\boldsymbol{\mu}_n = \sigma^{-2}\mathbf{K}_{zz}\mathbf{C}_n^{-1}\mathbf{K}_{zn}\mathbf{y}_n, \quad \boldsymbol{\Sigma}_n = \mathbf{K}_{zz}\mathbf{C}_n^{-1}\mathbf{K}_{zz}. \tag{238}$$

Substituting this optimum back into (235) gives the collapsed Titsias bound

$$\mathcal{L}_n(\mathbf{z}, \theta, \sigma^2) = \log \texttt{Normal}\left(\mathbf{y}_n \mid \mathbf{0}, \mathbf{Q}_{nn} + \sigma^2\mathbf{I}_{M_n}\right) - \frac{1}{2\sigma^2}\operatorname{tr}(\mathbf{K}_{nn} - \mathbf{Q}_{nn}). \tag{239}$$

The first term is the low-rank marginal likelihood and the second penalizes the discarded covariance.[28]

The PRISM ELBO is then just the sum over independent datasets,

$$\mathcal{L}(\mathbf{z}, \theta, \sigma^2) = \sum_{n=1}^{N} \mathcal{L}_n(\mathbf{z}, \theta, \sigma^2), \tag{240}$$

which is seen to depend only on the global parameters $(\mathbf{z}, \theta, \sigma^2)$.

Evaluating the bound In practice we whiten the basis so the local prior becomes isotropic. Let $\mathbf{K}_{zz} = \mathbf{L}\mathbf{L}^\top$ and define the amplitude coordinates

[28] The newer bounds of (Titsias, 2025) can also be used here. For BNGIF they did not yield better models in practice and were less efficient to batch, so we stayed with the original bound of (Titsias, 2009).

$$\mathbf{a}_n = \mathbf{L}^{-1}\mathbf{u}_n, \tag{241}$$

so that $\mathbf{a}_n \sim \texttt{Normal}(\mathbf{0}, \mathbf{I}_R)$. The corresponding feature map is

$$\varphi(t) = \mathbf{L}^{-1}\mathbf{k}_z(t), \quad \mathbf{k}_z(t) = (k_\theta(z_1, t), ..., k_\theta(z_R, t))^\top, \tag{242}$$

and the design matrix for dataset n is

$$\mathbf{\Phi}_n = \begin{pmatrix} \varphi(t_{n1})^\top \\ \vdots \\ \varphi\left(t_{nM_n}\right)^\top \end{pmatrix} \in \mathbb{R}^{M_n \times R}. \tag{243}$$

Then the sparse GP mean becomes a BLR mean,

$$\mathbf{y}_n = \mathbf{\Phi}_n \mathbf{a}_n + \varepsilon_n, \quad \varepsilon_n \sim \texttt{Normal}\left(\mathbf{0}, \sigma^2 \mathbf{I}_{M_n}\right). \tag{244}$$

Define

$$\mathbf{A}_n = \frac{\mathbf{\Phi}_n^\top}{\sigma} \in \mathbb{R}^{R \times M_n}, \quad \mathbf{B}_n = \mathbf{I}_R + \mathbf{A}_n \mathbf{A}_n^\top, \quad \mathbf{v}_n = \mathbf{A}_n \mathbf{y}_n \in \mathbb{R}^R. \tag{245}$$

Both terms in (239) are then computed efficiently from $\mathbf{A}_n$, $\mathbf{B}_n$, and $\mathbf{v}_n$, so the cost per dataset scales as $\mathcal{O}(R^2 M_n)$ and the $M_n \times M_n$ matrix $\mathbf{Q}_{nn} + \sigma^2 \mathbf{I}$ is never formed.

For the log marginal term, write $\mathbf{Q}_{nn} + \sigma^2 \mathbf{I} = \sigma^2\left(\mathbf{I}_{M_n} + \mathbf{A}_n^\top \mathbf{A}_n\right)$. By the matrix determinant lemma,

$$\log\det(\mathbf{Q}_{nn} + \sigma^2 \mathbf{I}) = M_n \log \sigma^2 + \log\det(\mathbf{B}_n). \tag{246}$$

The Woodbury identity gives

$$(\mathbf{Q}_{nn} + \sigma^2 \mathbf{I})^{-1} = \sigma^{-2}\left(\mathbf{I}_{M_n} - \mathbf{A}_n^\top \mathbf{B}_n^{-1} \mathbf{A}_n\right), \tag{247}$$

so the quadratic form is

$$\mathbf{y}_n^\top (\mathbf{Q}_{nn} + \sigma^2 \mathbf{I})^{-1} \mathbf{y}_n = \sigma^{-2}(\|\mathbf{y}_n\|^2 - \mathbf{v}_n^\top \mathbf{B}_n^{-1} \mathbf{v}_n). \tag{248}$$

For the trace correction, $\mathrm{tr}(\mathbf{Q}_{nn}) = \|\mathbf{\Phi}_n\|_F^2$, so

$$-\frac{1}{2\sigma^2}\ \mathrm{tr}(\mathbf{K}_{nn} - \mathbf{Q}_{nn}) = -\frac{1}{2\sigma^2}(\mathrm{tr}(\mathbf{K}_{nn}) - \|\mathbf{\Phi}_n\|_F^2). \tag{249}$$

Every quantity in (239) is thus computable from the global parameters and the current dataset alone.

3.4. Stochastic optimization

Because the datasets are independent, an unbiased estimate of (240) is available from any minibatch $\mathcal{B} \subset \{1, ..., N\}$,

$$\hat{\mathcal{L}} = \frac{N}{|\mathcal{B}|} \sum_{n \in \mathcal{B}} \mathcal{L}_n(\mathbf{z}, \theta, \sigma^2). \tag{250}$$

The minibatch unit is a *complete dataset*, not an individual observation. This is what keeps irregular grids and unequal lengths harmless: each dataset contributes one collapsed term, regardless of its internal geometry. In practice we optimize (250) with Adam following (Orvieto & Gower, 2025) at a learning rate of 10^{-3}. We found this stable and amenable to float32 optimization, which gives a substantial speed gain over float64.

The contrast with the ELBO of stochastic variational inference (Hensman et al., 2013, Eq. 3) is interesting here. There, stochastic optimization over very large datasets is obtained by introducing and maintaining an explicit variational state of size $\mathcal{O}(R^2)$ in order to have the objective factorize over datapoints (and allow more general likelihoods beyond (228)). We avoid that move here because the factorization is already over complete datasets. The local inducing posteriors are collapsed analytically and only the global parameters $(\mathbf{z}, \theta, \sigma^2)$ are updated.

3.5. Projection to amplitude space

Once $(\mathbf{z}, \theta, \sigma^2)$ are trained, PRISM maps any dataset of arbitrary length M_n to a Gaussian in the fixed amplitude space $\mathbb{R}^R$. At the fitted global parameters, the local inducing posterior is simply the optimal factor (238) evaluated for the new dataset:

$$\mathbf{u}_n \mid \mathbf{y}_n \sim \texttt{Normal}(\boldsymbol{\mu}_n, \boldsymbol{\Sigma}_n). \tag{251}$$

Passing to amplitude coordinates via $\mathbf{a}_n = \mathbf{L}^{-1}\mathbf{u}_n$ gives

$$\mathbf{a}_n \mid \mathbf{y}_n \sim \texttt{Normal}(\mathbf{m}_n, \mathbf{S}_n), \tag{252}$$

$$\mathbf{S}_n = \mathbf{B}_n^{-1}, \quad \mathbf{m}_n = \mathbf{B}_n^{-1}\mathbf{v}_n. \tag{253}$$

This reuses $\mathbf{B}_n$ and $\mathbf{v}_n$ from the collapsed bound, so projection is essentially for free.

PRISM therefore learns the shared basis (242) and returns the projected Gaussian coordinates

$$\mathcal{D}_R = \{(\mathbf{m}_n, \mathbf{S}_n)\}_{n=1}^N. \tag{254}$$

This is a collection of Gaussian blobs in a common $\mathbb{R}^R$. Together with the learned feature map $\boldsymbol{\varphi}(t)$, it defines the empirical amplitude-space density

$$p(\mathbf{a} \mid \mathcal{D}_R) = \frac{1}{N} \sum_{n=1}^{N} \texttt{Normal}(\mathbf{a} \mid \mathbf{m}_n, \mathbf{S}_n), \tag{255}$$

which is the object handed to downstream methods.

3.6. Surrogate modeling

Importantly, PRISM alone does not yet produce a good *global* surrogate model of the corpus. What it learns is the regularized basis (242) together with the Gaussian collection (254) of the observed data in that basis. This is already enough to define a set of *local* Gaussian surrogate posteriors. For dataset n, one can sample amplitudes

$$\mathbf{a}_n^* \sim \texttt{Normal}(\mathbf{m}_n, \mathbf{S}_n) \tag{256}$$

cheaply, push them through the synthesis map $\mathbf{\Phi}_n \mathbf{a}_n^*$, and thereby resample from the local surrogate posterior induced by the shared basis. The posterior covariances are what make this cheap resampling possible.

To obtain a better corpus-level surrogate model, extra structure must be learned on top of the empirical amplitude density $p(\mathbf{a} \mid \mathcal{D}_R)$ of (255). One may ignore the covariances and apply ordinary tools such as PCA, clustering, or Gaussian mixtures to the posterior means. Or one may keep the covariances and use errors-in-variables (uncertain-input) variants of those same ideas. For very large collections one generally also wants to quantize or compress the number of Gaussian blobs, for example by hierarchical clustering or by the latent-variable model of the next chapter. In this thesis we take the latter route and hand the amplitude-space object $\mathcal{D}_R$ off to q-GPLVM (Algorithm 5), which learns a more compact generative model in the same shared amplitude space.

3.7. Summary

PRISM learns a shared, global kernel-induced basis from a possibly large collection of independent variable-length time series without storing or updating per-dataset variational parameters. In terms of GP approximation, PRISM is a reduced-rank projection of the GP prior k_θ to a rank-R BLR. The infinite-dimensional function space is projected onto the span of the learned basis, and the remaining degrees of freedom are absorbed by the observation noise σ^2.

After training, each dataset is projected to a Gaussian amplitude posterior (252) in a common, R-dimensional amplitude space. The returned collection $\mathcal{D}_R$ of (254), together with the shared basis (242), can then be used as-is for cheap surrogate sampling or passed to more structured downstream models.

Algorithm 4. t-PRISM

The Gaussian likelihood in PRISM (Algorithm 3) is analytically convenient, but inappropriate when data contain outliers. For learning a pitch-track model from APLAWD (Dataset 2), for example, the observations are inter-GCI period measurements, where a single detector failure, silent gap, or irregular closure can create an isolated gross outlier. More generally, any functional series may contain observations that should be downweighted rather than fitted exactly.

To address this, we introduce t-PRISM here as the "robust" version of PRISM. It replaces the Gaussian likelihood by a Student-t while retaining the same collapsed basis-learning structure. Tractability is maintained by representing the Student-t as a Gaussian scale mixture with one local precision variable per observation. Conditional on those precisions the likelihood is Gaussian again, so the collapsed bound from Algorithm 3 applies with the single modification that the noise covariance becomes diagonally weighted. The result is again a minibatch algorithm over complete datasets, and like PRISM it outputs Gaussian amplitude posteriors in a shared space.

Figure 47 illustrates the principle. Under a Gaussian likelihood, the posterior mean is pulled toward the locally irregular periods because the model must explain every residual at full strength. t-PRISM instead preserves the smooth trend and incorporates outliers by reducing their corresponding local weights $w_m \ll 1$.

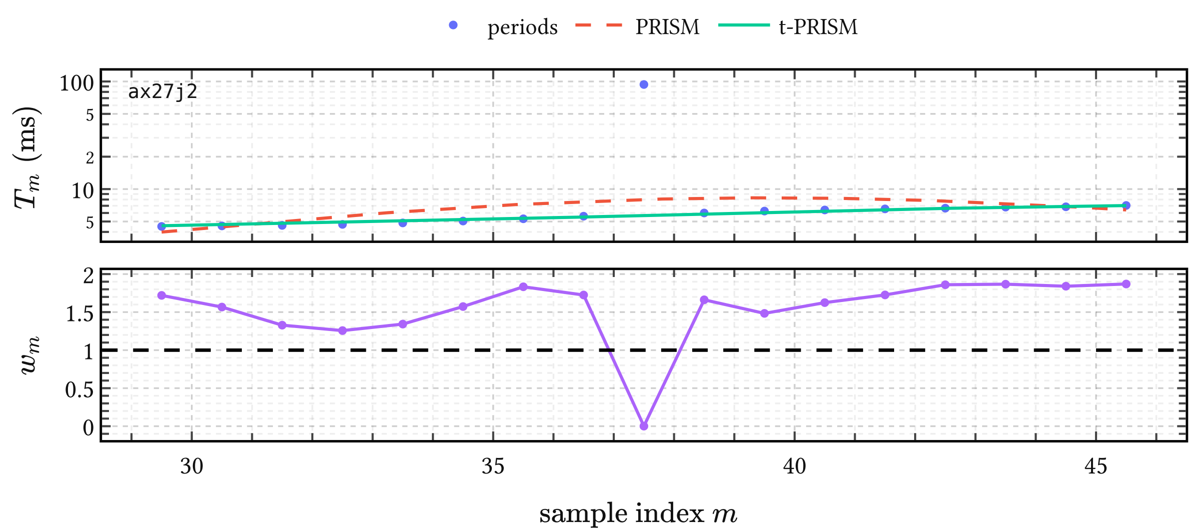


Figure 47 | **t-PRISM on a local APLAWD period track.** The top panel compares PRISM and t-PRISM on one excerpt of inter-GCI periods from the APLAWD pitch-track corpus, using the learned pitch track model of Dataset 2. The bottom panel shows the inferred precision weights $w_m = \mathbb{E}[\lambda_m]$. Weights $w_m \geq 1$ mark observations consistent with the local trend; smaller weights $w_m < 1$ signal outliers. Note that t-PRISM ignores the outlier at $m = 37.5$.

4.1. Student-t as a scale mixture

Recall the PRISM setup from Algorithm 3. Given our kernel of choice k_θ, we are presented with N independent datasets $\{(\mathbf{t}_n, \mathbf{y}_n)\}_{n=1}^{N}$, where dataset n has length M_n, latent function

$$f_n(.) \sim \texttt{GaussianProcess}(0, k_\theta(., .)), \tag{257}$$

and inducing variables $\mathbf{u}_n = f_n(\mathbf{z})$ at shared inducing locations $\mathbf{z}$. In PRISM this prior is paired with the Gaussian observation model

$$\mathbf{y}_n \mid f_n \sim \texttt{Normal}\left(f_n(\mathbf{t}_n), \sigma^2 \mathbf{I}_{M_n}\right), \tag{258}$$

as in (228). t-PRISM leaves this prior side untouched. The only change is that the homoscedastic Gaussian likelihood is replaced by a locally weighted one: we introduce one latent precision variable per observation and integrate it out to obtain a Student-t likelihood.

The Student-t likelihood with degrees of freedom $\nu > 0$ and scale σ^2 is recovered from the standard Normal-Gamma augmentation

$$\lambda_{nm} \sim \texttt{Gamma}(\nu/2, \nu/2), \quad y_{nm} \mid f_{nm}, \lambda_{nm} \sim \texttt{Normal}(f_{nm}, \sigma^2/\lambda_{nm}). \tag{259}$$

Marginalizing over λ_{nm} recovers $\texttt{StudentT}(y_{nm} \mid f_{nm}, \nu, \sigma^2)$ exactly. For dataset n, stack the per-sample precisions into

$$\mathbf{\Lambda}_n = \operatorname{diag}\left(\lambda_{n1}, ..., \lambda_{nM_n}\right). \tag{260}$$

The augmented likelihood is then Gaussian with diagonal but heteroscedastic noise covariance:

$$\mathbf{y}_n \mid f_n, \mathbf{\Lambda}_n \sim \texttt{Normal}(f_n(\mathbf{t}_n), \sigma^2 \mathbf{\Lambda}_n^{-1}). \tag{261}$$

Large λ_{nm} means that sample (n, m) is trusted; small λ_{nm} means that its residual is down-weighted. The prior mean of every λ_{nm} is one, so if the data contain no clear outliers the posterior expected precisions remain close to one. Then $\mathbf{W}_n \approx \mathbf{I}$ below, and the weighted formulas reduce continuously back to the PRISM expressions.

Figure 48 shows the intended fallback behavior. When there is no clear evidence for contamination, robustness costs essentially nothing: t-PRISM simply behaves like PRISM.

4.2. Variational family and objective

We introduce a mean-field variational family (Blei et al., 2017) over the latent precisions,

$$q_n(\mathbf{\Lambda}_n) = \prod_{m=1}^{M_n} q_{nm}(\lambda_{nm}), \quad q_{nm}(\lambda_{nm}) = \texttt{Gamma}(\alpha_{nm}, \beta_{nm}), \tag{262}$$

while the GP part is kept exactly as in the PRISM derivation. That is, for fixed precisions we still introduce a Gaussian factor $q_n(\mathbf{u}_n)$ over the inducing variables, use the exact GP

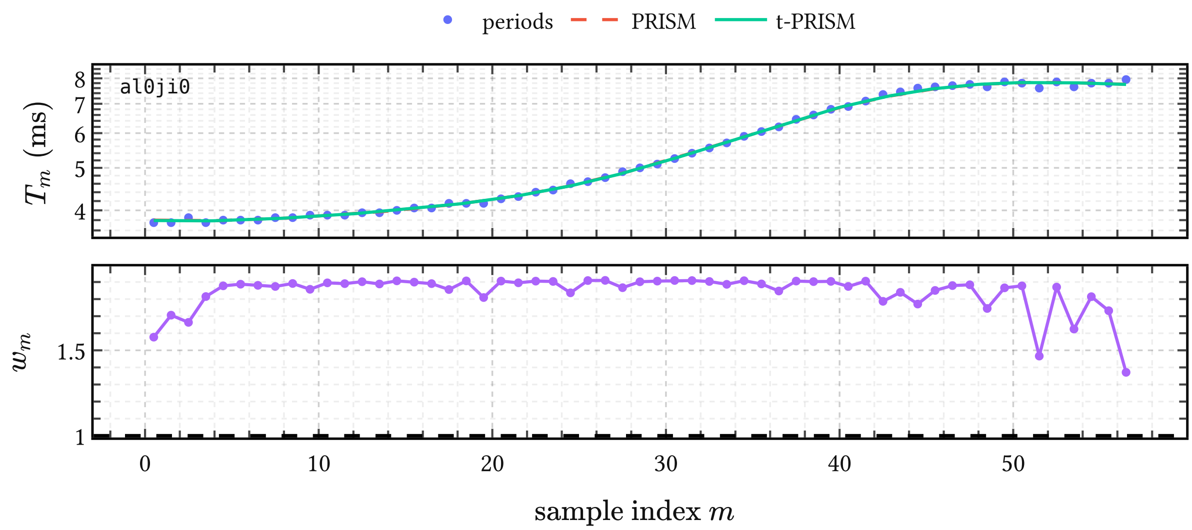


Figure 48 | **t-PRISM reducing to PRISM in the absence of outliers.** On this local APLAWD period track the observations are already consistent with a smooth trend. The PRISM and t-PRISM fits are nearly indistinguishable, and all inferred weights $w_m \geq 1$ and $O(1)$.

conditional $p(f_n \mid \mathbf{u}_n)$, and collapse the local Gaussian state analytically as in (235) and (239). The per-dataset ELBO before collapse is

$$\begin{aligned} \mathcal{L}_n^t = \mathbb{E}_{q_n}[\log p(\mathbf{y}_n \mid f_n, \mathbf{\Lambda}_n)] - D_{\mathrm{KL}}(q_n(\mathbf{u}_n) \parallel p(\mathbf{u}_n)) \\ - \sum_{m=1}^{M_n} D_{\mathrm{KL}}(q_{nm}(\lambda_{nm}) \parallel p(\lambda_{nm})). \end{aligned} \tag{263}$$

Define the expected precision and expected log-precision under q_{nm}:

$$\begin{aligned} w_{nm} &= \mathbb{E}[\lambda_{nm}] = \alpha_{nm}/\beta_{nm} \\ \ell_{nm} &= \mathbb{E}[\log \lambda_{nm}] = \psi(\alpha_{nm}) - \log \beta_{nm}, \end{aligned} \tag{264}$$

where ψ is the digamma function (MacKay & Bauman Peto, 1994).

4.3. Weighted collapsed variational bound

Given fixed expected precisions, write

$$\mathbf{W}_n = \mathrm{diag}\left(w_{n1}, ..., w_{nM_n}\right). \tag{265}$$

Conditional on $\mathbf{W}_n$, the likelihood is Gaussian with noise covariance $\sigma^2 \mathbf{W}_n^{-1}$. This is exactly the PRISM setting with weighted observation noise, so the collapsed bound applies again.

Define the weighted matrices

$$
\begin{aligned}
\mathbf{A}_n^W &= \frac{1}{\sigma}\mathbf{\Phi}_n^\top \mathbf{W}_n^{1/2} \in \mathbb{R}^{R\times M_n} \\
\mathbf{B}_n^W &= \mathbf{I}_R + \mathbf{A}_n^W (\mathbf{A}_n^W)^\top \in \mathbb{R}^{R\times R} \\
\mathbf{v}_n^W &= \mathbf{A}_n^W \mathbf{W}_n^{1/2} \mathbf{y}_n \in \mathbb{R}^R.
\end{aligned} \tag{266}
$$

In the unweighted case $\mathbf{W}_n = \mathbf{I}$ these reduce exactly to $\mathbf{A}_n$, $\mathbf{B}_n$, and $\mathbf{v}_n$ from (245).

The collapsed Gaussian contribution becomes

$$
\mathcal{L}_n^{\mathrm{coll}}(\mathbf{W}_n) = \log \mathtt{Normal}\big(\mathbf{y}_n \mid \mathbf{0}, \mathbf{Q}_{nn} + \sigma^2 \mathbf{W}_n^{-1}\big) - \frac{1}{2\sigma^2}\ \mathrm{Tr}(\mathbf{W}_n(\mathbf{K}_{nn} - \mathbf{Q}_{nn})). \tag{267}
$$

The weighted predictive moments needed below are

$$
\begin{aligned}
m_{nm} &= \varphi(t_{nm})^\top (\mathbf{B}_n^W)^{-1} \mathbf{v}_n^W, \\
v_{nm} &= k(t_{nm}, t_{nm}) - \varphi(t_{nm})^\top \Big(\mathbf{I} - (\mathbf{B}_n^W)^{-1}\Big) \varphi(t_{nm}).
\end{aligned} \tag{268}
$$

The only new part is the expected log-likelihood under the random precisions. At sample (n, m), conditioning on λ_{nm} gives a Gaussian variance σ^2/λ_{nm}. Its normalization therefore contributes

$$
-\frac{1}{2}\log\left(\frac{\sigma^2}{\lambda_{nm}}\right) = -\frac{1}{2}\log\sigma^2 + \frac{1}{2}\log\lambda_{nm}, \tag{269}
$$

so after expectation the log-precision enters through $\ell_{nm} = \mathbb{E}[\log \lambda_{nm}]$. The quadratic penalty is weighted by $w_{nm} = \mathbb{E}[\lambda_{nm}]$. Substituting the moments from (268) gives, up to constants,

$$
\mathbb{E}[\log p(\mathbf{y}_n \mid f_n, \mathbf{\Lambda}_n)] = -\frac{1}{2}\sum_{m=1}^{M_n}\Big(\log\sigma^2 - \ell_{nm} + \frac{w_{nm}}{\sigma^2}\big((y_{nm} - m_{nm})^2 + v_{nm}\big)\Big), \tag{270}
$$

and the full per-dataset t-PRISM ELBO is

$$
\begin{aligned}
\mathcal{L}_n^t(\mathbf{W}_n, \ell_n) = \mathcal{L}_n^{\mathrm{coll}}(\mathbf{W}_n) &+ \frac{1}{2}\sum_{m=1}^{M_n} \ell_{nm} \\
&- \sum_{m=1}^{M_n} D_{\mathrm{KL}}(q_{nm}(\lambda_{nm}) \parallel \mathtt{Gamma}(\nu/2, \nu/2)).
\end{aligned} \tag{271}
$$

Local CAVI updates for the precision variables Differentiating (271) gives the optimal values for tightening the ELBO:

$$
\alpha_{nm}^* = (\nu+1)/2, \quad \beta_{nm}^* = \frac{1}{2}\Big(\nu + \big((y_{nm} - m_{nm})^2 + v_{nm}\big)/\sigma^2\Big). \tag{272}
$$

The shape α is fixed once ν is chosen and does not depend on the data. In the deployed APLAWD model of this thesis we use the conservative choice $\nu = 1$, giving deliberately heavy tails for occasional gross detector failures or silent gaps. As noted in Dataset 2, a learned-ν extension was also tried, but did not improve held-out score over this simpler fixed choice. Only the rate β changes, increasing when the current residual $(y_{nm} - m_{nm})^2 + v_{nm}$ is large relative to σ^2. Large residuals therefore drive w_{nm} down in the next weighted collapse. That is the mechanism driving "robustness."

4.4. Training

The full t-PRISM objective is

$$\mathcal{L}^t = \sum_{n=1}^{N} \mathcal{L}_n^t. \tag{273}$$

For a minibatch $\mathcal{B}$ of complete datasets, the stochastic estimate is

$$\widehat{\mathcal{L}}^t = \frac{N}{|\mathcal{B}|} \sum_{n \in \mathcal{B}} \mathcal{L}_n^t. \tag{274}$$

For each dataset n in the minibatch:

1. initialize $w_{nm} = 1$ for all m,
2. repeat for a small fixed number of steps:
   - run the weighted collapsed GP computation with current $\mathbf{W}_n$, obtaining (m_{nm}, v_{nm}) via (268),
   - update $(\alpha_{nm}, \beta_{nm})$ via (272) and refresh (w_{nm}, ℓ_{nm}) via (264),
3. evaluate $\mathcal{L}_n^t$ via (271).

Then take a gradient step in the global parameters $(\mathbf{z}, \theta, \sigma^2, \nu)$. The Gamma parameters are local to the current minibatch pass and discarded afterwards. As in PRISM, no large per-dataset state persists across iterations. In the experiments of this thesis we use a fixed three inner CAVI steps, do not differentiate through those local updates, and optimize the global parameters with the same Adam setup and float32 arithmetic as in PRISM.

4.5. Projection to amplitude space

After training, each dataset is projected to a Gaussian amplitude posterior exactly as in Algorithm 3.5., now using the final converged weights from the inner loop. The posterior formulas are weighted versions of (252) and (253):

$$\begin{aligned} \mathbf{a}_n \mid \mathbf{y}_n &\sim \texttt{Normal}(\mathbf{m}_n, \mathbf{S}_n) \\ \mathbf{S}_n &= \left(\mathbf{B}_n^W\right)^{-1} \\ \mathbf{m}_n &= \left(\mathbf{B}_n^W\right)^{-1} \mathbf{v}_n^W. \end{aligned} \tag{275}$$

This again reuses quantities already computed during the inner loop. So t-PRISM returns the same interface object as PRISM,

$$\mathcal{D}_R = \{(\mathbf{m}_n, \mathbf{S}_n)\}_{n=1}^{N}, \tag{276}$$

but now obtained from a robust fit. The weight vectors $\{\mathbf{w}_n\}_{n=1}^{N}$ can be retained separately as auxiliary diagnostics when one wants local inlier scores in the original observation domain.

4.6. Summary

t-PRISM relaxes the Gaussian noise assumption of PRISM, which allows it to "robustly" learn a shared basis for a set of exemplars contaminated with outliers. This is achieved by replacing the Gaussian likelihood with a Student-t likelihood written as a Gaussian scale mixture.

Conditional on the local precision variables, the model is Gaussian again, so the same collapsed bound can be reused with weighted noise covariance $\sigma^2 \mathbf{W}_n^{-1}$. The local CAVI updates (272) then shrink the weights of observations whose residuals are too large for the current smooth fit. Training therefore remains a minibatch procedure over complete datasets, and the final output remains the same Gaussian collection $\mathcal{D}_R$ as in PRISM.

In this thesis the fitted weight vectors $\mathbf{w}_n$ are used both indirectly, through the robust amplitude posteriors, and directly, as local inlier scores on period trajectories. The motivation for developing t-PRISM came from needing to solve the following problems:

- learning a pitch-track model from the pitch-trajectory exemplars of APLAWD in Dataset 2;
- and segmenting pitch trajectories into voiced groups for the EGIFA and CommonVoice datasets (Dataset 1 and Dataset 3, respectively).

Algorithm 5. q-GPLVM

As we established in Algorithm 3, PRISM projects a collection of variable-length waveforms into Gaussians in a shared amplitude space $\mathbb{R}^R$. More precisely, PRISM learns the shared feature map $\boldsymbol{\varphi}(t)$ and returns the Gaussian collection

$$\mathcal{D}_R = \{(\mathbf{m}_n, \mathbf{S}_n)\}_{n=1}^{N} \tag{277}$$

where waveform n is projected from data space to amplitude space through the posterior

$$\mathbf{a}_n \mid \mathbf{y}_n \sim \texttt{Normal}(\mathbf{m}_n, \mathbf{S}_n). \tag{278}$$

This induces the empirical amplitude-space density

$$p(\mathbf{a} \mid \mathcal{D}_R) = \frac{1}{N} \sum_{n=1}^{N} \texttt{Normal}(\mathbf{a} \mid \mathbf{m}_n, \mathbf{S}_n), \tag{279}$$

which is not yet a compact global generative model. The density (255) remains tied to the training set, and scoring a new waveform against it still requires keeping all N components around.

q-GPLVM compresses (255) into a compact generative prior. It stands for *quantized Gaussian process latent variable model*. It first learns a Bayesian GPLVM in a lower-dimensional manifold space, then quantizes the aggregated manifold posterior into a small Gaussian mixture, and finally pushes that mixture forward through the learned GP map back to amplitude space. The result is a mixture of K Gaussians in $\mathbb{R}^R$ that summarizes the training distribution without reference to individual training waveforms. After composition with the PRISM synthesis map, this becomes a compact mixture of BLR models in data space. The key idea is that a nonlinear manifold model can be converted into a mixture of linear Gaussian models via moment matching, and that this mixture is a cheap surrogate model for waveforms learned by PRISM.

5.1. Prior art

This idea was first introduced by Nickisch & Rasmussen (2010), who showed that fitting a GMM in the latent space of a GPLVM and pushing the mixture through the GP map via the moment-matching formulas of Girard et al. (2002) yields a tractable mixture density in observation space. They also showed that this improved generalization capability compared to a GMM in the raw high-dimensional input space in the context of density scoring. The main difference here is that for q-GPLVM we "upgrade" their usage of vanilla MAP GPLVM and GMM to Bayesian GPLVM and XDGMM, respectively (Bovy et al., 2011; Titsias & Lawrence, 2010). This is done in the interest of retaining enough variability in the learned components and contributes to downstream uncertainty calibration.

5.2. Data, amplitude, and manifold space

q-GPLVM is an input-output pipeline with three levels.

Level 0: data space This is the space of observed waveforms. The input to PRISM is the collection

$$\mathcal{D} = \{(\mathbf{t}_n, \mathbf{y}_n)\}_{n=1}^{N}, \tag{280}$$

with waveform n sampled at its own grid $\mathbf{t}_n$ and values $\mathbf{y}_n$. After all compression steps, q-GPLVM returns a prior that again lives in this same variably-dimensioned space.

Level 1: amplitude space PRISM maps each dataset $(\mathbf{t}_n, \mathbf{y}_n) \in \mathcal{D}$ to a Gaussian in amplitude space $\mathbb{R}^R$ and collects those posteriors as $\mathcal{D}_R$ in (277). Its shared feature map $\boldsymbol{\varphi}(t)$ (242) implies the linear synthesis model

$$y(t) \approx \boldsymbol{\varphi}(t)^\top \mathbf{a}, \quad \mathbf{a} \sim \texttt{Normal}(\mathbf{0}, \mathbf{I}), \tag{281}$$

so Level 1 is summarized exactly by the density $p(\mathbf{a} \mid \mathcal{D}_R)$ of (255). This is a *linear* dimensionality reduction: variable-length waveforms become Gaussian amplitude coordinates in fixed dimension R.

Level 2: manifold space The Bayesian GPLVM introduces manifold coordinates $\mathbf{x}_n \in \mathbb{R}^Q$ with $Q \ll R$ and learns the posterior

$$\mathcal{D}_Q = \{(\boldsymbol{\mu}_n^x, \mathbf{S}_n^x)\}_{n=1}^{N}, \tag{282}$$

equivalently

$$q(\mathbf{X} \mid \mathcal{D}_Q) = \prod_{n=1}^{N} \texttt{Normal}(\mathbf{x}_n \mid \boldsymbol{\mu}_n^x, \mathbf{S}_n^x). \tag{283}$$

Here the covariance $\mathbf{S}_n^x$ is diagonal, as in the mean-field variational family of (Titsias & Lawrence, 2010). This is a *nonlinear* dimensionality reduction from amplitude space to manifold space. Quantization then replaces the aggregated posterior by a small mixture

$$q(\mathbf{x} \mid \mathcal{M}_Q) = \sum_{k=1}^{K} \pi_k \, \texttt{Normal}(\mathbf{x} \mid \mathbf{c}_k, \mathbf{C}_k), \qquad \sum_{k=1}^{K} \pi_k = 1, \tag{284}$$

which is later pushed back to Level 1 as the amplitude-space density $p(\mathbf{a} \mid \mathcal{M}_R)$ and then composed with PRISM via (313) to return to Level 0.

5.3. The Bayesian GPLVM

The theoretical foundation is the Bayesian GP latent variable model of Titsias & Lawrence (2010), which we now review. We refer to (Titsias & Lawrence, 2010) for the full derivations and details; here we record only the pieces needed later for quantization and push-forward.

Observations and notation PRISM returns the Gaussian amplitude collection $\mathcal{D}_R$. The Bayesian GPLVM, however, is formulated only for point observations in amplitude space, subject to homoscedastic observation noise. We therefore apply one global affine whitening transform

$$\tilde{\mathbf{a}} = \mathbf{W}(\mathbf{a} - \mathbf{b}), \tag{285}$$

with empirical center

$$\mathbf{b} = \frac{1}{N} \sum_{n=1}^{N} \mathbf{m}_n, \tag{286}$$

and a fixed global whitening matrix $\mathbf{W}$ chosen once from the PRISM amplitude cloud. To avoid a copious amount of tildes, we continue to write $\mathbf{a}_n$ for the transformed posterior mean and fit the standard BGPLVM to these point estimates.

Collect them into the amplitude matrix

$$\mathbf{A} = \begin{pmatrix} \mathbf{a}_1^\top \\ \vdots \\ \mathbf{a}_N^\top \end{pmatrix} \in \mathbb{R}^{N \times R}, \tag{287}$$

so that a_{nr} denotes the r-th amplitude coordinate of waveform n. This matrix plays the role of the observed data matrix in standard GPLVM usage.

Latent variable model For each amplitude coordinate $r = 1, ..., R$, introduce an independent latent function $g_r(\cdot)$ with shared GP prior

$$g_r(\cdot) \sim \texttt{GaussianProcess}(0, k(\cdot, \cdot)), \tag{288}$$

and an observation model

$$a_{nr} \mid g_r, \mathbf{x}_n \sim \texttt{Normal}(g_r(\mathbf{x}_n), \beta^{-1}), \tag{289}$$

where $\mathbf{x}_n \in \mathbb{R}^Q$ is the manifold coordinate of waveform n and β^{-1} is the learned isotropic GPLVM noise precision. The latent prior is isotropic:

$$p(\mathbf{x}_n) = \texttt{Normal}\big(\mathbf{0}, \mathbf{I}_Q\big). \tag{290}$$

The kernel is ARD squared exponential (MacKay, 1998),

$$k(\mathbf{x}, \mathbf{x}') = \sigma_g^2 \exp\left(-\frac{1}{2} \sum_{q=1}^{Q} \alpha_q \big(x_q - x_q'\big)^2\right), \tag{291}$$

with per-dimension inverse lengthscales α_q. When the model drives $\alpha_q \to 0$ during training, dimension q is effectively switched off, providing automatic selection of the intrinsic latent

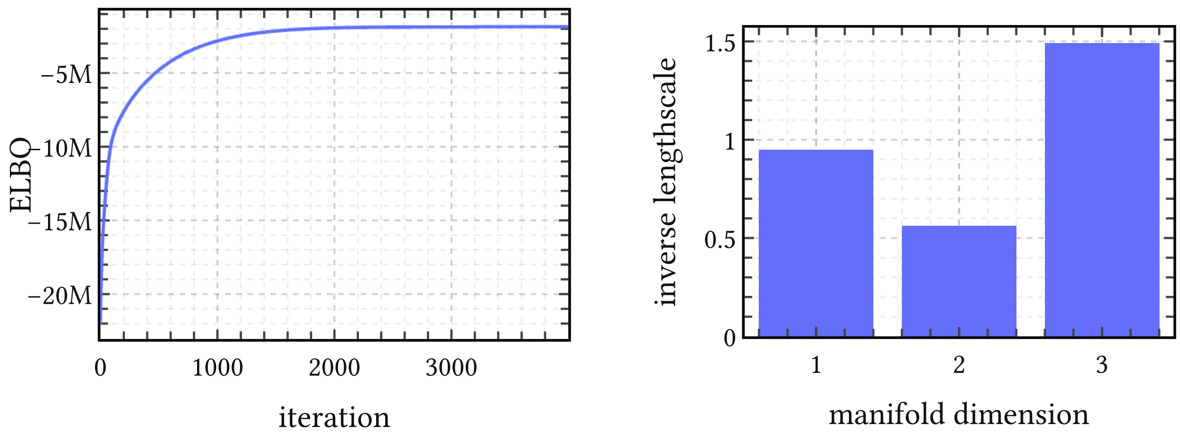


Figure 49 | **Representative objective and ARD on EGIFA.** Left: ELBO trace for one typical q-GPLVM run. Right: the learned inverse lengthscales in manifold space. The optimization stabilizes cleanly, and the ARD kernel keeps all three manifold dimensions active, though not equally: the third carries the sharpest variation and the second is the smoothest.

dimensionality. In the q-GPLVM fits to DGF waveforms we set $Q = 3$ and all three manifold dimensions remain active, though to different degrees as seen in Figure 49.

Variational inference Integrating out the manifold coordinates from the marginal likelihood is intractable because they enter the kernel matrix nonlinearly. Following (Titsias & Lawrence, 2010), introduce P inducing inputs $\mathbf{Z} \in \mathbb{R}^{P \times Q}$ in manifold space and optimize the factorized Gaussian posterior (282), with diagonal latent covariance

$$\mathbf{S}_n^x = \mathrm{diag}\left(S_{n1}^x, ..., S_{nQ}^x\right). \tag{292}$$

The integer P is the number of inducing points; importantly later it also bounds the rank of the structured covariance terms produced by push-forward. For these point observations, the collapsed variational bound has the standard form

$$\mathcal{L}^q = \tilde{\mathcal{L}}^q - D_{\mathrm{KL}}(q(\mathbf{X}) \parallel p(\mathbf{X})), \tag{293}$$

where $\tilde{\mathcal{L}}^q$ is the sparse-GP expected log-likelihood and the KL term is analytic. The inducing variables are again collapsed analytically, exactly as in Algorithm 3. The variational parameters $\{\boldsymbol{\mu}_n^x, \mathbf{S}_n^x\}_{n=1}^N$, the inducing inputs $\mathbf{Z}$, and the kernel and noise parameters $\left\{\sigma_g^2, \alpha_1, ..., \alpha_Q, \beta\right\}$ are all optimized jointly by gradient ascent on (293).

The Ψ statistics All tractable computation in the Bayesian GPLVM reduces to these expectations of kernel quantities under $q(\mathbf{x}_n)$:

$$
\begin{aligned}
\psi_0 &= \sum_{n=1}^{N} \mathbb{E}_{q(\mathbf{x}_n)}[k(\mathbf{x}_n, \mathbf{x}_n)], \\
[\mathbf{\Psi}_1]_{np} &= \mathbb{E}_{q(\mathbf{x}_n)}\left[k(\mathbf{x}_n, \mathbf{z}_p)\right], \\
[\mathbf{\Psi}_2]_{pp'} &= \sum_{n=1}^{N} \mathbb{E}_{q(\mathbf{x}_n)}\left[k(\mathbf{x}_n, \mathbf{z}_p) k(\mathbf{x}_n, \mathbf{z}_{p'})\right],
\end{aligned} \tag{294}
$$

with $\mathbf{\Psi}_1 \in \mathbb{R}^{N \times P}$ and $\mathbf{\Psi}_2 \in \mathbb{R}^{P \times P}$. Note that $\mathbf{\Psi}_2$ is *not* $\mathbf{\Psi}_1^\top \mathbf{\Psi}_1$: the integration couples the two kernel evaluations through the shared uncertain input $\mathbf{x}_n$.

For the ARD RBF kernel (291) with $q(\mathbf{x}_n) = \mathtt{Normal}(\boldsymbol{\mu}_n^x, \mathbf{S}_n^x)$, (Titsias & Lawrence, 2010) show that all three statistics admit closed forms. The first statistic is trivial: $\psi_0 = N\sigma_g^2$ since $k(\mathbf{x}, \mathbf{x}) = \sigma_g^2$ for any $\mathbf{x}$. For $\mathbf{\Psi}_1$,

$$
[\mathbf{\Psi}_1]_{np} = \sigma_g^2 \prod_{q=1}^{Q} \left(\alpha_q S_{nq}^x + 1\right)^{-1/2} \exp\left(-\frac{\alpha_q \left(\mu_{nq}^x - z_{pq}\right)^2}{2\left(\alpha_q S_{nq}^x + 1\right)}\right), \tag{295}
$$

which is a kernel evaluation at $\boldsymbol{\mu}_n^x$ with inflated lengthscales $\alpha_q^{-1} + S_{nq}^x$. When $\mathbf{S}_n^x \to \mathbf{0}$ (no input uncertainty), this recovers the standard kernel evaluation $k(\boldsymbol{\mu}_n^x, \mathbf{z}_p)$. $\mathbf{\Psi}_2$ is likewise explicit:

$$
[\mathbf{\Psi}_2]_{pp'} = \sigma_g^4 \sum_{n=1}^{N} \prod_{q=1}^{Q} \left(2\alpha_q S_{nq}^x + 1\right)^{-1/2} \exp\left(-\alpha_q \frac{\left(z_{pq} - z_{p'q}\right)^2}{4} - \alpha_q \frac{\left(\mu_{nq}^x - \frac{z_{pq} + z_{p'q}}{2}\right)^2}{2\alpha_q S_{nq}^x + 1}\right) \tag{296}
$$

Later, in the push-forward step, the quantized manifold components carry full covariances $\mathbf{C}_k$ rather than diagonal $\mathbf{S}_n^x$. The same Gaussian-integral mechanism extends to any Gaussian input covariance, so no new derivations are needed (Damianou et al., 2014).

5.4. Quantization

After training, the Bayesian GPLVM provides a variational posterior (282) for each training waveform. Define the empirical manifold density induced by $\mathcal{D}_Q$ as

$$
q\left(\mathbf{x} \mid \mathcal{D}_Q\right) = \frac{1}{N} \sum_{n=1}^{N} \mathtt{Normal}(\mathbf{x} \mid \boldsymbol{\mu}_n^x, \mathbf{S}_n^x). \tag{297}
$$

This is already a generative model in manifold space, but with N components it still encodes the entire training set, and is bound to generalize poorly. Quantization replaces it by the compact K-component approximation ($K \ll N$)

$$
q\left(\mathbf{x} \mid \mathcal{M}_Q\right) = \sum_{k=1}^{K} \pi_k \, \mathtt{Normal}(\mathbf{x} \mid \mathbf{c}_k, \mathbf{C}_k), \qquad \sum_{k=1}^{K} \pi_k = 1, \tag{298}
$$

where $\mathbf{c}_k \in \mathbb{R}^Q$ and $\mathbf{C}_k \in \mathbb{R}^{Q \times Q}$ are the mean and full covariance of component k. Full covariances are retained because local curvature and anisotropy in manifold space are precisely the structure we wish to preserve for downstream uncertainty calibration.

GMM with uncertain inputs A standard GMM fit to the posterior means $\{\boldsymbol{\mu}_n^x\}$ alone would ignore the latent uncertainty and overfit to noise in the mean estimates. The standard algorithm to take into account the posterior uncertainty in $\{\mathbf{S}_n^x\}$ is XDGMM, or extreme deconvolution Gaussian mixture models (Bovy et al., 2011). XDGMM is a variant of EM for Gaussian mixtures that accounts for known, heterogeneous Gaussian uncertainty on the inputs, and is easily fit to (297) by treating each posterior mean $\boldsymbol{\mu}_n^x$ as an uncertain observation with known covariance $\mathbf{S}_n^x$.

The output of the quantization step is the compact object

$$\mathcal{M}_Q = \{(\pi_k, \mathbf{c}_k, \mathbf{C}_k)\}_{k=1}^K, \tag{299}$$

which parameterizes the mixture $q(\mathbf{x} \mid \mathcal{M}_Q)$ in (298) and stands in for the entire training collection. On EGIFA, the learned manifold for $Q = 3$ is already visibly organized by open quotient, and the fitted ellipsoids with $K = 5$ then provide a compact cover of that structure, as shown in Figure 50.

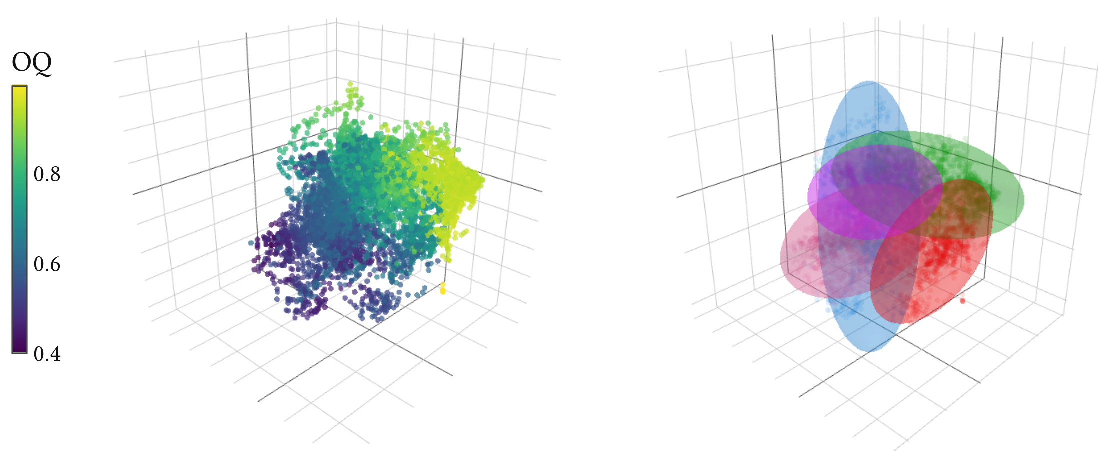


Figure 50 | **Manifold structure and quantization on EGIFA.** Left: the three-dimensional manifold space learned by q-GPLVM, with each waveform colored by open quotient. Right: the $K = 5$ fitted Gaussian ellipsoids used for quantization in that same manifold space (299). The left panel shows that open quotient varies smoothly over the manifold rather than appearing as disconnected clusters, while the right panel shows how XDGMM compresses the empirical mixture $q(\mathbf{x} \mid \mathcal{D}_Q)$ of (297) into the $K = 5$ approximation $q(\mathbf{x} \mid \mathcal{M}_Q)$ of (298).

5.5. Push-forward

We now have the quantized manifold mixture $q(\mathbf{x} \mid \mathcal{M}_Q)$ from (298) and a trained GP map from manifold space $\mathbb{R}^Q$ to amplitude space $\mathbb{R}^R$. Each component of $\mathcal{M}_Q$ must be pushed through that map. One can either sample from the manifold mixture and propagate those samples through the nonlinear GP directly (Figure 51), or construct the moment-matched Gaussian approximation in amplitude space first and then synthesize waveforms from that. The latter method allows deriving surrogate priors directly in data space.

Recovered inducing parameters During training, the inducing variables were eliminated analytically from the bound. For the push-forward we need them back. Since the GPLVM uses R independent GPs that share the same kernel but have independent function draws, the collapsed sparse GP posterior of (Titsias, 2009) gives a shared inducing covariance

$$\hat{\mathbf{S}}_u = \mathbf{K}_{ZZ}\left(\mathbf{K}_{ZZ} + \beta\boldsymbol{\Psi}_2^{\mathrm{tr}}\right)^{-1}\mathbf{K}_{ZZ} \in \mathbb{R}^{P\times P}, \tag{300}$$

which depends only on the kernel and the Ψ statistics, and is therefore the same for all output dimensions. The inducing means, however, depend on the observed amplitudes for each dimension r separately. Collecting them into the matrix $\widehat{\mathbf{M}} \in \mathbb{R}^{P\times R}$ gives

$$\widehat{\mathbf{M}} = \beta\,\mathbf{K}_{ZZ}\left(\mathbf{K}_{ZZ} + \beta\boldsymbol{\Psi}_2^{\mathrm{tr}}\right)^{-1}\left[\boldsymbol{\Psi}_1^{\mathrm{tr}}\right]^{\top}\mathbf{A}, \tag{301}$$

where $\boldsymbol{\Psi}_1^{\mathrm{tr}}$ and $\boldsymbol{\Psi}_2^{\mathrm{tr}}$ denote the trained versions of the statistics defined in (294) and $\mathbf{A} \in \mathbb{R}^{N\times R}$ is the amplitude matrix. Thus $\widehat{\mathbf{M}}$ encodes how the training amplitudes project onto the inducing basis, one column per output dimension.

Moment matching For one manifold component $\mathbf{x} \sim \mathtt{Normal}(\mathbf{c}, \mathbf{C})$, the induced distribution of the amplitude vector

$$\mathbf{a} = (g_1(\mathbf{x}), \ldots, g_R(\mathbf{x}))^{\top} \tag{302}$$

under the GP is non-Gaussian due to the nonlinearity of the map. The moment-matched Gaussian approximation $\mathtt{Normal}(\boldsymbol{\mu}_a, \mathbf{S}_a)$ whose first two moments equal those of the true push-forward was derived by Girard et al. (2002) for GP prediction at uncertain inputs, applied to GPLVM density modeling by Nickisch & Rasmussen (2010), and is standard in GP uncertainty propagation (Deisenroth & Rasmussen, 2011). We use the sparse-GP variant of those formulas, where the Ψ statistics of Damianou et al. (2014) replace the full-GP kernel expectations.

The required expectations are the same Ψ statistics of (294), now evaluated at the single GMM component $(\mathbf{c}, \mathbf{C})$ rather than summed over the training posteriors. Since the ARD RBF formula (295) remains valid for any Gaussian input, including one with full covariance $\mathbf{C}$, no new derivations are needed. We write $\psi_1(\mathbf{c}, \mathbf{C}) \in \mathbb{R}^P$ and $\boldsymbol{\Psi}_2(\mathbf{c}, \mathbf{C}) \in \mathbb{R}^{P\times P}$ for the per-component versions of these statistics.

Mean The sparse GP predictive mean at a fixed manifold point $\mathbf{x}$ is $\widehat{\mathbf{M}}^\top \mathbf{K}_{ZZ}^{-1} \mathbf{k}_Z(\mathbf{x})$, where $\mathbf{k}_Z(\mathbf{x}) = (k(\mathbf{x}, \mathbf{z}_1), ..., k(\mathbf{x}, \mathbf{z}_P))^\top$. Taking the expectation over $\mathbf{x} \sim \texttt{Normal}(\mathbf{c}, \mathbf{C})$ replaces $\mathbf{k}_Z(\mathbf{x})$ by $\psi_1(\mathbf{c}, \mathbf{C})$:

$$\boldsymbol{\mu}_a = \widehat{\mathbf{M}}^\top \mathbf{K}_{ZZ}^{-1} \psi_1(\mathbf{c}, \mathbf{C}) \in \mathbb{R}^R. \tag{303}$$

Covariance The law of total variance gives

$$\mathbf{S}_a = \underbrace{\mathbb{E}_{\mathbf{x}}[\texttt{Cov}(\mathbf{a} \mid \mathbf{x})]}_{\text{averaged predictive variance}} + \underbrace{\texttt{Cov}_{\mathbf{x}}(\mathbb{E}[\mathbf{a} \mid \mathbf{x}])}_{\text{input uncertainty}}. \tag{304}$$

The first term averages the GP predictive variance (which is the same for every amplitude coordinate r since all outputs share the kernel) under the input distribution. Given the latent function values at a fixed $\mathbf{x}$, the output coordinates are conditionally independent, so this term is isotropic:

$$\mathbb{E}_{\mathbf{x}}[\texttt{Cov}(\mathbf{a} \mid \mathbf{x})] = \gamma \, \mathbf{I}_R, \tag{305}$$

where

$$\gamma = \psi_0 + \beta^{-1} - \operatorname{Tr}(\mathbf{K}_{ZZ}^{-1} \boldsymbol{\Psi}_2(\mathbf{c}, \mathbf{C})) + \operatorname{Tr}\Big(\mathbf{K}_{ZZ}^{-1} \hat{\mathbf{S}}_u \mathbf{K}_{ZZ}^{-1} \boldsymbol{\Psi}_2(\mathbf{c}, \mathbf{C})\Big). \tag{306}$$

The four contributions are respectively: the prior variance at $\mathbf{x}$, the observation noise, minus the variance explained by the inducing inputs, plus the posterior uncertainty on those inducing inputs. Thus the isotropic term $\gamma \, \mathbf{I}_R$ collects averaged predictive uncertainty and observation noise. Cross-output correlations reappear only after integrating over inducing-variable uncertainty and latent uncertainty, which is exactly what the second term below captures.

The second term in (304) captures how the predictive mean varies as $\mathbf{x}$ moves over the component. It introduces cross-output correlations via

$$\texttt{Cov}_{\mathbf{x}}(\mathbb{E}[\mathbf{a} \mid \mathbf{x}]) = \widehat{\mathbf{M}}^\top \mathbf{K}_{ZZ}^{-1} (\boldsymbol{\Psi}_2(\mathbf{c}, \mathbf{C}) - \psi_1(\mathbf{c}, \mathbf{C}) \psi_1(\mathbf{c}, \mathbf{C})^\top) \mathbf{K}_{ZZ}^{-1} \widehat{\mathbf{M}}, \tag{307}$$

which has rank at most P.

Combining, the moment-matched covariance is

$$\mathbf{S}_a = \gamma \, \mathbf{I}_R + \widehat{\mathbf{M}}^\top \mathbf{K}_{ZZ}^{-1} (\boldsymbol{\Psi}_2(\mathbf{c}, \mathbf{C}) - \psi_1(\mathbf{c}, \mathbf{C}) \psi_1(\mathbf{c}, \mathbf{C})^\top) \mathbf{K}_{ZZ}^{-1} \widehat{\mathbf{M}}. \tag{308}$$

The push-forward covariance thus decomposes into an isotropic term, which reflects averaged GP uncertainty and observation noise, and a structured term, which carries the geometry induced by the manifold component.

The mixture weights The weights π_k of the manifold-space mixture $q\big(\mathbf{x} \mid \mathcal{M}_Q\big)$ are a prior on manifold space. Under the push-forward, a simple approximation is to retain those same

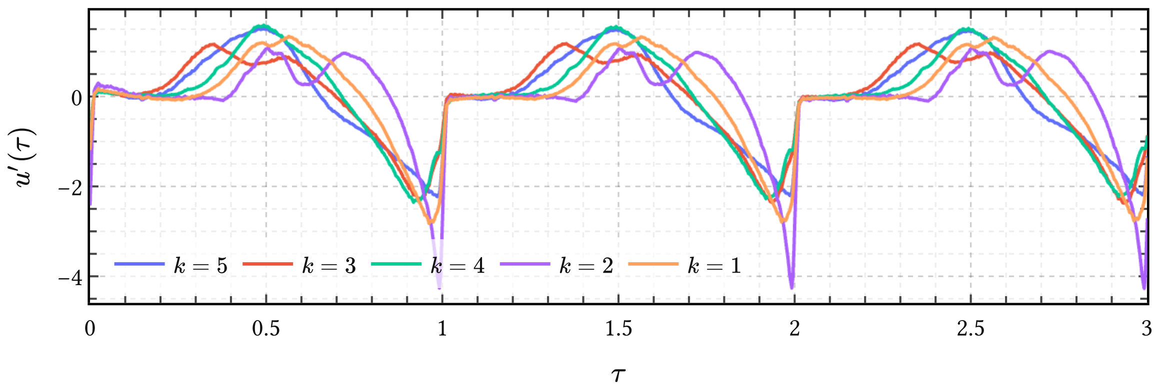


Figure 51 | **Direct samples from the quantized manifold mixture in data space** for the data in Figure 50. Here one first samples a component and manifold coordinate from $q(\mathbf{x} \mid \mathcal{M}_Q)$ in (298), then pushes that sample through the GPLVM map described in Algorithm 5.5.2. and finally through the PRISM synthesis map (281) to Level 0. This shows the nonlinear generative mechanism itself, before moment matching replaces it by a Gaussian approximation in amplitude space.

weights in amplitude space:[29] Define the corresponding compact amplitude-space object as

$$\mathcal{M}_R = \{(\pi_k, \boldsymbol{\mu}_k^a, \mathbf{S}_k^a)\}_{k=1}^K. \tag{309}$$

The induced amplitude-space density is then

$$p(\mathbf{a} \mid \mathcal{M}_R) \approx \sum_{k=1}^K \pi_k \, \texttt{Normal}(\mathbf{a} \mid \boldsymbol{\mu}_k^a, \mathbf{S}_k^a), \tag{310}$$

where $(\boldsymbol{\mu}_k^a, \mathbf{S}_k^a)$ is the moment-matched push-forward of the k-th component via (303) and (308).

5.6. Composing with PRISM

The density $p(\mathbf{a} \mid \mathcal{M}_R)$ of (310) lives in the PRISM amplitude space $\mathbb{R}^R$. To obtain a prior over observations on a grid $\mathbf{t}$, let $\boldsymbol{\Phi}(\mathbf{t}) \in \mathbb{R}^{M \times R}$ denote the design matrix obtained by evaluating the PRISM feature map (242) on that grid. Then PRISM synthesizes via the linear map

$$\mathbf{y} = \boldsymbol{\Phi}(\mathbf{t})\mathbf{a} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \texttt{Normal}(\mathbf{0}, \sigma^2 \mathbf{I}_M), \tag{311}$$

Because this step is linear, pushing a Gaussian in amplitude space through it preserves Gaussianity:

$$\mathbf{y} \mid \mathbf{t}, k, \mathcal{M}_R \sim \texttt{Normal}(\boldsymbol{\Phi}(\mathbf{t})\boldsymbol{\mu}_k^a, \quad \boldsymbol{\Phi}(\mathbf{t})\mathbf{S}_k^a \boldsymbol{\Phi}(\mathbf{t})^\top + \sigma^2 \mathbf{I}_M), \tag{312}$$

and the full prior over data space is

[29] More specialized analyses of pushing Gaussians through nonlinear maps, including higher-order corrections beyond moment matching, are given by (Zhang & Shin, 2021) and (Zhang & Shin, 2022).

$$p(\mathbf{y} \mid \mathbf{t}, \mathcal{M}_R) \approx \sum_{k=1}^{K} \pi_k \, \texttt{Normal}(\mathbf{y} \mid \mathbf{\Phi}(\mathbf{t})\boldsymbol{\mu}_k^a, \mathbf{\Phi}(\mathbf{t})\mathbf{S}_k^a\mathbf{\Phi}(\mathbf{t})^\top + \sigma^2\mathbf{I}_M). \tag{313}$$

This is a finite mixture of BLR models in data space, all sharing the same synthesis map $\boldsymbol{\varphi}(t)$ inherited from PRISM.

Samples from (313) are shown in Figure 52 for the same data from Figure 50 where $Q = 3$ and $K = 5$. It is seen that each component k effectively learns a DGF waveform type with variations in open quotient (OQ) and GCI spike intensity. Compared with Figure 51, these draws are visibly rougher because the nonlinear push-forward has been replaced by a Gaussian approximation in amplitude space, whose covariance includes the extra isotropic uncertainty term of (306) to compensate for the approximation.

Low-rank structure In amplitude space, (308) is the sum of a full-rank isotropic term $\gamma\,\mathbf{I}_R$ and a structured term of rank at most P. The latent dimension Q controls the geometry of that structured term through the kernel and the posterior over manifold positions, but it does not itself give a literal rank bound after passage through the inducing-feature map.

After composition with PRISM on a fixed grid $\mathbf{t}$, the data-space covariance becomes

$$\mathbf{\Phi}(\mathbf{t})\mathbf{S}_k^a\mathbf{\Phi}(\mathbf{t})^\top + \sigma^2\mathbf{I}_M = \sigma^2\mathbf{I}_M + \gamma\,\mathbf{\Phi}(\mathbf{t})\mathbf{\Phi}(\mathbf{t})^\top + \mathbf{\Phi}(\mathbf{t})\mathbf{S}_{\text{str}}\mathbf{\Phi}(\mathbf{t})^\top, \tag{314}$$

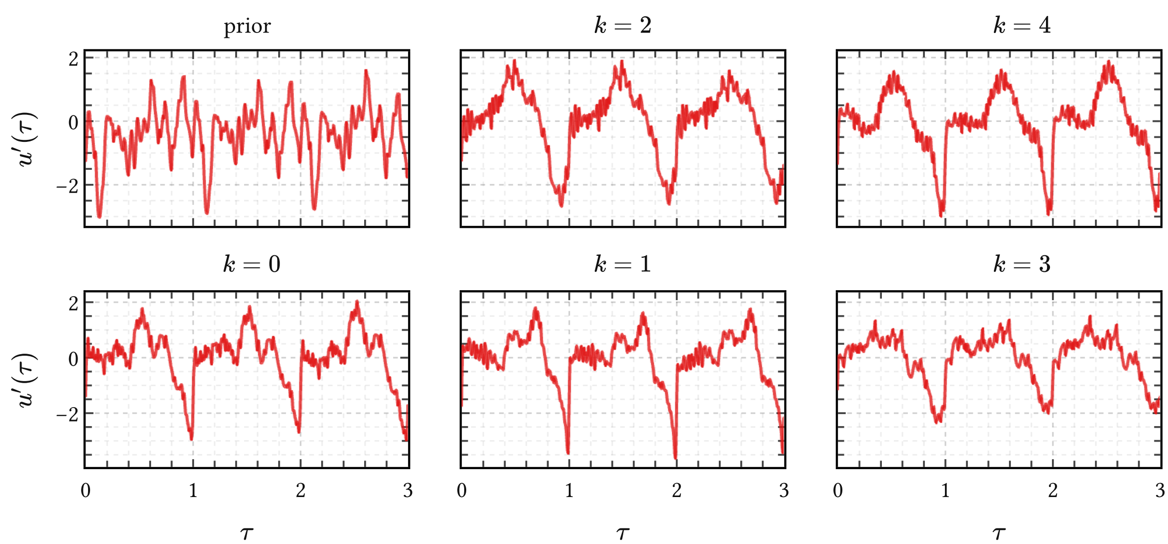


Figure 52 | **Samples from the moment-matched prior in data space** for the data in Figure 50. The "prior" panel at the top left draws from the PRISM prior (311) on the displayed grid, which are just generic combinations of the learned PRISM basis. The other panels condition on individual mixture components in (313) and have learned to model the density in amplitude space, each mixture component k modeling different types of DGF waveforms. Note how much more structured and DGF-like these are in comparison to the "prior" panel.

where $\mathbf{S}_{\text{str}}$ denotes the structured part of (308). Hence the only full-rank term in data space is the nugget $\sigma^2 \mathbf{I}_M$, while the nontrivial covariance structure remains low rank, with rank bounded by $\min(R, P)$.

So the dimensionality reduction story is this: PRISM reduces the original function space to amplitude space of dimension R; the GPLVM then organizes those amplitudes through a manifold of dimension Q; and the final waveform prior keeps only a low-rank structured covariance on top of the nugget. The entire computation required to produce that structure is amortized at training time.

5.7. Summary

q-GPLVM is a three-stage compression pipeline that converts the PRISM object $\mathcal{D}_R$ into a compact prior over waveforms.

1. Bayesian GPLVM discovers the low-dimensional manifold structure of the PRISM amplitude means, producing the posterior collection $\mathcal{D}_Q$ of (282) in a Q-dimensional manifold space.
2. XDGMM quantizes the aggregated posterior into the compact parameter set $\mathcal{M}_Q$ of (299), using the latent-space uncertainties $\mathbf{S}_n^x$ in an uncertain-inputs formulation.
3. Moment matching pushes each component of $\mathcal{M}_Q$ through the trained GP map, giving the compact amplitude-space object $\mathcal{M}_R$ of (309) and the density $p(\mathbf{a} \mid \mathcal{M}_R)$ of (310). Because PRISM synthesis is linear, this extends immediately to the analytically tractable data-space prior $p(\mathbf{y} \mid \mathbf{t}, \mathcal{M}_R)$ in (313).

The entire computation is amortized: once trained, only the small parameter sets $\mathcal{M}_Q$ and $\mathcal{M}_R$ must be retained. In this thesis PRISM followed by q-GPLVM is used to learn priors for DGF waveforms directly from many synthetic examples. These priors are mixtures of cheap BLR surrogates which are used as generative models in their own right or as kernel mixture priors in IKLP for GIF.

Bibliography

Airaksinen, M., Bäckström, T., & Alku, P. (2017). Quadratic Programming Approach to Glottal Inverse Filtering by Joint Norm-1 and Norm-2 Optimization. *IEEE/ACM Transactions on Audio, Speech, And Language Processing*, *25*(5), 929–939. https://doi.org/10.1109/TASLP.2016.2620718

Airaksinen, M., Raitio, T., Story, B., & Alku, P. (2014). Quasi Closed Phase Glottal Inverse Filtering Analysis With Weighted Linear Prediction. *IEEE/ACM Transactions on Audio, Speech, And Language Processing*, *22*(3), 596–607. https://doi.org/10.1109/TASLP.2013.2294585

Alku, P. (1992). *Glottal Wave Analysis with Pitch Synchronous Iterative Adaptive Inverse Filtering*. *11*.

Alku, P. (2011). Glottal Inverse Filtering Analysis of Human Voice Production — A Review of Estimation and Parameterization Methods of the Glottal Excitation and Their Applications. *Sadhana*, *36*(5), 623–650. https://doi.org/10.1007/s12046-011-0041-5

Alku, P., Bäckström, T., & Vilkman, E. (2002). Normalized Amplitude Quotient for Parametrization of the Glottal Flow. *The Journal of the Acoustical Society of America*, *112*(2), 701–710. https://doi.org/10.1121/1.1490365

Alku, P., Murtola, T., Malinen, J., Kuortti, J., Story, B., Airaksinen, M., Salmi, M., Vilkman, E., & Geneid, A. (2019). OPENGLOT – An Open Environment for the Evaluation of Glottal Inverse Filtering. *Speech Communication*, *107*, 38–47. https://doi.org/10.1016/j.specom.2019.01.005

Alku, P., Pohjalainen, J., Vainio, M., Laukkanen, A.-M., & Story, B. H. (2013). Formant Frequency Estimation of High-Pitched Vowels Using Weighted Linear Prediction. *The Journal of the Acoustical Society of America*, *134*(2), 1295–1313. https://doi.org/10.1121/1.4812756

Altamirano, M., & Tobar, F. (2022). Nonstationary Multi-Output Gaussian Processes via Harmonizable Spectral Mixtures. *Proceedings of The 25th International Conference on Artificial Intelligence and Statistics*, 3204–3218.

Alvarado, P. A., & Stowell, D. (2016). Gaussian Processes for Music Audio Modelling and Content Analysis. *Arxiv:1606.01039 [Cs, Stat]*.

Alzamendi, G. A., & Schlotthauer, G. (2017). Modeling and Joint Estimation of Glottal Source and Vocal Tract Filter by State-Space Methods. *Biomedical Signal Processing and Control*, *37*, 5–15. https://doi.org/10.1016/j.bspc.2016.12.022

Ananthapadmanabha, T. V., & Fant, G. (1982). Calculation of True Glottal Flow and Its Components. *Speech Communication*, *1*(3), 167–184. https://doi.org/10.1016/0167-6393(82)90015-2

Andrade-Miranda, G., Chatzipapas, K., Arias-Londoño, J. D., & Godino-Llorente, J. I. (2024, December). *GIRAFE: Glottal Imaging Dataset for Advanced Segmentation, Analysis, and Facilitative Playbacks Evaluation* (Issue arXiv:2412.15054). arXiv. https://doi.org/10.48550/arXiv.2412.15054

Antsaklis, P. J., & Michel, A. N. (2006). *Linear Systems*. Springer Science & Business Media.

Atal, B. S., & Hanauer, S. L. (1971). Speech Analysis and Synthesis by Linear Prediction of the Speech Wave. *The Journal of the Acoustical Society of America*, *50*(2B), 637–655.

Auvinen, H., Raitio, T., Airaksinen, M., Siltanen, S., Story, B. H., & Alku, P. (2014). Automatic Glottal Inverse Filtering with the Markov Chain Monte Carlo Method. *Computer Speech & Language*, *28*(5), 1139–1155. https://doi.org/10.1016/j.csl.2013.09.004

Becker, T., Jessen, M., & Grigoras, C. (2008,). Forensic Speaker Verification Using Formant Features and Gaussian Mixture Models. *Ninth Annual Conference of the International Speech Communication Association.*

Birkholz, P. (2013). Modeling Consonant-Vowel Coarticulation for Articulatory Speech Synthesis. *Plos One*, *8*(4), e60603.

Blei, D. M., Kucukelbir, A., & McAuliffe, J. D. (2017). Variational Inference: A Review for Statisticians. *Journal of the American Statistical Association*, *112*(518), 859–877. https://doi.org/10.1080/01621459.2017.1285773

Bleyer, I. R., Lybeck, L., Auvinen, H., Airaksinen, M., Alku, P., & Siltanen, S. (2017). Alternating Minimisation for Glottal Inverse Filtering. *Inverse Problems*, *33*(6), 65005. https://doi.org/10.1088/1361-6420/aa6eb8

Boersma, P. (2001). *Praat, a System for Doing Phonetics by Computer. 5*(9), 341–345.

Bonastre, J.-F., Kahn, J., Rossato, S., & Ajili, M. (2015). Forensic Speaker Recognition: Mirages and Reality. *S. Fuchs/d*, 255.

Bovy, J., Hogg, D. W., & Roweis, S. T. (2011). Extreme Deconvolution: Inferring Complete Distribution Functions from Noisy, Heterogeneous and Incomplete Observations. *The Annals of Applied Statistics*, *5*(2B). https://doi.org/10.1214/10-AOAS439

Bradbury, J., Frostig, R., Hawkins, P., Johnson, M. J., Leary, C., Maclaurin, D., & Wanderman-Milne, S. (2020). JAX: Composable Transformations of Python+ NumPy Programs, 2018. *URL Http://github. Com/google/jax*, *4*, 16.

Bretthorst, G. L. (1988). *Bayesian Spectrum Analysis and Parameter Estimation.*

Calvetti, D., & Somersalo, E. (2018). Inverse Problems: From Regularization to Bayesian Inference. *Wires Computational Statistics*, *10*(3), e1427. https://doi.org/10.1002/wics.1427

Cao, F., Vogel, A. P., Gharahkhani, P., & Renteria, M. E. (2025). Speech and Language Biomarkers for Parkinson's Disease Prediction, Early Diagnosis and Progression. *Npj Parkinson's Disease*, *11*(1), 57. https://doi.org/10.1038/s41531-025-00913-4

Chang-Sheng Yang, & Kasuya, H. (1998). Automatic Estimation of Formant and Voice Source Parameters Using a Subspace Based Algorithm. *Proceedings of the 1998 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP '98 (Cat. No.98CH36181)*, *2*, 941–944. https://doi.org/10.1109/ICASSP.1998.675421

Chen, C. J. (2013). *Acoustics of Human Voice Production.* 5.

Chen, C. J. (2016). *Elements of Human Voice*. WORLD SCIENTIFIC. https://doi.org/10.1142/9891

Chien, Y.-R., Mehta, D. D., Guenason, J., Zanartu, M., & Quatieri, T. F. (2017). Evaluation of Glottal Inverse Filtering Algorithms Using a Physiologically Based Articulatory Speech Synthesizer. *IEEE/ACM Transactions on Audio, Speech, And Language Processing*, *25*(8), 1718–1730. https://doi.org/10.1109/TASLP.2017.2714839

Childers, D. G., & Hu, H. T. (1994). Speech Synthesis by Glottal Excited Linear Prediction. *The Journal of the Acoustical Society of America*, *96*(4), 2026–2036. https://doi.org/10.1121/1.411319

Cho, Y., & Saul, L. (2009). Kernel Methods for Deep Learning. *Advances in Neural Information Processing Systems*, *22*.

Corcoran, P., Hensman, A., & Kirkpatrick, B. (2019). Glottal Flow Analysis in Parkinsonian Speech:. *Proceedings of the 12th International Joint Conference on Biomedical Engineering Systems and Technologies*, 116–123. https://doi.org/10.5220/0007259701160123

Cox, R. T. (1946). Probability, Frequency and Reasonable Expectation. *Physics*.

Cuevas, A., López, S., Mandic, D., & Tobar, F. (2021). Bayesian Autoregressive Spectral Estimation. *2021 IEEE Latin American Conference on Computational Intelligence (LA-CCI)*, 1–7. https://doi.org/10.1109/LA-CCI48322.2021.9769834

Damianou, A. C., Titsias, M. K., & Lawrence, N. D. (2014, September). *Variational Inference for Uncertainty on the Inputs of Gaussian Process Models* (Issue arXiv:1409.2287). arXiv.

de Boer, B. (2008). Acoustic Tubes with Maximal and Minimal Resonance Frequencies. *The Journal of the Acoustical Society of America*, *123*(5), 3732. https://doi.org/10.1121/1.2935231

Degottex, G. (2010). *Glottal Source and Vocal-Tract Separation* [Doctoral dissertation].

Degottex, G., Kane, J., Drugman, T., Raitio, T., & Scherer, S. (2014). COVAREP — A Collaborative Voice Analysis Repository for Speech Technologies. *2014 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 960–964. https://doi.org/10.1109/ICASSP.2014.6853739

Deisenroth, M. P., & Rasmussen, C. E. (2011). *PILCO: A Model-Based and Data-Efficient Approach to Policy Search.*

Deller, J. R. (1983). On the Time Domain Properties of the Two-Pole Model of the Glottal Waveform and Implications for LPC. *Speech Communication*, *2*(1), 57–63.

Deng, L., Cui, X., Pruvenok, R., Huang, J., Momen, S., Chen, Y., & Alwan, A. (2006). A Database of Vocal Tract Resonance Trajectories for Research in Speech Processing. *2006 IEEE International Conference on Acoustics Speech and Signal Processing Proceedings*, *1*, I–I.

Diaconis, P., & Freedman, D. (1999). Iterated Random Functions. *SIAM Review*, *41*(1), 45–76.

Doval, B., D'Alessandro, C., & Henrich, N. (2006). The Spectrum of Glottal Flow Models. *Acta Acustica United with Acustica*, *92*(6), 1026–1046.

Drugman, T., Alku, P., Alwan, A., & Yegnanarayana, B. (2019a, December). *Glottal Source Processing: From Analysis to Applications* (Issue arXiv:1912.12604). arXiv.

Drugman, T., Bozkurt, B., & Dutoit, T. (2011). Causal–Anticausal Decomposition of Speech Using Complex Cepstrum for Glottal Source Estimation. *Speech Communication*, *53*(6), 855–866. https://doi.org/10.1016/j.specom.2011.02.004

Drugman, T., Bozkurt, B., & Dutoit, T. (2019b). A Comparative Study of Glottal Source Estimation Techniques. *Arxiv:2001.00840 [Cs, Eess].*

Drugman, T., Thomas, M., Gudnason, J., Naylor, P., & Dutoit, T. (2012). Detection of Glottal Closure Instants From Speech Signals: A Quantitative Review. *IEEE Transactions on Audio, Speech, And Language Processing*, *20*(3), 994–1006. https://doi.org/10.1109/TASL.2011.2170835

Ekström, A. G., Moran, S., Sundberg, J., & Lameira, A. (2023). *PREQUEL: Supervised Phonetic Approaches to Analyses of Great Ape Quasi-Vowels.*

Fant, G. (1960). *Acoustic Theory of Speech Production.* Mouton.

Fant, G. (1979). Glottal Source and Excitation Analysis. *STL-QPSR*, *1*(1979), 85–107.

Fant, G. (1995). The LF-model Revisited. Transformations and Frequency Domain Analysis. *Speech Trans. Lab. Q. Rep., Royal Inst. Of Tech. Stockholm*, *2*(3), 40.

Fant, G., Liljencrants, J., & Lin, Q.-g. (1985). A Four-Parameter Model of Glottal Flow. *STL-QPSR*, *4*(1985), 1–13.

Flanagan, J. L. (1965). *Speech Analysis Synthesis and Perception.* Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-662-00849-2

Freixes, M., Joglar-Ongay, L., Socoró, J. C., & Alías-Pujol, F. (2023). Evaluation of Glottal Inverse Filtering Techniques on OPENGLOT Synthetic Male and Female Vowels. *Applied Sciences*, *13*(15), 8775. https://doi.org/10.3390/app13158775

Fu, Q., & Murphy, P. (2006). Robust Glottal Source Estimation Based on Joint Source-Filter Model Optimization. *IEEE Transactions on Audio, Speech and Language Processing*, *14*(2), 492–501. https://doi.org/10.1109/TSA.2005.857807

Fujisaki, H., & Ljungqvist, M. (1986). Proposal and Evaluation of Models for the Glottal Source Waveform. *ICASSP '86. IEEE International Conference on Acoustics, Speech, And Signal Processing*, *11*, 1605–1608. https://doi.org/10.1109/ICASSP.1986.1169239

Fulop, S. A. (2011). *Speech Spectrum Analysis*. Springer.

Fulop, S. A., & Disner, S. F. (2011). Examining the Voice Bar. *Proceedings of Meetings on Acoustics 162ASA*, *14*(1), 60002.

Gao, Y., Birkholz, P., & Li, Y. (2024). Articulatory Copy Synthesis Based on the Speech Synthesizer VocalTractLab and Convolutional Recurrent Neural Networks. *IEEE/ACM Transactions on Audio, Speech, And Language Processing*, *32*, 1845–1858. https://doi.org/10.1109/TASLP.2024.3372874

Garofolo, J. S., Lamel, L. F., Fisher, W. M., Pallett, D. S., Dahlgren, N. L., Zue, V., & Fiscus, J. G. (1993). TIMIT Acoustic-Phonetic Continuous Speech Corpus. *(No Title)*.

Ghosal, S., & van der Vaart, A. (2017). *Fundamentals of Nonparametric Bayesian Inference* (1st ed.). Cambridge University Press. https://doi.org/10.1017/9781139029834

Gibson, J. (2018). Entropy Power, Autoregressive Models, and Mutual Information. *Entropy*, *20*(10), 750. https://doi.org/10.3390/e20100750

Girard, A., Rasmussen, C., Candela, J. Q., & Murray-Smith, R. (2002). Gaussian Process Priors with Uncertain Inputs Application to Multiple-Step Ahead Time Series Forecasting. *Advances in Neural Information Processing Systems*, *15*.

Glorot, X., & Bengio, Y. (2010). Understanding the Difficulty of Training Deep Feedforward Neural Networks. *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics*, 249–256.

Gobl, C. (2017). Reshaping the Transformed LF Model: Generating the Glottal Source from the Waveshape Parameter Rd. *Interspeech 2017*, 3008–3012. https://doi.org/10.21437/Interspeech.2017-1140

Harrison, P. (2013). *Making Accurate Formant Measurements: An Empirical Investigation of the Influence of the Measurement Tool, Analysis Settings and Speaker on Formant Measurements* [Doctoral dissertation].

Hensman, J., Fusi, N., & Lawrence, N. D. (2013, September). *Gaussian Processes for Big Data* (Issue arXiv:1309.6835). arXiv. https://doi.org/10.48550/arXiv.1309.6835

Herbst, C. T., Lohscheller, J., Švec, J. G., Henrich, N., Weissengruber, G., & Fitch, W. T. (2014). Glottal Opening and Closing Events Investigated by Electroglottography and Super-High-

Speed Video Recordings. *Journal of Experimental Biology*, *217*(6), 955–963. https://doi.org/10.1242/jeb.093203

Holoien, T. W.-S., Marshall, P. J., & Wechsler, R. H. (2017). EmpiriciSN: Re-sampling Observed Supernova/Host Galaxy Populations Using an XD Gaussian Mixture Model. *The Astronomical Journal*, *153*, 249. https://doi.org/10.3847/1538-3881/aa68a1

Hornik, K. (1993). Some New Results on Neural Network Approximation. *Neural Networks*, *6*(8), 1069–1072.

Jaynes, E. T. (1984). Prior Information and Ambiguity in Inverse Problems. *Inverse Problems. An International Journal on the Theory and Practice of Inverse Problems, Inverse Methods and Computerized Inversion of Data*, *14*, 151–166.

Jaynes, E. T. (1991). Notes on Present Status and Future Prospects. In W. T. Grandy & L. H. Schick (Eds.), *Maximum Entropy and Bayesian Methods: Maximum Entropy and Bayesian Methods* (pp. 1–13). Springer Netherlands. https://doi.org/10.1007/978-94-011-3460-6_1

Jaynes, E. T. (2003). *Probability Theory: The Logic of Science*. Cambridge University Press.

Jopling, R. L. (2009). *Voice Initiation and Voice Offset Patterns in Normal Females: Investigated by High Speed Digital Imaging* [Doctoral dissertation].

Kadiri, S. R., Alku, P., & Yegnanarayana, B. (2021). Extraction and Utilization of Excitation Information of Speech: A Review. *Proceedings of the IEEE*, *109*(12), 1920–1941. https://doi.org/10.1109/JPROC.2021.3126493

Kazlauskaite, I., Ek, C. H., & Campbell, N. D. (2018). Gaussian Process Latent Variable Alignment Learning. *Arxiv Preprint Arxiv:1803.02603*.

Kent, R. D., & Vorperian, H. K. (2018). Static Measurements of Vowel Formant Frequencies and Bandwidths: A Review. *Journal of Communication Disorders*, *74*, 74–97. https://doi.org/10.1016/j.jcomdis.2018.05.004

Kingma, D. P., & Ba, J. L. (2015). *Adam: A Method for Stochastic Optimization*.

Klatt, D. H. (1986). Representation of the First Formant in Speech Recognition and in Models of the Auditory Periphery. *Canadian Acoustics*, *14*(3bis), 5–7.

Klatt, D. H., & Klatt, L. C. (1990). Analysis, Synthesis, and Perception of Voice Quality Variations among Female and Male Talkers. *The Journal of the Acoustical Society of America*, *87*(2), 820–857. https://doi.org/10.1121/1.398894

Knuth, K. H., & Skilling, J. (2012). Foundations of Inference. *Axioms*, *1*(1), 38–73. https://doi.org/10.3390/axioms1010038

Kominek, J., & Black, A. W. (2004,). The CMU Arctic Speech Databases. *Fifth ISCA Workshop on Speech Synthesis*.

Kreiman, J., Gerratt, B. R., & Antoñanzas-Barroso, N. (2007). Measures of the Glottal Source Spectrum. *Journal of Speech, Language, And Hearing Research*, *50*(3), 595–610. https://doi.org/10.1044/1092-4388(2007/042)

Krug, P. K., Stone, S., & Birkholz, P. (2021). Intelligibility and Naturalness of Articulatory Synthesis with VocalTractLab Compared to Established Speech Synthesis Technologies. *11th ISCA Speech Synthesis Workshop (SSW 11)*, 102–107. https://doi.org/10.21437/SSW.2021-18

Landau, H. J. (1983). The Inverse Problem for the Vocal Tract and the Moment Problem. *SIAM Journal on Mathematical Analysis*, *14*(5), 1019–1035. https://doi.org/10.1137/0514082

Le Besnerais, G., & Giovannelli, J.-F. (2010). Inverse Filtering and Other Linear Methods. In J. Idier (Ed.), *Bayesian Approach to Inverse Problems: Bayesian Approach to Inverse Problems* (pp. 80–116). ISTE. https://doi.org/10.1002/9780470611197.ch4

Lighthill, M. J. (1958). *An Introduction to Fourier Analysis and Generalised Functions*. Cambridge University Press.

Lindeberg, T. (2013). *Scale-Space Theory in Computer Vision* (Vol. 256). Springer Science & Business Media.

Little, M. A. (2011). Mathematical Foundations of Nonlinear, Non-Gaussian, and Time-Varying Digital Speech Signal Processing. In C. M. Travieso-González & J. B. Alonso-Hernández (Eds.), *Advances in Nonlinear Speech Processing: Vol. 7015. Advances in Nonlinear Speech Processing* (pp. 9–16). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-25020-0_2

Little, M. A., McSharry, P. E., Roberts, S. J., Costello, D. A., & Moroz, I. M. (2007). Exploiting Nonlinear Recurrence and Fractal Scaling Properties for Voice Disorder Detection. *Biomedical Engineering Online*, *6*(1), 23. https://doi.org/10.1186/1475-925X-6-23

Ma, A., Lau, K. K., & Thyagarajan, D. (2020). Voice Changes in Parkinson's Disease: What Are They Telling Us?. *Journal of Clinical Neuroscience*, *72*, 1–7. https://doi.org/10.1016/j.jocn.2019.12.029

Ma, C., Kamp, Y., & Willems, L. F. (1993). Robust Signal Selection for Linear Prediction Analysis of Voiced Speech. *Speech Communication*, *12*(1), 69–81. https://doi.org/10.1016/0167-6393(93)90019-H

MacKay, D. J. (1998). Introduction to Gaussian Processes. *NATO ASI Series F Computer and Systems Sciences*, *168*, 133–166.

MacKay, D. J. C. (2005). *Information Theory, Inference, and Learning Algorithms*. https://doi.org/10.1198/jasa.2005.s54

MacKay, D. J. C., & Bauman Peto, L. C. (1994). *A Hierarchical Dirichlet Language Model*. *1*(1), 1–19.

Markel, J. D., & Gray, A. H. (1976). *Linear Prediction of Speech: 12* (Vol. 12). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-66286-7

Maurer, D. (2016). *Acoustics of the Vowel.* Peter Lang.

Mehta, D. D., Rudoy, D., & Wolfe, P. J. (2012). Kalman-Based Autoregressive Moving Average Modeling and Inference for Formant and Antiformant Tracking. *The Journal of the Acoustical Society of America*, *132*(3), 1732–1746. https://doi.org/10.1121/1.4739462

Milenkovic, P. (1986). Glottal Inverse Filtering by Joint Estimation of an AR System with a Linear Input Model. *IEEE Transactions on Acoustics, Speech, And Signal Processing*, *34*(1), 28–42. https://doi.org/10.1109/TASSP.1986.1164778

Miller, R. L. (1959). Nature of the Vocal Cord Wave. *The Journal of the Acoustical Society of America*, *31*(6), 667–677. https://doi.org/10.1121/1.1907771

Moell, B., Aronsson, F. S., Östberg, P., & Beskow, J. (2025, March). *The Order in Speech Disorder: A Scoping Review of State of the Art Machine Learning Methods for Clinical Speech Classification* (Issue arXiv:2503.04802). arXiv. https://doi.org/10.48550/arXiv.2503.04802

Monahan, J. F. (1984). A Note on Enforcing Stationarity in Autoregressive-Moving Average Models. *Biometrika*, *71*(2), 403–404. https://doi.org/10.1093/biomet/71.2.403

Murphy, K. P. (2022). *Probabilistic Machine Learning: An Introduction.* MIT Press.

Narendra, N. P., Airaksinen, M., Story, B., & Alku, P. (2019). Estimation of the Glottal Source from Coded Telephone Speech Using Deep Neural Networks. *Speech Communication*, *106*, 95–104. https://doi.org/10.1016/j.specom.2018.12.002

Naylor, P. A., Kounoudes, A., Gudnason, J., & Brookes, M. (2007). Estimation of Glottal Closure Instants in Voiced Speech Using the DYPSA Algorithm. *IEEE Transactions on Audio, Speech and Language Processing*, *15*(1), 34–43. https://doi.org/10.1109/TASL.2006.876878

Neal, R. M. (1996). Priors for Infinite Networks. In *Bayesian Learning for Neural Networks: Bayesian Learning for Neural Networks* (pp. 29–53). Springer.

Nickisch, H., & Rasmussen, C. E. (2010, June). *Gaussian Mixture Modeling with Gaussian Process Latent Variable Models.*

O Cinneide, A. (2012). *Phase-Distortion-Robust Voice-Source Analysis.* https://doi.org/10.21427/D7FW3B

Orbanz, P., & Teh, Y. W. (2010). Bayesian Nonparametric Models. *Encyclopedia of Machine Learning*, *1*.

Orlikoff, R. F., Golla, M. E., & Deliyski, D. D. (2012). Analysis of Longitudinal Phase Differences in Vocal-Fold Vibration Using Synchronous High-Speed Videoendoscopy and Electroglottography. *Journal of Voice : Official Journal of the Voice Foundation*, *26*(6), 816.e13–816.e20. https://doi.org/10.1016/j.jvoice.2012.04.009

Orvieto, A., & Gower, R. M. (2025, December). *In Search of Adam's Secret Sauce* (Issue arXiv:2505.21829). arXiv. https://doi.org/10.48550/arXiv.2505.21829

Palaparthi, A., & Titze, I. R. (2020). Analysis of Glottal Inverse Filtering in the Presence of Source-Filter Interaction. *Speech Communication*. https://doi.org/10.1016/j.specom.2020.07.003

Pandey, G., & Dukkipati, A. (2014). *To Go Deep or Wide in Learning?*.

Perrotin, O., Feugère, L., & d'Alessandro, C. (2021). Perceptual Equivalence of the Liljencrants–Fant and Linear-Filter Glottal Flow Models. *The Journal of the Acoustical Society of America*, *150*(2), 1273–1285. https://doi.org/10.1121/10.0005879

Peterson, G. E., & Barney, H. L. (1952). Control Methods Used in a Study of the Vowels. *The Journal of the Acoustical Society of America*, *24*(2), 175–184.

Plumpe, M., Quatieri, T., & Reynolds, D. (1999). Modeling of the Glottal Flow Derivative Waveform with Application to Speaker Identification. *IEEE Transactions on Speech and Audio Processing*, *7*(5), 569–586. https://doi.org/10.1109/89.784109

Press, W. H., Teukolsky, S. A., Vetterling, W. T., & Flannery, B. P. (1992). *Numerical Recipes in C (2nd Ed.): The Art of Scientific Computing*. Cambridge University Press.

Rabiner, L. R., & Schafer, R. W. (2007). *Introduction to Digital Speech Processing* (Vol. 1). https://doi.org/10.1561/2000000001

Rabiner, L. R., Juang, B.-H., & Rutledge, J. C. (1993). *Fundamentals of Speech Recognition* (Vol. 14). PTR Prentice Hall Englewood Cliffs.

Rahimi, A., & Recht, B. (2008). Weighted Sums of Random Kitchen Sinks: Replacing Minimization with Randomization in Learning. In D. Koller, D. Schuurmans, Y. Bengio, & L. Bottou (Eds.), *Advances in Neural Information Processing Systems: Vol. 21. Advances in Neural Information Processing Systems*.

Ramsay, J. O., & Silverman, B. W. (1997). *Functional Data Analysis*. Springer.

Rao, A., & Ghosh, P. K. (2018). Glottal Inverse Filtering Using Probabilistic Weighted Linear Prediction. *IEEE/ACM Transactions on Audio, Speech, And Language Processing*, *27*(1), 114–124.

Rasmussen, C. E., & Williams, C. K. I. (2006). *Gaussian Processes for Machine Learning*. MIT Press.

Riutort-Mayol, G., Bürkner, P.-C., Andersen, M. R., Solin, A., & Vehtari, A. (2020). Practical Hilbert Space Approximate Bayesian Gaussian Processes for Probabilistic Programming. *Arxiv:2004.11408 [Stat]*.

Rosenberg, A. E. (1971). Effect of Glottal Pulse Shape on the Quality of Natural Vowels. *The Journal of the Acoustical Society of America*, *49*(2B), 583–590. https://doi.org/10.1121/1.1912389

Sasou, A., & Chen, Y. (2023). Comparison of GIF- and SSL-based Features in Pathological-voice Detection. *INTERSPEECH 2023*, 2893–2897. https://doi.org/10.21437/Interspeech.2023-1418

Sawada, H., Kameoka, H., Araki, S., & Ueda, N. (2012). Efficient Algorithms for Multichannel Extensions of Itakura-Saito Nonnegative Matrix Factorization. *2012 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 261–264. https://doi.org/10.1109/ICASSP.2012.6287867

Schleusing, O. (2012). *Multi-Parametric Source-Filter Separation of Speech and Prosodic Voice Restoration* [Technical report].

Schnupp, J., Nelken, I., & King, A. (2011). *Auditory Neuroscience: Making Sense of Sound*. MIT press.

Schoder, S., Falk, S., Wurzinger, A., Lodermeyer, A., Becker, S., & Kniesburges, S. (2024). A Benchmark Case for Aeroacoustic Simulations Involving Fluid-Structure-Acoustic Interaction Transferred from the Process of Human Phonation. *Acta Acustica*, *8*, 13. https://doi.org/10.1051/aacus/2024005

Schroeder, M. R. (1999). *Computer Speech: Recognition, Compression, Synthesis*. Springer-Verlag. https://doi.org/10.1007/978-3-662-03861-1

Serwy, R. D. (2017). *Hilbert Phase Methods for Glottal Activity Detection* [Doctoral dissertation].

Shi, L., Nielsen, J. K., Jensen, J. R., & Christensen, M. G. (2017). A Variational EM Method for Pole-Zero Modeling of Speech with Mixed Block Sparse and Gaussian Excitation. *2017 25th European Signal Processing Conference (EUSIPCO)*, 1784–1788. https://doi.org/10.23919/EUSIPCO.2017.8081516

Sivia, D. S., & Skilling, J. (2006). *Data Analysis: A Bayesian Tutorial* (2nd ed). Oxford University Press.

Skilling, J. (2005). Bayesics. *AIP Conference Proceedings*, *803*, 3–24.

Stevens, K. N. (2000). *Acoustic Phonetics*. MIT press.

Stoica, P., & Moses, R. L. (2005). *Spectral Analysis of Signals*. Pearson/Prentice Hall.

Sun, Y., Huang, Q., & Wu, X. (2022). Unsupervised Inference of Physiologically Meaningful Articulatory Trajectories with VocalTractLab. *Interspeech 2022*, 4661–4665. https://doi.org/10.21437/Interspeech.2022-659

Svec, J. G., Schutte, H. K., Chen, C. J., & Titze, I. R. (2021). Integrative Insights into the Myoelastic-Aerodynamic Theory and Acoustics of Phonation. Scientific Tribute to Donald G. Miller. *Journal of Voice*. https://doi.org/10.1016/j.jvoice.2021.01.023

Sørbye, S. H., & Rue, H. (2017). Penalised Complexity Priors for Stationary Autoregressive Processes. *Journal of Time Series Analysis*, *38*(6), 923–935. https://doi.org/10.1111/jtsa.12242

Taylor, A. M., & Reby, D. (2010). The Contribution of Source–Filter Theory to Mammal Vocal Communication Research. *Journal of Zoology*, *280*(3), 221–236. https://doi.org/10.1111/j.1469-7998.2009.00661.x

Thomas, M. R. P., Gudnason, J., & Naylor, P. A. (2009). *Data-Driven Voice Source Waveform Modelling*.

Titsias, M. (2009). Variational Learning of Inducing Variables in Sparse Gaussian Processes. *Proceedings of the Twelfth International Conference on Artificial Intelligence and Statistics*, 567–574.

Titsias, M. K. (2025, June). *New Bounds for Sparse Variational Gaussian Processes* (Issue arXiv:2502.08730). arXiv. https://doi.org/10.48550/arXiv.2502.08730

Titsias, M., & Lawrence, N. D. (2010). Bayesian Gaussian Process Latent Variable Model. *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics*, 844–851.

Titze, I. R. (1995). *Workshop on Acoustic Voice Analysis: Summary Statement*. National Center for Voice and Speech.

Titze, I. R. (2000). Principles of Voice Production (Second Printing). *Iowa City, IA: National Center for Voice and Speech.*

Titze, I. R., Baken, R. J., Bozeman, K. W., Granqvist, S., Henrich, N., Herbst, C. T., Howard, D. M., Hunter, E. J., Kaelin, D., Kent, R. D., Kreiman, J., Kob, M., Löfqvist, A., McCoy, S., Miller, D. G., Noé, H., Scherer, R. C., Smith, J. R., Story, B. H., … Wolfe, J. (2015). Toward a Consensus on Symbolic Notation of Harmonics, Resonances, and Formants in Vocalization. *The Journal of the Acoustical Society of America*, *137*(5), 3005–3007. https://doi.org/10.1121/1.4919349

Turner, R. E., & Sahani, M. (2014). Time-Frequency Analysis as Probabilistic Inference. *IEEE Transactions on Signal Processing*, *62*(23), 6171–6183. https://doi.org/10.1109/TSP.2014.2362100

Van Soom, M., & de Boer, B. (2019). A New Approach to the Formant Measuring Problem. *Proceedings*, *33*(1), 29. https://doi.org/10.3390/proceedings2019033029

Van Soom, M., & de Boer, B. (2020). Detrending the Waveforms of Steady-State Vowels. *Entropy*, *22*(3), 331. https://doi.org/10.3390/e22030331

Van Soom, M., & de Boer, B. (2021). A Weakly Informative Prior for Resonance Frequencies. *Physical Sciences Forum*, *3*(1), 2. https://doi.org/10.3390/psf2021003002

Van Soom, M., & De Boer, B. (2022,). Softly Constrained Agent-Based Models. *Joint Conference on Language Evolution*.

Veldhuis, R. (1998). A Computationally Efficient Alternative for the Liljencrants-Fant Model and Its Perceptual Evaluation. *The Journal of the Acoustical Society of America*, *103*(1), 566–571. https://doi.org/10.1121/1.421103

Verdolini, K., & Titze, I. R. (1995). The Application of Laboratory Formulas to Clinical Voice Management. *American Journal of Speech-Language Pathology*, *4*(2), 62–69. https://doi.org/10.1044/1058-0360.0402.62

Viterbi, A. (1967). Error Bounds for Convolutional Codes and an Asymptotically Optimum Decoding Algorithm. *IEEE Transactions on Information Theory*, *13*(2), 260–269. https://doi.org/10.1109/TIT.1967.1054010

Walker, J., & Murphy, P. (2005). Advanced Methods for Glottal Wave Extraction. *Nonlinear Analyses and Algorithms for Speech Processing: International Conference on Non-Linear Speech Processing, NOLISP 2005, Barcelona, Spain, April 19-22, 2005, Revised Selected Papers*, 139–149.

Wang, R., Zhang, Y., Ou, Z., & Hasegawa-Johnson, M. (2016). Use of Particle Filtering and MCMC for Inference in Probabilistic Acoustic Tube Model. *2016 IEEE Statistical Signal Processing Workshop (SSP)*, 1–5. https://doi.org/10.1109/SSP.2016.7551748

Whalen, D. H., Chen, W.-R., Shadle, C. H., & Fulop, S. A. (2022). Formants Are Easy to Measure; Resonances, Not so Much: Lessons from Klatt (1986). *The Journal of the Acoustical Society of America*, *152*(2), 933–941. https://doi.org/10.1121/10.0013410

Wilk, M. van der, Dutordoir, V., John, S. T., Artemev, A., Adam, V., & Hensman, J. (2020, March). *A Framework for Interdomain and Multioutput Gaussian Processes* (Issue arXiv:2003.01115). arXiv. https://doi.org/10.48550/arXiv.2003.01115

Williams, C. K. I. (1998). Computation with Infinite Neural Networks. *Neural Computation*, *10*(5), 1203–1216. https://doi.org/10.1162/089976698300017412

Wong, D., Markel, J., & Gray, A. (1979). Least Squares Glottal Inverse Filtering from the Acoustic Speech Waveform. *IEEE Transactions on Acoustics, Speech, And Signal Processing*, *27*(4), 350–355. https://doi.org/10.1109/TASSP.1979.1163260

Xuan, J., Lu, J., & Zhang, G. (2021, September). *Bayesian Transfer Learning: An Overview of Probabilistic Graphical Models for Transfer Learning* (Issue arXiv:2109.13233). arXiv.

Yokota, K., Harakawa, R., Baba, M., & Iwahashi, M. (2025,). *Physics-Informed Neural Networks for Speech Production*. arXiv. https://doi.org/10.48550/ARXIV.2511.00428

Yoshii, K., & Goto, M. (2013). Infinite Kernel Linear Prediction for Joint Estimation of Spectral Envelope and Fundamental Frequency. *2013 IEEE International Conference on Acoustics, Speech and Signal Processing*, 463–467. https://doi.org/10.1109/ICASSP.2013.6637690

Zhang, B., & Shin, Y. C. (2021). An Adaptive Gaussian Mixture Method for Nonlinear Uncertainty Propagation in Neural Networks. *Neurocomputing*. https://doi.org/10.1016/j.neucom.2021.06.007

Zhang, B., & Shin, Y. C. (2022). A Gaussian Mixture Filter with Adaptive Refinement for Nonlinear State Estimation. *Signal Processing*, *201*, 108677. https://doi.org/10.1016/j.sigpro.2022.108677

Zhang, Y., Zheng, X., & Xue, Q. (2020). A Deep Neural Network Based Glottal Flow Model for Predicting Fluid-Structure Interactions during Voice Production. *Applied Sciences*, *10*(2), 705. https://doi.org/10.3390/app10020705

Appendices

Appendix A. Regularization and priors

As its name suggests, GIF belongs to the class of *inverse problems* (Le Besnerais & Giovannelli, 2010). Inverse problems confront us when we have observations of some effects [noisy oservations of $s(t)$ in our case] of which we want to infer the causes $u(t)$ and $h(t)$ within the context of some model $s(t) = u(t) * h(t) * r(t)$.

In the applied sciences it is very often the case for inverse problems that the observed data are quite consistent with a broad and sometimes colorful class of possible solutions rather than a unique and logically deductible solution. From the viewpoint of pure mathematics therefore such inverse problems are often deemed 'ill-posed', but perhaps a more productive viewpoint[30] is that such problems are simply underdetermined.

Underdetermined problems are ubiquitous in a wide range of applications, from medical diagnosis (what could be causing the patient's persistent cough) to astronomy (what is perturbing the orbit of Uranus). Bayesian theory by itself does not require any particular structure on the "feasible set" of plausible solutions and as such underdetermined problems have no special status within it, nor pose any special challenge to it (Jaynes, 2003). In practice however, we would like to have a more well determined solution to underdetermined problems as our computing budgets (and patience) are limited. For example, it would most certainly help if we can already discard those solutions that lack to some degree a set of required properties we established beforehand. This is where *regularization* and *priors* come into play.

In general, *regularization* is the process of constraining the solution space of some problem to obtain a more stable and perhaps unique solution; in other words, to convert an underdetermined problem into something more 'well-determined' and, hopefully, more amenable to computational optimization. For example, ridge regression is a widely used regularization technique to numerically stabilize linear regression in high dimensions.[31]

In the Bayesian framework regularization is imposed simply through careful assignment of the probability distributions describing a specific problem – of which the bulk of the effort is traditionally concentrated on the prior probabilities, or *priors* (Sivia & Skilling, 2006). Assuming our prior information about the problem at hand is correct and the assignment of the priors accurately reflect that information (and we got away with the approximations made in our inference methods) we will have automatically implemented just the right degree of regularization for our problem; indeed, specially catered to the one data set we

[30] And closer to the historical origin of the epithet "mal posée" (Jaynes, 1984).

[31] Ridge regression (also known as ℓ^2 regularization) can be derived from Bayesian theory as standard linear regression with Gaussian priors on the amplitudes (Murphy, 2022, Sec. 11.3). This principled view allows us to go much further than numerical stabilization. We will show in the next chapters that ridge regression can (approximately) express intricate pieces of prior information such as definite signal polarity, the differentiability class to which a function belongs, and power constraints on waveforms.

happened to observe, because we express everything in terms of posterior probabilities that are conditioned on the observed data.

For simple inverse problems where intuition can see the answer clearly, Bayesian regularization leads to answers that comply with common sense (MacKay, 2005). This is simply because Bayesian probability theory is a formalized theory of plausible reasoning itself (Skilling, 2005), which humans undertake every day. But regularization in the Bayesian framework can go beyond our intuition: it will of course continue to work for complex problems that require considerable analysis, and this consistency makes it a valuable tool for attacking inverse problems (Calvetti & Somersalo, 2018).

Appendix B. Parametric, nonparametric, semiparametric?

We clarify some of the confusing terminology associated with Bayesian nonparametric models.

Parametric models are simply models with a fixed and prespecified amount of unknown parameters R, independent of the size of the dataset N.

Examples are linear regression and neural networks. In BNGIF the filter priors π_{PZ} and π_{AP} priors are parametric as they have a fixed amount of parameters R depending on their order K, similar to linear prediction (LP) methods. And of course the π_{LF} prior with $R = 4$ is parametric as well.

Nonparametric models in a Bayesian context are models for which the effective number of parameters $\tilde{R}$ is capable of adapting to N (Orbanz & Teh, 2010).

They are also referred to as infinite-dimensional parametric models, which is just a shorthand way of saying that they are parametric models which have been mathematically constructed to behave consistently under inference as $R \to \infty$. That is, given the data of size N, a nonparametric model roughly behaves as an ordinary parametric model with $\tilde{R} < \infty$ parameters, with the exception that $\tilde{R}$ is determined automatically from the data rather than specified beforehand.

Examples are Dirichlet process mixture models, for which the effective number of mixture components does not need to be known beforehand, and, of course, GPs, which are infinite-dimensional distributions over functions. In BNGIF the glottal-flow prior is nonparametric because it is modeled with a GP.[32]

The name 'nonparametric' is a bit of a misnomer because nonparametric models almost always have a set of Q hyperparameters which typically determine either the scales present in the data or the degree to which sparsity or smoothness is encouraged. In theory, inference of these Q hyperparameters is no different from inference with a parametric model of order $R = Q + \tilde{R}$ in which $\tilde{R}$ parameters have been marginalized over analytically.

Semiparametric models in a Bayesian context are models with both parametric and nonparametric components for which the latter is considered a nuisance parameter (Ghosal & van der Vaart, 2017, p. 368). This means that we are mainly interested in conclusions about the parametric component of the model, which require marginalizing over the nonparametric component.

In BNGIF the parametric and nonparametric components are very neatly divided into models for the filter $h(t)$ and the source $u(t)$, respectively, reflecting the qualitative difference between the information available from acoustic phonetics about the two. One might be

[32]Strictly, this is not fully nonparametric because it is only a reduced-rank GP and not a full-rank GP. Its capacity to adapt its complexity to growing amounts of data N is limited by the number of basisfunctions M; nonparametric behavior is recovered in the limit $M \to \infty$.

tempted to call BNGIF semiparametric for this reason if it weren't for the fact that the nonparametric component describing $u(t)$ is actually the main object of interest for the GIF problem. If anything the nuisance bit for GIF is in the parametric component describing $h(t)$, not the nonparametric component for $u(t)$.

Appendix C. JAX implementation of the Liljencrants-Fant model

```python
import jax
import jax.numpy as jnp
import jaxopt


def _get_initial_bracket(T0, tol, UPPER_FACTOR=1e3):
    lower = (1 / T0) * tol
    upper = (1 / T0) * UPPER_FACTOR
    return lower, upper


def _bisect_exponential_rate(
    f, tol, initial_bracket, maxiter, fail_val=jnp.nan, **kwargs
):
    lower, upper = initial_bracket
    bisec = jaxopt.Bisection(
        optimality_fun=f,
        lower=lower,
        upper=upper,
        maxiter=maxiter,
        tol=tol,
        check_bracket=False,
    )
    result = bisec.run(**kwargs)
    success = result.state.error < tol
    exponential_rate = result.params
    return jax.lax.cond(success, lambda: exponential_rate, lambda: fail_val)


def _bisect_epsilon(p, tol, initial_bracket, maxiter):
    def f(epsilon, p):
        LHS = epsilon * p["Ta"]
        RHS = 1 - jnp.exp(-epsilon * (p["T0"] - p["Te"]))
        return LHS - RHS

    return _bisect_exponential_rate(f, tol, initial_bracket, maxiter, p=p)


def _bisect_a(p, epsilon, tol, initial_bracket, maxiter):
    def f(a, p, epsilon):
        factor = 1 / (a**2 + (jnp.pi / p["Tp"]) ** 2)
        term1 = (
            jnp.exp(-a * p["Te"])
            * (jnp.pi / p["Tp"])
            / jnp.sin(jnp.pi * p["Te"] / p["Tp"])
        )
        term2 = a
        term3 = (
            jnp.pi / p["Tp"] * (1 / jnp.tan(jnp.pi * p["Te"] / p["Tp"]))
        )  # cotg() is 1/tan()
```

```python
        LHS = factor * (term1 + term2 - term3)

        RHS = (p["T0"] - p["Te"]) / (
            jnp.exp(epsilon * (p["T0"] - p["Te"])) - 1
        ) - 1 / epsilon

        return LHS - RHS

    # Assume that the bisection will only fail when `a -> 0`
    return _bisect_exponential_rate(
        f, tol, initial_bracket, maxiter, fail_val=0.0, p=p, epsilon=epsilon
    )


def _dgf(t, p, epsilon, a, **kwargs):
    """Implement the LF model per Doval et al. (2006) Section A1.4."""

    def rise(t):
        return (
            -jnp.exp(a * (t - p["Te"]))
            * jnp.sin(jnp.pi * t / p["Tp"])
            / jnp.sin(jnp.pi * p["Te"] / p["Tp"])
        )

    def decrease(t):
        return (
            -1
            / (epsilon * p["Ta"])
            * (
                jnp.exp(-epsilon * (t - p["Te"]))
                - jnp.exp(-epsilon * (p["T0"] - p["Te"]))
            )
        )

    return jnp.piecewise(
        t,
        [t < 0, (0 <= t) & (t < p["Te"]), (p["Te"] <= t) & (t <= p["T0"])],
        [0.0, rise, decrease, 0.0],
    )


def _nans_like(a):
    return jnp.full_like(a, jnp.nan)


def dgf(t, p, offset=0.0, tol=1e-6, initial_bracket=None, maxiter=100):
    """Calculate the glottal flow derivative (DGF) using the LF (Liljencrants-Fant) model

    This JAX implementation can be jitted and is differentiable with respect to `p`.
```

```
    Jitting this function will gain several a speedup of several orders of magnitude.

    The LF model [1,2] is nonzero only for values of `t` in `[offset, offset + T0]`, where
    `T0 = p['T0']`. If the LF parameters in `p` are inconsistent, such that the
    implicit equations cannot be solved, the function's output `u` will be all `nan`s.

    Args:
        `t` (array): Time points at which to evaluate the DGF `u(t)`
        `p` (dict): Dictionary with the LF parameters. It must have the following keys:
            `['T0', 'Te', 'Tp', 'Ta']`. `Ee` is always assumed to be 1.
        `offset` (scalar): The DGF waveform starts at this offset and is nonzero in
            `[offset, offset + T0]`.
        `tol`, `initial_bracket`, `maxiter` (scalar): Parameters controlling the bisection
            routine `_bisect_exponential_rate()` to solve the two equations implicit in
            the LF model.

    Returns:
        `u` (DeviceArray): The glottal flow derivative evaluated at times `t` if the LF
            parameters in `p` are consistent, otherwise an array of `nan`s shaped as `t`.

    References:
        [1] G. Fant, J. Liljencrants, and Q. Lin, "A four-parameter model of glottal flow",
            STL-QPSR, vol. 4, no. 1985, pp. 1–13, 1985.
        [2] Doval, Boris, Christophe d'Alessandro, and Nathalie Henrich. "The spectrum of
            glottal flow models." Acta acustica united with acustica 92.6 (2006): 1026-1046.
    """
    if initial_bracket is None:
        initial_bracket = _get_initial_bracket(p["T0"], tol)

    epsilon = _bisect_epsilon(
        p, tol=tol, initial_bracket=initial_bracket, maxiter=maxiter
    )
    a = _bisect_a(
        p, epsilon=epsilon, tol=tol, initial_bracket=initial_bracket, maxiter=maxiter
    )
    u = _dgf(t - offset, p=p, epsilon=epsilon, a=a)
    inconsistent_parameters = jnp.any(jnp.isnan(u))
```

```python
    return jax.lax.cond(inconsistent_parameters, lambda: _nans_like(u), lambda: u)


def consistent_lf_params(p):
    """Check whether the implicit LF equations can be solved given the LF parameters in `p`"""
    return ~jnp.any(jnp.isnan(dgf(0.0, p)))


def _select_keys(q, *keys):
    return {k: q[k] for k in keys if k in q}


def convert_lf_params(p, s, join=True):
    if s == "R -> generic":
        # Perrotin et al. (2021) Eq. (A1)
        Ra, Rk, Rg = p["Ra"], p["Rk"], p["Rg"]
        Oq = (1 + Rk) / (2 * Rg)
        am = 1 / (1 + Rk)
        Qa = Ra / (1 - Oq)  # Doval (2006) p. 5
        q = dict(Oq=Oq, am=am, Qa=Qa)
    elif s == "generic -> R":
        Oq, am, Qa = p["Oq"], p["am"], p["Qa"]
        Rk = 1 / am - 1
        Rg = (1 + Rk) / (2 * Oq)
        Ra = Qa * (1 - Oq)
        q = dict(Ra=Ra, Rk=Rk, Rg=Rg)
    elif s == "T -> R":
        # Fant (1994)
        T0, Te, Tp, Ta = p["T0"], p["Te"], p["Tp"], p["Ta"]
        Ra = Ta / T0
        Rk = (Te - Tp) / Tp
        Rg = T0 / (2 * Tp)
        q = dict(Ra=Ra, Rk=Rk, Rg=Rg)
    elif s == "R -> T":
        T0, Ra, Rk, Rg = p["T0"], p["Ra"], p["Rk"], p["Rg"]
        Ta = T0 * Ra
        Tp = T0 / (2 * Rg)
        Te = Tp * (1 + Rk)
        q = dict(Te=Te, Tp=Tp, Ta=Ta)
    elif s == "generic -> T":
        q = convert_lf_params(convert_lf_params(p, "generic -> R"), "R -> T")
        q = _select_keys(q, "Te", "Tp", "Ta")
    elif s == "T -> generic":
        q = convert_lf_params(convert_lf_params(p, "T -> R"), "R -> generic")
        q = _select_keys(q, "Oq", "am", "Qa")
    elif s == "Rd -> R":
        # Perrotin et al. (2021) Eq. (A1)
        # See also Fant (1994) for the meaning of these dimensionless parameters
```

```
        Rd = p["Rd"]
        Ra = (-1 + 4.8 * Rd) / 100
        Rk = (22.4 + 11.8 * Rd) / 100
        Rg = Rk * (0.5 + 1.2 * Rk) / (0.44 * Rd - 4 * Ra * (0.5 + 1.2 * Rk))
        q = dict(Ra=Ra, Rk=Rk, Rg=Rg)
    elif s == "Rd -> T":
        q = convert_lf_params(convert_lf_params(p, "Rd -> R"), "R -> T")
        q = _select_keys(q, "Te", "Tp", "Ta")
    else:
        raise ValueError(f"Unknown conversion: {s}")
    return {**p, **q} if join else q


def fant_params(s):
    """Generate typical LF model parameters for a male or female vowel

    Note that the unit of time is `msec`. Values are taken from [1, p. 121].

      [1] Fant, Gunnar. "The LF-model revisited. Transformations and frequency domain analysis."
          Speech Trans. Lab. Q. Rep., Royal Inst. of Tech. Stockholm 2.3 (1995): 40.
    """
    if s == "female vowel":
        return {"T0": 5.0, "Te": 3.25, "Tp": 2.5, "Ta": 0.3978873577297384}
    elif s == "male vowel":
        return {
            "T0": 8.333333333333334,
            "Te": 4.513888888888889,
            "Tp": 3.4722222222222228,
            "Ta": 0.22736420441699334,
        }
    else:
        raise ValueError(s)
```

Appendix D. More comments on the nonparametric glottal flow model

Some aspects of the nonparametric GFM derived in Chapter 4 are discussed in more detail here.

Infinite precision? In the GP view, increasing H increases the *Monte Carlo resolution* with which the finite neural network model samples the arc cosine kernel — the deviation from the limiting kernel vanishes as $\mathcal{O}\left(1/\sqrt{H}\right)$ by (122). The nonparametric limit replaces explicit random changepoints with their continuous Gaussian measure, and the prior over $u'(t)$ becomes a GP whose covariance is the arc cosine kernel restricted to the open phase.

There is a caveat though. We gained full rank — the prior now has support over an infinite-dimensional function space — but we lost the ability to represent true discontinuities. Arc cosine GP sample paths are continuous; they cannot jump.[33] This is the analogue of the amplitude prior constraining the ellipsoid in the finite-H case (footnote 17 of Chapter 4): there we traded a large discrete family of possible changepoint configurations for a continuous Gaussian prior that concentrates mass inside a smooth ellipsoid; here we trade a distribution over Poisson-style jump processes for a Gaussian prior that concentrates mass on smooth function classes. The best the arc cosine GP can do is approximate a sharp GCI with a steep ramp.

Whether this is a problem in practice is an empirical question. The closed-phase GCI is physiologically sharp but not infinitely so, and the DGF data from VocalTractLab are sampled at finite rate, so the distinction between "true jump" and "ramp steep enough that it spans one sample" may not be detectable.

Why this matters for the choice of modeled signal The continuity constraint has a direct implication for which signal we should model with the arc cosine GP. We model $u'(t)$ — the DGF — because the GCI appears as a sharp negative peak there, which is easier to detect and align than the integral $u(t)$. But if the arc cosine GP is continuous, then $u'(t)$ is smooth, which means $u(t)$ is differentiable — and real glottal flow does not have a differentiable closure.

One way around this is to model $u(t)$ directly with the arc cosine GP instead of $u'(t)$. Then the GP sample paths are continuous (which $u(t)$ should be) and discontinuities can be pushed into the *derivative* implicitly via the radiation factor: the combined vocal tract and radiation impulse response is modeled as the AR filter, and the combined source signal becomes $u(t) * r(t)$ rather than $u'(t)$ alone. This is, at bottom, what classical LPC does — it posits $s(t) = e(t) * h(t)$ and allows $h(t)$ to absorb the radiation effect $r(t)$, since differentiation corresponds to a zero in the transfer function, not a pole, so $h(t)$ can remain all-pole. From this viewpoint the present approach generalizes LPC in exactly that way.

[33] Specifically, sample paths of a GP with a kernel of degree d are d-times continuously differentiable with probability one (Rasmussen & Williams, 2006). For $d = 0$ this gives continuous but nowhere-differentiable paths; for $d \geq 1$, paths are smoother still. In either case there are no jumps.

The alternative of obtaining $u'(t)$ implicitly — from the spectral domain or from the AR pole structure — is also possible and may allow the sharp-closure structure to be recovered without committing the GP to model it directly.

Why not go infinite depth? Increasing depth has a different character from increasing width. For a deep GP with Gaussian kernels at each layer, letting the depth diverge drives the process toward one that is independent of its inputs (Diaconis & Freedman, 1999) — a known pathology in practice, where deep GPs tend to "forget" their inputs and produce nearly constant sample paths. Finite but large depth mitigates this, but one might note that infinite width carries a vaguely analogous cost: in the GP limit, the basis functions are frozen at their prior expectation and no longer adapt to the data as drawn features. What is learned is entirely encoded in the kernel hyperparameters. This makes kernel learning — and in particular the multiple-kernel structure of IKLP — the mechanism through which the model does acquire data-driven spectral features, rather than learning them through random feature adaptation.

Why not spline models? Piecewise spline models are well-studied and effective in low-dimensional nonparametric regression. The issue here is resolution dependence. As GP priors, spline kernels produce posterior means that are splines with knots fixed at the observed input locations (MacKay, 1998, p. 6): add more data, add more knots. The arc cosine kernel, by contrast, learns the effective number of hingepoints from the data through its hyperparameters, and this number can remain $\mathcal{O}(1)$ even as the number of observations grows. For glottal flow this matters: a DGF waveform sampled at 16 kHz and one sampled at 44 kHz should both require $\mathcal{O}(1)$ effective changepoints to describe the open phase, not a number proportional to the sampling rate.

Why not Lévy processes? These encode Poisson-style jumps, $\mathcal{O}(1)$ in number, which is exactly the right prior for a signal with a small number of hard discontinuities. The problem is inference: the posterior over a Lévy process is indexed by the jump locations, and marginalizing those out requires MCMC or particle methods that scale with the number of jumps. The GP approach achieves fast closed-form posterior inference precisely because everything — changepoints, amplitudes, timing — has been absorbed into the kernel. Trading exact jumps for steep smooth ramps is the price of that tractability.

Appendix E. On derivative time scales

The EGIFA and CommonVoice datasets present us with ground-truth $u(t)$ but we need $u'(t)$ for learning. Naive finite difference of $u(t)$,

$$u'_{\Delta}(t) = \frac{u(t+\Delta) - u(t)}{\Delta}, \qquad \Delta = \frac{1}{f_s}, \tag{315}$$

often yields spikes in $u'(t)$ which effectively defeat kernel modeling because the increased observation noise will prevent learning fine structure. Luckily, we are saved by the physiology of the speech production process.

Glottal closure is not an instantaneous discontinuity but a short transition extending over a finite time interval. High-speed imaging and synchronized EGG measurements show that opening and closing events occupy $O(0.1)$ ms durations and may exhibit small temporal offsets depending on how closure propagates along the folds (e.g. zipper-like anterior-posterior contact), as described by Herbst et al. (2014) and Orlikoff et al. (2012). Consequently, a derivative operator that reacts to arbitrarily small sample-scale fluctuations does not reflect the underlying physiology; it instead amplifies numerical artefacts or simulator-specific microstructure. A meaningful derivative therefore requires an explicit temporal scale reflecting the duration over which closure dynamics remain physically interpretable.

Since we already model $u(t)$ by Gaussian processes,[34] we are implicitly asserting that first and second moments alone carry most of the important information. Therefore, we propose to use a *Gaussian-smoothed derivative* to measure the slope of a *locally averaged signal*

$$u'_{\sigma}(t) := \partial_t (G_\sigma * u)(t), \qquad G_\sigma(t) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{t^2}{\sigma^2}\right), \tag{316}$$

rather than differentiating raw samples directly.[35] This type of differentiation results in another GP, since differentiation and convolution (smearing) are closed operators in GP space.

Selecting σ such that the effective smoothing width corresponds to approximately 0.1 ms (about four to five samples at $f_s = 44.1$ kHz) aligns the operator with the expected duration of glottal closure transitions. At smaller scales the derivative is dominated by high-frequency noise and discretization effects; at larger scales the transition itself becomes blurred. The resulting derivative can therefore be interpreted as a slope measured at the physically relevant temporal resolution rather than as a raw numerical difference, and this removes the occurence of large spikes in $u'(t)$ almost completely; an example is shown in Figure 53.

[34]More precisely, we model its derivative $u'(t)$ as a GP, but the integration in (8) preserves GP-ness.

[35]Gaussian smoothing also turns out to be the unique linear kernel that preserves causality in scale-space, meaning that increasing scale removes fine structure without introducing spurious extrema (Lindeberg, 2013).

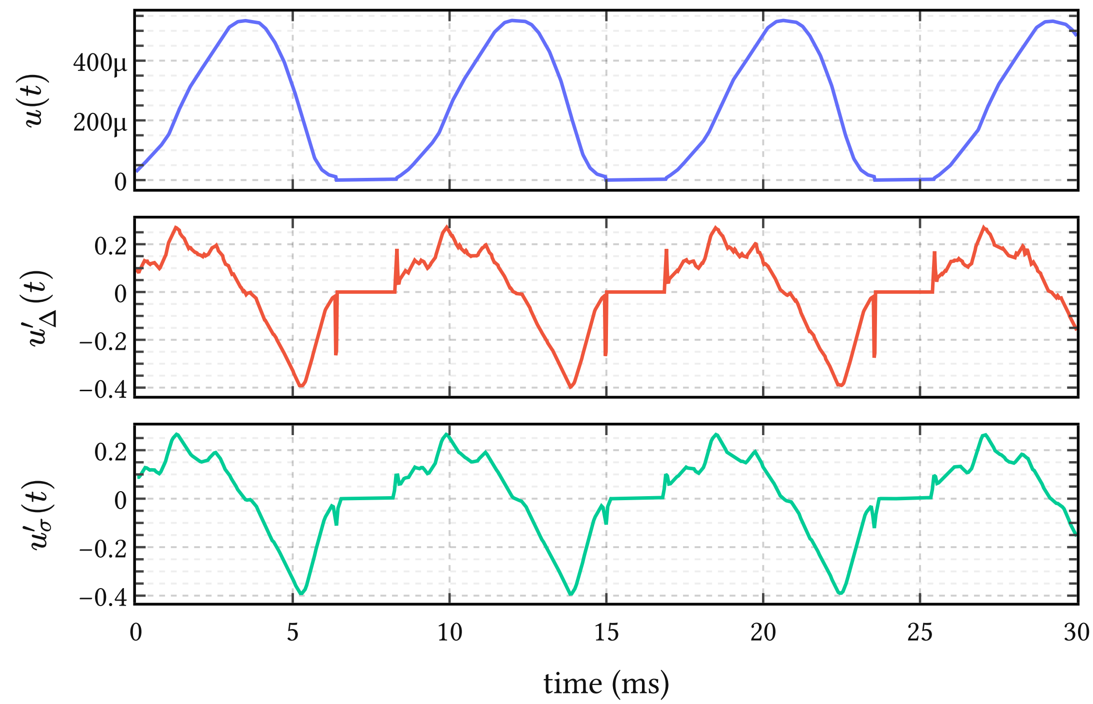


Figure 53 | **Smooth differentiation dampens spikes in $u'(t)$.** The top panel shows the ground truth glottal flow $u(t)$ extracted from a random EGIFA file. The middle panel contains its derivative $u'_\Delta(t)$ obtained by finite differences (315), and the bottom panel the Gaussian-smoothed derivative $u'_\sigma(t)$ proposed in (316). It can be seen that raw differentiation amplifies sample-scale fluctuations into large spikes, whereas the smoothed derivative retains the closure structure at the intended temporal resolution.

Appendix F. Predicting period-to-period jitter

Classical local jitter is defined as the expected absolute change in consecutive periods, normalized by the mean period,

$$\text{jitter} := \frac{\langle |T_{k+1} - T_k| \rangle}{\langle T_k \rangle}. \tag{317}$$

In the present framework, jitter is not introduced as an independent noise source. Instead, it emerges deterministically from a smooth stochastic model of phase warping. All variability originates from curvature of the time warp $t(\tau)$, induced by a Gaussian process prior on the log-period

$$g(\tau) = \log T(\tau). \tag{318}$$

Sampling the process at integer phases $\tau = k$ yields a stationary Gaussian sequence

$$g_k := g(k), \tag{319}$$

with mean μ, marginal variance σ^2, and lag-one correlation

$$\rho(\ell) := \frac{k_{\ell(1)}}{k_{\ell(0)}}, \tag{320}$$

which depends only on the kernel family and the lengthscale ℓ measured in cycles.

For the kernels considered here, the lag-one correlation is given by[36]

The observed periods are lognormal variables,

$$T_k = \exp(g_k), \tag{321}$$

and the pair (T_k, T_{k+1}) is bivariate lognormal. Because g_k and g_{k+1} have equal variance, the sum and difference variables

$$S = \frac{g_{k+1} + g_k}{2}, \quad D = g_{k+1} - g_k \tag{322}$$

are independent Gaussians, with

$$S \sim \text{Normal}\left(\mu, \sigma^2 \frac{1+\rho}{2}\right), \quad D \sim \text{Normal}(0, 2\sigma^2(1-\rho)). \tag{323}$$

Using the identity

[36] Matérn 1/2: $\rho(\ell) = \exp\left(-\frac{1}{\ell}\right)$. Matérn 3/2: $\rho(\ell) = \left(1 + \frac{\sqrt{3}}{\ell}\right) \exp\left(-\frac{\sqrt{3}}{\ell}\right)$. Matérn 5/2: $\rho(\ell) = \left(1 + \frac{\sqrt{5}}{\ell} + \frac{5}{3\ell^2}\right) \exp\left(-\frac{\sqrt{5}}{\ell}\right)$. Squared exponential: $\rho(\ell) = \exp\left(-\frac{1}{2\ell^2}\right)$.

$$|\exp(a) - \exp(b)| = 2\exp\left(\frac{a+b}{2}\right)\ |\sinh\left(\frac{a-b}{2}\right)|, \tag{324}$$

the expected absolute period difference factorizes as

$$\langle |T_{k+1} - T_k| \rangle = 2\langle \exp(S) \rangle \left\langle |\sinh\left(\frac{D}{2}\right)| \right\rangle. \tag{325}$$

The first factor evaluates to

$$\langle \exp(S) \rangle = \exp\left(\mu + \sigma^2 \frac{1+\rho}{4}\right), \tag{326}$$

while the second admits a closed-form expression in terms of the Gaussian error function. After normalization by $\langle T_k \rangle = \exp\left(\mu + \frac{\sigma^2}{2}\right)$, all dependence on μ cancels, yielding the exact prediction

$$\langle \text{jitter} \mid \ell \rangle = 4\Phi\left(\sigma\sqrt{\frac{1-\rho(\ell)}{2}}\right) - 2, \tag{327}$$

where Φ denotes the standard normal cumulative distribution function.

This expression constitutes a quantitative theory of jitter. Given the marginal log-period variability σ and the kernel-dependent correlation $\rho(\ell)$, it predicts the expected period-to-period jitter without further assumptions.

For small arguments, the Gaussian CDF admits the expansion

$$\Phi(x) \approx \frac{1}{2} + \frac{x}{\sqrt{2\pi}}, \tag{328}$$

which yields the asymptotic scaling law

$$\langle \text{jitter} \mid \ell \rangle \approx \left(2\frac{\sigma}{\sqrt{\pi}}\right)\sqrt{1-\rho(\ell)}, \tag{329}$$

recovering the intuitive result that jitter vanishes as $\rho(\ell) \to 1$, i.e. as the phase warp becomes locally affine.

To make the prediction concrete, we fix a representative adult pitch distribution with median period $T_0 \approx 6.7$ ms (150 Hz) and log-period standard deviation $\sigma = 0.25$. The table below reports the predicted mean jitter for several kernel families and lengthscales.

Normal adult jitter values around 0.5% are only attained for lengthscales of several tens of cycles under smooth kernels. Rougher kernels require substantially larger ℓ to suppress cycle-to-cycle variation. In this sense, ℓ admits a direct physiological interpretation as a cycle-to-cycle smoothness parameter governing the temporal regularity of voicing.

ℓ (cycles)	$\nu = 1/2$	$\nu = 3/2$	$\nu = 5/2$	$\nu = \infty$
5	12.0%	6.2%	5.0%	4.0%
10	8.7%	3.3%	2.6%	2.0%
20	6.2%	1.7%	1.3%	1.0%
40	4.4%	0.85%	0.64%	0.50%
80	3.1%	0.43%	0.32%	0.25%

Table 16 | **Predicted mean period-to-period jitter** as a function of lengthscale ℓ for Matérn kernels with $\nu \in \left\{\frac{1}{2}, \frac{3}{2}, \frac{5}{2}\right\}$ and the squared exponential kernel ($\nu = \infty$). Predictions use $\sigma = 0.25$ and are obtained from the exact closed-form expression

$$\langle \text{jitter} \mid \ell \rangle = 4\Phi\left(\sigma\sqrt{\frac{1-\rho(\ell)}{2}}\right) - 2. \tag{330}$$

This framework yields falsifiable predictions linking observed jitter statistics to latent smoothness of the underlying pitch trajectory, which can subsequently be tested by fitting the model to data.

Appendix G. Gauge invariance in glottal inverse filtering evaluation

Every evaluation metric carries an implicit assumption about what constitutes a meaningful difference between two signals. Spectral metrics make this assumption explicit: envelope-based measures deliberately ignore phase, ignore absolute timing, and ignore excitation fine structure. The question is whether source-domain evaluation — comparing an estimated DGF waveform against a ground-truth one — should be held to the same standard, or whether it is legitimate to compare raw waveforms sample by sample.

This chapter argues that it is not legitimate, and that doing so systematically penalizes algorithms for choices that are physically unobservable — choices about what physicists call the *gauge*. The argument is not a matter of taste. It follows directly from the structure of the source-filter model and from the physics of acoustic propagation. The practical consequence is a specific evaluation procedure, aligned at frame level with an affine correction, which we use throughout the experimental chapters.

G.1. The source-filter gauge group

The speech production model is convolutional:

$$y(t) = (u'(t) * h(t)) + \varepsilon(t), \tag{331}$$

where $u'(t)$ is the glottal flow derivative, $h(t)$ is the vocal tract impulse response including radiation, and $\varepsilon(t)$ is noise. The inverse filtering problem asks: given $y(t)$, recover $u'(t)$ and $h(t)$.

The trouble is that the factorization is not unique. Two transformations leave $y(t)$ invariant.

The first is *scale*: for any nonzero constant α, the pair $\left(\alpha u'(t), \frac{h(t)}{\alpha}\right)$ produces exactly the same observed signal as $(u'(t), h(t))$. Absolute source amplitude is therefore not recoverable from acoustics alone; it is determined by convention.

The second is *time-translation*: for any constant delay τ, the pair $(u'(t-\tau), h(t+\tau))$ again produces the same convolution. The two transformations together form a gauge group acting on (u', h): the observable signal depends only on the equivalence class

$$[(u', h)] = \{(\alpha\, u'(t-\tau),; h(t+\tau)/\alpha) \mid \alpha \in \mathbb{R} \setminus \{0\}, \tau \in \mathbb{R}\}. \tag{332}$$

Inverse filtering does not recover a unique (u', h) pair. It selects one representative from this class.

The implication for evaluation is immediate: different algorithms can legitimately choose different representatives while producing identical acoustic reconstructions. An evaluation that compares representatives directly — rather than comparing the equivalence classes they represent — is measuring gauge choice, not signal quality. This is not a hypothetical concern. The closed-phase LP-based algorithms (CPCA, SLP, WLP) anchor the filter by minimizing LP residual energy in the closed phase, which drives the gauge toward a particular delay convention. Complex cepstrum decomposition (CCD) (Drugman et al., 2011) does something entirely different: it separates the maximum-phase anticausal component of the cepstrum, which selects a different gauge altogether. The two families are not wrong in different ways; they are right in different gauges. Evaluating them against the same raw waveform reference therefore introduces a systematic bias.

G.2. Physical dispersion reinforces the ambiguity

The gauge ambiguity from the source-filter model is compounded by a physical one. The vocal tract is a resonant system with a frequency-dependent phase response. Its group delay

$$\tau_{g(\omega)} = -\frac{\mathrm{d}}{\mathrm{d}\omega} \arg H\left(e^{j\omega}\right) \tag{333}$$

is not constant across frequency: narrow formants store and release energy over extended time windows, smearing the temporal relationship between the excitation event and the observed pressure waveform. This smearing varies across vowels — close rounded vowels like /o/ and /u/ with their narrow first-formant bandwidths are empirically the most challenging cases for GIF (Chien et al., 2017) — and there is no single "correct" delay that compensates it.

The standard evaluation benchmark of Chien et al. (2017) accounts for this with a fixed 0.65 ms delay applied uniformly to all ground-truth waveforms, motivated by an assumed 22 cm radiation distance. This is a reasonable first-order correction, but it is a single number applied to a phenomenon that varies with vowel, pitch, voice quality, and fundamental frequency. More importantly, it corrects for the mean delay of one physical effect while leaving the gauge ambiguity from the source-filter decomposition entirely unaddressed. Applying a fixed shift does not constitute gauge-fixing; it shifts the entire evaluation into a particular gauge that may match the CPCA convention reasonably well but will systematically misalign with CCD estimates.

G.3. Per-cycle alignment is the wrong granularity

The same benchmark evaluates performance in a pitch-synchronous, per-cycle fashion: ground-truth and estimated waveforms are aligned within each glottal cycle separately, using glottal closure instants extracted from the synthesizer. This procedure is motivated by the desire to isolate the pulse shape from pitch-period-to-pitch-period variability.

The problem is that it introduces a reference frame — the glottal cycle — that is not implied by the generative model. The source-filter interaction in (331) occurs at frame scale: the filter $h(t)$ is estimated from a windowed segment spanning many cycles, and the gauge it selects is therefore a frame-level property, not a cycle-level one. Different LPC-based algorithms apply their weighting functions (the closed-phase window in CPCA, the Gaussian weights around glottal closure in SLP, the piecewise-linear weights in WLP) at frame scale, and the effective delay they induce drifts slowly across frames as the filter estimate adapts. Cycle-level alignment obscures this drift by correcting it separately at each cycle, which has the unintended effect of partially correcting for LPC-induced gauge variation on behalf of the algorithm.

Frame-level alignment is the granularity that matches the statistical assumptions of the model and the practical structure of the algorithms.

G.4. Affine alignment at frame level

The physically motivated evaluation procedure aligns each estimated frame to the corresponding ground-truth frame by fitting an affine map and a nonnegative integer lag jointly,

$$(a^*, b^*, \tau^*) = \underset{a,b,\tau}{\operatorname{argmin}} \| \mathbf{y}_{\text{est}[\tau:]} \cdot a + b - \mathbf{y}_{\text{true}[:n-\tau]} \|^2, \tag{334}$$

where $\tau \in \{0, 1, ..., \tau_{\text{max}}\}$ is a nonnegative integer lag measured in samples.[37] The scalar a absorbs scale and polarity; the scalar b absorbs any mean offset introduced by pre-emphasis or measurement convention. The lag τ handles the frame-level delay that varies across algorithms and physical conditions.

The aligned estimate is

$$\tilde{\mathbf{y}}_{\text{est}} = a^* \mathbf{y}_{\text{est}[\tau^*:]} + b^*, \tag{335}$$

defined on the overlap $[: n - \tau^*]$. Performance is then measured by centered cosine similarity between $\tilde{\mathbf{y}}_{\text{est}}$ and $\mathbf{y}_{\text{true}[:n-\tau^*]}$:

$$\text{score} = \frac{\langle \tilde{\mathbf{y}}_{\text{est}} - \overline{\tilde{\mathbf{y}}_{\text{est}}}, \mathbf{y}_{\text{true}} - \overline{\mathbf{y}}_{\text{true}} \rangle}{\| \tilde{\mathbf{y}}_{\text{est}} - \overline{\tilde{\mathbf{y}}_{\text{est}}} \| \| \mathbf{y}_{\text{true}} - \overline{\mathbf{y}}_{\text{true}} \|}. \tag{336}$$

Centering before taking the inner product removes the b degree of freedom redundantly, making the score invariant to the bias correction. Cosine similarity is a monotone function of RMSE after affine alignment and is bounded in $[-1, 1]$, which makes it convenient for visualization and for averaging across frames of different lengths.

The procedure is summarized in Table 17, alongside the corresponding treatment in the EGIFA benchmark (Chien et al., 2017) for comparison.

G.5. Summary

The source-filter model has a gauge group: scale and time-translation transformations that leave the observed signal invariant and therefore make the corresponding source properties unobservable from acoustics alone. Different inverse filtering algorithms choose different representatives from this group — LP-based methods anchor on closed-phase energy, CCD

[37]The lag is constrained to be nonnegative and to shift the estimate earlier in time, reflecting the physical direction of propagation delay: the acoustic observation $y(t)$ always lags the source, so a positive lag on the estimate corresponds to shifting it left to match the ground-truth timing. Allowing both directions would conflate physical delay correction with arbitrary time-warping.

Gauge source	Origin	EGIFA treatment	This work
Scale / polarity	Source-filter non-identifiability	Orthogonal projection per cycle	Affine fit per frame (334)
Absolute delay	Source-filter non-identifiability	Fixed 0.65 ms shift	Lag search per frame (334)
Dispersive group delay	Vocal tract resonances	Fixed 0.65 ms shift (partial)	Lag search per frame
Algorithmic gauge	LP vs. cepstral estimation	Not corrected	Corrected uniformly
Mean offset	Pre-emphasis / measurement	Not corrected	Affine fit (b term)

Table 17 | **Gauge sources in GIF evaluation** and their treatment in the EGIFA benchmark of Chien et al. (2017) versus this work. The EGIFA fixed delay of 0.65 ms addresses one physical effect; it does not address the gauge ambiguity from the source-filter decomposition or the algorithm-dependent gauge drift. Affine alignment at frame level handles all sources uniformly.

anchors on cepstral phase — and any evaluation that compares representatives directly rather than correcting for this choice conflates algorithmic style with algorithmic error.

The same argument applies to the fixed-delay and per-cycle alignment choices in the EGIFA benchmark (Chien et al., 2017). A fixed 0.65 ms delay corrects for one specific physical effect at one specific propagation distance; it does not address the gauge freedom of the model, and it does not adapt to the algorithm being evaluated. Per-cycle alignment introduces a reference frame that the generative model does not privilege and that partially absorbs algorithm-induced gauge drift.

Frame-level affine alignment with a lag search, followed by centered cosine similarity, addresses all of these uniformly. It is the evaluation procedure used throughout the experimental chapters of this thesis.

Appendix H. Related contributions

Van Soom & de Boer (2019) Measuring formant frequency $\mathbf{F}$ and bandwidth $\mathbf{B}$ from steady-state vowel waveforms, taking into account the low-frequency effects of the DGF $u'(t)$ during the open phase of the pitch period. Inference is implemented using a maximum-a-posteriori (MAP) approximation. In BNGIF these low-frequency effects are taken into account automatically.

Van Soom & de Boer (2020) Same as the above, but now using robust nested sampling inference instead of a MAP approximation. A new heuristic derivation of the model function from source-filter theory is added. In BNGIF variational inference is the preferred inference method.

Van Soom & De Boer (2022) Theoretical derivation of a new weakly informative prior for resonance frequencies. Shown to have superior performance in predicting steady-state vowel waveforms compared to the usual priors in inferring the pole frequency $\mathbf{x}$ and bandwidth $\mathbf{y}$, while being roughly equally (un)informative.